Existence and Mass Gap in Planar $SU(N)$ Yang–Mills Theory on $S^3 \times \mathbb{R}$ and $\mathbb{R}^4$

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ABSTACT: The large-N program initiated by 't Hooft has suggested for fifty years that planar Yang–Mills should admit a closed-string description. The central question is whether this expectation can be turned into a nonperturbative construction sufficiently precise to establish existence and a mass gap. In this work we give such a construction for the strict planar theory.

Table of Contents

1. Warped Dynamical Closed B-Model and Its Linearized Hilbert Space 8
1.1. Geometric data and field space 8
1.2. Closed-field curvature 9
1.3. Degree-dependent Riesz structure 9
1.4. Dynamical closed-string-field action 10
1.5. Vacuum and linearized theory 10
1.6. Gauss constraint 11
1.7. The independent ∂-closed condition 12
1.8. Linearized equation of motion 13
1.9. Degree-dependent adjoints 13
1.10. Variational definition of the physical operator 14
1.11. Physical one-particle Hilbert space 15
1.12. Quantization 16
2. SO(4)-Singlet Scalar Sector: Primitive and Lefschetz Blocks 16
2.1. Geometric setup 16
2.2 Radial weight 17
2.3. $SO(4)$-invariant 2, 1-forms 18
3. Primitive scalar block 19
3.1 Exact solution of the ∂-constraint 19
3.2 Dolbeault curvature 20
3.3 Boundary conditions 21
3.4 Exact zero mode 21
3.5 Primitive spectrum 22
4. Lefschetz scalar block 22
4.1 Ansatz 22
4.2 Reduced quadratic form 22
4.3 Strong radial equation 23
4.4 Endpoint behaviour 24
4.5 Lefschetz spectrum 25
5. Comparison of the two scalar singlet blocks 25

6. SO(4) Tensor Block and Tensor Eigenvalues 26
6.1 Tensor representation 26
6.2 Stenzel geometry in the tensor coordinates 27
6.3 Reduced Dolbeault operators 28
6.4 The ∂-constraint 29
6.5 Regularity at the Stenzel cap 30
6.6 One-sided tensor curvature 30
6.7 Warped tensor norm 31
6.8 Raw tensor eigenvalue problem 32
6.9 Conversion to the common scalar normalization 32
6.10 Tensor Rayleigh quotient in the common convention 34
6.11 Numerical discretization 34
6.12 Resolution convergence 35
6.13 First tensor eigenvalues 36
6.14 Comparison with the primitive scalar 36
6.15 Comparison with the Lefschetz scalar 36
6.16 Comparison of the low-lying spectra 37
7 SO(4) to ISO(3) Reduction of the Tensor Sector 37
7.1 Compact and flat tensor limits 37
7.2 The tensor contraction must not deform the Dolbeault complex 38
7.3 Tensor channels under the contraction 39
7.4 Chain-map expansion 39
7.6 Degree-wise Riesz transport 40
7.7 1PI interpretation 41
7.8. Expansion around the flat endpoint 42
7.9 Direct angular contribution 42
7.10 Green-function or 1PI correction 42
7.11 Riesz and horizontal correction 43
7.12 Resolution dependence of the flat endpoint 44
7.13 Flat-space tensor-to-scalar ratio 45
7.14 Why a moving chain map alone is insufficient 45
7.15 Physical interpretation of the reduction 46

7.16 The Riesz transport is necessarily nonlocal 46
7.17 Effective 1PI tensor operator 47
7.18 Final contracted tensor problem 47
8. Yang–Mills Theory on $S^3 \times \mathbb{R}$ 49
8.1. Yang–Mills theory on $S^3 \times \mathbb{R}$ 49
8.2. The gauge group and the orbit space of connections 51
8.3 The Hilbert–Schmidt metric on the space of connections 52
8.4 First variation: the transverse gauge slice 54
8.5 Second variation and the Faddeev–Popov operator 55
8.6 The fundamental modular region 56
8.7 Horizontal and vertical directions 57
8.8 The FMR and the Hamiltonian formulation 58
8.9 The 't Hooft normalization 59
8.10 Why the FMR is the appropriate nonperturbative starting point 60
9 Large-N Color Space as a Fuzzy Sphere and Its Hilbert-Space Limit 62
9.1. Purpose of the construction 62
9.2 The irreducible $SU(2)$ embedding 64
9.3 Fuzzy coordinates 65
9.4 Exact decomposition of the matrix algebra 66
9.5 Fuzzy spherical harmonics 67
9.6 The normalized Hilbert–Schmidt norm in harmonic variables 68
9.7 Ordinary spherical harmonics and the symbol map 69
9.8 Exact finite-N star product 70
9.9 Gauge connections in fuzzy harmonic variables 71
9.10 The Hilbert–Schmidt metric becomes an $S2$ integral 72
9.11 The large-N Hilbert-space limit 73
9.12 No-mode-loss statement 74
9.13 Large-N Yang–Mills connection space 75
9.14 Comparing smooth gauge algebra and Hilbert completion 76
10. Geometrization of the Planar Yang–Mills RG Data and Reconstruction of the Deformed Conifold 78
10.1. Purpose and logical status 78

10.2 The fixed-scale planar carrier 80
10.3 From the color sphere to cotangent directions 80
10.4 The RG scale supplies the cotangent radius 81
10.5 Explicit real realization of T^*S^3 82
10.6 Yang–Mills RG data as period data 83
10.7 The universal running coupling as a period matrix 84
10.8 Reconstruction of the dual period 85
10.9 From the RG monodromy to the local complex threefold 86
10.10 The holomorphic volume form 86
10.11 The RG data determine a complex structure on the source space 87
10.12 The Yang–Mills scalar action as the RG-conjugate source 87
10.13 Infinitesimal scalar sources and $(2,1)$-forms 88
10.14 From complex geometry to the Ricci-flat Kähler metric 88
10.15 Explicit Stenzel form 89
10.16. The RG scale as the conifold radial coordinate 90
10.17 The integrated RG source 91
10.18. The radial Stenzel Dirichlet form 91
10.19 Solution of the radial Poisson problem 92
10.20 Gauss-law interpretation of the RG source 93
10.21 Interpretation in terms of the FMR metric 94
10.22 Reconstruction theorem 94
10.23 Consequence for the subsequent master-field construction 96
11 Planar Single-Trace Observables and the Collective One-Particle Spectrum 97
11.1 Purpose 97
11.2 The physical planar field space and its geometric representation 97
11.3 Derivation of the full large-N master-field action 98
11.4 The physical mass operator 104
11.5 Why a collective mode defines a single-trace Yang–Mills observable 105
11.6 Channel decomposition of the single-trace observables 105
11.7 Expansion of the planar master-field fluctuations 106
11.8 Quantization and the one-particle states 106
11.9 Identification with planar glueball states 107

11.10 Operator overlaps and the pole expansion 107
11.11 Spectral meaning of the collective geometry 108
11.12 Consequence for the scalar sector 108
12 The Planar FMR Partition Function and the Gaussian Master-Field Measure 109
12.1 Strict planar master field and concentration 109
12.2 Exact strict-planar Gaussian scaling limit 110
12.3 Independent derivation from large-N factorization 111
12.4 Transport to the RG-completed geometric variables 112
12.5 The Euclidean covariance operator 113
12.6 Spectral representation of the covariance 113
12.7 Rigorous construction of the infinite-dimensional Gaussian measure 114
12.8 Identification of the planar FMR fluctuation measure 115
12.9 The planar generating functional 115
12.10 Relation to the original Yang–Mills partition function 116
12.11 Relation to the glueball spectrum 117
12.12 Summary proposition 117
13 Complex Polarization, Scalar Source Normalization, and the Ultraviolet Two-Point Functions 119
13.1 Purpose 119
13.2 The real scalar source-response phase space 119
13.3 Positive scalar kinetic operator 120
13.4 Complex structure from the polar decomposition 120
13.5 Relation to the RG complex structure 121
13.6 The induced Riesz metric 121
13.7 Source normalization between the master field and Yang–Mills 122
13.8 Transformation of the pole overlaps 123
13.9 Raw ultraviolet residue laws 124
13.10 Universal n^5 Yang–Mills residues 124
13.11 Continuous spectral density 125
13.12 Ultraviolet two-point function 126
13.13 The same result directly from the discrete tower 127
13.14 Primitive and Lefschetz correlators 127

13.15 Ultraviolet RG expansion of the scalar source 128
13.16 Numerical Verification of the Scalar Ultraviolet Residue Exponents 131
13.17 Numerical extraction of pole residues 132
13.18 Primitive scalar source 132
13.19 Primitive ultraviolet cutoff convergence 133
13.20 Lefschetz scalar source 134
13.21 Lefschetz resolution convergence 134
13.22 Asymptotically affine mass spectrum 135
13.23 Comparison with the independent analytic prediction 136
14 Mass Gap Theorem on $S^3 \times \mathbb{R}$ 136
14.1 Cohomology of the constrained $(2, \bullet)$ complex 136
14.2 Fixed Yang–Mills scale removes the unique cohomological zero mode 141
14.3 Triviality of the physical kernel 141
14.4 Coercive estimate and positive mass gap 142
14.5 Mass-gap theorem 148
15 $\mathbb{R}^4$ Mass Gap Theorem and Proof 149
15.1 Exact Cochain Construction in the Decompactification Limit 149
15.2 Flat-limit collective embedding 150
15.3 Removal of the cohomological obstruction 150
15.4. Exact recursive construction 151
15.5. Exact cochain theorem 151
15.6. Consequences 153
15.7. Bounded Homotopy and Graph-Norm Control 154
15.8. Fixed-period constrained complex 154
15.9. Graph norms 155
15.10. Closed range and the reduced Green operator 155
15.11. Boundedness of the homotopy 156
15.12. Application to the physical degree-one field 157
15.13. Graph-norm convergence of the exact cochain 157
15.14. Stability of the embedded physical subspace 158
15.15. Proposition 159
15.16. What has been established 160

15.17. Quadratic master-field pencil....160
15.18. Visible and complementary closed-string directions....161
15.19Uniform invertibility of the complement....162
15.20. Exact 1PI-Feshbach reduction....163
15.21. Effective degree-one Riesz metric....164
15.22. Degree-two Riesz transport....164
15.23. Expansion at the flat endpoint....165
15.24. Uniform $O(\varepsilon)^2$ control....166
15.25. Convergence of the inverse Riesz operator....167
15.26. Proposition....167
15.27. Interpretation....168
15.28 Full-complex 1PI Riesz regularity....168
15.29 Fixed endpoint splitting....169
15.30 Effective 1PI pencil....170
15.31 Uniform complementary resolvent bound....171
15.32. Stability for small ε....171
15.33. Spectral pencil....172
15.34. Endpoint adjoint graph norm....173
15.35. Green regularity....173
15.36. Structure of the Riesz perturbation....173
15.37. Convergence of the adjoint....175
15.38. Analytic closure proposition....175
15.39 Moving Gauss slice....176
15.40 Kinetic master-field operator....176
15.41 Quadratic-form formulation....177
15.42 Uniform closedness....177
15.43 Resolvent convergence....178
15.44 Euclidean master-field covariance....179
15.45 Proposition....179
15.46 Consequence for the mass-gap problem....180
15.47 $\mathbb{R}^4$ Mass-Gap Theorem....180
15.48 Endpoint coercivity....180

15.49 Finite-radius form estimates 181
15.50 Stability of the bottom of the spectrum 182
15.51 Decompactified master-field covariance 183
15.52 Restoration of the Yang–Mills scale 184
15.53 Large-R asymptotics 184
15.54 Theorem 185
15.55 Relation to the scalar singlet and tensor sectors 186
16 Existence Theorem 200
16.1 Reflection positivity of the Gaussian planar master-field process 200
16.2. Euclidean covariance kernel 200
16.3. Time reflection 201
16.4. Factorization of the reflected covariance 201
16.5. Reflection positivity of the full Gaussian measure 202
16.6 Spectral interpretation 203
16.7. Relation to the previously constructed Hilbert space 203
16.8. Reflection-positivity theorem 204
16.9 Equivariance of the collective map 205
16.10 The $SO(4)$ action on the Yang–Mills physical space 205
16.11 The $SO(4)$ action on the collective space 206
16.12 Conjugating the collective map 207
16.13 Differential intertwining 207
16.14 Source and period normalization 208
16.15 Use uniqueness 208
16.16 Proposition 209
16.17 Decompactification gives translations 209
16.18 Momentum decomposition of the decompactified master-field Hilbert space 210
16.19 Strongly continuous translation representation 211
16.20 Momentum generators 211
16.21 Why the measure is ordinary momentum measure 212
16.22 Physical constraints decompose fiberwise 212
16.23 Decomposition of the master-field operator 213
16.24 Compatibility with the Yang–Mills realization 214

16.25 Relativistic dispersion of the planar master-field modes 215
16.26 Decompactified Yang–Mills Hessian 215
16.27 Momentum-space form of the quadratic kernel 216
16.28 Gaussian master-field normalization 216
16.29 Fiber Hamiltonian 217
16.30 Covariance 218
16.31 Why this is not an additional assumption 218
16.32 Regularity and Temperedness of the Planar Schwinger Functions 219
16.33 The operator-valued Euclidean covariance 220
16.34 Continuity on Schwartz test functions 220
16.35 Position-space covariance 221
16.36 Gaussian Schwinger functions 222
16.37 Symmetry 222
16.38 Euclidean covariance 223
16.39 Positivity and continuity of the Gaussian characteristic functional 223
16.40 Connected Schwinger functions 224
16.41 Clustering and Uniqueness of the Vacuum 224
16.42 Spectral representation and the mass threshold 225
16.43 Exponential decay in Euclidean time 225
16.44 Full Euclidean separation 226
16.45 Clustering of the Gaussian field 227
16.46 Clustering of polynomial observables 227
16.47 Connected Schwinger functions 228
16.48 Cluster property and the vacuum sector 228
16.49 Direct spectral proof in the Gaussian Fock realization 229
16.50 Spectral gap above the vacuum 230
16.51 Osterwalder–Schrader Reconstruction 231
16.52 Positive-time functionals and the OS inner product 231
16.53 The one-particle OS space 232
16.54 Reconstruction of Euclidean time evolution 233
16.55 Spatial translations and rotations 234
16.56 Gaussian reconstruction and Fock space 234

16.57 Reconstructed Hamiltonian 234
16.58 Reconstruction of the field 235
16.59 Compatibility with the planar FMR realization 236
16.60 Existence Theorem for Planar Yang–Mills on $\mathbb{R}^4$ 237
17 String theory perspective 238
17.1 The Extended Open String-Field Theory 238
17.2 The extended string field 239
17.3 Gauge symmetry 240
17.4 Riesz structure on string-field space 240
17.5 The extended second-order action 241
17.6 Linearized physical operator 242
17.7 Topological A-model specialization 242
17.8 Yang–Mills as the extended A-model string-field theory 243
17.9 What is—and is not—being claimed 244
17.10 The Large-N Trace Sector and the Open–Closed String-Field Algebra 245
17.11 Cyclic words from open-string Chan–Paton factors 246
17.12 Single traces and multi-traces 246
17.13 Open–closed topological field theory 247
17.14 Why the present construction is stronger than the abstract open–closed statement 247
17.15 From the fibre algebra to the deformed conifold 248
17.16 The closed B-model field algebra 249
17.17 The master-field curvature 249
17.18 The constrained closed complex 250
17.19 Time-dependent open–closed map 250
17.20 Transport of the algebraic operations 251
17.21 Transport of the Riesz structure 251
17.22 The closed master-field action 252
17.23 Single strings, glueballs and traces 252
17.24 Relation to BCOV theory 253

Introduction

Yang–Mills theory occupies a peculiar position in modern mathematical physics. Its classical equations are simple to state, its perturbative quantum theory is extraordinarily successful, and the theory forms the non-abelian gauge-theoretic core of the Standard Model. Yet the mathematical construction of the four-dimensional quantum theory at strong coupling remains one of the central unresolved problems in quantum field theory [1].

The basic difficulty can be summarized in two questions. First, does four-dimensional pure Yang–Mills theory exist nonperturbatively as a mathematically well-defined quantum field theory? Second, does its physical spectrum possess a strictly positive mass gap,

$$M_{\mathrm{gap}} > 0,$$

despite the absence of an explicit mass parameter in the classical Lagrangian? These questions are closely related but logically distinct. The first concerns the construction of the quantum theory itself: its measure, Hilbert space, observables, positivity structure, and dynamics. The second concerns the infrared spectrum of that theory. A mass gap means that the vacuum is isolated and that every nontrivial physical excitation has strictly positive energy above it.

For four-dimensional pure Yang–Mills theory, this infrared scale is expected to be generated dynamically. Classically, the theory contains only a dimensionless gauge coupling. Quantum mechanically, renormalization replaces the bare coupling by a running coupling and generates the dimensional transmutation scale

$$\Lambda_{\mathrm{YM}}.$$

The physical expectation is therefore

$$M_{\mathrm{gap}} = c_{\mathrm{gap}}\Lambda_{\mathrm{YM}}, \qquad c_{\mathrm{gap}} > 0.$$

Explaining mathematically how a dimensionless microscopic interaction gives rise to a positive macroscopic mass scale is one of the deepest problems in nonperturbative quantum field theory.

Historical background

Yang and Mills [2] introduced non-abelian gauge theory in 1954. In the following decades, non-abelian gauge fields became the organizing principle of elementary-particle physics, but their strongly coupled dynamics remained poorly understood. A decisive development [3,4] came with the discovery of asymptotic freedom in the early 1970s. Pure non-abelian gauge theories become weakly coupled at short distances,

$$g_{\mathrm{YM}}(\mu) \longrightarrow 0 \qquad (\mu \longrightarrow \infty),$$

while the coupling grows toward the infrared. Perturbation theory therefore gives precise control in the ultraviolet but necessarily loses control at the scale where confinement and the mass gap are expected to emerge.

Wilson's lattice formulation [5] gave a nonperturbative regularization of gauge theory and provided the first compelling theoretical framework for confinement. Numerical lattice calculations [6] have since produced detailed evidence for a discrete glueball spectrum and a nonzero Yang–Mills mass scale. Nevertheless, numerical evidence does not by itself supply a continuum existence theorem or an analytic derivation of the mass gap. A second major development [7] was 't Hooft's large-N expansion. Writing

$$\lambda = g_{\mathrm{YM}}^2 N$$

and taking

$$N \to \infty$$

with λ fixed reorganizes the gauge-theory expansion according to the topology of two-dimensional surfaces. In double-line notation, a vacuum diagram of genus g is weighted schematically by

$$N^{2-2g}.$$

This observation suggested that large-N gauge theory should admit a string-theoretic description, with

$$g_s \sim \frac{1}{N}$$

playing the role of the closed-string coupling. This idea is one of the oldest and most persistent clues to the nonperturbative structure of Yang–Mills theory. Single-trace gauge-invariant operators [7] behave as natural single-string observables, while products of traces have the combinatorics of multi-string states. In the confining phase, the physical color-singlet excitations are glueballs, and at strict large N their interactions are suppressed. The expected planar spectrum therefore resembles the spectrum of a weakly coupled closed string theory.

What has remained difficult is not the qualitative string interpretation but its concrete realization. For four-dimensional non-supersymmetric Yang–Mills theory one would need a closed-string description that simultaneously incorporates asymptotic freedom, dimension transmutation, confinement, a discrete glueball spectrum and, ultimately,

$$M_{\mathrm{gap}} > 0.$$

Standard perturbative string backgrounds do not provide this structure automatically.

The large-N problem addressed in this work

The aim of the present work is to construct the strict planar theory directly. More precisely, we consider pure $SU(N)$ Yang–Mills theory and study the limit

$$N \to \infty.$$

The problem addressed here is therefore the following:

construct the gauge-invariant planar Yang–Mills theory as a well-defined quantum theory, identify its physical Hilbert space and Euclidean measure, and prove that the reconstructed Hamiltonian has a strictly positive mass gap.

The finite-N Yang–Mills existence and mass-gap problem is not claimed here. The result concerns the strict planar theory. Finite-N corrections are naturally organized by the closed-string coupling

$$g_s = \frac{1}{N}$$

and lie beyond the scope of the present work. The strategy is to treat the large-N master field not merely as an abstract limiting noncommutative random variable, but as a concrete closed string field. This requires combining several structures which are usually discussed separately. The resulting closed-string geometry is the deformed conifold

$$X = T^* S^3.$$

The Yang–Mills master field becomes a time-dependent closed B-model field

$$\mathcal{G}(t) \in \ker \partial \cap \Omega^{2,1}(X),$$

or equivalently a Kodaira–Spencer-type [8] polyvector field

$$\mu(t) \in PV^{1,1}(X).$$

Its nonlinear curvature is

$$\mathcal{F}(\mathcal{G}) = \bar{\partial}\mathcal{G} + \frac{1}{2}[\mathcal{G}, \mathcal{G}].$$

The physical master-field action takes the form

$$S[\mathcal{G}] = \frac{1}{2g_s^2} \int dt \left[\|D_t \mathcal{G}\|_h^2 - \|\mathcal{F}(\mathcal{G})\|_0^2 \right].$$

The crucial point is that the Yang–Mills renormalization-group flow does not appear merely as an external running parameter. It is geometrized as a radial dependence of the physical Riesz metric $\|\cdot\|_h$ on the closed string-field space. The resulting one-particle spectral operator is not the unrestricted Dolbeault Laplacian. After gauge reduction and imposition of the physical Gauss condition, it is the one-sided operator

$$K_h = h^{-1} \bar{\partial}^\dagger \bar{\partial}$$

on the constrained physical Hilbert space

$$\mathcal{H}^{(1)}_{\text{phys}} = \{\Psi : \partial\Psi = 0,\ \bar{\partial}^{\dagger}(h\Psi) = 0\}_{L^2_h}.$$

This distinction is essential for the spectral analysis. The function h is commonly known in physics literature as warping field or warp factor. It contains information about the energy-scale dependent reorganization of the gauge theory degrees of freedom in the fundamental modular region (FMR).

Open strings, Yang–Mills, and the closed master field

The string-theoretic origin of the construction may be summarized as follows. The open topological A-model [9] on N branes wrapped on

$$S^3 \subset T^*S^3$$

has a classical open string-field algebra quasi-isomorphic to

$$\Omega^{\bullet}(S^3, \mathrm{Mat}_N).$$

Extending this algebra by the de Rham complex of physical time,

$$\hat{\mathcal{A}} = \mathcal{A} \otimes \Omega^{\bullet}(\mathbb{R}_t),$$

and equipping it with the appropriate degree-wise Riesz structure produces a second-order one-dimensional sigma model whose degree-one truncation is Yang–Mills theory on

$$S^3 \times \mathbb{R}.$$

Thus the Yang–Mills connection is interpreted as a time-dependent open A-model string field. The large-N trace sector then supplies the bridge to closed strings. Single traces define cyclic open-string chains [10],

$$CC^{-}_{\bullet}(\mathcal{A}_{\text{open}}),$$

and the open–closed formalism associates a closed deformation sector to this cyclic-Hochschild structure. At the same time, the large-N Chan–Paton algebra admits a concrete geometric realization. Together with the renormalization-group direction it becomes the three-dimensional fibre of the deformed conifold. Schematically,

$$\mathfrak{su}(N) \longrightarrow \mathfrak{sdiff}(W),$$

and

$$S^3 \times W \longrightarrow T^*S^3.$$

The planar Yang–Mills master field is thereby realized as a closed B-model field. The abstract open–closed correspondence explains why a closed sector should emerge from the planar trace

algebra; the collective map constructed here gives its physically normalized Yang–Mills realization.

The mass gap

The central analytic result is a coercive estimate for the physical closed-string complex. After fixing the unique cohomological A-period direction associated with the Yang–Mills scale, one proves

$$\|\Psi\|_h^2 \le C\left\|\bar{\partial}\Psi\right\|_0^{2\cdot}$$

Equivalently,

$$\boxed{\left\|\bar{\partial}\Psi\right\|_0^2 \ge \Delta^2\|\Psi\|_h^2, \qquad \Delta > 0.}$$

Hence

$$\boxed{K_h \ge \Delta^2 I.}$$

Therefore

$$0 \notin \operatorname{spec} K_h$$

on the physical fixed-period Hilbert space. This establishes a strictly positive planar mass scale. The important issue is then whether this positivity survives the decompactification limit

$$S_R^3 \longrightarrow \mathbb{R}^3, \qquad R \to \infty.$$

A substantial part of the analysis is devoted to this question. The finite-radius open and closed complexes are related by an exact cochain map C_R, and the deformation away from the flat limit is controlled in graph norm. A bounded contracting homotopy, complementary resolvent estimates, and the 1PI-Feshbach reduction are used to prove convergence of the physical Riesz structures and adjoints. The mass-gap estimate is therefore stable under decompactification. Restoring dimensions,

$$\boxed{M_{\mathrm{gap}} = c_{\mathrm{gap}}\Lambda_{\mathrm{YM}}, \qquad c_{\mathrm{gap}} > 0.}$$

The constant c_{gap} is determined by the physical spectral problem; the coercive argument establishes its strict positivity rather than its numerical value.

Existence of the planar quantum theory

A positive spectral operator alone does not constitute a quantum field theory. The second part of the existence problem is therefore to construct the Euclidean theory and reconstruct its physical Hilbert space. At strict $N = \infty$, centered canonically normalized single-trace fluctuations become Gaussian. The Euclidean master-field covariance is

$$\boxed{\mathfrak{C}_E = (-\Delta_{\mathbb{R}^4} + M^2)^{-1},}$$

where

$$M^2 \geq M_{\text{gap}}^2 I.$$

The corresponding Gaussian measure is constructed on a nuclear test-function space by the Bochner–Minlos theorem. The resulting Schwinger functions are shown to satisfy:

Euclidean invariance, temperedness, permutation symmetry, reflection positivity, clustering.

Reflection positivity follows directly from

$$\mathfrak{C}_E(\tau, \sigma) = \frac{1}{2\omega} e^{-|\tau-\sigma|\omega}, \qquad \omega = \sqrt{\mathbf{P}^2 + M^2},$$

since for positive-time test functions,

$$\langle \Theta f, \mathfrak{C}_E f \rangle = \frac{1}{2} \left\| \int_0^\infty \omega^{-1/2} e^{-\tau\omega} f(\tau)\, d\tau \right\|^2 \geq 0.$$

The Osterwalder–Schrader reconstruction [11] therefore produces a positive-energy Hilbert space. Because the strict planar theory is Gaussian, this Hilbert space is

$$\boxed{\mathcal{H} = \Gamma_s\left(\mathcal{H}_{\text{phys}}^{(1)}\right).}$$

The one-particle Hamiltonian is

$$\boxed{H^{(1)} = \sqrt{\mathbf{P}^2 + M^2},}$$

and the full Hamiltonian is

$$\boxed{H = d\Gamma\left(\sqrt{\mathbf{P}^2 + M^2}\right).}$$

The vacuum is unique,

$$\ker H = \mathbb{C}\Omega,$$

and

$$\boxed{\operatorname{spec} H \subset \{0\} \cup [M_{\text{gap}}, \infty).}$$

Thus the mass gap appears simultaneously as the lower bound of the rest-frame master-field operator, the exponential clustering scale of the Euclidean theory, and the spectral gap above the reconstructed vacuum.

Relation to the fifty-year large-N program

The conceptual picture that emerges is close to the one suggested by the original large-N expansion, but with the string theory made explicit. The relation

$$g_s = \frac{1}{N}$$

is not merely a diagrammatic analogy. At $N = \infty$, the master field is classical on the closed-string side. The $1/N$ expansion then becomes the closed-string loop expansion,

$$\boxed{\frac{1}{N^2} \sim g_s^{2.}}$$

The strict planar theory constructed here is therefore the genus-zero quantum-field-theoretic background about which finite-N corrections may in principle be organized. Whether this expansion can be controlled uniformly down to small N is a separate problem. The present work makes no claim to solve the finite-N Yang–Mills existence and mass-gap problem. Its result is instead the construction of the $N = \infty$ theory which has historically been expected to underlie such an expansion.

Main result

The results of this work may be summarized in the following statement.

Within the planar FMR-master-field realization constructed in this work, pure $SU(N)$ Yang–Mills theory admits a strict $N = \infty$ limit on $\mathbb{R}^4$ whose gauge-invariant trace sector is represented by a constrained, warped, dynamical closed B-model field theory on the deformed conifold. The associated Euclidean master-field measure is reflection positive, Euclidean invariant, tempered and clustering, and hence admits Osterwalder–Schrader reconstruction. The reconstructed theory has a unique vacuum and a positive mass gap

$$M_{\mathrm{gap}} = c_{\mathrm{gap}} \Lambda_{\mathrm{YM}}, \qquad c_{\mathrm{gap}} > 0.$$

From the historical point of view, the result supplies a concrete realization of the string-theoretic structure anticipated by the large-N expansion: the planar trace sector becomes a closed-string field theory, while the dynamically generated Yang–Mills scale becomes geometric data of the closed-string master-field space. The remainder of the paper develops this construction in detail.

1. Warped Dynamical Closed B-Model and Its Linearized Hilbert Space

In this section we formally introduce the master field action and compute its scalar and tensor spectrum numerically. The purpose of the remainder of this work is then to derive this master field description from planar Yang-Mills, and to prove its mass gap and existence as a QFT.

1.1. Geometric data and field space

Let X be a non-compact Calabi–Yau threefold equipped with Kähler metric g_X, Kähler form J, and nowhere-vanishing holomorphic volume form

$$\Omega \in \Omega^{3,0}(X).$$

We denote by

$$\langle \alpha, \beta \rangle_0 = \int_X \alpha \wedge *_6 \overline{\beta}$$

the reference unwarped L^2-pairing induced by g_X. Let

$$h: X \longrightarrow \mathbb{R}_{>0}$$

be a smooth positive function. In the applications of interest h will be radial, but this restriction is not required for the local construction below. The dynamical degree-one field is a time-dependent Beltrami-field fluctuation

$$\mu(t) \in \Omega^{0,1}(X, T^{1,0}X).$$

Using contraction with the holomorphic volume form, we write

$$\mathcal{G}(t) = \mu(t) \lrcorner\, \Omega \in \Omega^{2,1}(X). \tag{1.1}$$

The admissible closed-field sector is restricted by

$$\partial \mathcal{G} = 0. \tag{1.2}$$

This condition is distinct from the gauge constraint derived below. It selects the closed subcomplex in which the dynamical field takes values. We also introduce a temporal gauge field

$$\mathcal{G}_0(t) \in \Omega^{2,0}(X).$$

The relevant part of the Dolbeault complex is therefore

$$\Omega^{2,0}(X) \xrightarrow{\overline{\partial}} \Omega^{2,1}(X) \xrightarrow{\overline{\partial}} \Omega^{2,2}(X).$$

It will be useful to distinguish the two adjacent maps by writing

$$Q_0: \Omega^{2,0} \to \Omega^{2,1}, \qquad Q_1: \Omega^{2,1} \to \Omega^{2,2},$$

with

$$Q_0 = Q_1 = \bar{\partial}$$

on their respective degrees. Since $\bar{\partial}^2 = 0$,

$$Q_1 Q_0 = 0. \tag{1.3}$$

This Hilbert-complex structure underlies the linearized constrained theory.

1.2. Closed-field curvature

The nonlinear integrability curvature of the closed field is

$$\mathcal{F}(\mathcal{G}) = \bar{\partial}\mathcal{G} + \frac{1}{2}[\mathcal{G}, \mathcal{G}], \tag{1.4}$$

where $[\cdot,\cdot]$ denotes the Schouten-type bracket inherited from the polyvector-field description.

Defining

$$D_{\mathcal{G}} = \bar{\partial} + [\mathcal{G}, \cdot],$$

the linear variation of the curvature takes the form

$$\delta\mathcal{F}(\mathcal{G}) = D_{\mathcal{G}}(\delta\mathcal{G}).$$

The temporal gauge field enters through the covariant time derivative

$$D_t\mathcal{G} = \dot{\mathcal{G}} - D_{\mathcal{G}}\mathcal{G}_0. \tag{1.5}$$

Thus $\mathcal{G}_0$ is nondynamical and acts as the temporal gauge variable of the theory.

1.3. Degree-dependent Riesz structure

A defining feature of the theory is that different degrees of the closed complex carry different Hilbert metrics.

The degree-one field

$$\mathcal{G} \in \Omega^{2,1}(X)$$

is equipped with the weighted inner product

$$\boxed{\langle \Psi, \Phi \rangle_h = \langle \Psi, B_h \Phi \rangle_0, \qquad B_h \Phi = h\Phi.} \tag{1.6}$$

Equivalently,

$$\|\Psi\|_h^2 = \int_X h\, |\Psi|^2 \, dV_X.$$

By contrast, the degree-two curvature

$$\mathcal{F}(\mathcal{G}) \in \Omega^{2,2}(X)$$

retains the unwarped norm

$$\|\mathcal{F}\|_0^2 = \int_X |\mathcal{F}|^2 \, dV_X.$$

Thus the multiplication operator h is not a uniform weight on the full Dolbeault complex. It modifies the kinetic metric of the degree-one field, whereas the norm of its curvature remains unwarped. This degree dependence is the essential origin of the one-sided spectral operator.

1.4. Dynamical closed-string-field action

We define the Lorentzian action

$$\boxed{S[\mathcal{G}, \mathcal{G}_0] = \frac{1}{2g_s^2} \int_{\mathbb{R}_t} dt \left[\|D_t \mathcal{G}\|_h^2 - \|\mathcal{F}(\mathcal{G})\|_0^2 \right].} \tag{1.7}$$

Explicitly,

$$S = \frac{1}{2g_s^2} \int dt \int_X \left[h \, D_t \mathcal{G} \wedge *_6 \overline{D_t \mathcal{G}} - \mathcal{F}(\mathcal{G}) \wedge *_6 \overline{\mathcal{F}(\mathcal{G})} \right].$$

The theory may therefore be regarded as a dynamical one-dimensional extension of the off-shell closed B-model field space. The internal Maurer–Cartan curvature remains the spatial curvature of the theory, while the originally static fields acquire physical time dependence.

In temporal gauge,

$$\mathcal{G}_0 = 0,$$

the action becomes

$$S[\mathcal{G}] = \frac{1}{2g_s^2} \int dt \left[\|\dot{\mathcal{G}}\|_h^2 - \|\mathcal{F}(\mathcal{G})\|_0^2 \right] \tag{1.8}$$

The important asymmetry is that the kinetic term of the $(2,1)$ field contains the weight h, whereas its $(2,2)$ curvature does not.

1.5. Vacuum and linearized theory

Consider the vacuum

$$\mathcal{G} = 0.$$

Write

$$\mathcal{G} = \varepsilon \psi + O(\varepsilon^2), \qquad \mathcal{G}_0 = \varepsilon a + O(\varepsilon^2).$$

Then

$$\mathcal{F}(\mathcal{G}) = \varepsilon \bar{\partial} \psi + O(\varepsilon^2),$$

and

$$D_t\mathcal{G} = \varepsilon\left(\dot{\psi} - \bar{\partial}a\right) + O(\varepsilon^2).$$

The quadratic action is therefore

$$S^{(2)}[\psi, a] = \frac{1}{2g_s^2}\int dt\left[\left\|\dot{\psi} - Q_0 a\right\|_h^2 - \|Q_1\psi\|_0^2\right] \tag{1.9}$$

Equivalently,

$$S^{(2)} = \frac{1}{2g_s^2}\int dt\left[\langle\dot{\psi} - Q_0 a,\, B_h\left(\dot{\psi} - Q_0 a\right)\rangle_0 - \langle Q_1\psi, Q_1\psi\rangle_0\right].$$

In particular, no factor of h multiplies the curvature norm. At the linearized level the gauge transformations are

$$\delta_\epsilon\psi = Q_0\epsilon, \qquad \delta_\epsilon a = \dot{\epsilon}.$$

Because

$$Q_1 Q_0 = 0,$$

the spatial curvature is gauge invariant:

$$Q_1(\psi + Q_0\epsilon) = Q_1\psi.$$

Hence the spatial quadratic form descends naturally to the gauge quotient.

1.6. Gauss constraint

The temporal field a appears without its own time derivative and therefore acts as a Lagrange multiplier. Variation of the quadratic action with respect to a gives

$$Q_0^\dagger B_h\left(\dot{\psi} - Q_0 a\right) = 0. \tag{1.10}$$

In temporal gauge,

$$a = 0,$$

this becomes

$$Q_0^\dagger B_h\dot{\psi} = 0.$$

For a nonzero-frequency mode

$$\psi(t) = e^{-iMt}\Psi,$$

we therefore obtain

$$\boxed{Q_0^\dagger B_h\Psi = 0.} \tag{1.11}$$

Since

$$Q_0 = \bar{\partial}, \qquad B_h = h,$$

the constraint reads

$$\bar{\partial}^{\dagger}(h\Psi) = 0 \tag{1.12}$$

This is the warped Gauss condition. It also has an immediate Hilbert-space interpretation. For every gauge direction $Q_0\epsilon$,

$$\langle Q_0\epsilon, \Psi\rangle_h = \langle \epsilon, Q_0^{\dagger}B_h\Psi\rangle_{0.}$$

Thus

$$Q_0^{\dagger}B_h\Psi = 0$$

is equivalent to

$$\Psi \perp_h \mathrm{Ran}\, Q_{0.}$$

Consequently the Gauss slice is

$$\boxed{\mathcal{G}_h = \ker\left(Q_0^{\dagger}B_h\right) = \left(\overline{\mathrm{Ran}\, Q_0}\right)^{\perp_h}.} \tag{1.13}$$

Whenever

$$Q_0^{\dagger}B_hQ_0$$

is invertible on the complement of its zero modes, the h-orthogonal projector onto this slice is

$$P_h = 1 - Q_0\left(Q_0^{\dagger}B_hQ_0\right)^{-1}Q_0^{\dagger}B_h \tag{1.14}$$

Thus the Gauss slice selects the representative of each gauge orbit orthogonal to the gauge directions with respect to the physical weighted metric.

1.7. The independent ∂-closed condition

The Gauss law must be distinguished from the restriction

$$\partial\Psi = 0. \tag{1.15}$$

The conditions

$$\partial\Psi = 0$$

and

$$\bar{\partial}^{\dagger}(h\Psi) = 0$$

have different origins. The former selects the admissible closed-field subcomplex, whereas the latter removes gauge directions according to the physical kinetic metric. The physical linearized configuration space is therefore

$$\mathcal{V}_h = \left\{\Psi : \partial\Psi = 0, \qquad \bar{\partial}^{\dagger}(h\Psi) = 0\right\} \tag{1.16}$$

with the appropriate operator domains and boundary conditions understood. Since

$$\partial\bar{\partial} + \bar{\partial}\partial = 0,$$

the curvature differential preserves the ∂-closed sector:

$$\partial\Psi = 0 \quad \Longrightarrow \quad \partial\big(\bar{\partial}\Psi\big) = 0.$$

Hence the closed subcomplex is stable under the linearized curvature differential.

1.8. Linearized equation of motion

Variation of the quadratic action with respect to ψ gives, in temporal gauge,

$$B_h\ddot{\psi} + Q_1^\dagger Q_1\psi = 0.$$

For

$$\psi(t) = e^{-iMt}\Psi,$$

we obtain

$$Q_1^\dagger Q_1\Psi = M^2 B_h\Psi \tag{1.17}$$

Using

$$Q_1 = \bar{\partial}, \qquad B_h = h,$$

this becomes

$$\boxed{\bar{\partial}^\dagger\bar{\partial}\Psi = M^2 h\Psi.} \tag{1.18}$$

The complete linearized normal-mode problem is therefore

$$\bar{\partial}^\dagger\bar{\partial}\Psi = M^2 h\Psi,$$

$$\bar{\partial}^\dagger(h\Psi) = 0,$$

$$\partial\Psi = 0.$$

The first equation is dynamical. The second is the Gauss constraint. The third selects the admissible closed-field subcomplex.

1.9. Degree-dependent adjoints

The degree-dependent Hilbert metrics provide a useful alternative description.
Consider

$$Q_1 : (\mathcal{H}_1, \langle\cdot,\cdot\rangle_h) \longrightarrow (\mathcal{H}_2, \langle\cdot,\cdot\rangle_0).$$

Its adjoint with respect to these physical metrics is

$$Q_{1,h}^\dagger = B_h^{-1} Q_1^\dagger. \tag{1.19}$$

By contrast,

$$Q_0 : (\mathcal{H}_0, \langle\cdot,\cdot\rangle_0) \longrightarrow (\mathcal{H}_1, \langle\cdot,\cdot\rangle_h)$$

has physical adjoint

$$Q_{0,h}^{\dagger} = Q_0^{\dagger} B_h. \tag{1.20}$$

The same degree-one Riesz operator therefore occurs on different sides of the two neighboring adjoints. The Gauss condition is simply

$$Q_{0,h}^{\dagger} \Psi = 0.$$

The physical spatial operator is

$$K_h = Q_{1,h}^{\dagger} Q_1 = B_h^{-1} Q_1^{\dagger} Q_1. \tag{1.21}$$

Hence, on smooth fields,

$$\boxed{K_h = h^{-1} \bar{\partial}^{\dagger} \bar{\partial}.} \tag{1.22}$$

It is important that

$$K_h \neq h^{-1} \bar{\partial}^{\dagger} h \bar{\partial}.$$

The latter would arise if the degree-two curvature norm were itself weighted by h. Likewise, the physical operator is not the full Dolbeault Laplacian

$$\Delta_{\bar{\partial}} = \bar{\partial}^{\dagger} \bar{\partial} + \bar{\partial} \bar{\partial}^{\dagger}.$$

The preceding gauge direction has already been removed by the Gauss constraint. There is therefore no need to dynamically penalize it by adding the second positive term. The spatial energy is simply

$$\mathfrak{q}_h[\Psi] = \left\| \bar{\partial} \Psi \right\|_0^2. \tag{1.23}$$

This is the origin of the one-sided spectral problem.

1.10. Variational definition of the physical operator

For functional-analytic purposes it is preferable to define the operator through its quadratic form.

Let

$$\mathcal{D}_h = D(\bar{\partial}) \cap \ker \partial \cap \ker(\bar{\partial}^{\dagger} h).$$

Equip this space with the weighted Hilbert norm

$$\|\Psi\|_h^2 = \int_X h \, |\Psi|^2 \, dV_X.$$

Define

$$\mathfrak{q}_h[\Psi, \Phi] = \langle \bar{\partial} \Psi, \bar{\partial} \Phi \rangle_0. \tag{1.24}$$

Assuming that this form is densely [12] defined and closed, the representation theorem for closed nonnegative quadratic forms determines a unique positive self-adjoint operator K_h satisfying

$$\mathfrak{q}_h[\Psi,\Phi] = \left\langle K_h^{\frac{1}{2}}\Psi, K_h^{\frac{1}{2}}\Phi \right\rangle_h . \tag{1.25}$$

On the smooth strong domain this operator is

$$K_h = h^{-1}\bar{\partial}^\dagger\bar{\partial}.$$

Thus the quadratic-form construction gives the rigorous definition of the one-sided Hamiltonian.

1.11. Physical one-particle Hilbert space

The full weighted space

$$L^2_h\Omega^{2,1}(X)$$

is not the physical linearized Hilbert space.
Define

$$L^2_h\Omega^{2,1}(X) = \left\{\Psi\colon \int_X h\,|\Psi|^2\, dV_X < \infty\right\}.$$

The physical tangent Hilbert space at the vacuum is

$$\mathcal{H}^{(1)}_{\text{phys}} = \overline{\left\{\Psi\colon \partial\Psi = 0,\ \bar{\partial}^\dagger(h\Psi) = 0\right\}}^{L^2_h}, \tag{1.26}$$

with regularity, operator-domain and asymptotic conditions understood. Equivalently,

$$\mathcal{H}^{(1)}_{\text{phys}} = \overline{\ker\partial \cap \mathcal{G}_h}^{L^2_h}. \tag{1.27}$$

Thus the physical one-particle Hilbert space is the weighted completion of the ∂-closed warped Gauss slice rather than the unrestricted space of $(2,1)$-forms. The mass operator is the positive self-adjoint operator K_h associated with the energy form (1.23). Whenever its physical resolvent is compact,

$$0 \le M_0^2 \le M_1^2 \le \cdots$$

and the lowest spectral value obeys

$$\boxed{M_0^2 = \inf_{\substack{\Psi\neq 0\\ \partial\Psi=0\\ \bar{\partial}^\dagger(h\Psi)=0}} \frac{\|\bar{\partial}\Psi\|_0^2}{\|\Psi\|_h^2}.} \tag{1.28}$$

This formula makes the one-sided nature particularly transparent.

1.12. Quantization

The one-particle Hilbert space of the linearized theory is

$$\mathcal{H}^{(1)} = \mathcal{H}^{(1)}_{\text{phys}}. \tag{1.29}$$

Bosonic quantization produces the symmetric Fock space

$$\mathcal{F}_{\text{lin}} = \Gamma_s\left(\mathcal{H}^{(1)}_{\text{phys}}\right) = \bigoplus_{n=0}^{\infty} \text{Sym}^n \mathcal{H}^{(1)}_{\text{phys}}. \tag{1.30}$$

If

$$K_h \Psi_n = M_n^2 \Psi_n, \qquad \|\Psi_n\|_h = 1,$$

then

$$a_n^\dagger |0\rangle$$

is the corresponding one-particle excitation with frequency M_n. The linearized theory can therefore be summarized by

$$\boxed{\begin{gathered} \Omega^{2,0} \xrightarrow{\bar\partial} \Omega^{2,1} \xrightarrow{\bar\partial} \Omega^{2,2}, \\ G_1 = hG_0, \qquad G_2 = G_0, \\ \mathcal{H}^{(1)}_{\text{phys}} = \overline{\ker\partial \cap \ker\left[\bar\partial^\dagger h\right]}^{L^2_h}, \\ K_h = h^{-1}\bar\partial^\dagger\bar\partial, \\ \mathcal{F}_{\text{lin}} = \Gamma_s\left(\mathcal{H}^{(1)}_{\text{phys}}\right). \end{gathered}}$$

2. SO(4)-Singlet Scalar Sector: Primitive and Lefschetz Blocks

2.1. Geometric setup

Let

$$X = T^*S^3 \tag{2.1}$$

be the deformed conifold equipped with its Ricci-flat Stenzel metric [13]. We use the standard radial coordinate

$$\tau \in [0, \infty).$$

The geometry [14] is of cohomogeneity one under

$$SO(4) \simeq SU(2)_L \times SU(2)_R,$$

with principal orbit

$$T^{1,1} \simeq \frac{SO(4)}{SO(2)}. \tag{2.2}$$

Choose an $SO(4)$-adapted unitary $(1,0)$ coframe

$$\zeta^0, \qquad \zeta^1, \qquad \zeta^2, \tag{2.3}$$

such that

$$\Omega = \zeta^0 \wedge \zeta^1 \wedge \zeta^2$$

is the holomorphic volume form. The Kähler form is

$$J = \frac{i}{2}\sum_{a=0}^{2} \zeta^a \wedge \bar{\zeta}^a.$$

The linearized closed field is taken in the constrained space

$$\mathcal{G} \in \ker \partial \subset \Omega^{2,1}(X).$$

In temporal gauge only the spatial field is retained, and the normal-mode equation is obtained from the one-sided quadratic form

$$\left\|\bar{\partial}\mathcal{G}\right\|_0^2$$

with weighted norm

$$\|\mathcal{G}\|_h^2 = \int_X h(\tau)|\mathcal{G}|^2 \, dV_6.$$

Thus the reduced spectral problem is

$$\bar{\partial}^\dagger \bar{\partial}\mathcal{G} = M^2 h(\tau)\mathcal{G}, \qquad \partial\mathcal{G} = 0.$$

No further interpretation of h is required in the present section; it is simply taken as a fixed positive radial function.

2.2 Radial weight

Define

$$u_0 = \left(\frac{2}{3}\right)^{3/2}$$

and

$$u(\tau) = u_0 \operatorname{sech} \tau. \tag{2.4}$$

Introduce the radial source function

$$S(\tau) = u(\tau)\log u\,(\tau) - u(\tau) - u_0 \log u_0 + u_{0.} \tag{2.5}$$

The Stenzel radial coefficient entering the reduced quadratic forms is

$$p(\tau) = (\sinh 2\,\tau - 2\tau)^{\frac{2}{3}}. \tag{2.6}$$

Define

$$Q(\tau) = \int_0^\tau S(s)\, ds.$$

The radial weight used below is

$$h(\tau) = \int_\tau^\infty \frac{Q(x)}{p(x)}\, dx. \tag{2.7}$$

Equivalently,

$$-\frac{d}{d\tau}[p(\tau)h'(\tau)] = S(\tau), \tag{2.8}$$

with

$$h(\infty) = 0, \qquad h'(0) = 0.$$

Numerically,

$$h(0) \simeq 0.13407856.$$

It is convenient to remove this overall scale and define

$$\hat{h}(\tau) = \frac{h(\tau)}{h(0)}.$$

The radial mass weight appearing in both scalar blocks is then

$$W(\tau) = \sinh^2 \tau\ \hat{h}(\tau). \tag{2.9}$$

Only the normalized profile $\hat{h}$ enters the dimensionless eigenvalues quoted below.

2.3. $SO(4)$-invariant $(2,1)$-forms

The $SO(4)$-singlet subspace of $\Omega^{2,1}(X)$ is spanned by three invariant forms,

$$\Phi_0 = \zeta^1 \wedge \zeta^2 \wedge \bar{\zeta}^0$$

$$\Phi_1 = \zeta^0 \wedge \zeta^1 \wedge \bar{\zeta}^1 + \zeta^0 \wedge \zeta^2 \wedge \bar{\zeta}^2,$$

$$\Phi_2 = \zeta^0 \wedge \zeta^1 \wedge \bar{\zeta}^2 - \zeta^0 \wedge \zeta^2 \wedge \bar{\zeta}^1. \tag{2.10}$$

The most general invariant field is therefore

$$\mathcal{G}_{\text{sing}} = f_0(\tau)\Phi_0 + f_1(\tau)\Phi_1 + f_2(\tau)\Phi_2. \tag{2.11}$$

The Lefschetz decomposition separates this space into two invariant blocks [15]. The forms Φ_0 and Φ_2 are primitive,

$$J \wedge \Phi_0 = 0, \qquad J \wedge \Phi_2 = 0.$$

By contrast,

$$\Phi_1 = -2i\, J \wedge \zeta^0,$$

so

$$J \wedge \Phi_1 \neq 0.$$

Hence

$$\mathcal{H}_{\text{sing}} = \mathcal{H}_{\text{P}} \oplus \mathcal{H}_{\text{L}}, \tag{2.12}$$

where

$$\mathcal{H}_{\text{P}} = \text{span}\{\Phi_0, \Phi_2\}$$

is the primitive block and

$$\mathcal{H}_{\text{L}} = \text{span}\{\Phi_1\}$$

is the Lefschetz block. Since the Stenzel metric, $h(\tau)$, ∂, and $\bar{\partial}$ preserve the $SO(4)$ symmetry and the Lefschetz decomposition, these two blocks can be treated independently.

3. Primitive scalar block

3.1 Exact solution of the ∂-constraint

A general primitive singlet would initially be written as

$$\mathcal{G}_{\text{P}} = f_0(\tau)\Phi_0 + f_2(\tau)\Phi_{2.} \tag{3.1}$$

The constraint

$$\partial \mathcal{G}_{\text{P}} = 0$$

relates the two functions. A convenient way to solve it identically is to introduce the invariant $(1,1)$-form

$$\eta_1 = \epsilon_{ijkl} z_i \bar{z}_j \, dz_k \wedge d\bar{z}_l$$

and define

$$\mathcal{G}_{\text{P}}[v] = \partial(v(\tau)\eta_1). \tag{3.2}$$

Because

$$\partial^2 = 0,$$

one has immediately

$$\partial \mathcal{G}_{\mathrm{P}}[v] = 0.$$

In the invariant basis this becomes

$$\mathcal{G}_{\mathrm{P}}[v] = v(\tau)\Phi_0 - q(\tau)\Phi_2, \tag{3.3}$$

where

$$q(\tau) = \cosh\tau\ v(\tau) + \frac{1}{2}\sinh\tau\ v'(\tau).$$

Thus the complete constrained primitive block contains only one independent radial function. Under contraction with Ω, the corresponding Beltrami tensor has the diagonal form

$$\mu_{\mathrm{P}} \sim \mathrm{diag}(v, q, q).$$

It can be decomposed into trace and invariant traceless parts,

$$\phi = \frac{v + 2q}{\sqrt{3}},$$

and

$$\sigma = \sqrt{\frac{2}{3}}(q - v).$$

The scalar field is therefore not represented by a pure trace polarization. The ∂-constraint fixes a nontrivial traceless completion.

3.2 Dolbeault curvature

Evaluation of $\bar{\partial}\mathcal{G}_{\mathrm{P}}$ in the invariant Stenzel frame shows that all components are controlled by the single radial combination

$$D_{\mathrm{P}}v = v'' + 4\coth\tau\ v' + 4v. \tag{3.4}$$

Thus

$$\bar{\partial}\mathcal{G}_{\mathrm{P}} \propto D_{\mathrm{P}}v.$$

After angular integration the weighted norm becomes

$$N_{\mathrm{P}}[v] = \int_0^\infty W(\tau)[|v|^2 + 2|q|^2]d\tau.$$

The spatial quadratic form is

$$Q_{\mathrm{P}}[v] = \frac{1}{4}\int_0^\infty p(\tau)\sinh^2\tau\,|D_{\mathrm{P}}v|^2 d\tau$$

in the common dimensionless spectral normalization used below. Consequently,

$$c_{\mathrm{P}}^2[v] = \frac{\frac{1}{4}\int_0^\infty p(\tau)\sinh^2\tau\,|v'' + 4\coth\tau\;v' + 4v|^2 d\tau}{\int_0^\infty W(\tau)\left[|v|^2 + 2\left|\cosh\tau\;v + \frac{1}{2}\sinh\tau\,v'\right|^2\right] d\tau}. \tag{3.5}$$

The corresponding weak eigenvalue equation is

$$\frac{1}{4}\int_0^\infty p(\tau)\sinh^2\tau\;(D_{\mathrm{P}}v)\overline{(D_{\mathrm{P}}w)}\,d\tau = c_{\mathrm{P}}^2\int_0^\infty W\left[v\bar{w} + 2q[v]\overline{q[w]}\right] d\tau \tag{3.6}$$

for every admissible test function w. Because D_{P} is second order, this is a fourth-order variational problem for v.

3.3 Boundary conditions

Regularity at the Stenzel cap requires an even expansion,

$$v(\tau) = v_0 + v_2\tau^2 + O(\tau^4),$$

and therefore

$$v'(0) = 0.$$

At infinity the field is required to be normalizable with respect to N_{P}. Numerically the interval is truncated to

$$0 \le \tau \le \tau_{\max},$$

with

$$v(\tau_{\max}) = 0, \qquad v'(\tau_{\max}) = 0,$$

and convergence is checked as $\tau_{\max}$ is increased.

3.4 Exact zero mode

The differential operator D_{P} possesses the regular analytic solution

$$v_A(\tau) = \frac{\tau\coth\tau - 1}{\sinh^2\tau}. \tag{3.7}$$

Direct differentiation gives

$$D_{\mathrm{P}}v_A = 0.$$

Therefore

$$Q_{\mathrm{P}}[v_A] = 0.$$

Near the cap,

$$v_A(\tau) = \frac{1}{3} - \frac{2}{15}\tau^2 + O(\tau^4),$$

so the solution is regular. This zero-energy direction is removed from the nonzero normal-mode tower. This zero mode corresponds to the variation of the A-period, which is fixed in our consideration.

3.5 Primitive spectrum

After removing the zero mode, the lowest primitive eigenvalue is

$$c_{\mathrm{P},0}^2 \simeq 0.4220071. \tag{3.8}$$

Using this state as the mass reference,

$$M_{\mathrm{P},0} = 1,$$

the first radial excitations are approximately

$$\frac{M_{\mathrm{P},n}}{M_{\mathrm{P},0}} \simeq 1,\ 1.566,\ 2.108,\ 2.638,\ 3.162, \ldots$$

All these states belong to the same $SO(4)$ singlet representation and differ only by their radial excitation number.

4. Lefschetz scalar block

4.1 Ansatz

The Lefschetz block is one-dimensional in invariant polarization space. Write

$$\mathcal{G}_{\mathrm{L}}[f] = f(\tau)\Phi_1. \tag{4.1}$$

The invariant form Φ_1 is separately ∂-closed, and therefore

$$\partial \mathcal{G}_{\mathrm{L}} = 0$$

for an arbitrary radial profile $f(\tau)$. Unlike the primitive block, no second polarization is required to complete the constraint.

4.2 Reduced quadratic form

The exact Stenzel form calculation gives a first-order radial combination

$$D_{\mathrm{L}} f = f' + \coth\tau\ f. \tag{4.2}$$

Define

$$r(\tau) = \frac{p'(\tau)}{p(\tau)}.$$

Since

$$p(\tau) = (\sinh 2\,\tau - 2\tau)^{2/3},$$

one finds

$$r(\tau) = \frac{8}{3}\frac{\sinh^2 \tau}{\sinh 2\,\tau - 2\tau}. \tag{4.3}$$

The Lefschetz potential is

$$V_{\mathrm{L}}(\tau) = \frac{1}{2}\left(\frac{p'}{p}\right)^2 \tag{4.4}$$

The reduced weighted norm is

$$N_{\mathrm{L}}[f] = \int_0^\infty W(\tau)|f(\tau)|^2\, d\tau.$$

The spatial quadratic form becomes

$$Q_{\mathrm{L}}[f] = \frac{1}{2}\int_0^\infty p(\tau)\left[|f' + \coth\tau\ f|^2 + \frac{1}{2}\left(\frac{p'}{p}\right)^2 |f|^2\right] d\tau. \tag{4.5}$$

Hence the Lefschetz Rayleigh quotient is

$$c_{\mathrm{L}}^2[f] = \frac{\frac{1}{2}\int_0^\infty p(\tau)\left[|f' + \coth\tau\ f|^2 + \frac{1}{2}\left(\frac{p'}{p}\right)^2 |f|^2\right] d\tau}{\int_0^\infty W(\tau)|f|^2 d\tau}. \tag{4.6}$$

This gives an ordinary second-order Sturm–Liouville problem.

4.3 Strong radial equation

For real f, define

$$b(\tau) = \coth\tau.$$

The weak eigenvalue equation is

$$\frac{1}{2}\int_0^\infty p[(D_{\mathrm{L}}f)(D_{\mathrm{L}}g) + V_{\mathrm{L}}fg]d\tau = c_{\mathrm{L}}^2\int_0^\infty Wfg\, d\tau.$$

Integrating by parts gives

$$\frac{1}{2}\left[-\frac{d}{d\tau}(pD_{\mathrm{L}}f) + p\coth\tau\ D_{\mathrm{L}}f + pV_{\mathrm{L}}f\right] = c_{\mathrm{L}}^2 Wf.$$

Equivalently,

$$-\frac{1}{2}\frac{d}{d\tau}[p(f' + \coth\tau\, f)] + \frac{1}{2}p\coth\tau\,(f' + \coth\tau\, f) + \frac{1}{4}p\left(\frac{p'}{p}\right)^2 f = c_{\mathrm{L}}^2 Wf.$$

This may be rewritten as

$$-\frac{1}{2}(pf')' + U_{\mathrm{L}}(\tau)f = c_{\mathrm{L}}^2 W f, \tag{4.7}$$

with

$$U_{\mathrm{L}} = \frac{1}{2}p\left[-\coth'\tau - \frac{p'}{p}\coth\tau + \coth^2\tau + \frac{1}{2}\left(\frac{p'}{p}\right)^2\right].$$

4.4 Endpoint behaviour

Near the cap,

$$p(\tau) \sim \left(\frac{4}{3}\right)^{\frac{2}{3}}\tau^2$$

Hence

$$\frac{p'}{p} = \frac{2}{\tau} + O(\tau),$$

and

$$V_{\mathrm{L}} = \frac{2}{\tau^2} + \frac{8}{15} + O(\tau^2).$$

Also,

$$\coth\tau = \frac{1}{\tau} + \frac{\tau}{3} + O(\tau^3).$$

The regular branch is selected by smoothness and finite six-dimensional norm. At large τ,

$$\frac{p'}{p} \longrightarrow \frac{4}{3}$$

and

$$\coth\tau \longrightarrow 1.$$

Consequently,

$$V_{\mathrm{L}} \longrightarrow \frac{8}{9}.$$

For the zero-mass asymptotic equation one obtains

$$f'' + \frac{4}{3}f' - \frac{5}{9}f = 0.$$

The two independent asymptotic behaviours are

$$f \sim e^{\tau/3},$$

and

$$f \sim e^{-5\tau/3}.$$

Normalizability selects the decaying branch.

4.5 Lefschetz spectrum

Solving the generalized Sturm–Liouville problem numerically gives the lowest Lefschetz eigenvalue

$$c^2_{\mathrm{L},0} \simeq 0.9981. \tag{4.8}$$

The first few eigenvalues are approximately

$$c^2_{\mathrm{L},n} \simeq 0.9981,\ 1.7792,\ 2.7771,\ 3.9935,\ 5.4302, \ldots$$

Correspondingly,

$$\frac{M_{\mathrm{L},n}}{M_{\mathrm{L},0}} \simeq 1,\ 1.335,\ 1.668,\ 2.000,\ 2.332, \ldots$$

5. Comparison of the two scalar singlet blocks

The primitive and Lefschetz sectors are both $SO(4)$-singlets, but their radial dynamics are different. The primitive block is

$$\mathcal{G}_{\mathrm{P}} = v\Phi_0 - q\Phi_2,$$

and gives a fourth-order radial variational problem. The Lefschetz block is

$$\mathcal{G}_{\mathrm{L}} = f\Phi_1,$$

and gives a second-order Sturm–Liouville problem. Their lowest eigenvalues in the common convention are

$$c^2_{\mathrm{P},0} \simeq 0.4220071$$

and

$$c^2_{\mathrm{L},0} \simeq 0.9981.$$

Therefore

$$\frac{M_{\mathrm{L},0}}{M_{\mathrm{P},0}} = \sqrt{\frac{c^2_{\mathrm{L},0}}{c^2_{\mathrm{P},0}}} \simeq 1.538. \tag{5.1}$$

Thus the $SO(4)$-singlet sector decomposes into two independent towers,

$$\mathcal{H}_{\mathrm{sing}} = \mathcal{H}_{\mathrm{P}} \oplus \mathcal{H}_{\mathrm{L}}.$$

The primitive tower begins at

$$c^2_{\mathrm{P},0} \simeq 0.4220,$$

whereas the Lefschetz tower begins at

$$c^2_{\mathrm{L},0} \simeq 0.9981.$$

Their separation follows directly from the Lefschetz decomposition of the invariant $(2,1)$-form space.

6. SO(4) Tensor Block and Tensor Eigenvalues

6.1 Tensor representation

We now consider the first non-scalar $SO(4)$ sector of the constrained closed field

$$\mathcal{G} \in \ker \partial \cap \Omega^{2,1}(X)$$

on the Stenzel deformed conifold

$$X = T^* S^{3.}$$

The lowest chiral tensor representation is

$$(j_L, j_R) = (2,0),$$

together with its conjugate representation

$$(j_L, j_R) = (0,2).$$

Since

$$SO(4) \simeq SU(2)_L \times SU(2)_R,$$

the $(2,0)$ representation has dimension five. Under the diagonal spatial rotation group it carries spin two. Its $(0,2)$ partner has the same radial spectrum, so it is sufficient to solve one chirality. Let V_2 denote the five-dimensional spin-two representation of $SU(2)_L$. On a principal orbit

$$T^{1,1} \simeq SO(4)/SO(2),$$

a homogeneous (p,q)-form transforming in this representation is an element of

$$\mathrm{Hom}_{SO(2)}(V_2, \Lambda^{p,q}\mathbb{C}^3).$$

The equivariance condition is

$$\sigma_{p,q}(H)P + P\rho_2(H) = 0, \tag{6.1}$$

where H generates the isotropy algebra $\mathfrak{so}(2)$, $\sigma_{p,q}$ is its action on (p,q)-forms, and ρ_2 is its action in the $(2,0)$ representation. Solving this condition gives

$$\dim \mathcal{V}^{2,0} = 3$$

$$\dim \mathcal{V}^{2,1} = 9$$

$$\dim \mathcal{V}^{3,1} = 3,$$

$$\dim \mathcal{V}^{2,2} = 9.$$

Thus the tensor $(2,1)$-block contains nine homogeneous radial coefficient functions before the ∂-constraint is imposed. Choose an orthonormal homogeneous basis

$$\Xi_A^{(2,1)}, \qquad A = 1, \dots, 9.$$

The general tensor field is then

$$\mathcal{G}_T(r,\Omega) = \sum_{A=1}^{9} q_A(r)\, \Xi_A^{(2,1)}(\Omega). \tag{6.2}$$

It is convenient in this calculation to use

$$r = \frac{\tau}{2}.$$

We collect the radial amplitudes into

$$q(r) = \begin{pmatrix} q_1(r) \\ \vdots \\ q_9(r) \end{pmatrix}.$$

6.2 Stenzel geometry in the tensor coordinates

Define

$$\mathcal{R}(r) = \frac{\sinh 4r - 4r}{8}.$$

The Stenzel metric functions used in the tensor frame are

$$a^2(r) = \mathcal{R}(r)^{\frac{1}{3}} \coth r \tag{6.3}$$

$$b^2(r) = \mathcal{R}(r)^{1/3} \tanh r,$$

and

$$c^2(r) = \frac{1}{3}\mathcal{R}(r)^{-2/3} \sinh^2(2r).$$

After integration over the homogeneous orbit, and up to a common angular normalization which cancels from the Rayleigh quotient, the radial volume density is

$$\mu(r) = \frac{\sinh^2(2r)}{3}. \tag{6.4}$$

Since

$$\tau = 2r,$$

the normalized radial weight introduced in the scalar calculation enters as

$$\hat{h}(2r).$$

The weighted measure is therefore

$$\mu_h(r) = \mu(r)\hat{h}(2r). \tag{6.5}$$

The connection coefficients entering the Dolbeault operator may be written as

$$q_a = \frac{1}{c}\frac{a'}{a}, \qquad q_b = \frac{1}{c}\frac{b'}{b}, \qquad q_c = \frac{1}{c}\frac{c'}{c}, \tag{6.6}$$

together with

$$\gamma_a = \frac{a^2 - b^2 - c^2}{2abc}$$

$$\gamma_b = \frac{b^2 - c^2 - a^2}{2abc},$$

and

$$\gamma_c = \frac{c^2 - a^2 - b^2}{2abc}.$$

These functions completely determine the Levi-Civita connection in the $SO(4)$-adapted moving frame.

6.3 Reduced Dolbeault operators

Let $P_A^{(p,q)}$ be the homogeneous matrices defining a basis of

$$\mathcal{V}^{p,q} = \mathrm{Hom}_{SO(2)}(V_2, \Lambda^{p,q}\mathbb{C}^3).$$

For a tangent direction X_α on the principal orbit, the covariant derivative of a homogeneous matrix takes the form

$$\nabla_\alpha P = \sigma_{p,q}(\Gamma_\alpha)P + \frac{1}{\ell_\alpha} P\rho_2(X_\alpha), \tag{6.7}$$

where

$$\ell_\alpha \in \{a, b, c\}$$

is the corresponding Stenzel frame length. The first term acts on the form indices. The second acts on the spin-two representation index. Consequently both ∂ and $\bar{\partial}$ reduce to first-order matrix-valued radial operators. For the tensor $(2,1)$-block,

$$\partial_T = C_1(r)\frac{d}{dr} + C_0(r), \tag{6.8}$$

where

$$C_1, C_0 : \mathbb{C}^9 \longrightarrow \mathbb{C}^3,$$

because the homogeneous $(3,1)$ target space has dimension three. Similarly,

$$\bar{\partial}_T = D_1(r)\frac{d}{dr} + D_0(r), \tag{6.9}$$

where

$$D_1, D_0 : \mathbb{C}^9 \longrightarrow \mathbb{C}^9,$$

because the homogeneous $(2,2)$ target space has dimension nine. The matrices

$$C_0, \quad C_1, \quad D_0, \quad D_1$$

contain the complete radial-connection and $SO(4)$-angular information. No phenomenological tensor mass term is introduced separately.

6.4 The ∂-constraint

The closed field space is

$$\ker \partial.$$

Therefore the tensor field must satisfy

$$\partial \mathcal{G}_T = 0.$$

Using the reduced radial operator gives

$$C_1(r)q'(r) + C_0(r)q(r) = 0. \tag{6.10}$$

This is the tensor analogue of the primitive scalar relation

$$q_S(\tau) = \cosh\tau \; v + \frac{1}{2}\sinh\tau \; v'.$$

In the scalar block, the constraint eliminated one of the two invariant polarizations and left a single radial function v. In the tensor block the constraint is matrix-valued. It couples the nine homogeneous tensor polarizations, and the constrained space retains several independent radial degrees of freedom. Let Q_T denote a basis of radial fields satisfying

$$C_1 q' + C_0 q = 0$$

together with the endpoint regularity conditions. Then every admissible tensor field can be written as

$$q = Q_T u. \tag{6.11}$$

Thus Q_T is the tensor analogue of explicitly solving the scalar ∂-constraint.

6.5 Regularity at the Stenzel cap

The principal orbit degenerates at $r = 0$. Consequently an arbitrary nine-component vector $q(0)$ does not define a smooth tensor field. Let

$$\mathcal{S}_0 \subset \mathbb{C}^9$$

be the subspace of homogeneous values that extend smoothly through the Stenzel bolt, and let

$$\mathcal{S}_1 \subset \mathbb{C}^9$$

be the corresponding subspace of allowed first derivatives. Smoothness requires

$$q(0) \in \mathcal{S}_0, \tag{6.12}$$

and

$$q'(0) \in \mathcal{S}_1. \tag{6.13}$$

This is the representation-valued tensor analogue of the primitive scalar condition

$$v'(0) = 0.$$

At large radius the normalizable tensor modes obey

$$q(r) \longrightarrow 0 \qquad (r \to \infty).$$

Numerically this is implemented at a sufficiently large radial cutoff $r_{\max}$ by

$$q(r_{\max}) = 0.$$

6.6 One-sided tensor curvature

For an admissible field

$$\mathcal{G}_T \in \ker \partial,$$

the spatial curvature entering the quadratic problem is

$$\bar{\partial}\mathcal{G}_T.$$

In radial coefficients,

$$\bar{\partial}\mathcal{G}_T \leftrightarrow D_1(r)q'(r) + D_0(r)q(r). \tag{6.14}$$

Define

$$\mathcal{D}_T q = D_1 q' + D_0 q. \tag{6.15}$$

The unwarped spatial quadratic form is then

$$E_T[q] = \int_0^\infty dr\, \mu(r)\, (\mathcal{D}_T q)^\dagger (\mathcal{D}_T q). \tag{6.16}$$

Expanding this expression gives

$$E_T[q] = \int_0^\infty dr \big[q'^\dagger A_{11} q' + 2\,\mathrm{Re}\big(q^\dagger A_{01} q'\big) + q^\dagger A_{00} q \big],$$

with

$$A_{11} = \mu D_1^\dagger D_1, \qquad A_{01} = \mu D_0^\dagger D_1, \qquad A_{00} = \mu D_0^\dagger D_0. \tag{6.17}$$

The tensor-specific angular and connection contributions therefore appear automatically through the matrices D_0 and D_1. This is why the tensor block is heavier than the scalar singlet already at finite S^3: the nontrivial $SO(4)$ representation contributes additional positive angular and spin-connection terms to the one-sided Dolbeault energy.

6.7 Warped tensor norm

The weighted tensor norm is

$$N_T[q] = \int_0^\infty dr\, \mu(r) \hat{h}(2r) q^\dagger q. \tag{6.18}$$

Using

$$q = Q_T u,$$

the reduced mass matrix is

$$B_T = Q_T^\dagger W_h Q_T, \tag{6.19}$$

where W_h represents multiplication by

$$\mu(r)\hat{h}(2r).$$

Similarly define

$$Y_T = \bar{\partial}_T Q_T.$$

The reduced one-sided stiffness matrix is

$$K_T = Y_T^\dagger W_g Y_T, \tag{6.20}$$

or explicitly,

$$K_T = \big(\bar{\partial}_T Q_T\big)^\dagger W_g \big(\bar{\partial}_T Q_T\big), \tag{6.21}$$

where W_g contains the unwarped radial measure $\mu(r)$. This is directly analogous to the primitive scalar numerator

$$\frac{1}{4}\int p(\tau)\sinh^2\tau\,|D_P v|^2 d\tau.$$

6.8 Raw tensor eigenvalue problem

The constrained tensor modes satisfy the generalized eigenvalue problem

$$K_T u_n = \lambda_n^{\mathrm{c}} B_T u_n. \tag{6.22}$$

Equivalently,

$$\lambda_n^{\mathrm{c}} = \frac{\int_0^\infty dr\,\mu(r)|D_1 q_n{}' + D_0 q_n|^2}{\int_0^\infty dr\,\mu(r)\hat{h}(2r)|q_n|^2}. \tag{6.23}$$

The quantity λ_n^{c} is the raw generalized eigenvalue in the metric convention used by the homogeneous tensor calculation. It must still be converted to the c_n^2 convention used in the scalar calculation.

6.9 Conversion to the common scalar normalization

The normalization conversion can be derived explicitly. The scalar Stenzel convention uses

$$K(\tau) = \frac{(\sinh 2\,\tau - 2\tau)^{\frac{1}{3}}}{2^{\frac{1}{3}}\sinh\tau}. \tag{6.24}$$

Define

$$F(\tau) = (\sinh 2\,\tau - 2\tau)^{\frac{1}{3}}. \tag{6.25}$$

Since

$$r = \frac{\tau}{2},$$

we have

$$\mathcal{R}(r) = \frac{\sinh 2\,\tau - 2\tau}{8},$$

and therefore

$$\mathcal{R}(r)^{1/3} = \frac{F(\tau)}{2}. \tag{6.26}$$

For example, in the tensor metric convention,

$$a^2_{\mathrm{code}} = \frac{F}{2}\coth\frac{\tau}{2}.$$

In the scalar Stenzel convention,

$$a^2_{\mathrm{St}} = K(\tau)\cosh^2\frac{\tau}{2}.$$

Using

$$\sinh\tau = 2\sinh\frac{\tau}{2}\cosh\frac{\tau}{2},$$

this becomes

$$a_{\mathrm{St}}^2 = \frac{F}{2^{4/3}}\coth\frac{\tau}{2}.$$

Hence

$$a_{\mathrm{code}}^2 = 2^{\frac{1}{3}}a_{\mathrm{St}}^2. \tag{6.27}$$

The same relation holds for the remaining frame components, so the complete metrics obey

$$g_{\mathrm{code}} = 2^{\frac{1}{3}}g_{\mathrm{St}}. \tag{6.28}$$

Now consider a constant metric rescaling

$$g' = s\,g.$$

In six dimensions,

$$dV_{g'} = s^3 dV_g.$$

For a k-form,

$$|\alpha_k|_{g'}^2 = s^{-k}|\alpha_k|_g^2,$$

so

$$\|\alpha_k\|_{g'}^2 = s^{3-k}\|\alpha_k\|_g^2.$$

The field $\mathcal{G}_T$ is a three-form. Therefore

$$\|\mathcal{G}_T\|_{g'}^2 = \|\mathcal{G}_T\|_g^2.$$

Its curvature

$$\bar{\partial}\mathcal{G}_T$$

is a four-form, and hence

$$\left\|\bar{\partial}\mathcal{G}_T\right\|_{g'}^2 = s^{-1}\left\|\bar{\partial}\mathcal{G}_T\right\|_g^{2.} \tag{6.29}$$

With

$$s = 2^{1/3},$$

we therefore obtain

$$\lambda_{\mathrm{c}} = 2^{-1/3}\lambda_{\mathrm{St}}.$$

Equivalently,

$$\lambda_{\mathrm{St}} = 2^{\frac{1}{3}}\lambda_{\mathrm{c}}. \tag{6.30}$$

The common spectral convention used in the scalar calculation is

$$c^2 = \frac{\lambda_{\mathrm{St}}}{3\,2^{1/3}}.$$

Substituting the relation above gives

$$c_T^2 = \frac{2^{1/3}\lambda_{\mathrm{c}}}{3\,2^{1/3}},$$

so finally

$$c_{T,n}^2 = \frac{\lambda_n^{\mathrm{c}}}{3}. \tag{6.31}$$

6.10 Tensor Rayleigh quotient in the common convention

The tensor eigenvalue in precisely the same normalization as the scalar block is therefore

$$c_T^2[q] = \frac{1}{3}\frac{\int_0^\infty dr\ \sinh^2(2r)\,|D_1 q' + D_0 q|^2}{\int_0^\infty dr\ \sinh^2(2r)\,\hat{h}(2r)|q|^2}. \tag{6.32}$$

The minimization is performed over smooth normalizable fields satisfying

$$C_1 q' + C_0 q = 0. \tag{6.33}$$

This is the complete one-sided tensor variational problem.

6.11 Numerical discretization

The noncompact interval is replaced by

$$\epsilon \le r \le r_{\max},$$

with

$$\epsilon \ll 1.$$

A Chebyshev-Lobatto grid is introduced,

$$r_i = \frac{\epsilon + r_{\max}}{2} + \frac{r_{\max} - \epsilon}{2}x_i,$$

where

$$x_i = -\cos\frac{\pi i}{N-1}.$$

The associated spectral differentiation matrix represents

$$\frac{d}{dr}.$$

At every radial point one evaluates

$$C_1(r_i), \qquad C_0(r_i), \qquad D_1(r_i), \qquad D_0(r_i)$$

from the exact Stenzel frame and the $(2,0)$ representation matrices. The global closure matrix is

$$\mathcal{C} = C_1 D_r + C_{0.}$$

The admissible tensor basis Q_T is obtained from the null space of $\mathcal{C}$, together with the smooth cap and ultraviolet boundary conditions. The reduced matrices are then

$$B_T = Q_T^\dagger W_h Q_T$$

and

$$K_T = \left(\bar{\partial} Q_T\right)^\dagger W_g \left(\bar{\partial} Q_T\right).$$

The generalized eigenproblem is

$$K_T u_n = \lambda_n^{\mathrm{c}} B_T u_n,$$

followed by the conversion

$$c_{T,n}^2 = \lambda_n^{\mathrm{c}} / 3.$$

6.12 Resolution convergence

For a representative ultraviolet endpoint

$$r_{\max} = 5,$$

the lowest tensor eigenvalue converges approximately as

N	$c_{T,0}^2$
28	2.3287
32	2.3203
40	2.3058
48	2.2980
56	2.2942
64	2.2923
72	2.2912

The sequence approaches a value close to

$$c_{T,0}^2 \simeq 2.29$$

The remaining cutoff dependence is comparatively small. For example, at fixed high resolution one finds values in the neighbourhood

$$c_{T,0}^2 \simeq 2.29$$

through the stable cutoff window. A conservative result is therefore

$$\boxed{c_{T,0}^2 \simeq 2.29.} \tag{6.35}$$

6.13 First tensor eigenvalues

At representative high radial resolution, the first few tensor eigenvalues are approximately

$$c_{T,n}^2 \simeq 2.291,\ 2.926,\ 3.430,\ 3.555,\ 4.034,\ 4.118, \ldots$$

Their masses relative to the tensor ground state are

$$\frac{M_{T,n}}{M_{T,0}} \simeq 1,\ 1.130,\ 1.224,\ 1.246,\ 1.327,\ 1.341, \ldots$$

The tensor spectrum is denser than the scalar towers because several independent constrained tensor polarizations occur inside the same $SO(4)$ representation.

6.14 Comparison with the primitive scalar

For the primitive scalar block,

$$c_{P,0}^2 = 0.4220071.$$

The tensor-to-primitive mass ratio is therefore

$$\frac{M_{T,0}}{M_{P,0}} = \sqrt{\frac{c_{T,0}^2}{c_{P,0}^2}}. \tag{6.36}$$

Using

$$c_{T,0}^2 \simeq 2.291$$

gives

$$\boxed{\frac{M_{T,0}}{M_{P,0}} \simeq 2.33.} \tag{6.37}$$

Thus the lowest tensor state on the uncontracted $SO(4)$-symmetric geometry is more than twice as heavy as the primitive scalar ground state. The first tensor states expressed in units of the primitive scalar ground-state mass are approximately

$$\frac{M_{T,n}}{M_{P,0}} \simeq 2.33,\ 2.63,\ 2.85,\ 2.90,\ 3.09,\ 3.12, \ldots$$

6.15 Comparison with the Lefschetz scalar

For the Lefschetz scalar,

$$c_{L,0}^2 \simeq 0.9981.$$

Hence

$$\frac{M_{T,0}}{M_{L,0}} = \sqrt{\frac{2.291}{0.9981}} \simeq 1.515. \tag{6.38}$$

The ordering of the three lowest blocks is therefore

$$M_{P,0} < M_{L,0} < M_{T,0}.$$

In units of the primitive scalar ground-state mass,

$$M_{P,0} : M_{L,0} : M_{T,0} \simeq 1 : 1.538 : 2.33$$

Their squared eigenvalues are

$$c_{P,0}^2 : c_{L,0}^2 : c_{T,0}^2 \simeq 0.4220 : 0.9981 : 2.291.$$

6.16 Comparison of the low-lying spectra

The first few eigenvalues of the three blocks are approximately

n	$c_{P,n}^2$	$c_{L,n}^2$	$c_{T,n}^2$
0	0.4220	0.9981	2.291
1	1.035	1.779	2.926
2	1.875	2.777	3.430
3	2.936	3.994	3.555
4	4.220	5.430	4.034

The tensor block therefore starts considerably higher than either scalar block. At the same time its higher levels are relatively closely spaced because the tensor sector contains several internal constrained polarizations rather than one single radial degree of freedom.

7 SO(4) to ISO(3) Reduction of the Tensor Sector

7.1 Compact and flat tensor limits

The tensor block obtained in the last section is constructed from the lowest chiral $SO(4)$ representations

$$(j_L, j_R) = (2, 0) \qquad \text{and} \qquad (j_L, j_R) = (0, 2).$$

On the compact geometry the tensor therefore carries a nontrivial $SO(4)$ harmonic. Its one-sided eigenvalue is considerably larger than that of the primitive scalar singlet,

$$c_{T,0}^2(S^3) \simeq 2.29,$$

whereas

$$c_{S,0}^2 = 0.4220071.$$

The large compact tensor eigenvalue originates from the angular and spin-connection contributions carried by the nontrivial $SO(4)$ representation. The flat limit is obtained through the Inönü-Wigner contraction [16]

$$SO(4) \longrightarrow ISO(3). \tag{7.1}$$

Introduce the rotation generators

$$J_i = \frac{1}{2}\epsilon_{ijk}M_{jk} \tag{7.2}$$

and define rescaled translation generators

$$P_i(\alpha) = \alpha M_{0i}, \tag{7.3}$$

where α is a dimensionless contraction parameter. The commutation relations become

$$[J_i, J_j] = \epsilon_{ijk}J_k,$$

$$[J_i, P_j] = \epsilon_{ijk}P_k,$$

$$[P_i, P_j] = \alpha^2\epsilon_{ijk}J_k.$$

Hence

$$\alpha \to 0 \quad \Longrightarrow \quad [P_i, P_j] \to 0,$$

and the algebra contracts to

$$\mathfrak{iso}(3) = \mathfrak{so}(3) \ltimes \mathbb{R}^3.$$

The parameter α therefore interpolates between a compact $SO(4)$ tensor harmonic and a flat Cartesian spin-two excitation.

7.2 The tensor contraction must not deform the Dolbeault complex

A direct contraction of the Stenzel Dolbeault differential would amount to replacing

$$\bar{\partial}_X$$

by an α-dependent operator

$$\bar{\partial}_\alpha.$$

This is not the appropriate construction. The complex geometry of the deformed conifold remains fixed. In particular,

$$\bar{\partial}_X^2 = 0$$

must hold for every value of the contraction parameter. The contraction is therefore implemented through a family of maps

$$C_\alpha$$

from the contracted tensor module into the fixed field space. The fundamental relation is

$$\bar{\partial}_X C_\alpha = C_\alpha d_\alpha, \tag{7.3}$$

where d_α is the differential of the contracted tensor complex. Thus

$$\boxed{\bar{\partial}_X \text{ is fixed, while } C_\alpha \text{ moves.}}$$

This distinction is essential. Direct scaling of the $SO(4)$ terms inside $\bar{\partial}_X$ produces a nonintegrable almost-complex structure and leaves a nonzero zeroth-order nilpotency defect. By contrast, the chain-map construction preserves

$$\bar{\partial}_X^2 = 0$$

exactly and places the contraction in the representation data rather than in the ambient complex geometry.

7.3 Tensor channels under the contraction

After reduction to the diagonal spatial group

$$SO(3) \subset SO(4),$$

the lowest tensor module contains three invariant channels,

$$\ell = 1, 2, 3.$$

The corresponding first-order eigenvalues are

$$\kappa_\ell = \frac{8 - \ell(\ell+1)}{2}. \tag{7.4}$$

Therefore

$$\kappa_1 = 3, \qquad \kappa_2 = 1, \qquad \kappa_3 = -2.$$

Their squared values are

$$\kappa_\ell^2 = 9,\ 1,\ 4.$$

The angular operator therefore contracts as

$$L_{\text{ang}}(\alpha) = \alpha^2\, \text{diag}(9,1,4). \tag{7.5}$$

The important feature is the factor α^2. At finite compact radius the tensor pays a nonzero angular cost, while in the flat limit $\alpha \to 0$ this compact harmonic contribution disappears. The tensor branch is therefore expected to become considerably lighter as

$$SO(4) \to ISO(3).$$

7.4 Chain-map expansion

Consider first the channel

$$\kappa_1 = 3.$$

Its angular eigenvalue is

$$\lambda_{\text{ang}} = 9\alpha^2.$$

The corresponding representative can be expanded as

$$\mathcal{G}_\alpha = \mathcal{G}_0 + 9\alpha^2\mathcal{G}_1 + (9\alpha^2)^2\mathcal{G}_2 + \cdots. \tag{7,6}$$

The absence of a term linear in α reflects the fact that the physical branch is even under

$$\alpha \to -\alpha.$$

Let K denote the constrained unwarped one-sided stiffness and let M denote the corresponding unwarped norm operator. The chain-map condition produces the hierarchy

$$K\mathcal{G}_0 = 0$$

$$K\mathcal{G}_1 = M\mathcal{G}_0$$

$$K\mathcal{G}_2 = M\mathcal{G}_1 \tag{7.7}$$

and so forth. Thus $\mathcal{G}_1$ is not an independent normal mode. It is the first compensating deformation of the representative required by the contraction. Formally,

$$\mathcal{G}_1 = K_\perp^{-1} M\mathcal{G}_0,$$

where the inverse is taken on the complement of the kernel. This is a Green-operator or chain-homotopy correction.

7.6 Degree-wise Riesz transport

Let $G_{21}(\alpha)$ denote the Riesz metric on the tensor $(2,1)$ field space and let $G_{22}(\alpha)$ denote the corresponding metric on the $(2,2)$ curvature space. The differential itself remains fixed,

$$\bar{\partial}_X : \Omega^{2,1} \longrightarrow \Omega^{2,2},$$

but its adjoint becomes

$$\bar{\partial}_\alpha^\sharp = G_{21}(\alpha)^{-1}\bar{\partial}_X^* G_{22}(\alpha). \tag{7.8}$$

Consequently the physical one-sided operator is

$$L_T(\alpha) = \bar{\partial}_\alpha^\sharp \bar{\partial}_X. \tag{7.9}$$

Equivalently, at the level of quadratic forms,

$$K_T(\alpha) = \left(\bar{\partial}_X C_\alpha\right)^\dagger G_{22}(\alpha)\left(\bar{\partial}_X C_\alpha\right),$$

whereas

$$B_T(\alpha) = C_\alpha^\dagger G_{21}(\alpha) C_\alpha.$$

The tensor eigenproblem is therefore

$$K_T(\alpha)u_n = c^2_{T,n}(\alpha)B_T(\alpha)u_n. \tag{7.10}$$

The scalar singlet does not participate in this tensor-specific transport. Its radial operator and its reference value remain

$$c^2_{S,0} = 0.4220071.$$

This degree-wise Riesz transport is what permits the tensor spectrum to move below the fixed-Stenzel variational floor without altering the ambient Dolbeault complex.

7.7 1PI interpretation

The same structure has a natural interpretation in the quadratic 1PI field theory. Let the quadratic inverse propagator be

$$\mathcal{A}_\alpha(p^2) = K_\alpha - p^2 B_\alpha. \tag{7.11}$$

Decompose the tensor field space into a distinguished low-energy tensor sector P and its orthogonal complement Q,

$$1 = P + Q.$$

Write

$$\Psi = \phi + \chi, \tag{7.12}$$

with

$$\phi = P\Psi, \qquad \chi = Q\Psi. \tag{7.13}$$

The quadratic field equation is

$$\mathcal{A}_\alpha(\phi + \chi) = 0.$$

The Q-equation gives

$$Q\chi = -(Q\mathcal{A}_\alpha Q)^{-1} Q\mathcal{A}_\alpha P\phi.$$

Substituting this solution back into the action produces the 1PI quadratic kernel [17,18]

$$\boxed{\mathcal{A}_{1\mathrm{PI}} = P\mathcal{A}_\alpha P - P\mathcal{A}_\alpha Q(Q\mathcal{A}_\alpha Q)^{-1} Q\mathcal{A}_\alpha P.} \tag{7.14}$$

The second term is the Feshbach or Schur-complement correction. This formula explains why the tensor contraction is not given simply by deleting the compact angular energy. The direct compact contribution raises the tensor eigenvalue, but coupling to source-invisible intermediate tensor components produces a nonlocal negative correction. The chain-homotopy equation

$$K\mathcal{G}_1 = M\mathcal{G}_0$$

is precisely the zero-momentum form of this resolvent structure. Thus the first compensator and the 1PI-Feshbach correction are two representations of the same mechanism.

7.8. Expansion around the flat endpoint

To determine the flat tensor branch, expand the one-sided tensor differential near

$$\alpha = 0.$$

Write

$$D_\alpha = D_0 + \alpha D_1. \tag{7.15}$$

The corresponding quadratic stiffness has the expansion

$$K_\alpha = K_0 + \alpha V + \alpha^2 K_{\text{ang}}, \tag{7.16}$$

where

$$V = D_0^\dagger W D_1 + D_1^\dagger W D_0,$$

and

$$K_{\text{ang}} = D_1^\dagger W D_1.$$

For the flat ground state v_0, the expectation value of the term linear in α is numerically

$$\langle v_0, V v_0 \rangle \sim 10^{-11}.$$

Thus

$$c_T^2(\alpha) = c_{T,\infty}^2 + s_T \alpha^2 + O(\alpha^4). \tag{7.17}$$

The tensor branch is therefore even in α, as expected from the contraction and the chain-map expansion.

7.9 Direct angular contribution

The first contribution to the α^2 coefficient is the direct angular energy,

$$s_{\text{ang}} = \frac{1}{3} \langle v_0, K_{\text{ang}} v_0 \rangle. \tag{7.18}$$

Numerically,

$$\boxed{s_{\text{ang}} = 1.538814.} \tag{7.19}$$

This is the positive energy that would be obtained if the compact angular operator were simply switched on while all other tensor components remained frozen. It is therefore the bare compactification cost of the tensor harmonic.

7.10 Green-function or 1PI correction

The off-diagonal operator V mixes the lowest tensor state with orthogonal tensor modes. Standard second-order perturbation theory gives

$$s_{\mathrm{hom}} = \frac{1}{3}\sum_{n\neq 0}\frac{|\langle v_n, V v_0\rangle|^2}{\lambda_0 - \lambda_n}. \tag{7.20}$$

Because

$$\lambda_n > \lambda_0,$$

the denominators are negative, so this term lowers the eigenvalue. Numerically,

$$\boxed{s_{\mathrm{hom}} = -0.428394.} \tag{7.21}$$

This is precisely the 1PI-Feshbach contribution obtained by integrating out the complementary tensor modes. Therefore, if the physical domain and Riesz metric were frozen, the total coefficient would be

$$s_{\mathrm{frozen}} = s_{\mathrm{ang}} + s_{\mathrm{hom}},$$

giving

$$\boxed{s_{\mathrm{frozen}} = 1.538814 - 0.428394 = 1.110420.} \tag{7.22}$$

The nonlocal 1PI correction thus removes a substantial part of the bare compact angular cost.

7.11 Riesz and horizontal correction

For each degree q, let G_{2q} denote the positive Riesz operator representing the physical inner product on $\Omega^{2,q}$ relative to a fixed reference pairing,

$$\langle \Psi, \Phi\rangle_{2q} = \langle \Psi, G_{2q}\Phi\rangle_0. \tag{7.23}$$

The corresponding adjoint of

$$\bar{\partial}\colon \Omega^{2,1} \to \Omega^{2,2}$$

is therefore

$$\bar{\partial}^{\sharp} = G_{21}^{-1}\bar{\partial}^{*}G_{22}. \tag{7.24}$$

By "Riesz transport" we mean the α-dependent transport of these degree-wise inner products, with the Dolbeault differential itself held fixed. The contraction also changes the physical Riesz realization of the tensor state. When the reduced tensor space is represented inside the larger homogeneous $(2,1)$ coefficient space, its Riesz-orthogonal representative depends on α. Denote the corresponding projector by H_α. Then the reduced tensor forms are more precisely

$$B_T(\alpha) = H_\alpha^{\dagger} G_{21}(\alpha) H_\alpha, \tag{7.25}$$

and

$$K_T(\alpha) = H_\alpha^{\dagger}\bar{\partial}_X^{\dagger} G_{22}(\alpha)\bar{\partial}_X H_\alpha. \tag{7.26}$$

The variation of this Riesz horizontal structure supplies an additional negative contribution to the α^2 coefficient. Numerically,

$$\boxed{s_{\mathrm{hor}} = -0.431934.} \tag{7.27}$$

Thus the complete coefficient becomes

$$s_T = s_{\mathrm{ang}} + s_{\mathrm{hom}} + s_{\mathrm{hor}},$$

or

$$\boxed{s_T = 1.538814 - 0.428394 - 0.431934 = 0.678486.} \tag{7.28}$$

The three contributions can therefore be summarized as

direct angular:	$+\,1.5388,$
1PI-Green correction:	$-\,0.4284,$
Riesz-horizontal correction:	$-\,0.4319,$
total:	$+\,0.6785.$

The compact tensor cost is therefore strongly renormalized before it enters the physical eigenvalue.

7.12 Resolution dependence of the flat endpoint

The numerical flat tensor endpoint is stable under increasing spectral resolution. The calculated values are

N	$c_{T,\infty}^2$	$\sqrt{c_{T,\infty}^2/c_{S,0}^2}$
48	0.954545	1.5040
56	0.952809	1.5026
64	0.952297	1.5022

while the corresponding α^2 slopes are approximately

$$0.6865, \qquad 0.6785, \qquad 0.6710.$$

The results therefore converge toward

$$c_{T,\infty}^2 \simeq 0.95$$

and

$$\frac{dc_T^2}{d(\alpha^2)}\Big|_{\alpha=0} \simeq 0.65 - 0.68.$$

A simple $1/N^2$ extrapolation of the tensor eigenvalue gives

$$\boxed{c_{T,\infty}^2 \simeq 0.94919.} \tag{7.29}$$

7.13 Flat-space tensor-to-scalar ratio

The primitive scalar singlet ground state is unaffected by the tensor-specific contraction and remains

$$c_{S,0}^2 = 0.4220071.$$

The tensor-to-scalar mass ratio in the flat limit is therefore

$$\frac{M_{T,0}}{M_{S,0}} = \sqrt{\frac{c_{T,\infty}^2}{c_{S,0}^2}}.$$

Using

$$c_{T,\infty}^2 = 0.94919$$

gives

$$\boxed{\frac{M_{T,0}}{M_{S,0}} = \sqrt{\frac{0.94919}{0.4220071}} \simeq 1.49974.} \tag{7.30}$$

Hence

$$\frac{M_{T,0}}{M_{S,0}} \simeq 1.50$$

The contraction therefore changes the tensor-scalar ratio from the heavy compact value

$$\frac{M_T}{M_S} \sim 2.33$$

to the flat value

$$\frac{M_T}{M_S} \sim 1.50$$

The primitive scalar ground state does not move. The reduction is entirely tensor-specific.

7.14 Why a moving chain map alone is insufficient

Suppose the closed quadratic form and its norm were both kept fixed,

$$K = K_0, \qquad B = B_0,$$

while only the embedded tensor subspace $\mathrm{Im}\, C_\alpha$ were allowed to move. Then the Rayleigh-Ritz principle implies that restricting a fixed quadratic form to a moving subspace cannot lower its smallest eigenvalue below the minimum of the full fixed problem. But the uncontracted one-sided tensor problem has a lowest value of approximately

$$c_T^2 \simeq 2.29,$$

whereas the flat tensor branch occurs near

$$c_{T,\infty}^2 \simeq 0.95.$$

Therefore the contraction cannot be represented solely by

$$\mathrm{Im}\, C_\alpha$$

inside a fixed Hilbert-space metric. The inner-product structure itself must move.

7.15 Physical interpretation of the reduction

The reduction from the compact tensor eigenvalue to the flat one consists of three conceptually distinct effects. First, the compact $SO(4)$ harmonic carries a positive angular energy. In the contraction

$$SO(4) \to ISO(3),$$

this contribution scales with

$$\alpha^2$$

and disappears in the flat limit. Second, the tensor harmonic is not an isolated one-component state. The contraction rotates it into additional closed-string tensor polarizations. Integrating out these intermediate components generates the negative 1PI-Feshbach correction

$$s_{\mathrm{hom}} \simeq -0.43.$$

Third, the contraction changes the degree-wise Riesz realization of the tensor complex. The physical reduced tensor representative therefore moves inside the homogeneous $(2,1)$-coefficient space, producing another negative correction

$$s_{\mathrm{hor}} \simeq -0.43.$$

Schematically,

$$\boxed{\text{compact angular cost} \longrightarrow \text{direct}} + \text{1PI homotopy} + \text{Riesz transport}.$$

Numerically,

$$\boxed{+1.54 - 0.43 - 0.43 \simeq +0.68.}$$

The physical contraction is therefore substantially softer than a naive removal or rescaling of the compact angular Casimir.

7.16 The Riesz transport is necessarily nonlocal

A purely local deformation of the 9×9 tensor Gram matrix is not sufficient. The compensator obeys

$$K\mathcal{G}_1 = M\mathcal{G}_0,$$

so formally

$$\mathcal{G}_1 = K_\perp^{-1} M \mathcal{G}_0.$$

The inverse

$$K_{\perp}^{-1}$$

is a Green operator. Consequently the contraction necessarily contains nonlocal radial information. The same structure appears in the 1PI correction,

$$-P\mathcal{A}Q(Q\mathcal{A}Q)^{-1}Q\mathcal{A}P.$$

Thus both the chain-map compensator and the effective Riesz correction involve resolvents of the tensor kinetic operator. The correct contracted tensor theory is therefore not obtained by a pointwise rescaling of the Stenzel coframe. It is an operator-valued transport on the tensor field space.

7.17 Effective 1PI tensor operator

Combining the preceding ingredients, the contracted quadratic tensor operator may be written schematically as

$$K_{\mathrm{eff}}(\alpha) = K_{\mathrm{flat}} + \alpha^2\left[K_{\mathrm{ang}} - V^{\dagger}(K_{\perp} - \lambda B)^{-1}V + \delta K_{\mathrm{Riesz}}\right] + O(\alpha^4). \tag{7.31}$$

The three terms inside the brackets correspond respectively to

$$K_{\mathrm{ang}}$$

for the direct compact angular contribution,

$$-V^{\dagger}(K_{\perp} - \lambda B)^{-1}V$$

for the 1PI-Green-function contribution, and

$$\delta K_{\mathrm{Riesz}}$$

for the degree-wise Riesz- or horizontal transport. Numerically, their contribution to the lowest eigenvalue is

$$1.54 - 0.43 - 0.43 \simeq 0.68.$$

This provides a quantitative target for an exact 1PI closed-string derivation of the contraction.

7.18 Final contracted tensor problem

The tensor contraction can therefore be summarized by the following sequence. The compact theory contains the chiral tensor block

$$(2,0) \oplus (0,2).$$

Introduce

$$P_i = \alpha M_{0i}.$$

Then

$$[P_i, P_j] = \alpha^2 \epsilon_{ijk} J_k,$$

and consequently

$$SO(4) \overset{\alpha \to 0}{\longrightarrow} ISO(3).$$

The ambient complex remains fixed,

$$\bar{\partial}_X^2 = 0.$$

The contraction is carried by the chain map,

$$\bar{\partial}_X C_\alpha = C_\alpha d_\alpha,$$

and by the degree-wise Riesz metrics,

$$G_{21}(\alpha), \qquad G_{22}(\alpha).$$

The one-sided adjoint is

$$\bar{\partial}_\alpha^\sharp = G_{21}(\alpha)^{-1} \bar{\partial}_X^* G_{22}(\alpha).$$

The tensor eigenproblem becomes

$$\bar{\partial}_\alpha^\sharp \bar{\partial}_X \mathcal{G}_T = c_T^2(\alpha) \mathcal{G}_T.$$

Near the flat endpoint,

$$c_T^2(\alpha) = c_{T,\infty}^2 + 0.67\, \alpha^2 + O(\alpha^4), \tag{7.32}$$

with

$$c_{T,\infty}^2 \simeq 0.94919.$$

Since the primitive scalar remains

$$c_{S,0}^2 = 0.4220071,$$

the resulting mass ratio is

$$\frac{M_{T,0}}{M_{S,0}} = \sqrt{\frac{0.94919}{0.4220071}} \simeq 1.49974 \simeq 1.50.$$

Thus the flat tensor-to-scalar ratio is obtained not by changing the Stenzel Dolbeault complex, but by the combined action of the $SO(4) \to ISO(3)$ contraction on the tensor representation, the chain homotopy of the embedding, and the degree-wise Riesz-1PI transport of the quadratic form. The numerical value

$$M_T / M_S \simeq 1.50$$

is a stable endpoint of the present contraction calculation. A complete analytic derivation still requires a fully cochain-exact tensor-only construction of the nonlocal Riesz transport that

reproduces the Green and horizontal corrections while leaving the primitive scalar block unchanged.

8. Yang–Mills Theory on $S^3 \times \mathbb{R}$

8.1. Yang–Mills theory on $S^3 \times \mathbb{R}$

We consider pure $SU(N)$ Yang–Mills theory on the Lorentzian spacetime

$$\mathcal{X}_4 = \mathbb{R}_t \times S^3_R,$$

where S^3_R denotes the round three-sphere of radius R. The compact spatial manifold is particularly convenient for the Hamiltonian formulation, since infrared boundary conditions at spatial infinity are absent and the physical configuration space can be formulated directly as the quotient of the space of spatial connections by the group of time-independent gauge transformations. The large-volume limit

$$R \to \infty$$

may subsequently be considered when relating the construction to Yang–Mills theory on $\mathbb{R}^{3,1}$.

Let

$$P \longrightarrow S^3$$

be a principal $SU(N)$-bundle. In the sector considered below we take the bundle to be topologically trivial and identify its adjoint bundle with

$$\mathrm{ad}\, P \simeq S^3 \times \mathfrak{su}(N). \tag{8.1}$$

A spatial gauge connection is then an $\mathfrak{su}(N)$-valued one-form,

$$A \in \Omega^1\big(S^3, \mathfrak{su}(N)\big). \tag{8.2}$$

The space $\mathcal{A}$ of connections is affine rather than linear. If

$$A_1, A_2 \in \mathcal{A},$$

then their difference is a well-defined element of

$$A_1 - A_2 \in \Omega^1\big(S^3, \mathfrak{su}(N)\big).$$

Consequently,

$$T_A\mathcal{A} \simeq \Omega^1\big(S^3, \mathfrak{su}(N)\big) \tag{8.3}$$

for every

$$A \in \mathcal{A}.$$

We use anti-Hermitian Lie-algebra conventions. Thus, for

$$X, Y \in \mathfrak{su}(N),$$

the expression

$$-\operatorname{Tr}(XY) \tag{8.4}$$

defines a positive invariant scalar product. The covariant derivative associated with a connection A is

$$D_A = d + [A, \cdot], \tag{8.5}$$

and its curvature is

$$F_A = dA + A \wedge A. \tag{8.6}$$

The Yang–Mills action is

$$S_{\mathrm{YM}} = -\frac{1}{2g_{\mathrm{YM}}^2} \int_{\mathcal{X}_4} \operatorname{Tr}(F \wedge *_4 F). \tag{8.7}$$

Equivalently, in local coordinates,

$$S_{\mathrm{YM}} = -\frac{1}{4g_{\mathrm{YM}}^2} \int_{\mathcal{X}_4} d^4x \sqrt{|g|}\, \operatorname{Tr}\left(F_{\mu\nu} F^{\mu\nu}\right).$$

The convention used here is important for what follows. The coupling g_{YM} has not been absorbed into the covariant derivative. Instead,

$$D = d + A,$$

and the coupling multiplies the field-space metric through the overall factor

$$\frac{1}{g_{\mathrm{YM}}^2}. \tag{8.8}$$

This convention makes the relation between the Yang–Mills kinetic term and the geometry of the space of connections particularly transparent. In temporal gauge,

$$A_0 = 0,$$

a four-dimensional connection is represented by a path

$$t \longmapsto A(t) \in \mathcal{A}.$$

The electric field is

$$F_{0i} = \dot{A}_i. \tag{8.9}$$

The Yang–Mills action then assumes Hamiltonian form, with kinetic term determined by the Hilbert–Schmidt norm of $\dot{A}$ and potential term determined by the spatial curvature F_A. Temporal gauge does not remove the full gauge symmetry. Time-independent transformations remain, and the equation obtained by varying A_0 before imposing temporal gauge yields Gauss' law,

$$D_A^i E_i = 0. \tag{8.10}$$

Thus the physical configuration variable is not A itself, but its equivalence class under the residual gauge group. The geometry of this quotient is the central subject of the present section.

8.2. The gauge group and the orbit space of connections

The gauge group is the infinite-dimensional group

$$\mathcal{G} = C^\infty\big(S^3, SU(N)\big). \tag{8.11}$$

It is important to distinguish $\mathcal{G}$ from the finite-dimensional structure group $SU(N)$. The latter acts in each fibre of the principal bundle, whereas an element $g \in \mathcal{G}$ assigns an $SU(N)$ matrix $g(x)$ to every point $x \in S^3$. The action of $\mathcal{G}$ on $\mathcal{A}$ is

$$A \mapsto A^g = g^{-1}Ag + g^{-1}dg. \tag{8.12}$$

Connections related in this way represent the same physical gauge field. The gauge orbit through A is

$$\mathcal{O}_A = \{A^g \mid g \in \mathcal{G}\}. \tag{8.13}$$

The classical configuration space of Yang–Mills theory is therefore the quotient

$$\mathcal{A}/\mathcal{G}. \tag{8.14}$$

A point of $\mathcal{A}/\mathcal{G}$ is not a connection, but an entire gauge orbit

$$\mathcal{O}_A.$$

Formally one may depict the construction as

$$\mathcal{G} \longrightarrow \mathcal{A} \longrightarrow \mathcal{A}/\mathcal{G}.$$

Locally, and away from connections with nontrivial stabilizers, this resembles a principal bundle whose fibres are gauge orbits. Globally the situation is more subtle. The action of $\mathcal{G}$ need not be free. Reducible connections may possess nontrivial stabilizer groups,

$$\mathrm{Stab}(A) = \{g \in \mathcal{G} \mid A^g = A\},$$

and the orbit space is consequently stratified rather than globally a smooth Hilbert manifold. For many analytical purposes one introduces a based gauge group. Choosing

$$x_0 \in S^3,$$

define

$$\mathcal{G}_0 = \{g \in \mathcal{G} \mid g(x_0) = 1\}. \tag{8.15}$$

This removes the residual constant gauge transformations and improves the freeness properties of the action. None of the local expressions below depends essentially on this choice, and we shall usually write $\mathcal{G}$ while understanding that an appropriate based or reduced gauge

group can be employed whenever necessary. For a fully functional-analytic formulation, the smooth spaces above should be replaced by Sobolev completions. For example, for

$$s > \frac{\dim S^3}{2} + 1 = \frac{5}{2},$$

one may take

$$\mathcal{A}^s = A_{\mathrm{ref}} + H^s\Omega^1\big(S^3, \mathfrak{su}(N)\big) \tag{8.16}$$

and

$$\mathcal{G}^{s+1} = H^{s+1}\big(S^3, SU(N)\big). \tag{8.17}$$

The Sobolev multiplication theorem then makes the gauge action sufficiently regular for the differential-geometric constructions below. We shall suppress these completions in the notation unless domain questions are relevant. The infinitesimal form of the gauge action is obtained by setting

$$g(t) = e^{t\omega}, \qquad \omega \in \Omega^0\big(S^3, \mathfrak{su}(N)\big).$$

Differentiation at $t = 0$ gives

$$\delta_\omega A = D_A\,\omega.$$

Consequently, the tangent space to the orbit through A is

$$T_A\mathcal{O}_A = \operatorname{im} D_A\,. \tag{8.18}$$

These are the vertical directions in field space. Formally, the tangent space to the physical quotient is therefore

$$T_{[A]}(\mathcal{A}/\mathcal{G}) \simeq \frac{\Omega^1\big(S^3, \mathfrak{su}(N)\big)}{\operatorname{im} D_A}. \tag{8.19}$$

The problem of gauge fixing is precisely the problem of selecting representatives of these equivalence classes. For an Abelian gauge theory a linear transversality condition essentially accomplishes this globally after the standard treatment of zero modes. For a non-Abelian gauge theory the corresponding global construction fails. Gribov showed [19] that conventional differential gauge conditions possess multiple gauge-equivalent solutions, and Singer [20] subsequently identified the global topological obstruction to constructing a single continuous gauge section over the full non-Abelian orbit space. The fundamental modular region provides a geometrically natural nonperturbative replacement for a globally unique linear gauge choice.

8.3 The Hilbert–Schmidt metric on the space of connections

The round metric on S^3 and the invariant scalar product on $\mathfrak{su}(N)$ induce a natural L^2-inner product on the tangent space of $\mathcal{A}$. For

$$a, b \in T_A\mathcal{A} \simeq \Omega^1\big(S^3, \mathfrak{su}(N)\big),$$

define

$$G_{\mathrm{HS}}(a,b) = -\int_{S^3} \mathrm{Tr}(a \wedge *_3 b). \tag{8.20}$$

Equivalently,

$$G_{\mathrm{HS}}(a,b) = -\int_{S^3} d^3x \sqrt{g}\, g^{ij}\, \mathrm{Tr}\big(a_i b_j\big).$$

For $a = b$, one has

$$\|a\|^2_{\mathrm{HS}} = -\int_{S^3} \mathrm{Tr}(a \wedge *_3 a) \geq 0. \tag{8.21}$$

In the mathematical gauge-theory literature this is usually called the L^2 norm of the connection. We shall also refer to it as the Hilbert–Schmidt norm because pointwise it is the Hilbert–Schmidt norm of the matrix-valued gauge field, followed by integration over space. This terminology will become useful when the large-N color algebra is represented as an infinite-dimensional Hilbert space. The metric is invariant under homogeneous adjoint transformations because of invariance of

$$\mathrm{Tr}(ab).$$

The affine term in the transformation of a connection means, however, that the norm

$$\|A\|^2_{\mathrm{HS}}$$

is not constant along a gauge orbit. This simple observation is the starting point for the construction of the fundamental modular region. For each

$$A \in \mathcal{A},$$

define the orbit functional

$$\mathcal{F}_A[g] = \frac{1}{2}\|A^g\|^2_{\mathrm{HS}}. \tag{8.22}$$

The gauge orbit is fixed and g is varied. Gauge fixing by norm minimization means searching for stationary points, and ultimately absolute minima, of this functional. The geometric interpretation is immediate. The origin

$$A = 0$$

defines a distinguished point of the affine coordinate system, and

$$\mathcal{F}_A[g]$$

measures the squared L^2 distance of a representative A^g from this origin. A minimum of $\mathcal{F}_A$ therefore picks a representative of the orbit that lies as close as possible to $A = 0$ in the Hilbert–Schmidt geometry. This is not merely a convenient gauge-fixing prescription. The same

Hilbert–Schmidt metric is precisely the metric appearing in the kinetic term of the Yang–Mills action,

$$\frac{1}{2g_{\mathrm{YM}}^2}\int dt\, G_{\mathrm{HS}}(\dot{A},\dot{A}). \tag{8.23}$$

Thus the metric used to define the modular region is also the basic field-space metric of the Hamiltonian theory.

8.4 First variation: the transverse gauge slice

Let

$$g(t) = e^{t\omega}.$$

At $t = 0$,

$$\frac{d}{dt}A^{g(t)}|_{t=0} = D_A\omega.$$

The first variation of the norm functional is therefore

$$\delta\mathcal{F}_A = G_{\mathrm{HS}}(A, D_A\omega). \tag{8.24}$$

Using invariance of the trace, the commutator contribution drops out and one obtains

$$-\int_{S^3} d^3x\,\sqrt{g}\ \mathrm{Tr}\big(A^i\nabla_i\omega\big).$$

After integration by parts,

$$\delta\mathcal{F}_A = \int_{S^3} d^3x\,\sqrt{g}\ \mathrm{Tr}\big[\big(\nabla^i A_i\big)\omega\big].$$

Thus stationarity for arbitrary ω requires

$$d^\dagger A = 0. \tag{8.25}$$

On flat space this is the familiar Coulomb or Landau condition,

$$\partial_i A_i = 0.$$

We denote the corresponding transverse gauge slice by

$$\Gamma = \{A \in \mathcal{A} \mid d^\dagger A = 0\} \tag{8.26}$$

Every absolute minimum of the norm functional necessarily lies in Γ. The converse is false. A transverse configuration need only be a stationary point of the norm functional and may represent a local minimum, a maximum, or a saddle point along its gauge orbit. This is the first manifestation of the Gribov problem. The equation

$$d^\dagger A = 0$$

does not globally pick exactly one representative of each non-Abelian gauge orbit.

8.5 Second variation and the Faddeev–Popov operator

The nature of a stationary point is determined by the second variation of $\mathcal{F}_A$ along gauge directions. At a transverse connection

$$A \in \Gamma,$$

the Hessian of the orbit functional takes the form

$$\delta^2 \mathcal{F}_A = \langle \omega, \mathcal{M}_A \omega \rangle,$$

where

$$\mathcal{M}_A = -d^\dagger D_A \tag{8.27}$$

is the Faddeev–Popov operator in Coulomb/Landau gauge. In local coordinates,

$$\mathcal{M}_A = -\nabla^i D_{A,i}.$$

On a transverse configuration this operator is formally self-adjoint with respect to the scalar L^2 product

$$-\int_{S^3} d^3x \sqrt{g}\ \mathrm{Tr}(\omega_1 \omega_2). \tag{8.28}$$

The precise operator

$$-d^\dagger D_A$$

should be distinguished from the positive operator

$$D_A^\dagger D_A.$$

The two coincide only in special situations. For the norm-minimization construction of the Landau-Coulomb fundamental modular region,

$$-d^\dagger D_A$$

is the actual orbit Hessian and hence the relevant Faddeev–Popov operator. A stationary point is a local minimum whenever

$$\mathcal{M}_A \geq 0.$$

This motivates the definition of the first Gribov region,

$$\Omega = \{A \in \Gamma \mid \mathcal{M}_A \geq 0\}. \tag{8.29}$$

Its boundary,

$$\partial\Omega,$$

is the first Gribov horizon. Generically it is characterized by the appearance of a zero eigenvalue of the Faddeev–Popov operator,

$$\lambda_{\min}(\mathcal{M}_A) = 0.$$

Inside Ω, the norm functional is locally convex along infinitesimal gauge directions. Restriction to Ω therefore removes stationary points with a negative orbit Hessian. It does not, however, completely remove Gribov copies. Distinct gauge-equivalent configurations can both belong to the first Gribov region because both can be local minima of the orbit norm. The distinction between local and global minimization is therefore essential.

8.6 The fundamental modular region

The fundamental modular region [21] is defined by absolute rather than merely local minimization. For each gauge orbit

$$\mathcal{O}_A,$$

consider

$$\inf_{g\in\mathcal{G}} \|A^g\|_{\mathrm{HS}}^2.$$

A representative

$$A_{\min} \in \mathcal{O}_A$$

belongs to the fundamental modular region if

$$\|A_{\min}\|_{\mathrm{HS}}^2 \le \left\|A_{\min}^g\right\|_{\mathrm{HS}}^2 \qquad \forall g \in \mathcal{G}. \tag{8.30}$$

Thus

$$\Lambda = \{A \in \mathcal{A} \mid \|A\|_{\mathrm{HS}} \le \|A^g\|_{\mathrm{HS}} \quad \forall g \in \mathcal{G}\}. \tag{8.31}$$

Because an absolute minimum is necessarily stationary,

$$\Lambda \subset \Gamma,$$

and because its Hessian cannot possess a negative direction,

$$\Lambda \subset \Omega. \tag{8.32}$$

The hierarchy is therefore

$$\Lambda \subset \Omega \subset \Gamma \subset \mathcal{A}. \tag{8.33}$$

The physical meaning of the nested regions is worth emphasizing:

Γ: stationary representatives,

Ω: local minima along gauge orbits,

Λ: absolute minima along gauge orbits.

Dell'Antonio and Zwanziger [22] proved, in an appropriate L^2 completion, that gauge orbits are closed and that the Hilbert norm attains an absolute minimum on every orbit. In particular, each orbit admits a representative lying within the region bounded by the Gribov horizon. This result supplies the mathematical basis for interpreting the fundamental modular region

as a global gauge-reduced representative space. One should nevertheless not identify Λ naively with a smooth copy of

$$\mathcal{A}/\mathcal{G}.$$

Absolute minima need not be unique. If two gauge-equivalent configurations attain the same absolute minimum, they represent distinct points on the boundary of a chosen fundamental domain that must be identified. Schematically,

$$\mathcal{A}/\mathcal{G} \simeq \Lambda/\sim_{\partial\Lambda}, \tag{8.34}$$

where $\sim_{\partial\Lambda}$ denotes the residual identifications of gauge-equivalent boundary points. This phenomenon is familiar from ordinary finite-dimensional quotient spaces. A fundamental domain can contain one interior representative of each generic orbit while equivalent representatives meet along its boundary. The FMR should be understood in this sense. The significance of the construction is that the nonperturbative physical configuration space is encoded not by imposing a local differential equation alone, but by a global variational condition,

$$A \text{ is a physical representative} \iff A \text{ absolutely minimizes } \|A^g\|_{\mathrm{HS}} \text{ on its orbit.}$$

This makes the Hilbert–Schmidt geometry intrinsic to the global gauge reduction.

8.7 Horizontal and vertical directions

The norm-minimization picture can also be understood infinitesimally in terms of a decomposition of tangent directions. The vertical space at A is

$$V_A = \operatorname{im} D_A. \tag{8.35}$$

A tangent vector

$$a \in T_A\mathcal{A}$$

is orthogonal to every vertical vector if

$$G_{\mathrm{HS}}(a, D_A\omega) = 0 \qquad \forall\omega.$$

Hence

$$D_A^\dagger a = 0. \tag{8.36}$$

Whenever the relevant elliptic decomposition is valid,

$$\Omega^1\big(S^3, \mathfrak{su}(N)\big) = \operatorname{im} D_A \oplus \ker D_A^\dagger,$$

up to possible zero modes and stabilizer directions. Thus

$$H_A = \ker D_A^\dagger \tag{8.37}$$

is the Hilbert–Schmidt orthogonal complement to the gauge orbit. It provides a local horizontal model for the tangent space of the orbit space,

$$T_{[A]}(\mathcal{A}/\mathcal{G}) \simeq H_A. \tag{8.38}$$

This local orthogonal decomposition should not be confused with the global Coulomb condition

$$d^\dagger A = 0.$$

The latter is the stationarity condition for a connection itself under minimization of its orbit norm. The former describes horizontal fluctuations relative to a fixed background A. Both arise from the same Hilbert–Schmidt geometry but play different roles. The Faddeev–Popov operator controls the solvability of the corresponding horizontal projection. Given a general variation a, one seeks a gauge parameter ω such that

$$a + D_A\omega$$

obeys the desired gauge-fixing condition. The equation for ω involves the inverse of the Faddeev–Popov operator. Consequently, the projection becomes singular when $\mathcal{M}_A$ develops a zero mode, explaining geometrically why the Gribov horizon is a natural boundary of a local gauge coordinate system. This observation will become important later when the corresponding horizontal projection is transported to the large-N geometric representation.

8.8 The FMR and the Hamiltonian formulation

The FMR is especially natural in the Hamiltonian formulation. In temporal gauge, the kinematical wave functional is formally

$$\Psi[A].$$

Gauss' law requires physical states to be invariant under gauge transformations connected to the identity,

$$\Psi[A^g] = \Psi[A].$$

Thus a physical wave functional is fundamentally a function on the quotient

$$\mathcal{A}/\mathcal{G}$$

rather than on $\mathcal{A}$. One may therefore equivalently regard the state as a function defined on a fundamental modular region subject to the appropriate boundary identifications,

$$\Psi_{\text{phys}} : \Lambda/\sim \longrightarrow \mathbb{C}. \tag{8.39}$$

This makes the FMR directly relevant to the nonperturbative Schrödinger representation of Yang–Mills theory. Hamiltonian approaches to the Gribov problem exploit precisely the fact that the true configuration space is a quotient with nontrivial global geometry rather than an unconstrained linear space. The Yang–Mills kinetic energy is formally a Laplace-type operator on this configuration space. At the classical level the relevant Riemannian structure is inherited from the kinetic term,

$$\frac{1}{g_{\mathrm{YM}}^2} G_{\mathrm{HS}}. \tag{8.40}$$

Consequently both gauge reduction and quantum propagation depend on the same underlying metric. This point is essential for the later construction. The FMR is not introduced merely to eliminate a redundancy in notation. It provides a global representative of the gauge-reduced configuration space on which the Hilbert–Schmidt metric descends to the physical directions. The quantum kinetic operator must ultimately be understood relative to this reduced metric. The appearance of the coupling constant in this metric is particularly important. In the convention

$$D = d + A,$$

the gauge transformation law

$$A^g = g^{-1}Ag + g^{-1}dg$$

is independent of the coupling, while

$$g_{\mathrm{YM}}^{-2}$$

appears as an overall coefficient multiplying the metric of the connection space. Thus the renormalization of the Yang–Mills coupling may be viewed geometrically as a scale dependence of the Riemannian metric on the space of gauge connections. At a renormalization scale μ,

$$\boxed{G_{\mathrm{phys}}(\mu) = \frac{1}{g_{\mathrm{YM}}^2(\mu)} G_{\mathrm{HS}}.} \tag{8.41}$$

This observation does not alter the definition of the gauge orbit or of the FMR at a fixed scale: multiplication of a positive norm by a positive scalar does not change the location of its minima. It does, however, change the physical Riesz metric that converts quadratic forms into operators and therefore becomes relevant when the gauge-reduced theory is represented spectrally.

8.9 The 't Hooft normalization

For the large-N limit it is useful to separate the universal planar normalization from the running physical metric. Define the 't Hooft coupling

$$\lambda = g_{\mathrm{YM}}^2 N \tag{8.42}$$

and the normalized color trace

$$\mathrm{tr}_N = \frac{1}{N}\mathrm{Tr}. \tag{8.43}$$

The action can then be written schematically as

$$S_{\mathrm{YM}} = N^2 \frac{1}{2\lambda} \int \mathrm{tr}_N (F \wedge * F). \tag{8.44}$$

The overall factor N^2 measures the number of planar color degrees of freedom. After factoring out this universal normalization, the natural normalized metric is

$$\frac{1}{\lambda} G_{\mathrm{HS}}^{(N)}(a,b), \tag{8.45}$$

where

$$G_{\mathrm{HS}}^{(N)}(a,b) = -\int_{S^3} \mathrm{tr}_N (a \wedge *_3 b).$$

Thus the running field-space Riesz structure in the planar normalization is controlled by

$$\mathcal{R}_{\mathrm{YM}}(\mu) \propto \frac{1}{\lambda(\mu)}. \tag{8.46}$$

Again, one must distinguish this statement from a claim that $1/\lambda$ is by itself a coordinate-independent observable on field space. A canonical rescaling of the gauge potential can move factors of the coupling between the quadratic and interaction terms. Our convention is instead adapted to the geometry of connections: A transforms as an actual connection,

$$A^g = g^{-1} A g + g^{-1} dg,$$

and the coupling therefore appears in the metric rather than in the affine gauge transformation law. This is the convention in which the FMR construction and the subsequent large-N geometric representation fit most naturally together.

8.10 Why the FMR is the appropriate nonperturbative starting point

Perturbation theory expands around

$$A = 0$$

and effectively treats a sufficiently small neighbourhood of the vacuum as though the local gauge condition supplied unique coordinates. In this regime the global structure of the orbit space is largely invisible. Nonperturbatively this is insufficient. The existence of Gribov copies means that a local gauge condition such as

$$d^\dagger A = 0$$

does not globally parametrize physical gauge fields. The actual physical configuration space is the orbit space $\mathcal{A}/\mathcal{G}$, and the FMR provides a variational realization of that quotient based on the same Hilbert–Schmidt metric that appears in the Yang–Mills kinetic term. Three aspects of this construction are particularly important for the analysis to follow. First, the FMR is defined before any perturbative expansion. The condition

$$\|A\|_{\mathrm{HS}} \leq \|A^g\|_{\mathrm{HS}} \qquad \forall g \in \mathcal{G}$$

is global along the gauge orbit. It therefore retains information that is invisible to the local Faddeev–Popov construction around

$$A = 0.$$

Second, the FMR is intrinsically metric. Its definition depends on the Hilbert–Schmidt geometry of connection space. The same geometry determines the physical kinetic term. Consequently, a representation that preserves the Hilbert–Schmidt pairing on the gauge-reduced tangent space preserves more than a set of field configurations: it preserves the quadratic geometric structure from which the kinetic operator is constructed. Third, the distinction between the FMR and the first Gribov region separates global gauge reduction from positivity of the local orbit Hessian. The Gribov region Ω is characterized by

$$\mathcal{M}_A \geq 0,$$

whereas the FMR Λ is characterized by absolute minimization. A representation of the full nonperturbative gauge-reduced theory should therefore ultimately act on the physical quotient represented by Λ, rather than merely on the linearized transverse gauge slice. For the present work this leads to the following viewpoint: the fundamental object of planar Yang--Mills theory is the Hilbert-space geometry of the FMR-reduced connection space. The subsequent sections will construct a large-N representation of this geometry. The first step will be to resolve the $SU(N)$ color algebra in a complete fuzzy-spherical-harmonic basis.

In the limit

$$N \to \infty,$$

the finite upper angular-momentum cutoff recedes to infinity, producing a complete L^2 color space rather than a truncation of the gauge degrees of freedom. The second step will be to identify how the normalized Hilbert–Schmidt metric, including its running factor

$$\frac{1}{\lambda(\mu)},$$

is represented in the resulting geometric description. It is useful to emphasize what is and is not being assumed at this stage. The construction of

$$\mathcal{A}, \qquad \mathcal{G}, \qquad \mathcal{A}/\mathcal{G}, \qquad \Omega, \qquad \Lambda$$

is standard non-Abelian gauge theory. No open-closed-string correspondence is required. In particular, the FMR is defined entirely by the action of the gauge group on the space of connections and the Hilbert–Schmidt metric,

$$(\mathcal{A}, \mathcal{G}, G_{\mathrm{HS}}).$$

The new content developed later will concern a representation of this already-defined gauge-reduced metric space. We may summarize the standard geometry schematically as

$$\mathcal{A} \longrightarrow \mathcal{A}/\mathcal{G} \simeq \Lambda/\sim_{\partial\Lambda},$$

with

$$\Lambda \subset \Omega \subset \Gamma \subset \mathcal{A}.$$

The differential geometry of the orbit is controlled infinitesimally by

$$\delta_\omega A = D_A \omega,$$

and its local stability under norm minimization is controlled by

$$\mathcal{M}_A = -d^\dagger D_A.$$

The physical field-space metric, in the connection convention used here, is

$$\frac{1}{g_{\mathrm{YM}}^2(\mu)} G_{\mathrm{HS}},$$

or, after normalized large-N scaling,

$$\frac{1}{\lambda(\mu)} G_{\mathrm{HS}}^{(N)}.$$

These equations provide the starting point for the large-N construction developed below.

9 Large-N Color Space as a Fuzzy Sphere and Its Hilbert-Space Limit

9.1. Purpose of the construction

The previous section formulated pure $SU(N)$ Yang–Mills theory in terms of the space of spatial connections

$$\Omega^1\big(S^3, \mathfrak{su}(N)\big),$$

acted upon by the gauge group

$$C^\infty\big(S^3, SU(N)\big),$$

and identified the fundamental modular region as the set of absolute minima of the Hilbert–Schmidt norm along gauge orbits. The physical configuration space is represented by the quotient

$$\mathcal{A}_N/\mathcal{G}_N \simeq \Lambda_N/\sim_{\partial\Lambda_N}.$$

The purpose of the present section is to construct a geometric representation of the color degrees of freedom which remains complete as

$$N \to \infty.$$

The construction has two logically distinct components. First, at every finite N, the matrix algebra may be decomposed exactly into irreducible spherical-harmonic sectors under a distinguished

$$SU(2) \subset SU(N).$$

No degrees of freedom are discarded in this step. The$N^2 - 1$ generators of $\mathfrak{su}(N)$ are simply rewritten in a basis $\hat{Y}_{\ell m}^{(N)}$, with

$$1 \leq \ell \leq N-1, \qquad -\ell \leq m \leq \ell.$$

Second, the upper angular-momentum cutoff

$$\ell_{\max} = N - 1$$

moves to infinity when

$$N \to \infty.$$

With the normalized Hilbert–Schmidt metric, the resulting Hilbert-space completion is naturally identified with the space of square-integrable zero-average functions on a two-sphere,

$$\overline{\mathfrak{su}(\infty)}^{\mathrm{HS}} \simeq L_0^2(S^2). \tag{9.1}$$

Here

$$L_0^2(S^2) = \left\{ f \in L^2(S^2) \;\middle|\; \int_{S^2} f \; d\mu = 0 \right\}. \tag{9.2}$$

The absence of the constant harmonic is the large-N remnant of the tracelessness condition defining $SU(N)$ rather than $U(N)$. The resulting sphere will be referred to as the color sphere,

$$S^2_{\mathrm{col}}.$$

At finite N, this terminology denotes a noncommutative or fuzzy representation of the color matrix algebra. At $N = \infty$, the Hilbert-space completion becomes an ordinary L^2 space on

$$S^2_{\mathrm{col}}.$$

It is important from the outset to distinguish two kinds of large-N statements. The Hilbert-space statement

$$\mathfrak{su}(N) \longrightarrow L_0^2(S^2) \tag{9.3}$$

is obtained simply from completeness and orthogonality of the fuzzy harmonic basis. The stronger algebraic statement [23]

$$\frac{1}{i\hbar_N}[A, B] \longrightarrow \{a, b\}_{S^2} \tag{9.4}$$

is a semiclassical statement about suitably regular sequences of matrices and requires additional control of the large-N limit. The latter will be discussed below but is not required

for the basic Hilbert–Schmidt isometry. This distinction allows the complete color Hilbert space to be retained without assuming that every sequence of ultraviolet matrix modes possesses a uniform classical Poisson limit.

9.2 The irreducible $SU(2)$ embedding

Let

$$N = 2j + 1, \qquad j = \frac{N-1}{2}.$$

Consider the irreducible spin-j representation V_j of $SU(2)$. Its dimension is

$$\dim V_j = 2j + 1 = N.$$

Choosing an orthonormal basis

$$\{|j, m\rangle\}_{m=-j}^{j}, \tag{9.5}$$

the usual angular-momentum operators

$$J_1, \qquad J_2, \qquad J_3$$

satisfy

$$[J_a, J_b] = i\epsilon_{abc} J_c \tag{9.6}$$

and

$$\sum_{a=1}^{3} J_a^2 = j(j+1)\mathbf{1}_N. \tag{9.7}$$

Because these operators act irreducibly on an N-dimensional complex vector space, they define an embedding

$$\iota_N : \mathfrak{su}(2) \hookrightarrow \mathfrak{su}(N). \tag{9.8}$$

This is the irreducible, or principal, $SU(2)$ embedding relevant for the fuzzy two-sphere. The choice of such an embedding does not introduce an additional physical color direction. Any two irreducible N-dimensional unitary representations of $SU(2)$ are unitarily equivalent. Consequently, two such embeddings are related by conjugation,

$$J_a \longmapsto U J_a U^{-1}, \qquad U \in SU(N),$$

up to an irrelevant central phase in the corresponding $U(N)$ representation. Since the normalized Hilbert–Schmidt metric is invariant under conjugation,

$$\mathrm{tr}_N\left(A^\dagger B\right),$$

the particular representative of the principal embedding does not affect the color Hilbert-space metric. The fuzzy-sphere decomposition should therefore be regarded as a choice of coordinates in color space rather than an explicit breaking of the $SU(N)$ gauge symmetry.

9.3 Fuzzy coordinates

Define normalized matrix coordinates

$$\hat{x}_a^{(N)} = \frac{J_a}{\sqrt{j(j+1)}}. \tag{9.9}$$

Then

$$\sum_{a=1}^{3} \left(\hat{x}_a^{(N)}\right)^2 = \mathbf{1}_N.$$

Thus the matrices obey the analogue of the defining equation of a unit two-sphere,

$$x_1^2 + x_2^2 + x_3^2 = 1.$$

They fail to commute,

$$\left[\hat{x}_a^{(N)}, \hat{x}_b^{(N)}\right] = i\hbar_N \epsilon_{abc} \hat{x}_c^{(N)}, \tag{9.10}$$

where

$$\hbar_N = \frac{1}{\sqrt{j(j+1)}}. \tag{9.11}$$

Since

$$j = \frac{N-1}{2},$$

we have

$$\hbar_N \sim \frac{2}{N}$$

as $N \to \infty$. Therefore

$$\hbar_N \to 0.$$

The matrix coordinates commute in this limit, while their quadratic constraint remains fixed. This is the elementary origin of the classical sphere underlying the fuzzy-sphere construction. The effective noncommutativity parameter is therefore proportional to N^{-1}. The large-N limit plays the role of a semiclassical limit in color space. For later use, define the adjoint $SU(2)$ action on a matrix

$$M \in \mathrm{End}\left(V_j\right)$$

by

$$M \longmapsto [J_a, M].$$

The corresponding quadratic Casimir is

$$\mathcal{C}_2 M = \sum_{a=1}^{3} [J_a, [J_a, M]]. \tag{9.12}$$

This operator will provide the finite-N analogue of the spherical Laplacian.

9.4 Exact decomposition of the matrix algebra

The key representation-theoretic fact is the decomposition

$$\mathrm{End}(V_j) \simeq V_j \otimes V_j^*. \tag{9.13}$$

Because $SU(2)$ representations are self-dual,

$$V_j^* \simeq V_j,$$

and the Clebsch–Gordan decomposition gives

$$V_j \otimes V_j \simeq \bigoplus_{\ell=0}^{2j} V_\ell.$$

Since

$$2j = N - 1,$$

we obtain the exact decomposition [24]

$$\mathrm{End}(V_j) \simeq \bigoplus_{\ell=0}^{N-1} V_\ell\,. \tag{9.14}$$

Each spin-ℓ representation occurs exactly once. A basic dimension check gives

$$\sum_{\ell=0}^{N-1} (2\ell + 1) = N^{2.}$$

Indeed,

$$1 + 3 + 5 + \cdots + (2N - 1) = N^{2.}$$

Thus the harmonic decomposition contains exactly the correct number of degrees of freedom to span the complete complex matrix algebra. The $\ell = 0$ representation is generated by the identity matrix,

$$\mathbb{C}\mathbf{1}_N.$$

Consequently the traceless matrix algebra is

$$\mathfrak{sl}_N(\mathbb{C}) \simeq \bigoplus_{\ell=1}^{N-1} V_\ell\,. \tag{9.15}$$

The compact real form $\mathfrak{su}(N)$ is obtained by imposing the anti-Hermiticity condition. Its real dimension is N^2-1. Hence, as an $SU(2)$-module,

$$\mathfrak{su}(N) \simeq \bigoplus_{\ell=1}^{N-1} V_\ell . \tag{9.16}$$

This equation is the central finite-N statement. The fuzzy-spherical-harmonic expansion is not a truncation of $\mathfrak{su}(N)$. It is an exact decomposition of all N^2-1 color generators.

9.5 Fuzzy spherical harmonics

For each

$$0 \le \ell \le N-1, \qquad -\ell \le m \le \ell,$$

choose matrices

$$\hat{Y}_{\ell m}^{(N)} \in \mathrm{Mat}_N(\mathbb{C}) \tag{9.17}$$

forming an irreducible tensor operator of angular momentum ℓ under the adjoint action of the embedded $SU(2)$. They are characterized by

$$\left[J_3, \hat{Y}_{\ell m}^{(N)}\right] = m\hat{Y}_{\ell m}^{(N)},$$

and

$$\left[J_\pm, \hat{Y}_{\ell m}^{(N)}\right] = \sqrt{(\ell \mp m)(\ell \pm m + 1)}\, \hat{Y}_{\ell, m\pm 1}^{(N)}.$$

Equivalently,

$$\sum_{a=1}^{3} \left[J_a, \left[J_a, \hat{Y}_{\ell m}^{(N)}\right]\right] = \ell(\ell+1)\hat{Y}_{\ell m}^{(N)}. \tag{9.18}$$

This is exactly the eigenvalue equation satisfied by ordinary spherical harmonics,

$$-\Delta_{S^2} Y_{\ell m} = \ell(\ell+1) Y_{\ell m}. \tag{9.19}$$

The fuzzy harmonics may be constructed explicitly from Clebsch–Gordan coefficients, but their transformation law and orthogonality are sufficient for most of our purposes. We choose their normalization so that

$$\mathrm{tr}_N \left[\left(\hat{Y}_{\ell m}^{(N)}\right)^\dagger \hat{Y}_{\ell' m'}^{(N)}\right] = \delta_{\ell\ell'}\delta_{mm'}, \tag{9.20}$$

where

$$\mathrm{tr}_N = \frac{1}{N}\mathrm{Tr}.$$

The identity mode may then be chosen as

$$\hat{Y}_{00}^{(N)} = \mathbf{1}_N.$$

The traceless modes are precisely those with

$$\ell \geq 1.$$

Every complex $N \times N$ matrix admits the unique expansion

$$M = \sum_{\ell=0}^{N-1} \sum_{m=-\ell}^{\ell} M_{\ell m} \, \hat{Y}_{\ell m}^{(N)}. \tag{9.21}$$

For a traceless matrix,

$$M_{00} = 0,$$

so

$$M = \sum_{\ell=1}^{N-1} \sum_{m=-\ell}^{\ell} M_{\ell m} \, \hat{Y}_{\ell m}^{(N)}.$$

The coefficients follow directly from orthogonality,

$$M_{\ell m} = \mathrm{tr}_N \left[\left(\hat{Y}_{\ell m}^{(N)} \right)^{\dagger} M \right]. \tag{9.22}$$

For an anti-Hermitian

$$M \in \mathfrak{su}(N),$$

the coefficients satisfy the corresponding reality condition relating m and $-m$. One may either work directly with a real anti-Hermitian harmonic basis or, as we shall generally do, work in the complexification and impose the real form at the end.

9.6 The normalized Hilbert–Schmidt norm in harmonic variables

Let

$$M = \sum_{\ell m} M_{\ell m} \, \hat{Y}_{\ell m}^{(N)}.$$

Using orthogonality,

$$\mathrm{tr}_N \left(M^{\dagger} M \right) = \sum_{\ell, m} |M_{\ell m}|^2. \tag{9.23}$$

For traceless matrices,

$$\mathrm{tr}_N \left(M^{\dagger} M \right) = \sum_{\ell=1}^{N-1} \sum_{m=-\ell}^{\ell} |M_{\ell m}|^2. \tag{9.24}$$

Thus the normalized Hilbert–Schmidt geometry becomes an ordinary Euclidean ℓ^2 geometry on the harmonic coefficients. This fact is particularly important for the fundamental modular region. The nonlinear complexity of the gauge orbit is not removed by the harmonic basis, but

the metric used to measure distance along that orbit becomes diagonal in the color harmonic labels.

The change of color coordinates therefore separates two structures:

$$\boxed{\text{metric} \quad \longrightarrow \quad \text{diagonal,}}$$

while

$$\boxed{\text{gauge orbit} \quad \longrightarrow \quad \text{nonlinear mode coupling.}}$$

This separation will be central in taking the large-N limit.

9.7 Ordinary spherical harmonics and the symbol map

Let

$$d\mu = \frac{d\Omega}{4\pi}$$

denote the normalized area measure on the unit sphere,

$$\int_{S^2} d\mu = 1.$$

Choose spherical harmonics

$$Y_{\ell m}(\Omega)$$

normalized with respect to this measure,

$$\int_{S^2} d\mu \, \overline{Y_{\ell m}} Y_{\ell' m'} = \delta_{\ell \ell'} \delta_{m m'}.$$

In particular,

$$Y_{00} = 1.$$

Define the finite-N harmonic symbol map

$$\sigma_N : \mathrm{Mat}_N(\mathbb{C}) \longrightarrow \mathcal{H}_N(S^2) \tag{9.25}$$

by

$$\sigma_N\left(\hat{Y}_{\ell m}^{(N)}\right) = Y_{\ell m}. \tag{9.26}$$

Here

$$\mathcal{H}_N(S^2) = \mathrm{span}\{Y_{\ell m} : 0 \le \ell \le N-1\}.$$

Thus, if

$$M = \sum_{\ell=0}^{N-1} \sum_{m} M_{\ell m} \hat{Y}_{\ell m}^{(N)},$$

then

$$\sigma_N(M) = \sum_{\ell=0}^{N-1} \sum_{m} M_{\ell m}\, Y_{\ell m}(\Omega). \tag{9.27}$$

Since both bases are orthonormal,

$$\mathrm{tr}_N\big(M^\dagger K\big) = \int_{S^2} d\mu\, \overline{\sigma_N(M)}\sigma_N(K). \tag{9.28}$$

This identity is exact at every finite N.

It follows that

$$\sigma_N \colon (\mathrm{Mat}_N(\mathbb{C}), \langle\cdot,\cdot\rangle_{\mathrm{HS}}) \longrightarrow \mathcal{H}_N(S^2) \tag{9.29}$$

is a unitary isomorphism of finite-dimensional Hilbert spaces. Restricting to traceless matrices gives

$$\sigma_N \colon \mathfrak{sl}_N(\mathbb{C}) \longrightarrow \mathcal{H}_N^0(S^2), \tag{9.30}$$

where

$$\mathcal{H}_N^0(S^2) = \mathrm{span}\{Y_{\ell m} \colon 1 \le \ell \le N-1\}.$$

The constant mode is absent because

$$\mathrm{tr}_N\, M = 0.$$

This provides a precise meaning to the statement that the $SU(N)$ color algebra can be represented by functions on a fuzzy sphere. At finite N, however, ordinary pointwise multiplication of the symbols does not reproduce matrix multiplication [25]. The image is a Hilbert-space representation, but the algebra structure remains noncommutative.

9.8 Exact finite-N star product

The matrix product can be transported through the symbol map. For

$$f, g \in \mathcal{H}_N(S^2),$$

define

$$f \star_N g = \sigma_N[\sigma_N^{-1}(f)\sigma_N^{-1}(g)]. \tag{9.31}$$

Then

$$(\mathcal{H}_N(S^2), \star_N) \tag{9.32}$$

is exactly isomorphic as an associative algebra to

$$\mathrm{Mat}_N(\mathbb{C}).$$

Thus

$$\sigma_N(MK) = \sigma_N(M) \star_N \sigma_N(K).$$

Associativity follows immediately from matrix associativity,

$$f \star_N (g \star_N h) = (f \star_N g) \star_N h.$$

Products of fuzzy harmonics take the form

$$\hat{Y}^{(N)}_{\ell_1 m_1} \hat{Y}^{(N)}_{\ell_2 m_2} = \sum_{\ell,m} C^{(N),\ell m}_{\ell_1 m_1, \ell_2 m_2} \hat{Y}^{(N)}_{\ell m},$$

where the coefficients can be expressed through Clebsch–Gordan coefficients and $6j$-symbols. The corresponding symbol product is

$$Y_{\ell_1 m_1} \star_N Y_{\ell_2 m_2} = \sum_{\ell,m} C^{(N),\ell m}_{\ell_1 m_1, \ell_2 m_2} Y_{\ell m}.$$

Thus the finite-N fuzzy sphere contains all matrix multiplication information. The normalized trace becomes the canonical integral of this noncommutative algebra. Since the identity maps to the constant function 1,

$$\mathrm{tr}_N M = \int_{S^2} d\mu \, \sigma_N(M). \tag{9.33}$$

Consequently,

$$\mathrm{tr}_N(MK) = \int_{S^2} d\mu \, [\sigma_N(M) \star_N \sigma_N(K)]. \tag{9.34}$$

Cyclicity of the matrix trace gives

$$\int f \star_N g \, d\mu = \int g \star_N f \, d\mu.$$

The finite-N matrix algebra has therefore been rewritten as a noncommutative function algebra carrying a canonical integration functional. No approximation has entered.

9.9 Gauge connections in fuzzy harmonic variables

A spatial $SU(N)$ gauge connection may now be expanded as

$$A_i(x) = \sum_{\ell=1}^{N-1} \sum_{m=-\ell}^{\ell} A_{i;\ell m}(x) \hat{Y}^{(N)}_{\ell m}. \tag{9.35}$$

Likewise an infinitesimal gauge parameter takes the form

$$\omega(x) = \sum_{\ell=1}^{N-1} \sum_{m} \omega_{\ell m}(x) \hat{Y}^{(N)}_{\ell m}.$$

All $N^2 - 1$ color degrees of freedom occur in these sums. The gauge variation

$$\delta_\omega A_i = \partial_i \omega + [A_i, \omega]$$

becomes

$$\delta A_{i;\ell m} = \partial_i \omega_{\ell m} + \sum_{\ell_1 m_1} \sum_{\ell_2 m_2} f^{(N),\ell m}_{\ell_1 m_1, \ell_2 m_2} A_{i;\ell_1 m_1} \omega_{\ell_2 m_2},$$

where

$$f^{(N),\ell m}_{\ell_1 m_1, \ell_2 m_2}$$

are the structure constants of $\mathfrak{su}(N)$ expressed in the fuzzy harmonic basis. Thus changing to spherical color variables does not Abelianize the theory. The non-Abelian interaction is transferred into nonlinear couplings between color harmonics. Applying the symbol map gives

$$a_i^{(N)}(x, \Omega) = \sigma_N\big(A_i(x)\big),$$

or explicitly,

$$a_i^{(N)}(x, \Omega) = \sum_{\ell=1}^{N-1} \sum_{m} A_{i;\ell m}(x) Y_{\ell m}(\Omega). \tag{9.36}$$

Similarly,

$$\epsilon^{(N)} = \sigma_N(\omega).$$

The gauge transformation becomes

$$\delta a_i^{(N)} = \partial_i \epsilon^{(N)} + \left[a_i^{(N)}, \epsilon^{(N)}\right]_{\star_N}, \tag{9.37}$$

where

$$[f, g]_{\star_N} = f \star_N g - g \star_N f.$$

This equation is exactly equivalent to the matrix gauge transformation. Therefore a finite-N Yang–Mills field can be viewed equivalently as a one-form on S^3 whose color dependence is described by a noncommutative function on a fuzzy two-sphere. We emphasize that S^2_{col} is not an additional physical spatial dimension at this stage. It is a representation of the color matrix algebra.

9.10 The Hilbert–Schmidt metric becomes an S^2 integral

The normalized Hilbert–Schmidt metric on the space of connections is

$$G_{\mathrm{HS},N}(a, b) = -\int_{S^3} \mathrm{tr}_N(a \wedge *_3 b).$$

Expand

$$a_i = \sum_{\ell m} a_{i;\ell m} \hat{Y}^{(N)}_{\ell m}$$

and similarly for b. Orthogonality gives

$$G_{\mathrm{HS},N}(a,b) = \int_{S^3} d^3x\,\sqrt{g}\,g^{ij}\sum_{\ell=1}^{N-1}\sum_{m}\overline{a_{i;\ell m}}\,b_{j;\ell m},$$

with the obvious modification when anti-Hermitian rather than complexified fields are used. Using the symbol map, the exact finite-N identity becomes

$$G_{\mathrm{HS},N}(a,b) = \int_{S^3} d^3x\,\sqrt{g}\,g^{ij}\int_{S^2} d\,\mu(\Omega)\,\overline{a_i^{(N)}(x,\Omega)}b_j^{(N)}(x,\Omega). \tag{9.38}$$

Thus

$$\boxed{G_{\mathrm{HS},N} \cong G_{L^2(S^3\times S^2)}} \tag{9.39}$$

on the truncated harmonic subspace

$$1 \le \ell \le N-1.$$

This identity is stronger than an asymptotic relation. With the normalization chosen above, it is exact for every N. The matrix trace in the Yang–Mills kinetic term has therefore already become an integral over the color sphere before taking the large-N limit. What remains finite-N is the angular-momentum cutoff and the noncommutative star product.

9.11 The large-N Hilbert-space limit

We now take the large-N limit at the level of Hilbert spaces.

For every N, define

$$\mathcal{H}_N^0 = \mathrm{span}\{Y_{\ell m} : 1 \le \ell \le N-1\} \subset L_0^2(S^2).$$

These spaces satisfy

$$\mathcal{H}_N^0 \subset \mathcal{H}_{N+1}^{0.}$$

Moreover,

$$\bigcup_{N=2}^{\infty}\mathcal{H}_N^0$$

is precisely the space of finite spherical-harmonic sums with vanishing constant component. Because the ordinary spherical harmonics form a complete orthonormal basis of

$$L^2(S^2),$$

the closure is

$$\overline{\bigcup_{N=2}^{\infty}\mathcal{H}_N^0} = L_0^2(S^2). \tag{9.40}$$

Therefore the Hilbert-space inductive limit of the normalized color spaces is $L_0^2\big(S_{\mathrm{col}}^2\big)$. Equivalently,

$$\boxed{\overline{\mathfrak{su}(\infty)}^{\mathrm{HS}} \simeq L_0^2(S^2).} \tag{9.41}$$

This is the precise meaning of the statement that at $N = \infty$ no color harmonics are discarded.

At finite N,

$$1 \leq \ell \leq N - 1.$$

At infinite N,

$$1 \leq \ell < \infty. \tag{9.42}$$

Every fixed angular-momentum sector survives. The finite-N upper cutoff does not become a finite cutoff in the limiting theory. It recedes to infinity.

9.12 No-mode-loss statement

The preceding observation can be formulated as a simple proposition.

Proposition 9.1

Let

$$P_N : L_0^2(S^2) \longrightarrow \mathcal{H}_N^0$$

denote the orthogonal projector onto harmonics with

$$1 \leq \ell \leq N - 1.$$

Then for every

$$f \in L_0^2(S^2),$$

one has

$$\|f - P_N f\|_{L^2} \longrightarrow 0 \tag{9.43}$$

as

$$N \to \infty.$$

Proof

Write

$$f = \sum_{\ell=1}^{\infty} \sum_{m=-\ell}^{\ell} f_{\ell m}\, Y_{\ell m}.$$

Parseval's theorem gives

$$\sum_{\ell=1}^{\infty}\sum_{m}|f_{\ell m}|^2<\infty.$$

Moreover,

$$P_N f=\sum_{\ell=1}^{N-1}\sum_{m} f_{\ell m}\,Y_{\ell m}.$$

Hence

$$\|f-P_N f\|_{L^2}^2=\sum_{\ell=N}^{\infty}\sum_{m}|f_{\ell m}|^2.$$

Because the coefficient sequence is square summable, the tail converges to zero. Therefore

$$\|f-P_N f\|_{L^2}\to 0.$$

This proposition is elementary but conceptually important. It shows that the $N=\infty$ theory is not obtained by selecting a finite collection of low color harmonics. Rather, the complete infinite harmonic tower is recovered in the Hilbert completion.

9.13 Large-N Yang–Mills connection space

The finite-N connection space in harmonic representation is

$$\mathcal{A}_N\simeq\Omega^1(S^3)\otimes\mathcal{H}_N^0.$$

At the Hilbert-space level, the large-N limit therefore becomes

$$\mathcal{A}_\infty\simeq\Omega^1(S^3)\,\widehat{\otimes}\,L_0^2\big(S^2_{\mathrm{col}}\big). \tag{9.44}$$

More precisely, after Sobolev completion in the spatial variables one may write

$$H^s\Omega^1(S^3)\,\widehat{\otimes}\,L_0^2\big(S^2_{\mathrm{col}}\big). \tag{9.45}$$

A large-N connection is thus represented by

$$A_i(x,\Omega)=\sum_{\ell=1}^{\infty}\sum_{m=-\ell}^{\ell}A_{i;\ell m}(x)Y_{\ell m}(\Omega), \tag{9.46}$$

with square-integrability condition

$$\int_{S^3}d^3x\,\sqrt{g}\,g^{ij}\sum_{\ell m}\overline{A_{i;\ell m}}\,A_{j;\ell m}<\infty.$$

Equivalently,

$$\boxed{\int_{S^3\times S^2}d^3x\,d\mu(\Omega)\,\sqrt{g}\,g^{ij}\overline{A_i(x,\Omega)}A_j(x,\Omega)<\infty.} \tag{9.47}$$

Thus the large-N Hilbert–Schmidt completion has acquired a continuous S^2 color variable. Again, this statement concerns the Hilbert-space representation.

9.14 Comparing smooth gauge algebra and Hilbert completion

The sections above give two complementary large-N descriptions of the color degrees of freedom. At the level of the normalized Hilbert–Schmidt metric, the color Hilbert space completes to

$$\overline{\mathfrak{su}(\infty)}^{\mathrm{HS}} \simeq L_0^2\big(S^2_{\mathrm{col}}\big).$$

At the level of the smooth gauge algebra, the semiclassical fuzzy-sphere limit gives instead [26]

$$C_0^\infty\big(S^2_{\mathrm{col}}\big) \simeq \mathfrak{ham}\big(S^2_{\mathrm{col}}\big) = \mathfrak{sdiff}\big(S^2_{\mathrm{col}}\big), \tag{9.48}$$

where

$$C_0^\infty(S^2) = \left\{ f \in C^\infty(S^2) : \int_{S^2} f \, d\mu = 0 \right\}.$$

These statements describe the same large-N color space at two different levels. The smooth space is the dense algebraic core on which the Poisson bracket and gauge transformations are defined, while $L_0^2(S^2)$ is its Hilbert completion with respect to the normalized Hilbert–Schmidt norm.

A finite-N spatial Yang–Mills connection is

$$A = A_i(x)\, dx^i \in \Omega^1(S^3) \otimes \mathfrak{su}(N).$$

Under the fuzzy-sphere symbol map it may be expanded as

$$A_i^{(N)}(x, \Omega) = \sum_{\ell=1}^{N-1} \sum_{m=-\ell}^{\ell} A_{i;\ell m}(x)\, Y_{\ell m}(\Omega).$$

As $N \to \infty$, the upper harmonic cutoff recedes to infinity. On the smooth core one therefore obtains

$$\mathcal{A}_\infty^\infty = \Omega^1(S^3) \otimes C_0^\infty\big(S^2_{\mathrm{col}}\big). \tag{9.49}$$

Using the identification of zero-average smooth functions with Hamiltonian vector fields,

$$f \mapsto X_f, \qquad \iota_{X_f} \omega_{S^2} = df,$$

this may equivalently be written as

$$\boxed{\mathcal{A}_\infty^\infty \simeq \Omega^1\Big(S^3, \mathfrak{sdiff}\big(S^2_{\mathrm{col}}\big)\Big).} \tag{9.50}$$

Thus the strict large-N Yang–Mills connection is naturally represented as a one-form on S^3 valued in the Lie algebra of smooth area-preserving vector fields on the color sphere. The corresponding large-N gauge parameter is

$$\varepsilon \in C^\infty(S^3) \otimes C_0^\infty\left(S^2_{\mathrm{col}}\right).$$

With the normalized large-N matrix bracket chosen so that

$$\frac{1}{i\hbar_N}[A_N, B_N] \longrightarrow \{a, b\}_{S^2}, \qquad \hbar_N \sim N^{-1},$$

the finite-N gauge transformation

$$\delta_\varepsilon A_i = \partial_i \varepsilon + [A_i, \varepsilon]_{(N)}$$

has the smooth large-N limit

$$\boxed{\delta_\varepsilon A_i = \partial_i \varepsilon + \{A_i, \varepsilon\}_{S^2}.} \tag{9.51}$$

Likewise, the Yang–Mills curvature becomes

$$\boxed{F_{ij} = \partial_i A_j - \partial_j A_i + \left\{A_i, A_j\right\}_{S^2}.} \tag{9.52}$$

This is precisely the curvature of an

$$\mathrm{SDiff}(S^2)$$

gauge theory over S^3. The non-Abelian color algebra has therefore not disappeared in the planar limit. Rather, the matrix commutator has become the classical Poisson bracket on the color sphere. Equivalently, introducing the Hamiltonian vector field

$$X_{A_i} = \omega_{S^2}^{-1}(d_{S^2}A_i, \cdot),$$

one may regard the connection as

$$A_i(x, \Omega) \quad \leftrightarrow \quad X_{A_i}(x, \Omega) \in \mathfrak{sdiff}(S^2),$$

with

$$\left[X_{A_i}, X_{A_j}\right] = X_{\{A_i, A_j\}}.$$

Hence the large-N color dependence has acquired a smooth classical geometric interpretation. The Hilbert-space completion is obtained independently from the normalized Hilbert–Schmidt metric. After an appropriate Sobolev completion in the S^3 variables,

$$\mathcal{A}_\infty^{\mathrm{Hilb}} = H^s\Omega^1(S^3) \,\widehat{\otimes}\, L_0^2\left(S^2_{\mathrm{col}}\right). \tag{9.53}$$

The smooth field space is dense in this completion,

$$\mathcal{A}_\infty^\infty \subset \mathcal{A}_\infty^{\mathrm{Hilb}}, \tag{9.54}$$

and provides a common invariant core for the differential and Poisson-algebraic operations entering the classical large-N theory.
A large-N connection may therefore be represented as

$$A_i(x,\Omega) = \sum_{\ell=1}^{\infty} \sum_{m=-\ell}^{\ell} A_{i;\ell m}(x) Y_{\ell m}(\Omega),$$

with Hilbert norm

$$\|A\|_{\mathrm{HS}}^2 = \int_{S^3\times S^2} d^3x\, d\mu(\Omega)\, \sqrt{g}\, g^{ij}\, \overline{A_i(x,\Omega)} A_j(x,\Omega).$$

The two descriptions may therefore be summarized as

smooth large-N gauge algebra:	$\mathcal{A}_\infty^\infty \simeq \Omega^1\big(S^3, \mathfrak{sdiff}(S^2)\big),$
Hilbert-space completion:	$\overline{\mathcal{A}_\infty^\infty}^{\mathrm{HS}} \simeq H^s\Omega^1(S^3) \mathbin{\widehat{\otimes}} L_0^2(S^2).$

The distinction will be important below. The smooth $\mathrm{SDiff}(S^2)$ realization is the natural starting point for the geometric and Poisson description of the planar gauge theory, while the L^2 completion supplies the physical Hilbert structure required for the fundamental modular region, the Riesz metric, and the later spectral analysis. Thus the large-N limit has two simultaneous effects: the finite fuzzy color sphere becomes a smooth classical symplectic sphere, and the full infinite tower of color harmonics is retained in the corresponding Hilbert completion.

10. Geometrization of the Planar Yang–Mills RG Data and Reconstruction of the Deformed Conifold

10.1. Purpose and logical status

Chapter 9 produced two complementary large-N structures directly from Yang–Mills theory. First, after gauge reduction to the fundamental modular region and planar normalization, the physical field-space metric is

$$G_{\mathrm{YM},\mu} = \frac{1}{\lambda(\mu)} G_{\mathrm{HS}}.$$

Second, the large-N color degrees of freedom admit both a smooth Poisson-algebra realization and a Hilbert-space completion. On the smooth core,

$$C_0^\infty\big(S^2_{\mathrm{col}}\big) \simeq \mathfrak{sdiff}\big(S^2_{\mathrm{col}}\big),$$

where the rescaled matrix commutator becomes the Poisson bracket,

$$\frac{1}{i\hbar_N}[A_N, B_N] \longrightarrow \{a, b\}_{S^2}, \qquad \hbar_N \sim N^{-1}.$$

The corresponding large-N gauge field is therefore

$$A \in \Omega^1(S^3) \otimes C_0^\infty\left(S^2_{\mathrm{col}}\right) \simeq \Omega^1\left(S^3, \mathfrak{sdiff}\left(S^2_{\mathrm{col}}\right)\right).$$

At the level of the normalized Hilbert–Schmidt metric, this smooth field space has completion

$$\overline{\mathfrak{su}(\infty)}^{\mathrm{HS}} \simeq L^2_0\left(S^2_{\mathrm{col}}\right),$$

and hence

$$\mathcal{A}_\infty^{\mathrm{Hilb}} \simeq H^s\Omega^1(S^3) \,\widehat{\otimes}\, L^2_0\left(S^2_{\mathrm{col}}\right).$$

Thus the same large-N color degrees of freedom possess a smooth classical description, required for the Poisson and geometric constructions below, and a Hilbert completion, required for the physical field-space metric. At a fixed renormalization scale, the smooth collective variables depend on

$$x \in S^3, \qquad \Omega \in S^2_{\mathrm{col}},$$

so that their finite-dimensional geometric carrier is

$$\mathcal{Y}_5 \simeq S^3 \times S^2_{\mathrm{col}}. \tag{10.1}$$

This space is the carrier of the large-N collective variables; it should not be confused with the full fundamental modular region, which remains an infinite-dimensional quotient of the space of connections.The purpose of this section is to show that the Yang–Mills renormalization-group data supply the additional structure required to complete this five-dimensional Poisson carrier to the six-dimensional deformed conifold. No string-theoretic duality is assumed in this construction. Instead, we make a mathematical geometric realization of the Yang–Mills RG data and derive its consequences. The construction will proceed through the chain

$$\boxed{S^3 \times S^2_{\mathrm{col}}}$$
$$\Downarrow$$
$$\boxed{\text{RG radial completion}}$$
$$\Downarrow$$
$$\boxed{T^*S^3}$$
$$\Downarrow$$
$$\boxed{\text{Yang-Mills period data}}$$
$$\Downarrow$$
$$\boxed{\text{Picard-Lefschetz complex structure}}$$
$$\Downarrow$$
$$\boxed{\text{deformed conifold}}$$
$$\Downarrow$$
$$\boxed{\text{Stenzel geometry and RG Riesz weight } h.}$$

The distinction between assumptions and deductions will be kept explicit.

10.2 The fixed-scale planar carrier

The large-N connection space obtained in the previous section is

$$\mathcal{A}_\infty^\infty = \Omega^1(S^3) \otimes C_0^\infty\left(S^2_{\mathrm{col}}\right) \simeq \Omega^1\left(S^3, \mathfrak{sdiff}\left(S^2_{\mathrm{col}}\right)\right).$$

A smooth field has the expansion

$$A_i(x,\Omega) = \sum_{\ell=1}^{\infty} \sum_{m=-\ell}^{\ell} A_{i;\ell m}(x) Y_{\ell m}(\Omega),$$

where smoothness means that the coefficients decay sufficiently rapidly in ℓ, while square summability gives the corresponding Hilbert completion. Its normalized Hilbert–Schmidt metric is

$$G_{\mathrm{HS}}(a,b) = \int_{S^3 \times S^2_{\mathrm{col}}} d\,V_{S^3}\, d\mu_{S^2}\, g^{ij} \overline{a_i(x,\Omega)} b_j(x,\Omega).$$

At renormalization scale μ, the physical planar metric is therefore

$$G_{\mathrm{YM},\mu}(a,b) = \frac{1}{\lambda(\mu)} G_{\mathrm{HS}}(a,b). \tag{10.2}$$

The underlying Hilbert space does not change with μ. What changes is its Riesz metric. Consequently, the RG defines a one-parameter family

$$\left\{\mathcal{H}_{\mathrm{YM}}, G_{\mathrm{YM},\mu}\right\}_\mu. \tag{10.3}$$

It is useful to regard this as a Hilbert bundle over the RG trajectory. The five-dimensional space

$$S^3 \times S^2_{\mathrm{col}}$$

therefore represents the collective angular carrier at one fixed scale. The RG flow supplies one additional real parameter.

10.3 From the color sphere to cotangent directions

The appearance of S^2_{col} has a natural geometric interpretation. At a point

$$x \in S^3,$$

the unit sphere in the three-dimensional cotangent space is

$$S(T_x^* S^3) \simeq S^{2.}$$

We therefore impose the following equivariant identification of the large-N color sphere with the unit cotangent directions,

$$S^2_{\mathrm{col}} \cong S(T^*_x S^3). \tag{10.4}$$

This identification is to be understood $SO(4)$-equivariantly. The $SO(4)$ isometry group of the round three-sphere acts transitively on S^3, while the stabilizer of a point is $SO(3)$; the latter acts in the standard way on the unit two-sphere in the tangent or cotangent space. Globally, the bundle of unit cotangent directions is

$$S^*S^3.$$

Because S^3 is parallelizable,

$$S^*S^3 \simeq S^3 \times S^2$$

as a smooth manifold. Its homogeneous description is

$$S^*S^3 \simeq \frac{SO(4)}{SO(2)}. \tag{10.5}$$

Equivalently, using

$$SO(4) \simeq \frac{SU(2)_L \times SU(2)_R}{\mathbb{Z}_2},$$

one obtains the standard homogeneous space

$$T^{1,1} = \frac{SU(2)_L \times SU(2)_R}{U(1)} \simeq S^*S^3. \tag{10.6}$$

Thus the product

$$S^3 \times S^2_{\mathrm{col}}$$

should not subsequently be viewed as carrying the product metric. Its $SO(4)$-equivariant geometric realization is the homogeneous five-manifold $T^{1,1}$. This is exactly the principal-orbit geometry occurring in the six-dimensional Stenzel construction.

10.4 The RG scale supplies the cotangent radius

The color sphere determines only the angular direction of a cotangent vector. A general nonzero cotangent vector can be written as

$$p = r\,\hat{p}, \qquad r > 0, \qquad \hat{p} \in S(T^*_x S^3).$$

Hence

$$T^*S^3 \setminus S^3 \simeq (0,\infty) \times S^*S^3.$$

Using the identification above,

$$T^*S^3 \setminus S^3 \simeq (0,\infty) \times T^{1,1}. \tag{10.7}$$

The Yang–Mills RG trajectory is itself one-dimensional. We therefore identify its monotonic parameter with the cotangent radius up to reparametrization,

$$r = r(\mu).$$

Equivalently, introducing a convenient dimensionless radial coordinate,

$$\tau = \tau(\mu). \tag{10.8}$$

This statement does not yet determine the function $\mu(\tau)$. It identifies only the unique invariant geometric direction available for representing the RG flow. The angular S^2_{col} together with this radial variable forms an $\mathbb{R}^3$ fibre:

$$[0,\infty) \times S^2/(\{0\} \times S^2) \simeq \mathbb{R}^3.$$

Therefore the smooth completion at the infrared endpoint is

$$S^3 \times \mathbb{R}^3 \simeq T^*S^3. \tag{10.9}$$

This is the precise sense in which the additional RG coordinate completes the fixed-scale space. It is worth distinguishing this from spacetime. The physical Yang–Mills spacetime remains

$$S^3 \times \mathbb{R}_t.$$

The newly constructed six-dimensional space is the geometric carrier of the large-N color and RG data.

10.5 Explicit real realization of T^*S^3

The cotangent-bundle topology can be exhibited directly in the complex quadric that will emerge below.

Let

$$X_a = \left\{ z \in \mathbb{C}^4 \;\middle|\; \sum_{i=1}^4 z_i^2 = a \right\}, \qquad a > 0.$$

Write

$$z = x + iy, \qquad x, y \in \mathbb{R}^4.$$

The defining equation is equivalent to

$$x \cdot y = 0,$$

and

$$|x|^2 - |y|^2 = a.$$

Hence

$$|x| = \sqrt{a + |y|^2}.$$

Define

$$q = \frac{x}{|x|} \in S^3.$$

Because

$$x \cdot y = 0,$$

one has

$$q \cdot y = 0,$$

so y is a tangent, and by the round metric also a cotangent, vector at q. Conversely,

$$(q, p) \in T^*S^3$$

determines

$$z(q,p) = \sqrt{a + |p|^2}\, q + ip. \tag{10.10}$$

Indeed,

$$z(q,p) \cdot z(q,p) = a.$$

Thus

$$X_a \simeq T^*S^3 \tag{10.11}$$

as a smooth manifold. At fixed $|p| \neq 0$, the level set is the unit-cotangent orbit $T^{1,1}$. This gives an explicit realization of the topology inferred above from the large-N color and RG variables.

10.6 Yang–Mills RG data as period data

We now introduce the additional structure that determines the complex geometry. The construction is based on two identifications. Let A denote the compact three-cycle and B its symplectic dual,

$$A \cap B = 1.$$

Let Ω be a holomorphic three-form whose periods are

$$\Pi_A = \int_A \Omega\,, \qquad \Pi_B = \int_B \Omega. \tag{10.12}$$

We assume that the compact period represents the RG-invariant Yang–Mills scale,

$$\boxed{\Pi_A = c_A \Lambda_{\mathrm{YM}}^3}. \tag{10.13}$$

We further identify the cutoff entering the noncompact B-period with the renormalization scale,

$$\boxed{M_B = c_\mu \mu^3}. \tag{10.14}$$

The constants c_A and c_μ account for the fixed conversion of field-theory units into the units used by the geometric realization. Only their ratio will matter below. Define

$$a := \Pi_A.$$

The natural dimensionless RG coordinate is therefore

$$u = \frac{a}{M_B}. \tag{10.15}$$

Up to normalization,

$$u \sim \left(\frac{\Lambda_{\mathrm{YM}}}{\mu}\right)^{3.}$$

The infrared region corresponds to u of order one, whereas

$$u \longrightarrow 0$$

is the asymptotically free ultraviolet region.

10.7 The universal running coupling as a period matrix

The Yang–Mills action in planar normalization contains

$$\frac{1}{\lambda(\mu)}$$

as the source multiplying the gauge kinetic operator. The universal ultraviolet running [27,28] therefore has the logarithmic form

$$\frac{1}{\lambda(\mu)} = b_0 \log \frac{\mu}{\Lambda_{\mathrm{YM}}} + \cdots.$$

Equivalently,

$$\frac{1}{\lambda} = -\frac{b_0}{3} \log u + \cdots.$$

Introduce, if desired, the complexified source

$$\tau_{\mathrm{YM}} = \frac{\theta}{2\pi} + i\kappa \frac{1}{\lambda}.$$

We now require the period matrix of the geometric realization to reproduce this complexified RG source,

$$\boxed{\tau_{\mathrm{per}} := \frac{\partial \Pi_B}{\partial \Pi_A} = \tau_{\mathrm{YM}},} \tag{10.16}$$

with the normalization κ chosen consistently with the normalization of A and B. For the universal logarithmic part one may choose the normalization such that

$$\tau_{\mathrm{per}}(a; M_B) = \frac{1}{2\pi i} \log \frac{a}{M_B} + \tau_{\mathrm{reg}}. \tag{10.17}$$

For positive $a/M_B < 1$,

$$\mathrm{Im}\, \tau_{\mathrm{per}} = \frac{1}{2\pi} \log \frac{M_B}{a} + \mathrm{Im}\, \tau_{\mathrm{reg}},$$

and therefore

$$\boxed{\operatorname{Im}\tau_{\mathrm{per}} \propto \frac{1}{\lambda(\mu)}.} \tag{10.18}$$

This is the central RG-period identification. It does not require a pre-existing string interpretation. It is a geometric representation of the Yang–Mills running coupling.

10.8 Reconstruction of the dual period

Integrating

$$\frac{\partial \Pi_B}{\partial a} = \frac{1}{2\pi i}\log\frac{a}{M_B} + \tau_{\mathrm{reg}}(a)$$

gives

$$\Pi_B(a) = \frac{a}{2\pi i}\left(\log\frac{a}{M_B} - 1\right) + \Pi_B^{\mathrm{hol}}(a). \tag{10.19}$$

The analytic term depends on the choice of regularization and on the holomorphic coordinate used on the RG parameter space. The logarithmic term is universal. Now analytically continue once around

$$a = 0,$$

so that

$$a \longmapsto e^{2\pi i}a.$$

Then

$$\log a \longmapsto \log a + 2\pi i,$$

and hence

$$\Pi_B \longmapsto \Pi_B + \Pi_A. \tag{10.20}$$

Therefore the integral homology basis transforms as

$$A \longmapsto A, \qquad B \longmapsto B + A. \tag{10.21}$$

The monodromy matrix is

$$T = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}. \tag{10.22}$$

This is the rank-one Picard–Lefschetz monodromy [29] of a simple vanishing cycle. Picard–Lefschetz theory precisely relates such monodromy to vanishing cycles in a complex degeneration. Thus the logarithmic Yang–Mills RG law has produced an integral Picard–Lefschetz variation of Hodge structure.

10.9 From the RG monodromy to the local complex threefold

We now impose one geometric regularity assumption: the infrared degeneration is a single nondegenerate degeneration whose vanishing cycle is

$$A \simeq S^{3.}$$

For a nondegenerate holomorphic critical point, the holomorphic Morse lemma brings the local defining function to quadratic normal form [30]. Consequently, after a holomorphic redefinition of the modulus, the family is locally

$$X_a = \{(z_1, z_2, z_3, z_4) \in \mathbb{C}^4 \mid z_1^2 + z_2^2 + z_3^2 + z_4^2 = a\}. \tag{10.23}$$

At

$$a = 0$$

one obtains the ordinary conifold singularity. For

$$a \neq 0$$

one obtains its smoothing, the deformed conifold. Thus the complex geometry has not been postulated independently. Within the assumed class of one-parameter simple S^3 degenerations, it is reconstructed from

$$\boxed{\text{Yang-Mills logarithmic RG monodromy} + \text{the compact } S^3 \text{ cycle.}}$$

The previously obtained real manifold T^*S^3 is therefore equipped with the complex structure inherited from the affine quadric.

10.10 The holomorphic volume form

The smooth quadric X_a has trivial canonical bundle. Its natural holomorphic three-form is obtained by the Poincaré residue,

$$\Omega_a = \mathrm{Res}_{X_a} \frac{dz_1 \wedge dz_2 \wedge dz_3 \wedge dz_4}{z_1^2 + z_2^2 + z_3^2 + z_4^2 - a}. \tag{10.24}$$

For example, on the patch $z_4 \neq 0$,

$$\Omega_a = \frac{dz_1 \wedge dz_2 \wedge dz_3}{2z_4}.$$

The overall normalization of Ω_a remains free. That normalization is fixed by the Yang–Mills identification

$$\int_A \Omega_a = c_A \Lambda_{\mathrm{YM}}^3. \tag{10.25}$$

The scaling behaviour is already correct. Under

$$z_i = a^{1/2} w_i,$$

the residue form scales as

$$\Omega_a \propto a,$$

and therefore

$$\Pi_A(a) \propto a.$$

Thus the deformation coordinate itself is proportional to the compact period.

10.11 The RG data determine a complex structure on the source space

The period system also explains why the Yang–Mills scalar source sector acquires a complex structure. The real two-dimensional period space spanned by A and B carries the symplectic intersection form

$$\omega(A, B) = 1.$$

Let

$$\tau = \tau_1 + i\tau_2, \qquad \tau_2 > 0.$$

The compatible complex structure on the corresponding real symplectic plane is

$$J_\tau = \frac{1}{\tau_2}\begin{pmatrix} -\tau_1 & -|\tau|^2 \\ 1 & \tau_1 \end{pmatrix}. \tag{10.26}$$

A direct calculation gives

$$J_\tau^2 = -1. \tag{10.26}$$

Thus the period matrix does not merely measure a coupling. It defines the complex structure of the real source-response plane. Because

$$\tau_{\mathrm{per}} = \tau_{\mathrm{YM}},$$

the Yang–Mills RG source itself determines this complex structure.

10.12 The Yang–Mills scalar action as the RG-conjugate source

In the connection normalization used throughout this work, the planar Yang–Mills action contains

$$S_{\mathrm{YM}} = N^2 \frac{1}{4\lambda(\mu)} \int d^4x \ \mathrm{tr}_N F_{\mu\nu} F^{\mu\nu} + \cdots.$$

Therefore

$$\frac{\partial S_{\mathrm{YM}}}{\partial(1/\lambda)} = \frac{N^2}{4} \int d^4x \ \mathrm{tr}_N F_{\mu\nu} F^{\mu\nu}. \tag{10.27}$$

The scalar operator

$$\mathcal{O}_+ = \mathrm{tr}_N F_{\mu\nu} F^{\mu\nu} \tag{10.28}$$

is consequently the operator conjugate to motion in the inverse-coupling direction. Including the theta angle provides the second real scalar source,

$$\mathcal{O}_- \sim \mathrm{tr}_N\, F \wedge F.$$

The pair

$$\left(\frac{1}{\lambda}, \theta\right)$$

therefore forms the real scalar source plane underlying the complex coordinate τ_{YM}. Since

$$\tau_{\mathrm{YM}} = \tau_{\mathrm{per}},$$

variations of the scalar Yang–Mills source are mapped to variations of the reconstructed complex structure. Schematically,

$$\boxed{\delta\left(\frac{1}{\lambda}, \theta\right) \longrightarrow \delta\tau_{\mathrm{per}} \longrightarrow \delta J.} \tag{10.29}$$

This gives a purely Yang–Mills origin for the complex structure of the scalar source space.

10.13 Infinitesimal scalar sources and $(2,1)$-forms

Let a be the reconstructed complex-structure modulus. Then

$$\Omega = \Omega(a).$$

An infinitesimal variation satisfies [31]

$$\frac{\partial \Omega}{\partial a} = k_a \Omega + \chi_a,$$

where the nontrivial deformation component has Hodge type

$$\chi_a \in \Omega^{2,1}(X). \tag{10.30}$$

Thus the chain

$$\boxed{\text{Yang--Mills scalar source} \longrightarrow \text{RG period variation} \longrightarrow \text{complex-structure variation} \longrightarrow (2,1)\text{-form}}$$

is obtained before introducing any closed-string interpretation. The later appearance of $(2,1)$-form collective fields is therefore compatible with, and naturally predicted by, the RG geometry itself.

10.14 From complex geometry to the Ricci-flat Kähler metric

The reconstruction so far has determined

$$X_a \simeq T^* S^3,$$

its integrable complex structure, and its holomorphic volume form. To obtain a Riemannian metric one requires compatibility with the real symplectic geometry. Let ω be an $SO(4)$-

invariant Kähler form in the cotangent symplectic class. The real symplectic and holomorphic volume structures are not independent. In the strict planar limit the normalized matrix trace becomes the invariant Liouville measure of the large-N Hamiltonian color algebra, while the identification of the universal Yang–Mills RG source with the period matrix determines the complex structure and the holomorphic three-form Ω. Since the collective map preserves the normalized cyclic pairing, these two volume forms represent the same planar trace in the real and complex polarizations. Moreover, the scale dependence of $1/\lambda(\mu)$ is encoded separately in the radial Riesz field h and therefore does not introduce an additional density in the reference geometric volume. Consequently the two invariant top forms can differ only by an overall positive normalization constant C, so that

$$\frac{\omega^3}{3!} = C\, i\, \Omega \wedge \overline{\Omega}, \tag{10.31}$$

where $C > 0$ fixes the overall normalization. Since

$$d\omega = 0, \qquad d\Omega = 0,$$

this condition implies vanishing Ricci form. Indeed [31], for a Kähler metric,

$$\rho_{\mathrm{Ric}} = -i\partial\bar{\partial} \log \frac{\omega^3/3!}{i\Omega \wedge \overline{\Omega}}.$$

If the ratio is constant, then

$$\rho_{\mathrm{Ric}} = 0. \tag{10.32}$$

Hence the compatible metric is Ricci-flat and Kähler. Within the complete $SO(4)$-invariant cohomogeneity-one class on T^*S^3, with a smooth S^3 bolt at the origin, this selects the Stenzel geometry up to the overall scale fixed by the A-period. The Stenzel construction gives precisely such complete invariant Ricci-flat Kähler metrics on cotangent bundles of spheres and, more generally, compact symmetric spaces. No string equation has been used here. The metric follows from the reconstructed complex structure together with symplectic and volume compatibility.

10.15 Explicit Stenzel form

Introduce the standard radial coordinate τ by

$$\sum_{i=1}^{4} |z_i|^2 = |a| \cosh \tau\,. \tag{10.33}$$

The infrared S^3 lies at

$$\tau = 0,$$

while

$$\tau \to \infty$$

is the asymptotic conical region. Define

$$K(\tau) = \frac{(\sinh 2\,\tau - 2\tau)^{1/3}}{2^{1/3}\sinh\tau}.$$

Up to the overall length scale fixed by a, the Stenzel metric may be written in the standard $T^{1,1}$-adapted basis as

$$ds_6^2 = C_a K(\tau)\left[\frac{1}{3K(\tau)^3}(d\tau^2 + (g^5)^2) + \cosh^2\frac{\tau}{2}((g^3)^2 + (g^4)^2) + \sinh^2\frac{\tau}{2}((g^1)^2 + (g^2)^2)\right]. \tag{10.34}$$

The unequal radial coefficients show explicitly that the principal orbit is a squashed $T^{1,1}$, not a product metric on $S^3 \times S^2$. At $\tau = 0$, the S^2-type directions collapse smoothly while the S^3 remains of finite size. This is precisely the infrared boundary condition required by the cotangent-bundle construction.

10.16. The RG scale as the conifold radial coordinate

The defining relation

$$\sum_i |z_i|^2 = |a|\cosh\tau$$

now fixes the previously unspecified relation between the RG variable and the radial coordinate. Let the local ultraviolet scale be represented by

$$M(\tau) \propto \sum_i |z_i|^2.$$

Since

$$a \propto \Lambda_{\mathrm{YM}}^3,$$

the dimensionless RG variable becomes

$$u(\tau) = \frac{a}{M(\tau)}.$$

Therefore

$$\boxed{u(\tau) = u_0\,\mathrm{sech}\,\tau\,,} \tag{10.35}$$

where u_0 absorbs the fixed normalization constants. Thus

$$\tau = 0$$

corresponds to the infrared endpoint,

$$u(0) = u_0,$$

whereas

$$\tau \to \infty$$

gives

$$u(\tau) \to 0,$$

the ultraviolet limit. The relation

$$u(\tau) = u_0 \operatorname{sech} \tau$$

is therefore no longer an independent radial ansatz. It follows from combining the RG scale identification with the radial geometry of the deformed conifold.

10.17 The integrated RG source

The universal period matrix contains the logarithm

$$\tau_{\text{per}} \propto \log u.$$

We now define its infrared-subtracted primitive by

$$S(u) = \int_{u_0}^{u} \log s \; ds. \tag{10.36}$$

Evaluating the integral gives

$$S(u) = u \log u - u - u_0 \log u_0 + u_0. \tag{10.37}$$

Thus

$$S(u_0) = 0,$$

and

$$\frac{dS}{du} = \log u. \tag{10.38}$$

Consequently S is the integrated local response of the period matrix, and hence of the universal inverse running coupling. After inserting

$$u = u(\tau),$$

we obtain the radial source

$$S(\tau) = u(\tau) \log u\,(\tau) - u(\tau) - u_0 \log u_0 + u_{0.} \tag{10.39}$$

This quantity should be distinguished from the running coupling itself. The hierarchy is

$$\boxed{\frac{1}{\lambda} \;\leftrightarrow\; \log u \;\longrightarrow\; S(u).}$$

The first quantity is the local RG susceptibility; the second is its period representation; the third is the accumulated primitive that will source the radial Riesz response.

10.18. The radial Stenzel Dirichlet form

For $SO(4)$-invariant functions, the Stenzel geometry reduces the relevant radial Dirichlet form to

$$\mathfrak{d}[f,g] = \int_0^\infty p(\tau) f'(\tau) \overline{g'(\tau)}\, d\tau, \tag{10.40}$$

where

$$p(\tau) = (\sinh 2\,\tau - 2\tau)^{2/3}.$$

The factor $p(\tau)$ is the radial geometric coefficient determined by the Stenzel metric. It should not be identified with the full six-dimensional volume. Rather, it is the coefficient with which the radial gradient enters the reduced invariant energy. The RG source defines the linear functional

$$\mathcal{L}_S[\varphi] = \int_0^\infty S(\tau)\varphi(\tau)\, d\tau.$$

On an appropriate radial Sobolev space with regularity at $\tau = 0$ and decay at infinity, suppose this functional is continuous with respect to the Dirichlet norm. The Riesz representation theorem then determines a unique function h satisfying

$$\mathfrak{d}[h,\varphi] = \mathcal{L}_S[\varphi] \tag{10.41}$$

for every admissible test function φ. Integration by parts gives the weak equation

$$\boxed{-\frac{d}{d\tau}[p(\tau)h'(\tau)] = S(\tau).} \tag{10.42}$$

This is the precise mathematical sense in which the Yang–Mills RG source generates the radial weight.

10.19 Solution of the radial Poisson problem

Regularity at the Stenzel tip requires

$$h'(0) = 0.$$

The ultraviolet normalization is taken to be

$$h(\infty) = 0.$$

Define

$$Q(\tau) = \int_0^\tau S\,(s)\, ds. \tag{10.43}$$

Integrating the Poisson equation once gives

$$-p(\tau)h'(\tau) = Q(\tau). \tag{10.44}$$

Hence

$$-h'(\tau) = \frac{Q(\tau)}{p(\tau)}. \tag{10.45}$$

A second integration gives

$$h(\tau) = \int_{\tau}^{\infty} \frac{Q(x)}{p(x)}\, dx. \tag{10.46}$$

Thus the complete sequence is

$$\boxed{\frac{1}{\lambda} \longrightarrow \log u \longrightarrow S \longrightarrow Q \longrightarrow h.}$$

The warp-Riesz function h is therefore the Green-function response of the reconstructed Stenzel geometry to the integrated Yang–Mills RG source.

10.20 Gauss-law interpretation of the RG source

The preceding one-dimensional equation admits a useful differential-form formulation which does not require interpreting the resulting form as a physical string-theory flux. Let ν_5 be the normalized $SO(4)$-invariant five-form on the principal orbit $T^{1,1}$, chosen such that

$$\int_{T^{1,1}} \nu_5 = 1.$$

Define the RG five-form

$$F_5^{\mathrm{RG}} = Q(\tau)\nu_5. \tag{10.47}$$

Its flux through the constant-τ orbit is

$$\int_{T_\tau^{1,1}} F_5^{\mathrm{RG}} = Q(\tau). \tag{10.48}$$

Because

$$Q'(\tau) = S(\tau),$$

one has

$$dF_5^{\mathrm{RG}} = J_6^{\mathrm{RG}}, \tag{10.49}$$

with

$$J_6^{\mathrm{RG}} = S(\tau)\, d\tau \wedge \nu_5. \tag{10.50}$$

For two radial slices,

$$\tau_1 < \tau_2,$$

Stokes' theorem gives

$$\int_{T_{\tau_2}^{1,1}} F_5^{\mathrm{RG}} - \int_{T_{\tau_1}^{1,1}} F_5^{\mathrm{RG}} = \int_{\tau_1}^{\tau_2} J_6^{\mathrm{RG}}. \tag{10.51}$$

Thus $S(\tau)d\tau$ measures the local source inserted between neighbouring RG slices, while $Q(\tau)$ measures the total source accumulated from the infrared endpoint to scale τ. Within the present construction, this gives a precise geometric meaning to the reorganization of the planar color degrees of freedom under RG flow. The radial equation

$$-ph' = Q$$

then states that the local Riesz field is obtained by distributing this accumulated source through the radial geometry.

10.21 Interpretation in terms of the FMR metric

The original Yang–Mills information entering the construction was the scale-dependent FMR metric

$$G_{\mathrm{YM},\mu} = \frac{1}{\lambda(\mu)} G_{\mathrm{HS}}.$$

At any single value of μ, multiplication by the positive scalar $\lambda(\mu)^{-1}$ does not change the location of the minima defining the FMR. It does, however, change the Riesz metric on its physical tangent space. The full RG trajectory therefore contains more information than any one fixed-scale FMR slice.
Geometrizing the scale converts the family

$$\left\{\frac{1}{\lambda(\mu)} G_{\mathrm{HS}}\right\}_{\mu}$$

into the radial geometry described above. The five-dimensional $T^{1,1}$ orbit represents the planar configuration carrier at a fixed scale, while the source between neighbouring orbits records how the physical metric changes as modes are reorganized along the RG trajectory. The function h is then the integrated geometric response to this redistribution.

In this sense,

$$\boxed{\text{perturbative RG data}}$$

is promoted to

$$\boxed{\text{nonlocal geometric data on the RG-completed FMR representation.}}$$

This does not amount to a prior construction of the full nonperturbative Yang–Mills measure. Rather, it provides a geometric completion of the exact large-N color carrier using the RG information that is independently known.

10.22 Reconstruction theorem

The construction in this chapter may be summarized as follows.

Theorem 10.1 — RG reconstruction of the deformed conifold

Consider strict planar Yang–Mills theory on S^3, reduced to the fundamental modular region, with normalized Hilbert–Schmidt metric

$$G_{\mathrm{YM},\mu} = \lambda(\mu)^{-1} G_{\mathrm{HS}},$$

and with the large-N color algebra and Hilbert-space limit constructed in Chapter 9. The large-N Poisson carrier together with the one-dimensional Yang–Mills RG trajectory has the $SO(4)$-equivariant geometric realization

$$S^2_{\mathrm{col}} \simeq S(T^*_x S^3),$$

and its smooth radial completion is

$$X \simeq T^* S^3.$$

Normalize the compact and noncompact period coordinates by

$$\Pi_A = c_A \Lambda^3_{\mathrm{YM}}, \qquad M_B = c_\mu \mu^3,$$

where c_A, c_μ are fixed conversion constants. These relations fix units and do not constitute independent dynamical assumptions. The only additional physical identification is that the period matrix of the reconstructed complex geometry equals the complexified inverse 't Hooft coupling,

$$\boxed{\tau_{\mathrm{per}} := \frac{\partial \Pi_B}{\partial \Pi_A} = \tau_{\mathrm{YM}} = \frac{\theta}{2\pi} + i\,\kappa\,\lambda(\mu)^{-1}.} \tag{10.52}$$

Then the universal Yang–Mills RG logarithm determines the singular part of the dual period,

$$\tau_{\mathrm{per}} = \frac{1}{2\pi i} \log \frac{\Pi_A}{M_B} + \tau_{\mathrm{reg}},$$

and hence

$$\Pi_B = \frac{\Pi_A}{2\pi i}\left(\log \frac{\Pi_A}{M_B} - 1\right) + \Pi_B^{\mathrm{hol}}.$$

Analytic continuation around $\Pi_A = 0$ gives the Picard–Lefschetz monodromy

$$\Pi_B \mapsto \Pi_B + \Pi_A,$$

so the infrared endpoint is a simple rank-one degeneration with vanishing cycle

$$A \simeq S^3.$$

The corresponding local complex threefold is therefore, up to holomorphic reparametrization of its modulus, the deformed conifold

$$\boxed{X_a = \left\{ z \in \mathbb{C}^4 : \sum_{i=1}^{4} z_i^2 = a \right\}, \qquad a \propto \Lambda^3_{\mathrm{YM}}.}$$

Its holomorphic three-form is the residue form

$$\Omega_a = \mathrm{Res}_{X_a} \frac{dz_1 \wedge dz_2 \wedge dz_3 \wedge dz_4}{\sum_i z_i^2 - a},$$

with normalization fixed by the Yang–Mills A-period. The normalized large-N cyclic trace determines the real symplectic volume of the Poisson realization, while the period construction determines the complex volume form. Since the collective realization preserves this cyclic pairing, the two descriptions of the same planar trace satisfy

$$\frac{\omega^3}{3!} = C\, i\, \Omega \wedge \bar{\Omega}, \qquad C > 0.$$

Consequently the compatible $SO(4)$-invariant Kähler metric is Ricci-flat. Within the complete cohomogeneity-one class on T^*S^3 with smooth S^3 bolt, this selects the Stenzel metric up to the overall scale fixed by Π_A. Finally, the physical Yang–Mills Riesz family

$$G_{\mathrm{YM},\mu} = \lambda(\mu)^{-1} G_{\mathrm{HS}}$$

is geometrized along the RG radial direction. Writing

$$u(\tau) = u_0 \,\mathrm{sech}\, \tau,$$

the universal logarithmic period response determines

$$S(u) = u \log u - u - u_0 \log u_0 + u_0,$$

and the radial Riesz field h is uniquely determined by

$$-\big(p(\tau)h'(\tau)\big)' = S(\tau), \qquad h'(0) = 0, \qquad h(\infty) = 0,$$

with

$$p(\tau) = (\sinh 2\,\tau - 2\tau)^{2/3}.$$

Therefore the strict-planar Yang–Mills color algebra, Hilbert–Schmidt/Riesz structure and RG flow, together with the single identification

$$\boxed{\tau_{\mathrm{per}} = \tau_{\mathrm{YM}}},$$

reconstruct

$$\boxed{(T^*S^3, J, \Omega, g_{\mathrm{St}}, h).}$$

10.23 Consequence for the subsequent master-field construction

The significance of the reconstruction is that the deformed conifold is not introduced as an independent dual background. The logical direction is instead

$$\boxed{\text{Yang-Mills} \longrightarrow \text{large-}N \text{ color sphere} \longrightarrow \text{RG period geometry} \longrightarrow \text{deformed conifold.}}$$

The complex structure is generated by the Yang–Mills source-response data, while the radial weight is generated by its RG flow. Infinitesimal scalar-source variations consequently define $(2,1)$-form directions,

$$\delta\Omega^{(2,1)} \in \Omega^{2,1}(T^*S^3),$$

and the physical running metric becomes a radially varying Riesz structure. Only after these structures have been reconstructed does one introduce the corresponding closed collective field theory. The later spectral problem

$$\bar{\partial}^\dagger \bar{\partial}\mathcal{G} = M^2 h\mathcal{G}, \qquad \partial\mathcal{G} = 0,$$

is therefore formulated on a geometry whose topology, complex structure, invariant metric and radial Riesz weight have all been fixed by the large-N color and RG construction rather than assumed independently.

11 Planar Single-Trace Observables and the Collective One-Particle Spectrum

11.1 Purpose

In Chapter 10 we have reconstructed from the large-N Yang–Mills color and renormalization-group data the geometric carrier

$$X = T^*S^3,$$

together with its complex structure, Stenzel metric, and radial Riesz weight $h(\tau)$. We now have to establish the physical meaning of the normal modes of this geometry from the Yang–Mills point of view. Two statements are required.

First, a collective mode must define a gauge-invariant single-trace observable of the planar theory.

Second, the normal modes of the quadratic FMR action must correspond to the one-particle intermediate states appearing in the spectral representation of the two-point functions of these observables.

Neither statement requires a string-theoretic interpretation. They follow from the cyclic structure of the normalized matrix trace, the large-N collective map, and the spectral decomposition of the quadratic FMR action.

11.2 The physical planar field space and its geometric representation

Let

$$\mathcal{M}_{\mathrm{FMR}} = \Lambda_\infty / \sim_{\partial\Lambda_\infty}$$

denote the planar gauge-reduced configuration space represented by the fundamental modular region. At the vacuum, its physical tangent space will be denoted

$$\mathcal{H}_{\mathrm{YM}}^{\mathrm{phys}} = T_{[0]}\mathcal{M}_{\mathrm{FMR}}.$$

The construction in Chapter 9 and 10 gives a geometric realization

$$C_\infty : \mathcal{H}_{\mathrm{YM}}^{\mathrm{phys}} \longrightarrow \mathcal{H}_{\mathrm{coll}}, \tag{11.1}$$

where

$$\mathcal{H}_{\mathrm{coll}} \subset \ker \partial \cap L_h^2 \Omega^{2,1}(X).$$

The map C_∞ is not merely a correspondence between mode labels. It transports the physical quadratic structures. The Yang–Mills Riesz metric

$$\frac{1}{\lambda(\mu)} G_{\mathrm{HS}}$$

has been geometrized as the weighted inner product

$$G_h(\Psi, \Phi) = \int_X h(\tau)\, \Psi \wedge *_6 \overline{\Phi}.$$

Thus

$$G_{\mathrm{YM}}(a, b) = G_h(C_\infty a, C_\infty b). \tag{11.1}$$

Likewise, the spatial quadratic form of the planar FMR theory is represented by the geometric differential acting on the collective fields. In the reconstructed complex geometry this differential is represented by $\bar{\partial}$. Hence, on the physical planar sector,

$$C_\infty Q_{\mathrm{YM}} = \bar{\partial} C_\infty. \tag{11.2}$$

These two relations are sufficient to transport the quadratic Yang–Mills action into its geometric form.

11.3 Derivation of the full large-N master-field action

We now derive the nonlinear planar master-field action directly from Yang–Mills theory on the gauge-reduced configuration space.
Let

$$A(t) \in \Omega^1\big(S^3, \mathfrak{su}(N)\big)$$

denote the spatial gauge connection and let

$$A_0(t) \in \Omega^0\big(S^3, \mathfrak{su}(N)\big)$$

denote the temporal component. Before choosing temporal gauge, the electric and magnetic curvatures are

$$E_A = \dot{A} - D_A A_0,$$

and

$$F_A = d_{S^3}A + \frac{1}{2}[A, A].$$

After extracting the universal planar normalization,

$$\mathcal{N}_\infty \sim N^2,$$

the Yang–Mills action may be written schematically as

$$S_{\mathrm{YM}} = \frac{\mathcal{N}_\infty}{2}\int dt\,\left[\|E_A\|^2_{\mathrm{YM}} - \|F_A\|^2_{\mathrm{YM}}\right], \tag{11.3}$$

where the physical spatial configuration variable is the gauge orbit represented by the fundamental modular region. The temporal field A_0 is nondynamical and enforces Gauss' law. At finite N, the color algebra is represented by the fuzzy sphere. On the smooth large-N core,

$$\frac{1}{i\hbar_N}[A_N, B_N] \longrightarrow \{a, b\}_{S^2}, \qquad \hbar_N \sim N^{-1},$$

so that

$$\mathfrak{su}(N) \quad \xrightarrow{N\to\infty} \quad C_0^\infty\left(S^2_{\mathrm{col}}\right) \simeq \mathfrak{sdiff}\left(S^2_{\mathrm{col}}\right).$$

The planar spatial connection is therefore represented by

$$A \in \Omega^1\left(S^3, \mathfrak{sdiff}\left(S^2_{\mathrm{col}}\right)\right).$$

Likewise,

$$A_0 \in C^\infty\left(S^3, \mathfrak{sdiff}\left(S^2_{\mathrm{col}}\right)\right).$$

The finite-N covariant derivative and curvature consequently have the smooth planar limits

$$D_A \longrightarrow d_{S^3} + \{A, \cdot\}_{S^2},$$

$$F_A \longrightarrow d_{S^3}A + \frac{1}{2}\{A, A\}_{S^2}.$$

Thus the non-Abelian matrix structure has not disappeared. It has become the classical Poisson algebra of Hamiltonian vector fields on the color sphere. The RG construction of Chapter 10 identifies the color sphere equivariantly with the unit cotangent directions,

$$S^2_{\mathrm{col}} \simeq S(T^*S^3),$$

while the one-dimensional Yang–Mills RG trajectory supplies the cotangent radius. The resulting smooth completion is $X = T^*S^3$. The Yang–Mills period data then determine the complex structure J of X, the holomorphic volume form

$$\Omega \in \Omega^{3,0}(X),$$

the compatible Stenzel metric g_{St}, and the radial Riesz weight h. Hence these structures are properties of the RG-completed planar Yang–Mills variables rather than independently introduced background data. The complex structure converts the real $\mathrm{SDiff}(S^2)$-valued gauge field into a Beltrami-type polyvector field

$$\mu \in PV^{1,1}(X) = \Omega^{0,1}(X, T^{1,0}X).$$

Similarly, the temporal gauge field becomes a degree-zero polyvector

$$\phi_0 \in PV^{1,0}(X).$$

Since the large-N color vector fields are Hamiltonian and therefore divergence-free, the resulting polyvectors lie in the divergence-free subcomplex associated with the Calabi–Yau volume form,

$$\mathrm{div}_\Omega\, \mu = 0, \qquad \mathrm{div}_\Omega\, \phi_0 = 0. \tag{11.4}$$

Contraction with Ω gives the corresponding differential forms

$$\mathcal{G} = \mu \lrcorner\, \Omega \in \Omega^{2,1}(X), \tag{11.5}$$

and

$$\mathcal{A}_0 = \phi_0 \lrcorner\, \Omega \in \Omega^{2,0}(X). \tag{11.6}$$

The Calabi–Yau divergence identity

$$(\mathrm{div}_\Omega\, \mu) \lrcorner\, \Omega = \partial(\mu \lrcorner\, \Omega) \tag{11.7}$$

therefore gives

$$\partial \mathcal{G} = 0,$$

and analogously

$$\partial \mathcal{A}_0 = 0.$$

The smooth large-N gauge algebra is thus represented by the divergence-free polyvector complex, or equivalently by the corresponding ∂-closed differential-form complex. The Poisson bracket on the color variables is transported by the RG-generated complex polarization to the Schouten bracket of polyvector fields. Consequently the planar Yang–Mills covariant differential becomes

$$D_\mu = \bar{\partial} + [\mu, \cdot]_{\mathrm{Sch}}. \tag{11.8}$$

In particular, the spatial Yang–Mills curvature is carried to the Maurer–Cartan curvature

$$\mathcal{F}(\mu) = \bar{\partial}\mu + \frac{1}{2}[\mu, \mu]_{\mathrm{Sch}}. \tag{11.9}$$

After contraction with the holomorphic volume form we write

$$\mathcal{F}(\mathcal{G}) = \bar{\partial}\mathcal{G} + \frac{1}{2}[\mathcal{G}, \mathcal{G}]_\Omega, \tag{11.10}$$

where the bracket on forms is the bracket induced from the Schouten bracket,

$$[\mathcal{G}_1, \mathcal{G}_2]_\Omega := [\mu_1, \mu_2]_{\mathrm{Sch}} \lrcorner\, \Omega, \qquad \mathcal{G}_i = \mu_i \lrcorner\, \Omega. \tag{11.11}$$

Thus

$$d_{S^3} A + \frac{1}{2}\{A, A\}_{S^2} \quad \longrightarrow \quad \bar{\partial}\mathcal{G} + \frac{1}{2}[\mathcal{G}, \mathcal{G}]_\Omega. \tag{11.12}$$

The temporal component is transported in the same way. The Yang–Mills electric field

$$E_A = \dot{A} - D_A A_0$$

becomes the covariant time derivative

$$D_t \mu = \dot{\mu} - \bar{\partial}\phi_0 - [\mu, \phi_0]_{\mathrm{Sch}}, \tag{11.13}$$

or, in the $(2,\cdot)$-form representation,

$$D_t \mathcal{G} = \dot{\mathcal{G}} - \bar{\partial}\mathcal{A}_0 - [\mathcal{G}, \mathcal{A}_0]_\Omega. \tag{11.14}$$

We have therefore obtained the full nonlinear Yang–Mills curvature complex in geometric variables:

$$E_A \longrightarrow D_t \mathcal{G},$$

$$F_A \longrightarrow \mathcal{F}(\mathcal{G}) = \bar{\partial}\mathcal{G} + \frac{1}{2}[\mathcal{G}, \mathcal{G}]_\Omega.$$

The remaining ingredient is the metric. The normalized matrix trace becomes integration over the color sphere in the large-N representation. The RG completion then geometrizes the scale-dependent physical FMR metric

$$G_{\mathrm{YM},\mu} = \frac{1}{\lambda(\mu)} G_{\mathrm{HS}}$$

into the radial Riesz structure determined by h. For degree-one master-field variations this gives

$$\langle \alpha, \beta \rangle_h = \int_X h\, \alpha \wedge *_{\mathrm{St}}\, \overline{\beta}. \tag{11.15}$$

In particular,

$$\|E_A\|^2_{\mathrm{YM}} \longrightarrow \|D_t \mathcal{G}\|^2_h. \tag{11.16}$$

The adjacent curvature degree carries the reference Stenzel pairing,

$$\langle U, V \rangle_0 = \int_X U \wedge *_{\mathrm{St}}\, \overline{V},$$

so that

$$\|F_A\|^2_{\mathrm{YM}} \longrightarrow \|\mathcal{F}(\mathcal{G})\|^2_0. \tag{11.17}$$

The degree dependence of these two pairings is essential. The RG-generated weight h is the Riesz metric of the dynamical degree-one field, whereas the curvature degree retains the unwarped Stenzel norm. Substitution into the Yang–Mills action therefore gives the full nonlinear planar master-field action

$$\boxed{S_\infty[\mathcal{G},\mathcal{A}_0]=\frac{\mathcal{N}_\infty}{2}\int dt\left[\left\|\dot{\mathcal{G}}-\bar{\partial}\mathcal{A}_0-[\mathcal{G},\mathcal{A}_0]_\Omega\right\|_h^2-\left\|\bar{\partial}\mathcal{G}+\frac{1}{2}[\mathcal{G},\mathcal{G}]_\Omega\right\|_0^2\right],} \tag{11.18}$$

subject to

$$\partial\mathcal{G}=0,\qquad \partial\mathcal{A}_0=0.$$

Equivalently,

$$\boxed{S_\infty=\frac{\mathcal{N}_\infty}{2}\int dt\left[\|D_t\mathcal{G}\|_h^2-\|\mathcal{F}(\mathcal{G})\|_0^2\right].} \tag{11.19}$$

This is the large-N master-field formulation of Yang–Mills theory on the gauge-reduced configuration space. It has been obtained from the Yang–Mills action by three successive operations:

$$\boxed{\begin{array}{c}\text{matrix color algebra}\\ \Downarrow\ N\to\infty\\ \mathrm{SDiff}\left(S^2_{\mathrm{col}}\right)\\ \Downarrow\ \text{RG completion}\\ T^*S^3\\ \Downarrow\ J,\Omega\\ \text{divergence-free polyvector complex}\\ \Downarrow\ \lrcorner\,\Omega\\ \partial\text{-closed }(2,\cdot)\text{-form complex.}\end{array}}$$

The nonlinear information of the microscopic gauge theory has thereby been reorganized into the geometry

$$\boxed{X,\qquad J,\qquad \Omega,\qquad g_{\mathrm{St}},\qquad h,}$$

together with the induced Schouten/Maurer–Cartan structure of the collective field. The temporal field $\mathcal{A}_0$ has no independent kinetic term. Variation with respect to it gives the nonlinear Gauss constraint. Writing

$$D_{\mathcal{G}}=\bar{\partial}+[\mathcal{G},\cdot]_\Omega, \tag{11.20}$$

this constraint takes the covariant form

$$\boxed{D_{\mathcal{G}}^{\dagger,h}D_t\mathcal{G}=0,} \tag{11.21}$$

with the adjoint determined by the degree-one weighted Riesz metric. For the remainder of the spectral analysis it is convenient to choose temporal gauge,

$$\mathcal{A}_0=0. \tag{11.22}$$

The action then becomes

$$\boxed{S_\infty[\mathcal{G}] = \frac{\mathcal{N}_\infty}{2}\int dt\left[\left\|\dot{\mathcal{G}}\right\|_h^2 - \left\|\bar{\partial}\mathcal{G} + \frac{1}{2}[\mathcal{G},\mathcal{G}]_\Omega\right\|_0^2\right], \qquad \partial\mathcal{G} = 0.} \tag{11.23}$$

Temporal gauge does not remove Gauss' law. The constraint obtained by varying $\mathcal{A}_0$ before gauge fixing must continue to be imposed on the physical field space. Because the collective map transports the physical gauge-reduced metric and gauge directions, the image of every Yang–Mills FMR representative is stationary with respect to closed-string gauge variations. Hence the closed-field image lies in the warped Gauss slice,

$$D_{\mathcal{G}}^{\dagger,h} D_t \mathcal{G} = 0.$$

The converse need not hold: the warped Gauss condition is only the local stationarity condition, whereas the FMR is defined by global absolute minimization along the full gauge orbit. In particular, for a degree-one closed master field

$$\mathcal{G} \in \ker\partial \cap \Omega^{2,1}(X),$$

the physical norm is

$$\|\mathcal{G}\|_h^2 = \int_X h(\tau)\,\mathcal{G}\wedge *_6\,\overline{\mathcal{G}},$$

where $*_6$ is the Hodge star of the Stenzel metric. So the closed-side analogue of the Yang–Mills fundamental modular region is

$$\boxed{\mathfrak{F}_{\mathrm{cl}} = \left\{\mathcal{G} \,:\, \|\mathcal{G}\|_h \le \left\|\mathcal{G}^\Lambda\right\|_h \quad \text{for every gauge-equivalent } \mathcal{G}^\Lambda\right\}.} \tag{11.24}$$

That is, one chooses the absolute norm-minimizing representative on every closed-string gauge orbit. At the linearized level around a background $\mathcal{G}_\star$, write

$$\mathcal{G} = \mathcal{G}_\star + \Psi. \tag{11.25}$$

The infinitesimal gauge transformation is schematically

$$\delta_\Lambda \Psi = D\Lambda, \tag{11.26}$$

with, at the simplest background,

$$D = \bar{\partial}, \tag{11.27}$$

or more generally the background-deformed differential. Then consider

$$F_\Psi(\Lambda) = \left\|\Psi + \bar{\partial}\Lambda\right\|_h^2. \tag{11.28}$$

The first derivative at $\Lambda = 0$ is

$$\frac{d}{ds}\left\|\Psi + s\bar{\partial}\Lambda\right\|_h^2\big|_{s=0} = 2\,\mathrm{Re}\left\langle\bar{\partial}\Lambda, \Psi\right\rangle_h. \tag{11.29}$$

Using the physical adjoint,

$$\left\langle \bar{\partial}\Lambda, \Psi \right\rangle_h = \left\langle \Lambda, \bar{\partial}^\dagger h \Psi \right\rangle,$$

so stationarity for every Λ gives

$$\bar{\partial}^\dagger h \Psi = 0. \tag{11.30}$$

11.4 The physical mass operator

Define the nonnegative quadratic form

$$\mathfrak{q}_h[\Psi, \Phi] = \left\langle \bar{\partial}\Psi, \bar{\partial}\Phi \right\rangle_0 \tag{11.31}$$

on the physical collective Hilbert space. The Hilbert product is

$$\langle \Psi, \Phi \rangle_h = G_h(\Psi, \Phi). \tag{11.32}$$

Assuming that $\mathfrak{q}_h$ is densely defined and closed, the representation theorem for nonnegative quadratic forms determines a unique positive self-adjoint operator K_h such that

$$\mathfrak{q}_h[\Psi, \Phi] = \left\langle K_h^{1/2}\Psi, K_h^{1/2}\Phi \right\rangle_h.$$

On its smooth strong domain,

$$\boxed{K_h = h^{-1}\bar{\partial}^\dagger \bar{\partial}.} \tag{11.33}$$

The normal modes therefore satisfy

$$K_h \Psi_{nI} = M_{nI}^2 \Psi_{nI}. \tag{11.34}$$

Here I denotes the complete set of angular and polarization quantum numbers. Because the reconstructed geometry and the radial weight $h(\tau)$ are $SO(4)$-invariant,

$$[K_h, SO(4)] = 0.$$

The physical Hilbert space consequently decomposes into irreducible $SO(4)$ sectors,

$$\mathcal{H}_{\text{coll}} = \widehat{\bigoplus_I} \mathcal{H}_I. \tag{11.35}$$

We choose an orthonormal eigenbasis satisfying

$$G_h\left(\Psi_{nI}, \Psi_{mJ}\right) = \int_X h(\tau)\, \Psi_{nI} \wedge *_6\, \overline{\Psi_{mJ}} = \delta_{nm}\delta_{IJ}. \tag{11.36}$$

Different irreducible $SO(4)$ channels cannot mix. If the same irreducible representation appears with multiplicity, the remaining finite-dimensional multiplicity space can itself be diagonalized. Thus the decomposition

$$I = (s, \ell, m, \alpha)$$

diagonalizes the quadratic problem channel by channel.

11.5 Why a collective mode defines a single-trace Yang–Mills observable

We now establish the operator correspondence. Let $\mathcal{A}_{\mathrm{YM}}$ denote the algebra generated by the gauge-covariant Yang–Mills letters

$$F_{\mu\nu}, \qquad D_\rho F_{\mu\nu}, \qquad D_{\rho_1} D_{\rho_2} F_{\mu\nu}, \ldots.$$

At finite N, a normalized gauge-invariant single-trace observable has the form

$$\mathcal{O}[a_1, \ldots, a_k] = \mathrm{tr}_N(a_1 a_2 \cdots a_k). \tag{11.37}$$

The trace obeys

$$\mathrm{tr}_N(a_1 a_2 \cdots a_k) = \mathrm{tr}_N(a_2 \cdots a_k a_1).$$

Therefore every single-trace operator defines a cyclic multilinear functional [34]. The single-trace observable sector is consequently naturally embedded in the cyclic complex of the Yang–Mills field algebra,

$$\mathcal{O}_{1\mathrm{tr}} \subset CC_{\mathrm{cyc}}(\mathcal{A}_{\mathrm{YM}}). \tag{11.38}$$

The large-N color representation preserves this structure because the normalized matrix trace becomes the invariant integral over the color sphere,

$$\mathrm{tr}_N \longrightarrow \int_{S^2_{\mathrm{col}}} d\,\mu.$$

Consequently the cyclic trace pairing survives the planar limit. Now let

$$\Psi \in \mathcal{H}_{\mathrm{coll}}$$

be a collective mode. Using the Riesz pairing it defines the linear functional

$$J_\Psi(\delta\mathcal{G}) = G_h(\Psi, \delta\mathcal{G}).$$

Pulling this functional back through the collective map gives

$$\mathfrak{O}_\Psi(a) = G_h(\Psi, C_\infty a). \tag{11.39}$$

Since C_∞ preserves the normalized cyclic color pairing, $\mathfrak{O}_\Psi$ lies in the planar single-trace cyclic sector. Thus

$$\Psi \longmapsto \mathfrak{O}_\Psi \in CC^{1\mathrm{tr}}_{\mathrm{cyc}}(\mathcal{A}_{\mathrm{YM}}). \tag{11.40}$$

This is the required map from collective modes to Yang–Mills observables.

11.6 Channel decomposition of the single-trace observables

Because the collective map is $SO(4)$-equivariant, a mode transforming in an irreducible representation I defines a Yang–Mills observable transforming in the same channel. We denote it by

$$\mathcal{O}_I(t).$$

If P_I denotes the equivariant projector onto channel I, then

$$\mathcal{O}_I = \mathfrak{O}_{P_I\Psi}.$$

For inequivalent irreducible representations,

$$I \neq J,$$

$SO(4)$ invariance implies

$$\langle \mathcal{O}_I(t)\mathcal{O}_J(0)\rangle = 0. \tag{11.40}$$

Thus the same representation decomposition which diagonalizes the collective mass operator also diagonalizes the planar two-point functions. This establishes the physical meaning of the channel label I.

11.7 Expansion of the planar master-field fluctuations

The eigenfunctions

$$\{\Psi_{nI}\}$$

form an orthonormal spectral basis of the physical linearized collective field space. Consequently,

$$\delta\mathcal{G}(t) = \sum_{n,I} q_{nI}\,(t)\Psi_{nI}. \tag{11.41}$$

The coefficients

$$q_{nI}(t)$$

are the coordinates of the planar fluctuation in this basis. Using orthogonality,

$$\left\|\dot{\delta\mathcal{G}}\right\|_h^2 = \sum_{n,I} |\dot{q}_{nI}|^2,$$

and

$$\left\|\bar{\partial}\delta\mathcal{G}\right\|_0^2 = \sum_{n,I} M_{nI}^2\,|q_{nI}|^2.$$

Substituting into the quadratic action gives

$$\boxed{S_{\mathrm{YM}}^{(2)} = \frac{\mathcal{N}_\infty}{2} t[|\dot{q}_{nI}(t)|^2 - M_{nI}^2|q_{nI}(t)|^2].} \tag{11.42}$$

The quadratic planar theory therefore becomes an infinite collection of decoupled harmonic oscillators.

11.8 Quantization and the one-particle states

Introduce canonically normalized variables

$$Q_{nI} = \sqrt{\mathcal{N}_\infty}\, q_{nI}.$$

Canonical quantization gives creation and annihilation operators satisfying

$$\left[a_{nI}, a_{mJ}^\dagger\right] = \delta_{nm}\delta_{IJ}.$$

The corresponding one-particle states are

$$|n, I\rangle = a_{nI}^\dagger |0\rangle. \tag{11.43}$$

They have rest energy

$$E_{nI} = M_{nI}.$$

Thus every normal mode Ψ_{nI} defines one elementary one-particle excitation of the planar FMR theory.

11.9 Identification with planar glueball states

The original theory is pure Yang–Mills. Its physical states are gauge-invariant color singlets. The observables $\mathcal{O}_I$ constructed above are normalized gauge-invariant single traces. Large-N factorization implies

$$\langle \mathcal{O}_1 \mathcal{O}_2 \rangle = \langle \mathcal{O}_1 \rangle \langle \mathcal{O}_2 \rangle + O(N^{-2}).$$

More generally, the coupling of a normalized single-trace operator to multiparticle states is suppressed by powers of $1/N$. Therefore, at strict planar order, a single-trace operator creates an elementary one-particle color-singlet excitation, while products of single traces create the corresponding multiparticle states. For pure Yang–Mills, these elementary color-singlet excitations are the glueballs. Hence

$$\boxed{\Psi_{nI} \leftrightarrow |n, I\rangle_{\text{glueball}}.} \tag{11.44}$$

The normal-mode spectrum of K_h is therefore the planar one-particle glueball spectrum, channel by channel.

11.10 Operator overlaps and the pole expansion

The operator $\mathcal{O}_I$ can couple to every radial excitation carrying the same quantum numbers. At linear order,

$$\mathcal{O}_I(\omega) = \sum_n f_{nI}\, Q_{nI}(\omega) + O(N^{-1}), \tag{11.45}$$

where

$$f_{nI} = \langle 0 | \mathcal{O}_I | n, I \rangle \tag{11.46}$$

is the operator-state overlap. The oscillator propagator is

$$\langle Q_{nI}(\omega)Q_{mJ}(-\omega)\rangle = \frac{i\delta_{nm}\delta_{IJ}}{\omega^2 - M_{nI}^2 + i\epsilon}.$$

Consequently,

$$\langle \mathcal{O}_I(\omega)\mathcal{O}_J(-\omega)\rangle_{\text{conn}} = \delta_{IJ}\sum_n \frac{i|f_{nI}|^2}{\omega^2 - M_{nI}^2 + i\epsilon} + O(N^{-2}). \tag{11.47}$$

After Wick rotation,

$$G_I^{(2)}(\omega_E) = \sum_n \frac{|f_{nI}|^2}{\omega_E^2 + M_{nI}^2} + O(N^{-2}). \tag{11.48}$$

Therefore the masses M_{nI} computed from the collective geometric eigenvalue problem are precisely the pole masses of the planar Yang–Mills single-trace two-point functions whenever

$$f_{nI} \neq 0.$$

11.11 Spectral meaning of the collective geometry

The complete correspondence may therefore be summarized as

$$\text{FMR physical fluctuation} \longrightarrow \Psi_{nI} \longrightarrow \mathcal{O}_I \longrightarrow |n, I\rangle \longrightarrow M_{nI}.$$

Equivalently,

$$\text{Spec } K_h = \text{planar one-particle glueball spectrum}$$

in the single-trace channels with nonzero overlap. The correspondence contains more information than the eigenvalues alone. Each collective eigenmode determines

$$(I,\, M_{nI},\, f_{nI}),$$

namely its Yang–Mills quantum numbers, its pole mass and its coupling to the corresponding single-trace operator.

11.12 Consequence for the scalar sector

The scalar sector is particularly important because its source is already directly tied to the RG construction. The Yang–Mills action contains

$$\frac{1}{\lambda}\operatorname{tr} F_{\mu\nu}F^{\mu\nu},$$

and therefore

$$\frac{\partial S_{\text{YM}}}{\partial(1/\lambda)} \propto \int d^4x \ \operatorname{tr} F_{\mu\nu}F^{\mu\nu}.$$

Thus $\operatorname{tr} F^2$ is the single-trace scalar operator conjugate to motion along the running-coupling direction. The preceding RG reconstruction identified this source direction with variation of the period matrix and hence with variation of the reconstructed complex structure. The scalar

collective modes therefore have a particularly direct source-level relation to Yang–Mills. They necessarily couple to scalar single-trace operators. Hence the two $SO(4)$-singlet towers found in the geometric spectral problem must correspond to scalar glueball channels,

$$\Psi^{(1)}_{n,\mathrm{scalar}} \leftrightarrow |n, 0^{P_1 C_1}\rangle, \tag{11.49}$$

and

$$\Psi^{(2)}_{n,\mathrm{scalar}} \leftrightarrow |n, 0^{P_2 C_2}\rangle. \tag{11.50}$$

The remaining task is therefore no longer to establish that these modes are glueballs. That follows from the single-trace operator-state correspondence above. What remains is to determine their discrete quantum numbers

$$P, \qquad C,$$

and thereby identify which two scalar Yang–Mills glueball towers they represent. This provides an independent spectral test of the proposed identification between Yang–Mills RG data and the reconstructed complex-structure geometry.

12 The Planar FMR Partition Function and the Gaussian Master-Field Measure

12.1 Strict planar master field and concentration

The chapters above have derived the full large-N gauge-reduced Yang–Mills action in the RG-completed geometric variables,

$$S_\infty[\mathcal{G}, \mathcal{A}_0] = \frac{\mathcal{N}_\infty}{2} \int dt \big[\|D_t \mathcal{G}\|_h^2 - \|\mathcal{F}(\mathcal{G})\|_0^2 \big],$$

where

$$\mathcal{N}_\infty \sim N^2,$$

$$D_t \mathcal{G} = \dot{\mathcal{G}} - D_{\mathcal{G}} \mathcal{A}_0,$$

$$\mathcal{F}(\mathcal{G}) = \bar{\partial} \mathcal{G} + \frac{1}{2} [\mathcal{G}, \mathcal{G}]_\Omega.$$

The field $\mathcal{G}$ is therefore not an additional effective degree of freedom introduced after taking the planar limit. It is the RG-completed geometric representative of the gauge-reduced Yang–Mills master field. Let $\mathcal{G}_\star$ denote a vacuum solution of the full planar equations of motion, together with the corresponding Gauss constraint. Equivalently,

$$\delta S_\infty[\mathcal{G}_\star] = 0$$

on the physical gauge-reduced field space. The overall factor N^2 implies that the unrescaled physical measure becomes concentrated on the planar master-field state. This statement also follows directly from large-N factorization [34]. For normalized single-trace observables O_i,

$$\langle O_1 \cdots O_k \rangle_c = O\left(N^{2-2k}\right) \tag{12.1}$$

in the conventional normalized variables, and in particular

$$\mathrm{Var}(O) = O(N^{-2}).$$

Hence, the variance of every normalized single-trace observable vanishes in the strict planar state

$$O \longrightarrow \langle O \rangle_\infty. \tag{12.2}$$

We denote the corresponding planar master-field configuration by $\mathcal{G}_\star$. This statement concerns the strict $N = \infty$ state itself and does not require weak convergence of finite-N configuration-space probability measures. The nontrivial quantum sector of the strict planar theory consists of canonically normalized fluctuations about this master field. Their natural planar scaling is

$$\mathcal{G} - \mathcal{G}_\star = O(N^{-1}). \tag{12.3}$$

We therefore introduce the canonically rescaled fluctuation

$$\Phi = N(\mathcal{G} - \mathcal{G}_\star). \tag{12.4}$$

This distinction is essential. The master field $\mathcal{G}_\star$ contains the nonperturbatively reorganized large-N Yang–Mills background, while Φ describes the finite quantum fluctuations about that background.

12.2 Exact strict-planar Gaussian scaling limit

Because the full nonlinear action has already been derived, the Gaussian nature of the strict planar fluctuation theory follows directly from its N-scaling.

Write

$$\mathcal{G} = \mathcal{G}_\star + \frac{1}{N}\Phi. \tag{12.5}$$

Since $\mathcal{G}_\star$ solves the planar equations of motion, the term linear in Φ vanishes. Expanding the gauge-reduced action gives

$$S_{\mathrm{FMR}}\left[\mathcal{G}_\star + \frac{\Phi}{N}\right] = S_{\mathrm{FMR}}[\mathcal{G}_\star] + \frac{1}{2N^2} H_\star[\Phi,\Phi] + \frac{1}{3!\,N^3} V_\star^{(3)}[\Phi^3] + \frac{1}{4!\,N^4} V_\star^{(4)}[\Phi^4] + \cdots,$$

where $H_\star$ is the physical Hessian of the derived master-field action and $V_\star^{(k)}$ denote its higher variations. Multiplying by the universal planar factor N^2 gives

$$N^2 S_{\mathrm{FMR}} = N^2 S_{\mathrm{FMR}}[\mathcal{G}_\star] + \frac{1}{2} H_\star[\Phi,\Phi] + \frac{1}{3!\,N} V_\star^{(3)}[\Phi^3] + \frac{1}{4!\,N^2} V_\star^{(4)}[\Phi^4] + \cdots. \tag{12.6}$$

$$S_{\mathrm{fluct}}^{N=\infty}[\Phi] := \left[N^2 S_{\mathrm{FMR}} - N^2 S_{\mathrm{FMR}}[\mathcal{G}_\star]\right]_{N=\infty}. \tag{12.7}$$

Thus the strict $N = \infty$ fluctuation theory is exactly quadratic. All interaction vertices between canonically normalized master-field fluctuations vanish at strict $N = \infty$. The corresponding probability measure will be constructed directly from this quadratic form in Section 12.7. Formally

$$d\gamma_\infty(\Phi) \propto \exp\left[-\frac{1}{2} H_\star[\Phi, \Phi]\right]. \tag{12.8}$$

For the vacuum used in the spectral analysis, and after choosing temporal gauge only after deriving Gauss' law, the Hessian is

$$H_\star[\Phi, \Phi] = \int dt \left[\left\|\dot{\Phi}\right\|_h^2 + \left\|\bar{\partial}_{\mathcal{G}_\star}\Phi\right\|_0^2\right]$$

in Euclidean signature, where

$$\bar{\partial}_{\mathcal{G}_\star} = \bar{\partial} + [\mathcal{G}_\star, \cdot]_\Omega$$

is the linearization of the Maurer–Cartan curvature about the master-field background. For the reference vacuum

$$\mathcal{G}_\star = 0,$$

this reduces to

$$H_0[\Phi, \Phi] = \int dt \left[\left\|\dot{\Phi}\right\|_h^2 + \left\|\bar{\partial}\Phi\right\|_0^2\right]. \tag{12.9}$$

The physical fluctuation field additionally obeys

$$\partial\Phi = 0$$

and the linearized warped Gauss constraint. Therefore the strict planar theory is not a free microscopic Yang–Mills theory. Rather, the Stenzel geometry, complex structure and radial Riesz weight h retain the nonperturbative information of the microscopic theory, while interactions among the canonically normalized collective excitations are $1/N$-suppressed and vanish at strict $N = \infty$.

12.3 Independent derivation from large-N factorization

The Gaussian character can also be derived directly from planar counting. Let

$$\mathcal{O}_i = \frac{1}{N}\mathrm{Tr}(\cdots)$$

be normalized single-trace observables. The connected k-point functions obey the standard planar scaling

$$\langle \mathcal{O}_1 \cdots \mathcal{O}_k \rangle_c = O\big(N^{2-2k}\big). \tag{12.10}$$

Define the centered, canonically scaled fluctuations

$$\hat{\mathcal{O}}_i = N(\mathcal{O}_i - \langle \mathcal{O}_i \rangle). \tag{12.11}$$

Their connected cumulants satisfy

$$\begin{aligned} \big\langle \hat{\mathcal{O}}_1 \hat{\mathcal{O}}_2 \big\rangle_c &= O(1), \\ \big\langle \hat{\mathcal{O}}_1 \cdots \hat{\mathcal{O}}_k \big\rangle_c &= O\big(N^{2-k}\big), \qquad k \geq 3. \end{aligned}$$

Hence

$$\lim_{N\to\infty} \big\langle \hat{\mathcal{O}}_1 \cdots \hat{\mathcal{O}}_k \big\rangle_c = 0, \qquad k \geq 3. \tag{12.12}$$

Thus every connected strict-planar cumulant of canonically normalized single-trace fluctuations above second order vanishes. The strict-planar cumulant functional is therefore quadratic. This is precisely the cumulant structure of a centered Gaussian field. The independent Bochner–Minlos construction below produces the Gaussian measure with covariance determined by the planar Hessian. Large-N factorization therefore provides an independent consistency check of the strict-planar Gaussian master-field measure; it is not being used here as a theorem of weak convergence of finite-N measures.

large-N factorization $\quad\Longrightarrow\quad$ Gaussian master-field fluctuations.

Importantly, this does not say that finite-N Yang–Mills is free. The interactions are encoded in $1/N$-suppressed higher cumulants.

12.4 Transport to the RG-completed geometric variables

Let

$$C_\infty$$

be the collective map constructed in the preceding sections and let

$$\mathcal{C} = DC_\infty|_{A_\star}$$

denote its differential at the planar vacuum. The rescaled physical fluctuation is mapped to

$$\Psi = \mathcal{C}a \tag{12.13}$$

The geometric reconstruction established that the physical FMR Riesz metric is transported to

$$\langle \Psi, \Phi \rangle_h = \int_X h(\tau)\Psi \wedge *_6 \overline{\Phi}.$$

Likewise, the physical spatial quadratic form becomes

$$\mathfrak{q}_h[\Psi] = \big\|\bar{\partial}\Psi\big\|_0^{2.}$$

Therefore the Euclidean quadratic action for the planar fluctuation field is

$$S_E^{(2)}[\Psi] = \frac{1}{2}\int_{\mathbb{R}} dt_E \left[\left\|\partial_{t_E}\Psi\right\|_h^2 + \left\|\bar{\partial}\Psi\right\|_0^2\right] \tag{12.14}$$

This is precisely the Euclidean continuation of the large-N collective action obtained previously.

12.5 The Euclidean covariance operator

Let

$$K_h = h^{-1}\bar{\partial}^\dagger\bar{\partial}$$

be the positive self-adjoint spatial operator on the physical collective Hilbert space.

Then

$$\left\|\bar{\partial}\Psi\right\|_0^2 = \langle\Psi, K_h\Psi\rangle_h.$$

Consequently,

$$S_E^{(2)} = \frac{1}{2}\langle\Psi, \mathcal{A}_E\Psi\rangle,$$

Hence

$$\mathcal{A}_E = -\partial_{t_E}^2 + K_h \tag{12.5}$$

is a nonnegative self-adjoint Euclidean operator. Its inverse, understood by spectral calculus on the chosen infrared-regular test-function space, defines the covariance form

$$\mathcal{C}(f,g) = \langle f, \mathcal{A}_E^{-1}g\rangle.$$

Chapter 14 will establish the stronger global coercive estimate

$$K_h \geq \Delta^2 I, \qquad \Delta > 0,$$

which removes the possible infrared singularity and proves that the covariance is massive. Its inverse

$$\mathfrak{C} = \mathcal{A}_E^{-1} \tag{12.16}$$

is the covariance operator of the limiting Gaussian process. Thus the FMR construction does not merely imply that the limiting process is Gaussian. It determines its covariance.

12.6 Spectral representation of the covariance

Let

$$K_h\Psi_{nI} = M_{nI}^2\Psi_{nI},$$

with

$$\left\langle\Psi_{nI}, \Psi_{mJ}\right\rangle_h = \delta_{nm}\delta_{IJ}.$$

Then

$$\mathfrak{C}(\omega_E) = \sum_{n,I} \frac{|\Psi_{nI}\rangle\langle\Psi_{nI}|}{\omega_E^2 + M_{nI}^2}. \tag{12.17}$$

In Euclidean time,

$$\mathfrak{C}(t_E, t_E{}') = \sum_{n,I} \frac{e^{-M_{nI}|t_E - t_E{}'|}}{2M_{nI}} |\Psi_{nI}\rangle\langle\Psi_{nI}|. \tag{12.18}$$

Hence the same mass spectrum obtained from the geometric Sturm–Liouville problem determines the covariance of the planar master-field fluctuations.

12.7 Rigorous construction of the infinite-dimensional Gaussian measure

The formal expression

$$D\Psi\, e^{-S_E^{(2)}[\Psi]}$$

should not be interpreted as an infinite-dimensional Lebesgue measure. Instead choose a nuclear test-function space

$$\mathcal{E} \subset L^2\big(\mathbb{R}_{t_E}\big) \,\widehat{\otimes}\, \mathcal{H}_{\text{phys}},$$

for example a Schwartz space in Euclidean time with values in the smooth domain

$$\mathcal{H}_\infty = \bigcap_{m=0}^{\infty} D\,(K_h^m).$$

For

$$f, g \in \mathcal{E},$$

define

$$\mathfrak{C}(f, g) = \langle f, \mathcal{A}_E^{-1} g\rangle. \tag{12.19}$$

Here

$$\mathcal{A}_E = -\partial_{t_E}^2 + K_h, \qquad \mathfrak{C} = \mathcal{A}_E^{-1}.$$

This bilinear form is positive:

$$\mathfrak{C}(f, f) \geq 0.$$

Assuming continuity on $\mathcal{E}$, the functional

$$\chi(f) = \exp\left[-\frac{1}{2}\mathfrak{C}(f, f)\right] \tag{12.20}$$

is continuous and positive definite. By the Bochner–Minlos [35] theorem it therefore defines a unique centered Gaussian probability measure

$$\gamma_{\mathbb{C}} \tag{12.21}$$

on the distributional dual [36]

$$\mathcal{E}'.$$

Its characteristic functional is

$$\boxed{\int_{\mathcal{E}'} e^{i\Psi(f)}\, d\gamma_{\mathbb{C}}(\Psi) = \exp\left[-\frac{1}{2}\mathbb{C}(f,f)\right].} \tag{12.22}$$

This is the rigorous infinite-dimensional replacement for the formal path-integral expression. The typical field sampled from $\gamma_{\mathbb{C}}$ need not lie in the smooth Hilbert space itself; it generally belongs to a larger distributional completion. This is standard for Gaussian quantum fields.

12.8 Identification of the planar FMR fluctuation measure

Let $\tilde{\nu}_N^{\mathrm{FMR}}$ denote the probability law of the centered and rescaled FMR variable

$$\Phi = N(\mathcal{G}_N - \mathcal{G}_\star).$$

The Gaussian measure $\gamma_{\mathcal{C}}$ constructed above is not introduced as a weak limit of finite-N probability measures. It is the probability measure of the strict $N = \infty$ fluctuation theory itself.

The exact strict-planar Euclidean action is

$$S_{E,\infty}^{(2)}[\Psi] = \frac{1}{2}\langle \Psi, \mathcal{A}_E \Psi\rangle, \qquad \mathcal{A}_E = -\partial_{t_E}^2 + K_h. \tag{12.23}$$

The corresponding covariance form

$$\mathbb{C}(f,g) = \langle f, \mathcal{A}_E^{-1} g\rangle$$

determines, by the Bochner–Minlos theorem, the unique centered Gaussian measure $\gamma_{\mathbb{C}}$ constructed in Section 12.7. We therefore identify the strict-planar FMR fluctuation measure with

$$\boxed{\gamma_{\mathrm{FMR}}^{N=\infty} = \gamma_{\mathbb{C}}.} \tag{12.24}$$

Let

$$\mathcal{C}: \mathcal{H}_{\mathrm{YM,phys}} \longrightarrow \mathcal{H}_{\mathrm{coll,phys}}$$

denote the collective Hilbert-space identification established in Chapter 9 and 10. Then the Yang–Mills and geometric realizations of the strict-planar measure are related by pushforward,

$$\mathcal{C}_* \gamma_{\mathrm{YM},\infty} = \gamma_{\mathbb{C}}.$$

Equivalently, when $\mathcal{C}$ is represented as an isomorphism on the physical image,

$$\gamma_{\mathrm{YM},\infty} = (\mathcal{C}^{-1})_* \gamma_{\mathbb{C}} \tag{12.25}.$$

This is an identification internal to the strict $N = \infty$ theory. No assertion is made that a sequence of finite-N FMR probability measures converges weakly to $\gamma_{\mathbb{C}}$. Such a convergence theorem is neither assumed nor required for the construction of the strict-planar quantum theory. The identification is independently consistent with planar factorization, since $\gamma_{\mathbb{C}}$ has vanishing connected cumulants of every order $k \geq 3$, exactly as required for canonically normalized strict-planar single-trace fluctuations.

12.9 The planar generating functional

The normalized generating functional of the strict-planar fluctuation process is now immediate. For a real Euclidean source $J \in \mathcal{E}$,

$$\mathcal{Z}_\infty[J] = \int_{\mathcal{E}'} e^{\Psi(J)}\, d\gamma_{\mathbb{C}}(\Psi). \tag{12.26}$$

Gaussian integration gives

$$\frac{\mathcal{Z}_\infty[J]}{\mathcal{Z}_\infty[0]} = \exp\left[\frac{1}{2}\langle J, \mathbb{C}J\rangle\right]. \tag{12.27}$$

Therefore

$$\boxed{W_\infty[J] = \log\frac{\mathcal{Z}_\infty[J]}{\mathcal{Z}_\infty[0]} = \frac{1}{2}\langle J, \mathbb{C}J\rangle.} \tag{12.28}$$

All connected cumulants follow by differentiation.

In particular,

$$\frac{\delta^2 W_\infty}{\delta J\, \delta J} = \mathbb{C}, \tag{12.29}$$

whereas

$$\frac{\delta^k W_\infty}{\delta J^k} = 0, \qquad k \geq 3. \tag{12.30}$$

This agrees exactly with the large-N factorization result derived in Section 12.3.

12.10 Relation to the original Yang–Mills partition function

The Gaussian functional describes the nontrivial fluctuations around the planar saddle. For comparison with the usual large-N expansion, at finite regulator the formal saddle expansion of the finite-N partition function has the asymptotic form

$$\boxed{Z_{N,L} \sim e^{-N^2 S_{\mathrm{FMR}}[\mathcal{G}_\star]} Z_L^{\mathrm{Gauss}}[1 + O(N^{-2})]} \tag{12.31}$$

At finite cutoff,

$$Z_L^{\mathrm{Gauss}} \propto \left(\det \mathcal{A}_{E,L}\right)^{-1/2},$$

for a real physical field, with the obvious modification for a complex polarization. The absolute determinant is regulator dependent and is not needed for the physical spectral statements. The invariant object is the normalized generating functional

$$\frac{Z[J]}{Z[0]},$$

whose strict planar value is given by (12.27). Hence two different orders in the planar expansion should be distinguished:

$$-N^2 S_{\mathrm{FMR}}[\mathcal{G}_\star] \tag{12.32}$$

is the leading master-field contribution to the vacuum free energy, whereas $\gamma_{\mathfrak{C}}$ describes the finite centered fluctuations about that master field.

12.11 Relation to the glueball spectrum

For a normalized single-trace operator $\mathcal{O}_I$, define the finite fluctuation

$$\hat{\mathcal{O}}_I = N(\mathcal{O}_I - \langle \mathcal{O}_I \rangle).$$

Through the operator map derived in Chapter 11, $\hat{\mathcal{O}}_I$ is a linear functional of the Gaussian field Ψ.

Therefore

$$\left\langle \hat{\mathcal{O}}_I(\omega_E)\hat{\mathcal{O}}_I(-\omega_E)\right\rangle = \sum_n \frac{|f_{nI}|^2}{\omega_E^2 + M_{nI}^2}. \tag{12.33}$$

The glueball poles are therefore the poles of the covariance of the Gaussian master-field process. Thus

$$\text{FMR Hessian} \leftrightarrow K_h \leftrightarrow \mathfrak{C} = (-\partial_t^2 + K_h)^{-1} \leftrightarrow \text{planar glueball propagator}.$$

This identifies the spectral construction with the directly constructed probabilistic strict-planar formulation.

12.12 Summary proposition

The result may be summarized as follows.

Proposition 12.1 — Strict-planar Gaussian FMR master-field measure.

Let $\mathcal{H}_{\mathrm{phys}}^{(1)}$ be the gauge-reduced physical one-particle Hilbert space obtained from the strict-planar FMR/master-field construction, and let K_h be the nonnegative self-adjoint spatial operator associated with

$$q_h[\Psi] = \left\|\bar{\partial}\Psi\right\|_0^2$$

and the warped physical Hilbert product

$$\langle \Psi, \Phi \rangle_h.$$

Let

$$\mathcal{A}_E = -\partial_{t_E}^2 + K_h$$

and choose a nuclear test-function space

$$\mathcal{E} \subset L^2(\mathbb{R}_{t_E}) \widehat{\otimes} \mathcal{H}_{\text{phys}}^{(1)}$$

on which the covariance form

$$\mathfrak{C}(f, g) = \langle f, \mathcal{A}_E^{-1} g \rangle$$

is continuous.

Then the exact strict-$N = \infty$ Euclidean fluctuation action is

$$S_{E,\infty}^{(2)}[\Psi] = \frac{1}{2} \langle \Psi, \mathcal{A}_E \Psi \rangle.$$

The Bochner–Minlos theorem determines a unique centered Gaussian probability measure

$$\gamma_{\mathfrak{C}}$$

on $\mathcal{E}'$, with characteristic functional

$$\int_{\mathcal{E}'} e^{i\Psi(f)} \, d\gamma_{\mathfrak{C}}(\Psi) = \exp\left[-\frac{1}{2} \langle f, \mathcal{A}_E^{-1} f \rangle \right]. \tag{12.34}$$

This measure is the Euclidean FMR fluctuation measure of the strict-planar Yang–Mills master field. Its normalized generating functional is

$$\frac{\mathcal{Z}_\infty[J]}{\mathcal{Z}_\infty[0]} = \exp\left[\frac{1}{2} \langle J, \mathcal{A}_E^{-1} J \rangle \right]. \tag{12.35}$$

Consequently,

$$\frac{\delta^2 W_\infty}{\delta J \, \delta J} = \mathcal{A}_E^{-1},$$

whereas

$$\frac{\delta^k W_\infty}{\delta J^k} = 0, \qquad k \geq 3. \tag{12.36}$$

Thus all connected strict-planar cumulants above second order vanish. This agrees independently with large-N factorization. Under the collective Hilbert-space identification established in the preceding chapters, $\gamma_{\mathfrak{C}}$ is the geometric realization of the strict-planar Yang–Mills FMR fluctuation measure, and its covariance determines the complete leading one-particle glueball spectrum. No weak convergence of finite-N Yang–Mills probability measures is assumed or required.

13 Complex Polarization, Scalar Source Normalization, and the Ultraviolet Two-Point Functions

13.1 Purpose

The Chapter 10 and 11 established that the two real scalar source directions of planar Yang–Mills form a polarized symplectic source-response space. At the level of the universal RG variables, the compact and dual periods define a canonical pair

$$(\Pi_A, \Pi_B),$$

with symplectic pairing

$$A \cap B = 1,$$

while the period matrix

$$\tau_{\mathrm{RG}} = \frac{\partial \Pi_B}{\partial \Pi_A}$$

determines a compatible complex structure. We now extend this complex structure from the universal RG plane to the complete scalar fluctuation space. The result fixes the previously undetermined relative source normalization between the raw master-field variables and the canonically normalized Yang–Mills scalar sources. In particular, we will derive the operator-valued factors

$$L^{1/2}, \qquad L^{-1/2}, \tag{13.1}$$

show that they transform the two apparently different raw residue laws

$$Z^{\mathrm{raw}}_{\mathrm{P},n} \sim n^3, \qquad Z^{\mathrm{raw}}_{\mathrm{Lef},n} \sim n^7 \tag{13.2}$$

into the common Yang–Mills behaviour

$$Z^{\mathrm{YM}}_n \sim n^5, \tag{13.3}$$

and finally show that this produces the expected nonlocal ultraviolet behaviour

$$\Pi(Q^2) \sim Q^4 \log Q^2 \tag{13.4}$$

up to local counterterms and the logarithmic corrections generated by asymptotic freedom.

13.2 The real scalar source-response phase space

Let $\mathcal{H}_{\mathrm{s}}$ denote the real Hilbert space of scalar collective fluctuations after the gauge quotient and all regularity conditions have been imposed. The scalar source-response space is the real phase space

$$\mathcal{P}_{\mathrm{s}} = \mathcal{H}_{\mathrm{s}} \oplus \mathcal{H}_{\mathrm{s}}. \tag{13.5}$$

We denote an element by $X = (q, p)$, where q and p are the two conjugate real source-response directions. The canonical symplectic form is

$$\omega(X, Y) = \langle q, p' \rangle_0 - \langle p, q' \rangle_0, \tag{13.6}$$

for

$$X = (q, p), \qquad Y = (q', p').$$

With respect to the reference Hilbert pairing, the symplectic operator is therefore

$$\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{13.6a}$$

This is the infinite-dimensional extension of the A/B-period plane reconstructed from the Yang–Mills RG data.

13.3 Positive scalar kinetic operator

Let L denote the positive self-adjoint scalar mass-squared operator obtained from the Hessian of the planar master-field action. Thus

$$L\Psi_n = M_n^2 \Psi_n. \tag{13.7}$$

We work on the massive physical scalar space, so that any genuine zero mode associated with variation of the background A-period has already been removed. Hence $L > 0$. The spectral theorem therefore defines unambiguously all real fractional powers

$$L^\alpha, \qquad \alpha \in \mathbb{R}.$$

In particular, $L^{1/2}$ is the positive frequency operator. The canonical quadratic scalar Hamiltonian can be written as

$$H_s(q, p) = \frac{1}{2}[\langle p, p \rangle_0 + \langle q, Lq \rangle_0]. \tag{13.8}$$

Its Hamiltonian flow is generated by

$$\mathcal{A} = \begin{pmatrix} 0 & 1 \\ -L & 0 \end{pmatrix}.$$

One immediately finds

$$-\mathcal{A}^2 = \begin{pmatrix} L & 0 \\ 0 & L \end{pmatrix}.$$

Because $L > 0$,

$$(-\mathcal{A}^2)^{1/2} = \begin{pmatrix} L^{1/2} & 0 \\ 0 & L^{1/2} \end{pmatrix}.$$

13.4 Complex structure from the polar decomposition

The positive compatible complex structure associated with the Hamiltonian flow is obtained from its polar decomposition. Define

$$J_L = -\mathcal{A}(-\mathcal{A}^2)^{-1/2}. \tag{13.9}$$

Explicitly,

$$J_L = \begin{pmatrix} 0 & -L^{-1/2} \\ L^{1/2} & 0 \end{pmatrix}. \tag{13.10}$$

Its square is

$$J_L^2 = \begin{pmatrix} -L^{-1/2}L^{1/2} & 0 \\ 0 & -L^{1/2}L^{-1/2} \end{pmatrix} = -1.$$

Hence

$$J_L^2 = -1. \tag{13.11}$$

Thus J_L equips the real scalar source-response space with a genuine complex structure. This is not an independently postulated structure. It is the unique positive polarization obtained by combining the canonical symplectic form with the positive scalar Hamiltonian. The occurrence of the square root is therefore forced by the polar decomposition of the physical Hamiltonian flow.

13.5 Relation to the RG complex structure

The previous RG construction already produced a complex structure on the universal two-dimensional period plane. The collective map was required to preserve both the canonical symplectic structure and its positive polarization. Consequently, the operator-valued complex structure J_L is the spectral extension of the previously established RG complex structure from the universal period plane to the complete scalar fluctuation space. Indeed, on an eigenmode satisfying

$$L\Psi_n = M_n^2\Psi_n,$$

one obtains

$$J_{L,n} = \begin{pmatrix} 0 & -M_n^{-1} \\ M_n & 0 \end{pmatrix}. \tag{13.12}$$

Each scalar normal mode therefore carries an ordinary two-dimensional complex structure exchanging its two real source-response components. The background RG complex structure and the scalar fluctuation complex structure are consequently not independent:

$$J_{\mathrm{RG}} \longrightarrow J_L. \tag{13.13}$$

13.6 The induced Riesz metric

A symplectic form and a compatible positive complex structure determine a Riemannian metric by

$$G_L(X,Y) = \omega(X, J_L Y).$$

Using the symplectic operator defined in (13.6a) and the complex structure (13.10),

$$\Omega J_L = \begin{pmatrix} L^{1/2} & 0 \\ 0 & L^{-1/2} \end{pmatrix}.$$

Therefore

$$G_L = \begin{pmatrix} L^{1/2} & 0 \\ 0 & L^{-1/2} \end{pmatrix}. \tag{13.14}$$

Equivalently,

$$\|X\|^2_{G_L} = \langle q, L^{1/2} q \rangle_0 + \langle p, L^{-1/2} p \rangle_0. \tag{13.15}$$

This is the crucial result. The two conjugate real scalar directions necessarily carry opposite half-powers of the same positive operator. Thus the factors

$$L^{1/2} \quad \text{and} \quad L^{-1/2}$$

are not additional phenomenological source dressings. They are the two diagonal components of the Riesz map generated by the compatible complex structure.

13.7 Source normalization between the master field and Yang–Mills

Let

$$b = \begin{pmatrix} b_{\mathrm{P}} \\ b_{\mathrm{Lef}} \end{pmatrix}$$

denote the raw scalar boundary/source coordinates appearing naturally in the master-field variational problem.

Let

$$j = \begin{pmatrix} j_{\mathrm{P}} \\ j_{\mathrm{Lef}} \end{pmatrix}$$

denote the canonically normalized Yang–Mills scalar sources. The physical source covector is obtained from the scalar source vector by the Riesz map associated with G_L. Hence

$$b = \mathcal{R}_L j, \qquad \mathcal{R}_L = \begin{pmatrix} L^{1/2} & 0 \\ 0 & L^{-1/2} \end{pmatrix}. \tag{13.16}$$

We choose the orientation of the RG phase space such that the primitive scalar direction is the first Darboux component and the Lefschetz direction the second. Reversing this orientation interchanges the two labels but does not alter the argument. Thus

$$b_{\mathrm{P}} = L^{1/2} j_{\mathrm{P}}, \tag{13.17}$$

while

$$b_{\mathrm{Lef}} = L^{-1/2} j_{\mathrm{Lef}}. \tag{13.18}$$

If

$$W_{\mathrm{YM}}[j] = W_{\mathrm{raw}}[b(j)],$$

then differentiation gives

$$\mathcal{O}_{\mathrm{P}}^{\mathrm{YM}} = L^{1/2}\mathcal{O}_{\mathrm{P}}^{\mathrm{raw}},$$

and

$$\mathcal{O}_{\mathrm{Lef}}^{\mathrm{YM}} = L^{-1/2}\mathcal{O}_{\mathrm{Lef}}^{\mathrm{raw}}.$$

Therefore

$$\boxed{\mathcal{O}_{\mathrm{P}}^{\mathrm{YM}} = L^{1/2}\mathcal{O}_{\mathrm{P}}^{\mathrm{raw}}, \qquad \mathcal{O}_{\mathrm{Lef}}^{\mathrm{YM}} = L^{-1/2}\mathcal{O}_{\mathrm{Lef}}^{\mathrm{raw}}.} \tag{13.19}$$

The relative half-power normalization is therefore derived from the scalar Kähler structure.

13.8 Transformation of the pole overlaps

Let

$$f_{\mathrm{P},n}^{\mathrm{raw}} = \langle 0|\mathcal{O}_{\mathrm{P}}^{\mathrm{raw}}|n, \mathrm{P}\rangle$$

and

$$f_{\mathrm{Lef},n}^{\mathrm{raw}} = \langle 0|\mathcal{O}_{\mathrm{Lef}}^{\mathrm{raw}}|n, \mathrm{Lef}\rangle.$$

Since

$$L\big|n, I\rangle = M_{I,n}^2\big|n, I\rangle,$$

equation (13.19) gives

$$f_{\mathrm{P},n}^{\mathrm{YM}} = M_{\mathrm{P},n} f_{\mathrm{P},n}^{\mathrm{raw}}, \tag{13.20}$$

and

$$f_{\mathrm{Lef},n}^{\mathrm{YM}} = M_{\mathrm{Lef},n}^{-1} f_{\mathrm{Lef},n}^{\mathrm{raw}}. \tag{13.21}$$

The pole residues are

$$Z_{I,n} = \big|f_{I,n}\big|^{2.}$$

Hence

$$\boxed{Z_{\mathrm{P},n}^{\mathrm{YM}} = M_{\mathrm{P},n}^2 Z_{\mathrm{P},n}^{\mathrm{raw}},} \tag{13.22}$$

while

$$\boxed{Z_{\mathrm{Lef},n}^{\mathrm{YM}} = M_{\mathrm{Lef},n}^{-2} Z_{\mathrm{Lef},n}^{\mathrm{raw}}.} \tag{13.23}$$

No ultraviolet matching assumption has entered these equations. They follow from the complex structure and its Riesz map.

13.9 Raw ultraviolet residue laws

The independent asymptotic analysis of the two scalar radial equations gives different raw source exponents. For the primitive sector, the UV source and response roots differ by

$$\Delta\sigma_{\mathrm{P}} = \frac{2}{3},$$

Giving $\nu_{\mathrm{P}} = 1$ and consequently

$$Z_{\mathrm{P},n}^{\mathrm{raw}} \sim n^3. \tag{13.24}$$

For the Lefschetz sector,

$$\Delta\sigma_{\mathrm{Lef}} = 2,$$

so that

$$\nu_{\mathrm{Lef}} = 3$$

and

$$Z_{\mathrm{Lef},n}^{\mathrm{raw}} \sim n^7. \tag{13.25}$$

These asymptotic laws were obtained directly from the two radial UV problems rather than from the desired Yang–Mills answer. Both Sturm–Liouville towers have asymptotically linear masses,

$$M_{I,n} = \kappa_I n + o(n), \tag{13.26}$$

with channel-dependent constants $\kappa_I > 0$. Therefore

$$M_{I,n}^2 \sim n^2. \tag{13.27}$$

13.10 Universal n^5 Yang–Mills residues

Combining (13.22), (13.24), and (13.27),

$$Z_{\mathrm{P},n}^{\mathrm{YM}} \sim n^2 n^{3.}$$

Hence

$$Z_{\mathrm{P},n}^{\mathrm{YM}} \sim n^5. \tag{13.28}$$

Likewise,

$$Z_{\mathrm{Lef},n}^{\mathrm{YM}} \sim n^{-2} n^7,$$

so

$$Z_{\mathrm{Lef},n}^{\mathrm{YM}} \sim n^5. \tag{13.29}$$

Thus the same complex polarization which pairs the two real scalar source directions also removes the apparent mismatch between their raw ultraviolet residues. The result is

$$\boxed{Z_{\mathrm{P},n}^{\mathrm{YM}} \sim Z_{\mathrm{Lef},n}^{\mathrm{YM}} \sim n^5.} \tag{13.30}$$

The two scalar towers therefore acquire the same ultraviolet source dimension after conversion from raw master-field coordinates to the physical Yang–Mills polarization. This common n^5 envelope is precisely the one required for four-dimensional scalar operators of engineering dimension four. The earlier source analysis found the same target from the Yang–Mills spectral density.

13.11 Continuous spectral density

Consider either scalar tower and write

$$s_n = M_n^2.$$

From

$$M_n \sim \kappa n$$

one obtains

$$s_n \sim \kappa^2 n^2,$$

and

$$\frac{ds_n}{dn} \sim 2\kappa^2 n. \tag{13.31}$$

The discrete spectral density is

$$\rho(s) = \sum_n Z_n \, \delta(s - s_n).$$

For large n, where the levels may be replaced by a smooth envelope,

$$Z_n \simeq \rho(s_n) \frac{ds_n}{dn}.$$

Therefore

$$\rho(s_n) \simeq \frac{Z_n}{ds_n/dn}.$$

Using

$$Z_n \sim An^5,$$

we obtain

$$\rho(s_n) \sim \frac{An^5}{2\kappa^2 n} = \frac{A}{2\kappa^2} n^4.$$

Since

$$n^2 \sim \frac{s}{\kappa^2},$$

it follows that

$$\rho(s) \sim C\, s^2, \tag{13.32}$$

where

$$C = \frac{A}{2\kappa^6}$$

up to subleading corrections. Thus the n^5 residue law is exactly equivalent to the four-dimensional dimension-four spectral density

$$\rho(s) \propto s^2.$$

13.12 Ultraviolet two-point function

The Euclidean scalar two-point function has the spectral representation [37,38]

$$\Pi(Q^2) = \sum_n \frac{Z_n}{Q^2 + M_n^2}. \tag{13.33}$$

In the ultraviolet the high-lying states dominate the nonlocal part, and the pole sum may be replaced by the continuous spectral density,

$$\Pi(Q^2) \simeq \int_0^\infty d\, s\, \frac{\rho(s)}{Q^2 + s}.$$

Using

$$\rho(s) \sim C s^2,$$

gives

$$\Pi_{\mathrm{UV}}(Q^2) \sim C \int_0^\infty d\, s\, \frac{s^2}{s + Q^2}.$$

Introduce a spectral cutoff Λ_s. Then

$$\frac{s^2}{s + Q^2} = s - Q^2 + \frac{Q^4}{s + Q^2},$$

and hence

$$\Pi_{\mathrm{UV}}(Q^2; \Lambda_s) \sim C \left[\frac{1}{2}\Lambda_s^2 - Q^2\Lambda_s + + Q^4 \log\frac{\Lambda_s + Q^2}{Q^2}\right]$$

The first two terms are polynomial in momentum and therefore correspond to local contact terms. The $\log \Lambda_s$ part multiplying Q^4 is removed by the local dimension-eight source counterterm. After renormalization, only the universal nonanalytic part remains:

$$\Pi_{\mathrm{ren}}(Q^2) = C\, Q^4 \log\frac{\mu^2}{Q^2} + P_2(Q^2) + o(Q^4 \log Q^2), \tag{13.34}$$

where

$$P_2(Q^2) = c_0 + c_1 Q^2 + c_2 Q^4$$

is scheme dependent. Equivalently, disregarding the convention-dependent overall sign,

$$\Pi_{\text{ren}}(Q^2) \sim Q^4 \log Q^2. \tag{13.35}$$

This is the characteristic nonlocal ultraviolet behaviour [39] of a four-dimensional scalar operator of dimension four.

13.13 The same result directly from the discrete tower

The logarithm may also be seen directly from the pole sum. Using

$$M_n^2 \simeq \kappa^2 n^2, \qquad Z_n \simeq A n^5,$$

we have

$$\Pi(Q^2) \sim A \sum_n \frac{n^5}{Q^2 + \kappa^2 n^2}.$$

At large Q, Euler–Maclaurin replacement of the asymptotic part of the sum gives

$$\Pi(Q^2) \sim A \int^{\infty} dn \, \frac{n^5}{Q^2 + \kappa^2 n^2}.$$

Setting $s = \kappa^2 n^2$ yields

$$dn = \frac{ds}{2\kappa\sqrt{s}},$$

and therefore

$$n^5 dn = \frac{s^2}{2\kappa^6} \, ds.$$

Hence

$$\Pi(Q^2) \sim \frac{A}{2\kappa^6} \int^{\infty} ds \, \frac{s^2}{Q^2 + s}. \tag{13.36}$$

This is exactly the spectral integral obtained above and therefore produces the same

$$Q^4 \log Q^2$$

nonlocal term. Thus the logarithm follows directly from the asymptotic glueball tower.

13.14 Primitive and Lefschetz correlators

Applying the result separately to the two scalar towers gives

$$\Pi_{\text{P}}^{\text{YM}}(Q^2) = C_{\text{P}} Q^4 \log \frac{\mu^2}{Q^2} + P_{\text{P}}(Q^2) + \cdots, \tag{13.37}$$

and

$$\Pi_{\mathrm{Lef}}^{\mathrm{YM}}(Q^2) = C_{\mathrm{Lef}} Q^4 \log\frac{\mu^2}{Q^2} + P_{\mathrm{Lef}}(Q^2) + \cdots. \tag{13.38}$$

The coefficients

$$C_{\mathrm{P}}, \qquad C_{\mathrm{Lef}}$$

need not be equal, because they depend on the absolute source normalization and on the asymptotic spacing of the two towers. What is universal is the power of momentum and the logarithmic nonanalyticity:

$$\Pi_{\mathrm{P}}, \Pi_{\mathrm{Lef}} \propto Q^4 \log Q^2. \tag{13.39}$$

Thus both scalar master-field towers have the same ultraviolet operator dimension after the complex-structure Riesz normalization.

13.15 Ultraviolet RG expansion of the scalar source

The preceding sections determine the nonlocal power-law part of the scalar two-point function from the planar master-field geometry. After the Riesz normalization implied by the scalar complex polarization, the ultraviolet spectral density has the dimension-four form

$$\rho_{\mathrm{geom}}(s) \sim s^2,$$

or equivalently, for asymptotically linear masses $M_n \sim n$,

$$Z_n^{\mathrm{geom}} \sim n^5. \tag{13.40}$$

The familiar perturbative logarithms of Yang–Mills theory have a different origin. They arise from the ultraviolet expansion of the running scalar source in the natural weak-coupling coordinate of the RG flow. The noncompact B-period is cut off at the ultraviolet scale μ, with

$$M_B = c_\mu \, \mu^3. \tag{13.41}$$

In the asymptotic region of the deformed conifold, this scale is identified with the large radial coordinate up to normalization,

$$r \propto \mu. \tag{13.42}$$

The compact period is fixed by the Yang–Mills infrared scale,

$$\Pi_A = c_A \, \Lambda_{\mathrm{YM}}^3, \tag{13.43}$$

so that the dimensionless global RG variable is

$$u = \frac{\Pi_A}{M_B} \propto \left(\frac{\Lambda_{\mathrm{YM}}}{\mu}\right)^3. \tag{13.44}$$

In the Stenzel coordinate this variable is

$$u(\tau) = u_0 \operatorname{sech} \tau.$$

Thus

$$\tau \to \infty, \qquad r \to \infty, \qquad \mu \to \infty, \qquad u \to 0$$

all describe the same ultraviolet end, but they play different roles. The variables r and μ are the natural asymptotic UV coordinates, whereas u provides a global dimensionless parametrization of the RG trajectory extending from the ultraviolet boundary through the throat to the infrared cap. The Yang–Mills scalar source is the source conjugate to the running coupling direction. In the conventions used here, the natural running source coordinate is proportional to the inverse 't Hooft coupling,

$$\chi(\mu) := \frac{1}{\lambda(\mu)}.$$

The RG reconstruction identifies this quantity with the appropriate period-matrix coordinate of the reconstructed complex geometry,

$$\tau_{\mathrm{per}}(\mu) \propto \chi(\mu) = \frac{1}{\lambda(\mu)}, \tag{13.45}$$

up to fixed normalization and the complexified θ-angle contribution. The ultraviolet point corresponds to

$$\lambda = 0,$$

and therefore to

$$\chi = \infty.$$

The perturbative beta function determines the asymptotic expansion of $\chi(\mu)$ in the far-ultraviolet region,

$$\mu \to \infty.$$

Let

$$\mu \frac{d\lambda}{d\mu} = -\beta_0 \lambda^2 - \beta_1 \lambda^3 - \beta_2 \lambda^4 - \cdots.$$

Then

$$\chi = \lambda^{-1}$$

satisfies

$$\frac{d\chi}{d \log \mu} = \beta_0 + \frac{\beta_1}{\chi} + \frac{\beta_2}{\chi^2} + \cdots. \tag{13.46}$$

Introducing

$$L = \log \frac{\mu}{\Lambda_{\mathrm{YM}}},$$

one obtains the ultraviolet expansion

$$\chi(\mu) = \beta_0 L + \frac{\beta_1}{\beta_0} \log L + O\left(\frac{\log L}{L}\right). \tag{13.47}$$

Equivalently,

$$\frac{1}{\lambda(\mu)} = \beta_0 \log \frac{\mu}{\Lambda_{\mathrm{YM}}} + \frac{\beta_1}{\beta_0} \log \log \frac{\mu}{\Lambda_{\mathrm{YM}}} + \cdots. \tag{13.48}$$

The log log term and the higher inverse-logarithmic corrections are therefore not additional structures added to the bulk spectral problem. They are the weak-coupling ultraviolet expansion of the same RG source coordinate that is represented geometrically by the period matrix. Using the asymptotic identification

$$r \propto \mu,$$

the same expansion may be written as

$$\frac{1}{\lambda(r)} = \beta_0 \log \frac{r}{\Lambda_r} + \frac{\beta_1}{\beta_0} \log \log \frac{r}{\Lambda_r} + \cdots, \qquad r \to \infty. \tag{13.49}$$

Only if desired may this be re-expressed in terms of the global variable u. Since

$$u \propto \left(\frac{\Lambda_{\mathrm{YM}}}{\mu}\right)^3,$$

one has asymptotically

$$\log \frac{\mu}{\Lambda_{\mathrm{YM}}} = -\frac{13}{\log u} + \text{const},$$

and therefore

$$\frac{1}{\lambda(u)} = -\frac{\beta_0}{3} \log u + \frac{\beta_1}{\beta_0} \log[-\log u] + \cdots.$$

This form is secondary, however. The physically natural perturbative expansion is the expansion in the UV cutoff/radial variable $\mu \sim r$. Let the canonically normalized Yang–Mills scalar source be related to the geometrically natural source by

$$J_{\mathrm{YM}}(Q) = Z_J(Q)\, J_{\mathrm{geom}}(Q),$$

where the source normalization $Z_J(Q)$ is determined by the running inverse coupling according to the source convention fixed above. The Yang–Mills correlator is then obtained from the geometric correlator by the corresponding squared source Jacobian. Hence the ultraviolet two-point function separates naturally into

$$\boxed{\text{geometric power law} \times \text{running source normalization.}}$$

The first factor gives the dimension-four spectral envelope,

$$\rho_{\mathrm{geom}}(s) \sim s^2,$$

while the second generates the perturbative logarithmic dressing

$$\log Q\,, \qquad \log\log Q\,, \qquad \frac{\log\log Q}{\log Q}, \qquad \ldots.$$

The geometrical interpretation is now clear. The perturbative loop expansion is a local asymptotic expansion about the far-ultraviolet boundary

$$r, \mu \to \infty, \qquad \lambda \to 0.$$

It therefore tests only the asymptotic end of the reconstructed geometry. It constrains the ultraviolet behaviour of the period matrix and the source normalization, but it does not probe the full radial continuation. By contrast, the global variable

$$u(\tau) = u_0 \operatorname{sech} \tau$$

parametrizes the entire RG trajectory from $u = 0$ at the ultraviolet boundary to $u = u_0$ at the infrared endpoint. The extended radial region and the smooth deformed-conifold cap therefore encode information that is invisible to a finite-order weak-coupling expansion around $\lambda = 0$. The logical structure is thus

global planar RG/period geometry
⇓
deformed conifold with g_{St} and h
⇓
far-UV region $r \sim \mu \to \infty$
⇓
weak-coupling expansion in $\lambda(\mu)$
⇓
$\log \mu\,,\ \log\log \mu\,,\ \ldots.$

Agreement with the perturbative logarithmic structure therefore tests the ultraviolet boundary behaviour of the reconstructed master-field geometry, whereas the discrete glueball spectrum probes the full radial geometry, including the long planar throat and the nonperturbative infrared cap.

13.16 Numerical Verification of the Scalar Ultraviolet Residue Exponents

The preceding section derived the ultraviolet residue laws

$$Z_{\mathrm{P},n}^{\mathrm{raw}} \sim n^3, \qquad Z_{\mathrm{L},n}^{\mathrm{raw}} \sim n^7$$

from the asymptotic source-response structure of the primitive and Lefschetz scalar equations. These powers were obtained analytically from the ultraviolet indicial roots. They therefore did not rely on fitting the numerical spectrum. We now perform an independent numerical verification by computing the actual pole residues of the discretized boundary two-point

functions. The purpose of this calculation is twofold. First, it checks that the finite-dimensional spectral problem reproduces the source-response scaling derived analytically. Second, it provides an independent numerical test of the subsequent $L^{\pm 1/2}$ source normalization derived from the complex polarization.

13.17 Numerical extraction of pole residues

Let the generalized eigenvalue problem in either scalar channel be written as

$$Ku_n = \lambda_n B u_n,$$

with eigenvectors normalized according to

$$u_n^\dagger B u_m = \delta_{nm}.$$

The corresponding physical masses satisfy

$$M_n^2 \propto \lambda_n.$$

To compute a source residue, one must retain the ultraviolet source degree of freedom rather than impose a homogeneous boundary condition from the beginning. We therefore split the finite-element degrees of freedom into interior and boundary components,

$$u = \begin{pmatrix} u_I \\ u_B \end{pmatrix},$$

with

$$K = \begin{pmatrix} K_{II} & K_{IB} \\ K_{BI} & K_{BB} \end{pmatrix}, \qquad B = \begin{pmatrix} B_{II} & B_{IB} \\ B_{BI} & B_{BB} \end{pmatrix}.$$

For a chosen boundary source vector s, the coupling of the n-th normalizable eigenmode to this source is

$$f_n^{\mathrm{raw}} = u_n^\dagger (K_{IB} - \lambda_n B_{IB}) s. \tag{13.50}$$

The corresponding pole residue is

$$Z_n^{\mathrm{raw}} = |f_n^{\mathrm{raw}}|^2. \tag{13.51}$$

This is the finite-dimensional Schur-complement expression for the residue of the corresponding Dirichlet-to-Neumann boundary kernel. An overall constant normalization of s rescales every residue by the same constant and therefore has no effect on the ultraviolet exponent.

13.18 Primitive scalar source

The primitive scalar field is

$$\mathcal{G}_{\mathrm{P}}[v] = v(\tau)\Phi_0 - q(\tau)\Phi_2,$$

where

$$q(\tau) = \cosh\tau\ v(\tau) + \frac{1}{2}\sinh\tau\, v'(\tau).$$

Its radial differential operator is

$$D_{\mathrm{P}} v = v'' + 4\coth\tau\ v' + 4v.$$

Because the primitive variational problem is fourth order, cubic Hermite finite elements are used. Both v and v' are therefore explicit degrees of freedom at the ultraviolet boundary. The analytic UV analysis gives the non-normalizable source branch

$$v_{\mathrm{src}} \sim e^{-4\tau/3}.$$

Hence

$$\frac{v'_{\mathrm{src}}}{v_{\mathrm{src}}} = -\frac{4}{3}. \tag{13.52}$$

The numerical source direction can therefore be represented by

$$s_{\mathrm{P}} = \begin{pmatrix} 1 \\ -\frac{4}{3} \end{pmatrix} \tag{13.53}$$

The exact A-period zero mode is removed before analyzing the physical tower.

13.19 Primitive ultraviolet cutoff convergence

The residue exponent is extracted by fitting

$$\log Z_n = \alpha_{\mathrm{P}} \log M_n + c.$$

Because the numerical domain terminates at a finite value $\tau_{\max}$, it is important to test the exponent as this artificial UV wall is moved outward. Using the same intermediate spectral window, the measured exponent moves systematically toward the analytic prediction:

$\tau_{\max}$	α_{P}
18	2.7-2.8
20	2.8-2.9
22	≈ 3.0
24	≈ 3.1

The finite-$\tau_{\max}$ correction therefore has precisely the expected form: at smaller radial cutoffs the source and response branches have not yet separated completely, whereas increasing $\tau_{\max}$ drives the measured exponent toward

$$\boxed{\alpha_{\mathrm{P}} = 3.}$$

A more local diagnostic is the logarithmic slope

$$\alpha_n = \frac{\log Z_{n+1} - \log Z_{n-1}}{\log M_{n+1} - \log M_{n-1}}. \tag{13.54}$$

At the largest cutoff tested, the local slope develops a broad plateau around 3, with representative values decreasing from approximately

3.2

for the lower modes to values extremely close to

3.0

for the higher modes before finite-element cutoff effects become important. The numerical calculation therefore independently supports

$$\boxed{Z_{\mathrm{P},n}^{\mathrm{raw}} \sim M_n^{3.}} \tag{13.55}$$

13.20 Lefschetz scalar source

For the Lefschetz scalar,

$$\mathcal{G}_{\mathrm{L}} = f(\tau)\Phi_1,$$

the radial problem is second order. The existing correlator calculation retains the ultraviolet Dirichlet value of f as the external source and solves the generalized eigenvalue problem for the normalizable interior modes. Taking a unit boundary source,

$$s_{\mathrm{L}} = 1,$$

the coupling to the n-th mode is

$$f_{\mathrm{L},n}^{\mathrm{raw}} = u_n^{\dagger}(K_{IB} - \lambda_n B_{IB}). \tag{13.56}$$

Thus

$$Z_{\mathrm{L},n}^{\mathrm{raw}} = \left|f_{\mathrm{L},n}^{\mathrm{raw}}\right|^2. \tag{13.57}$$

Unlike the analytic indicial argument, this calculation extracts the residues directly from the finite-dimensional boundary response function.

13.21 Lefschetz resolution convergence

For the Lefschetz problem, a stable intermediate asymptotic window is found at approximately

$$n = 3, \ldots, 10$$

for

$$\tau_{\max} = 12.$$

A fit of the form

$$Z_{\mathrm{L},n}^{\mathrm{raw}} \propto M_n^{\alpha_{\mathrm{L}}}$$

gives approximately

$$\alpha_{\mathrm{L}} \simeq 7.15$$

at one resolution and

$$\alpha_{\mathrm{L}} \simeq 7.14$$

after increasing the finite-element resolution. The change in the exponent is therefore only of order

$$10^{-2}.$$

This shows that the extracted power is numerically stable under refinement of the discretization. The result is consistent with

$$\boxed{\alpha_{\mathrm{L}} = 7.} \tag{13.58}$$

At larger values of $\tau_{\max}$, the clean asymptotic window moves to higher mode number, as expected. This makes a simultaneous $(\tau_{\max}, N_{\mathrm{elem}})$ extrapolation more delicate than in the primitive channel. We therefore distinguish two statements:

primitive: cutoff and local-slope convergence toward 3,

whereas

Lefschetz: clear resolution-stable behaviour near 7.

A full two-parameter ultraviolet extrapolation of the Lefschetz residues would strengthen the latter further.

13.22 Asymptotically affine mass spectrum

Strictly speaking, the numerical fits above determine

$$Z_n \sim M_n^{\alpha}.$$

To convert this into a statement about powers of the excitation number n, one must independently establish

$$M_n \sim n.$$

The high-energy eigenvalues are indeed extremely accurately described by an affine spectral law

$$M_n = a(n + \delta) + o(1). \tag{13.59}$$

The constant δ is the usual finite spectral phase shift. For example, in stable ultraviolet windows the linear fits of M_n against n have coefficients of determination very close to unity,

$$R^2 > 0.9999.$$

The offset δ is important at moderate n. A naive fit of Z_n directly against the integer n therefore biases the exponent. The correct finite-n diagnostic is first

$$Z_n \sim M_n^{\alpha},$$

and only subsequently

$$M_n \simeq a(n+\delta).$$

Since

$$n+\delta \sim n$$

for large n, the numerical results imply

$$\boxed{Z_{\mathrm{P},n}^{\mathrm{raw}} \sim n^3, \qquad Z_{\mathrm{L},n}^{\mathrm{raw}} \sim n^7.} \tag{13.60}$$

13.23 Comparison with the independent analytic prediction

This agreement is nontrivial because the two calculations use different information. The analytic calculation begins with the UV differential equations and their source-response roots,

$$\sigma_s - \sigma_r,$$

and derives the Bessel index and hence the residue exponent. Thus

$$\text{UV indicial equations} \longrightarrow 3,\ 7.$$

The numerical calculation instead uses the full discretized operator and the boundary Schur complement,

$$\text{finite-element eigenproblem + boundary source} \longrightarrow Z_n,$$

and determines the exponent by fitting the actual computed pole residues. Thus independently,

$$Z_{\mathrm{P}}^{\mathrm{raw}} \sim M^3, \qquad Z_{\mathrm{L}}^{\mathrm{raw}} \sim M^7. \tag{13.61}$$

The numerical calculation therefore provides an independent check both of the UV asymptotic analysis and of the implementation of the scalar boundary source.

14 Mass Gap Theorem on $S^3 \times \mathbb{R}$

14.1 Cohomology of the constrained $(2, \bullet)$ complex

The physical master-field complex is not the unrestricted Dolbeault complex. Its degree-one fields satisfy

$$\partial\Psi = 0, \qquad \Psi \in \Omega^{2,1}(X),$$

and the gauge differential acts by

$$D_0 = \bar{\partial}$$

on ∂-closed $(2,0)$-forms. We therefore consider the complex

$$\mathcal{C}^q(X) = \{\alpha \in \Omega^{2,q}(X) : \partial\alpha = 0\}, \qquad D = \bar{\partial}. \tag{14.1}$$

Since

$$\partial\bar{\partial} + \bar{\partial}\partial = 0,$$

the differential preserves $\ker \partial$. Its degree-one cohomology is

$$H^1_{\bar{\partial}}(X) := \frac{\{\Psi \in \Omega^{2,1}(X) : \partial\Psi = 0,\ \bar{\partial}\Psi = 0\}}{\{\bar{\partial}\Lambda : \Lambda \in \Omega^{2,0}(X),\ \partial\Lambda = 0\}}. \tag{14.2}$$

We now identify this cohomology exactly.

Lemma 14.1 — Constrained Dolbeault cohomology of the deformed conifold

Let

$$X = X_a = \left\{ z \in \mathbb{C}^4 : \sum_{i=1}^{4} z_i^2 = a \right\}, \qquad a \neq 0,$$

be the deformed conifold. Then

$$H^1_{\bar{\partial}}(X) \simeq H^3_{\mathrm{dR}}(X;\mathbb{C}) \simeq \mathbb{C}. \tag{14.3}$$

Under this isomorphism the unique nontrivial class is detected by its period over the compact three-cycle

$$A \simeq S^3.$$

Consequently,

$$\boxed{[\Psi]_{\bar{\partial}} = 0 \quad \Leftrightarrow \quad \int_A \Psi = 0.} \tag{14.4}$$

Proof

We proceed in three steps.

Natural map to de Rham cohomology

Let

$$\Psi \in \Omega^{2,1}(X)$$

satisfy

$$\partial\Psi = 0, \qquad \bar{\partial}\Psi = 0.$$

Then

$$d\Psi = (\partial + \bar{\partial})\Psi = 0.$$

Hence Ψ determines a de Rham class

$$[\Psi]_{\mathrm{dR}} \in H^3_{\mathrm{dR}}(X;\mathbb{C}).$$

If

$$\Psi = \bar{\partial}\Lambda, \qquad \partial\Lambda = 0,$$

then

$$\Psi = \bar{\partial}\Lambda = \left(\partial + \bar{\partial}\right)\Lambda = d\Lambda.$$

Therefore constrained-exact fields are de Rham exact. Thus there is a well-defined map

$$\boxed{\iota\colon H^1_{\bar{\partial}}(X) \longrightarrow H^3_{\mathrm{dR}}(X;\mathbb{C}), \qquad [\Psi]_{\partial} \mapsto [\Psi]_{\mathrm{dR}}.} \tag{14.5}$$

We next prove that this map is injective.

Step 2. Injectivity

Suppose

$$[\Psi]_{\mathrm{dR}} = 0.$$

The deformed conifold X_a is a smooth affine hypersurface in $\mathbb{C}^4$; hence it is a Stein manifold. By Cartan's theorem B and the Dolbeault theorem,

$$H^{2,1}_{\bar{\partial}}(X) \simeq H^1\left(X, \Omega^2_{\mathrm{hol}}\right) = 0.$$

Therefore every $\bar{\partial}$-closed $(2,1)$-form is globally $\bar{\partial}$-exact. Thus there exists

$$\beta \in \Omega^{2,0}(X)$$

such that

$$\Psi = \bar{\partial}\beta. \tag{14.6}$$

This is not yet sufficient, because the gauge complex requires the primitive to satisfy

$$\partial\beta = 0.$$

Define

$$\Theta := \partial\beta.$$

Since

$$\partial\Psi = 0,$$

we have

$$\bar{\partial}\Theta = \bar{\partial}\partial\beta = -\partial\bar{\partial}\beta = -\partial\Psi = 0.$$

Furthermore,

$$\partial\Theta = 0.$$

Thus

$$\Theta \in \Omega^{3,0}(X)$$

is a holomorphic closed three-form. Now

$$d\beta = \partial\beta + \bar{\partial}\beta = \Theta + \Psi.$$

Taking de Rham classes gives

$$[\Theta]_{\mathrm{dR}} + [\Psi]_{\mathrm{dR}} = 0.$$

By assumption,

$$[\Psi]_{\mathrm{dR}} = 0,$$

and hence

$$[\Theta]_{\mathrm{dR}} = 0.$$

Because X is Stein, the holomorphic de Rham theorem says that the cohomology of the global holomorphic de Rham complex

$$\mathcal{O}(X) \xrightarrow{\partial} \Omega^1_{\mathrm{hol}}(X) \xrightarrow{\partial} \Omega^2_{\mathrm{hol}}(X) \xrightarrow{\partial} \Omega^3_{\mathrm{hol}}(X)$$

computes the ordinary complex de Rham cohomology. Therefore a holomorphic closed three-form with trivial de Rham class is holomorphically exact. There exists

$$\gamma \in \Omega^{2,0}_{\mathrm{hol}}(X)$$

such that

$$\Theta = \partial\gamma. \tag{14.7}$$

Set

$$\Lambda := \beta - \gamma.$$

Because γ is holomorphic,

$$\bar{\partial}\gamma = 0.$$

Hence

$$\bar{\partial}\Lambda = \bar{\partial}\beta - \bar{\partial}\gamma = \Psi.$$

Moreover,

$$\partial\Lambda = \partial\beta - \partial\gamma = \Theta - \Theta = 0.$$

We have therefore obtained

$$\boxed{\Psi = \bar{\partial}\Lambda, \qquad \partial\Lambda = 0.} \tag{14.8}$$

Thus Ψ is exact in the constrained complex, and

$$[\Psi]_\partial = 0.$$

Hence ι is injective.

$$\boxed{H^1_\partial(X) \hookrightarrow H^3_{\mathrm{dR}}(X;\mathbb{C}).} \tag{14.9}$$

Step 3. The image is one-dimensional

Topologically,

$$X = T^*S^3.$$

The cotangent fibres are contractible, so X deformation retracts onto the zero section,

$$X \simeq S^3.$$

Therefore

$$H^3_{\mathrm{dR}}(X;\mathbb{C}) \simeq H^3_{\mathrm{dR}}(S^3;\mathbb{C}) \simeq \mathbb{C}. \tag{14.10}$$

Thus the target of ι is one-dimensional. The master-field equations already exhibit a regular zero-energy $(2,1)$-form Ψ_A corresponding to infinitesimal variation of the compact A-period. It satisfies

$$\partial\Psi_A = 0, \qquad \bar{\partial}\Psi_A = 0,$$

and

$$\int_A \Psi_A \neq 0. \tag{14.11}$$

Therefore

$$[\Psi_A]_{\mathrm{dR}} \neq 0.$$

Hence the image of ι is nonzero. Since its target is one-dimensional and ι is injective,

$$H^1_\partial(X) \simeq H^3_{\mathrm{dR}}(X;\mathbb{C}) \simeq \mathbb{C}. \tag{14.12}$$

Finally, since $A \simeq S^3$ generates

$$H_3(X;\mathbb{Z}),$$

the period map

$$\mathrm{Per}_A : H^3_{\mathrm{dR}}(X;\mathbb{C}) \longrightarrow \mathbb{C}, \qquad [\Psi] \mapsto \int_A \Psi,$$

is an isomorphism. Combining the two maps gives

$$H^1_\partial(X) \xrightarrow{\simeq} \mathbb{C}, \qquad [\Psi]_\partial \mapsto \int_A \Psi. \tag{14.13}$$

Therefore

$$\boxed{\int_A \Psi = 0 \quad \Leftrightarrow \quad [\Psi]_\partial = 0.} \tag{14.14}$$

This proves the lemma. □

14.2 Fixed Yang–Mills scale removes the unique cohomological zero mode

The physical meaning of the preceding result is immediate. The compact period is identified with the RG-invariant Yang–Mills scale,

$$\Pi_A = c_A \Lambda_{\mathrm{YM}}^3.$$

The spectral theory is evaluated at fixed Yang–Mills theory, hence at fixed Λ_{YM}. Therefore physical fluctuations obey

$$\delta\Lambda_{\mathrm{YM}} = 0 \tag{14.15}$$

and consequently

$$\delta\Pi_A = 0. \tag{14.16}$$

For a master-field fluctuation Ψ,

$$\delta\Pi_A[\Psi] = \int_A \Psi.$$

The Lemma 14.1 therefore implies

$$\delta\Pi_A[\Psi] = 0 \quad \Longrightarrow \quad [\Psi]_\partial = 0. \tag{14.17}$$

Thus every ∂- and $\bar{\partial}$-closed fluctuation compatible with the fixed Yang–Mills background is exact in the constrained gauge complex:

$$\Psi = \bar{\partial}\Lambda, \qquad \partial\Lambda = 0. \tag{14.18}$$

Such a field is an infinitesimal gauge direction and represents the zero vector in the gauge-reduced physical configuration space. The only nontrivial cohomological zero-energy direction is therefore the variation of the compact A-period itself,

$$\boxed{H^1_\partial(X) = \mathrm{span}\{[\Psi_A]\}.} \tag{14.19}$$

That direction changes Λ_{YM}, and hence moves between different Yang–Mills theories rather than describing a physical excitation of a fixed theory. It must therefore be excluded from the glueball Hilbert space. This also explains conceptually the exact primitive zero mode found earlier: it is not a massless glueball. It is the tangent vector to the one-dimensional family of backgrounds parameterized by the compact period.

14.3 Triviality of the physical kernel

We can now finish the kernel argument without any channel-by-channel analysis.

Let

$$\Psi \in D(K_h)$$

satisfy

$$K_h \Psi = 0.$$

Since K_h is represented by the nonnegative quadratic form,

$$0 = \langle \Psi, K_h \Psi \rangle_h = \left\| \bar{\partial} \Psi \right\|_0^{2.}$$

Hence

$$\bar{\partial} \Psi = 0.$$

The physical field space already imposes

$$\partial \Psi = 0.$$

Therefore

$$[\Psi]_{\partial} \in H^1_{\partial}(X).$$

At fixed Yang–Mills scale,

$$\delta \Pi_A[\Psi] = 0,$$

so by Lemma 14.1,

$$[\Psi]_{\partial} = 0.$$

Hence

$$\Psi = \bar{\partial} \Lambda, \qquad \partial \Lambda = 0.$$

But exact elements are gauge directions and vanish in the gauge-reduced Hilbert space. Therefore

$$\ker K_h \,|_{\mathcal{H}_{\text{phys}}} = \{0\}. \tag{14.20}$$

Equivalently, if one represents the quotient by the warped Gauss slice, exact directions lie in

$$\overline{\text{Ran } D_0},$$

while the Gauss slice is

$$\ker D_0^{\dagger,h} = \left(\overline{\text{Ran } D_0} \right)^{\perp}.$$

A field belonging to both spaces must vanish. Thus the same conclusion follows directly in the Hilbert-space realization.

$$\boxed{\text{Ran } D_0 \cap \ker D_0^{\dagger,h} = \{0\}} \tag{14.21}$$

after quotienting the usual stabilizer directions.

14.4 Coercive estimate and positive mass gap

We now prove directly that the physical planar master-field operator possesses a strictly positive lower spectral bound.

Let

$$X = T^*S^3$$

be the deformed conifold equipped with the Stenzel metric. Consider the constrained complex

$$\mathcal{C}^q(X) = \ker \partial \cap \Omega^{2,q}(X), \qquad D_q = \bar{\partial}.$$

Thus

$$D_0 : \mathcal{C}^0 \longrightarrow \mathcal{C}^1, \qquad D_1 : \mathcal{C}^1 \longrightarrow \mathcal{C}^2,$$

with

$$D_1 D_0 = 0.$$

The degree-one physical field is represented by

$$\Psi \in \mathcal{C}^1,$$

modulo gauge transformations

$$\Psi \sim \Psi + D_0 \Lambda.$$

The spatial energy is

$$q[\Psi] = \|D_1 \Psi\|_0^2 = \left\|\bar{\partial}\Psi\right\|_0^2. \tag{14.22}$$

Because $D_1 D_0 = 0$, this form is gauge invariant. The physical norm is instead the RG-weighted norm

$$\|\Psi\|_h^2 = \int_X h|\Psi|^2 \, dV_{\mathrm{St}} \, . \tag{14.23}$$

The physical representative of each gauge orbit is the h-orthogonal one,

$$D_0^{\dagger,h}\Psi = 0. \tag{14.24}$$

Ellipticity of the constrained complex

We first verify that the complex is elliptic at degree one.

Let

$$\xi \neq 0$$

be a cotangent vector and write

$$\xi = \xi^{1,0} + \xi^{0,1}. \tag{14.25}$$

The principal symbol of ∂ is exterior multiplication by $\xi^{1,0}$, while that of $\bar{\partial}$ is exterior multiplication by $\xi^{0,1}$.

Define

$$E_q(\xi) = \ker(\xi^{1,0} \wedge : \Lambda^{2,q} \to \Lambda^{3,q}). \tag{14.26}$$

The symbol complex is

$$E_0(\xi) \xrightarrow{\xi^{0,1}\wedge} E_1(\xi) \xrightarrow{\xi^{0,1}\wedge} E_2(\xi). \tag{14.27}$$

Suppose

$$\beta \in E_1(\xi)$$

and

$$\xi^{0,1} \wedge \beta = 0.$$

By exactness of the exterior Koszul complex generated by $\xi^{0,1}$,

$$\beta = \xi^{0,1} \wedge \alpha \tag{14.28}$$

for some

$$\alpha \in \Lambda^{2,0}.$$

Since $\beta \in E_1(\xi)$,

$$0 = \xi^{1,0} \wedge \beta = -\xi^{0,1} \wedge \xi^{1,0} \wedge \alpha. \tag{14.29}$$

But

$$\xi^{1,0} \wedge \alpha \in \Lambda^{3,0},$$

and exterior multiplication by the nonzero $(0,1)$-form $\xi^{0,1}$ is injective on $\Lambda^{3,0}$. Therefore

$$\xi^{1,0} \wedge \alpha = 0. \tag{14.30}$$

Hence

$$\alpha \in E_0(\xi),$$

and thus

$$\ker \sigma(D_1) = \operatorname{im} \sigma(D_0). \tag{14.31}$$

Therefore the constrained complex is elliptic at degree one. Equivalently, the first-order operator

$$P = D_1 \oplus D_0^\dagger \tag{14.32}$$

is elliptic on $\mathcal{C}^1$.

Sobolev-Hodge estimate

We work in an admissible Lockhart–McOwen weighted Sobolev domain, with weight chosen away from the indicial spectrum of the asymptotic operator, such that P is Fredholm and the corresponding domain embeds continuously into the unweighted space $W^{1,2}(X)$. Standard elliptic estimates on asymptotically conical manifolds then give, modulo the finite-dimensional kernel of P,

$$\|\Psi\|_{W^{1,2}} \leq C(\|P\Psi\|_{L^2} + \|\Pi_{\ker P}\Psi\|_{L^2}).$$

Since X has real dimension six, the critical Sobolev embedding

$$W^{1,2}(X) \hookrightarrow L^3(X)$$

implies, after adjusting the constant,

$$\|\Psi\|^2_{L^3(X)} \leq C_{\mathrm{AC}}\Big(\|P\Psi\|^2_{L^2(X)} + \|\Pi_{\ker P}\Psi\|^2_{L^2(X)}\Big).$$

By the Fredholm Hodge decomposition on this admissible domain, each gauge class in the finite-energy sector admits an unweighted Coulomb representative Ψ_0, unique modulo the finite-dimensional kernel, satisfying

$$D_0^\dagger \Psi_0 = 0.$$

For this representative,

$$\|P\Psi_0\|^2_{L^2} = \|D_1\Psi_0\|^2_{L^2}$$

The Fredholm harmonic kernel represents the constrained degree-one cohomology on the chosen admissible domain. By Lemma 14.1,

$$H^1(\mathcal{C}^\cdot, D) \simeq \mathbb{C}[\Psi_A],$$

and this class is detected by the compact A-period,

$$[\Psi] \longmapsto \delta\Pi_A[\Psi] = \int_A \Psi.$$

After fixing the normalization of Ψ_A, the norm of the harmonic projection is therefore equivalent to the absolute value of this period. Hence the Sobolev–Hodge estimate takes the form

$$\|\Psi_0\|^2_{L^3(X)} \leq C_{\mathrm{AC}}\Big(\|D_1\Psi_0\|^2_{L^2(X)} + |\delta\Pi_A[\Psi_0]|^2\Big). \tag{14.34}$$

The finite-dimensional term is thus precisely the cohomological obstruction to inversion of the constrained elliptic complex. Since the Yang–Mills theory is considered at fixed RG-invariant scale,

$$\Pi_A = c_A \Lambda^3_{\mathrm{YM}},$$

physical fluctuations obey

$$\delta\Pi_A = 0.$$

Therefore, on the physical fixed-A-period sector,

$$\|\Psi_0\|^2_{L^3(X)} \leq C_{\mathrm{AC}} \|D_1 \Psi_0\|^2_{L^2(X)}. \tag{14.35}$$

Since $D_1 = \bar{\partial}$,

$$\|\Psi_0\|^2_{L^3} \leq C_{\mathrm{AC}} \left\|\bar{\partial} \Psi_0\right\|^2_0. \tag{14.36}$$

Integrability of the Yang–Mills Riesz weight

We next use the ultraviolet behaviour of the RG-generated weight. In the asymptotically conical radial coordinate,

$$dV_{\mathrm{St}} \sim r^5 \, dr \, dV_{T^{1,1}}, \tag{14.37}$$

while

$$h(r) \sim c_h \frac{\log r}{r^4}. \tag{14.38}$$

Therefore

$$h(r)^3 dV_{\mathrm{St}} \sim c_h^3 \frac{(\log r)^3}{r^{12}} r^5 \, dr \, dV_{T^{1,1}} = c_h^3 \frac{(\log r)^3}{r^7} \, dr \, dV_{T^{1,1}}.$$

Since

$$\int_R^\infty \frac{(\log r)^3}{r^7} \, dr < \infty,$$

the ultraviolet end contributes a finite amount. At the Stenzel cap h is smooth and finite, and every compact region has finite volume. Hence

$$\boxed{h \in L^3(X, dV_{\mathrm{St}}).} \tag{14.39}$$

Define

$$H_3 := \|h\|_{L^3(X)} < \infty.$$

Comparison of weighted and unweighted representatives

Let

$$[\Psi]$$

be a physical gauge class. Let Ψ_h denote its warped Gauss representative,

$$D_0^{\dagger,h} \Psi_h = 0.$$

Because this representative is orthogonal to all gauge directions with respect to the h-metric, it minimizes the weighted norm on its gauge orbit. Indeed, for every admissible Λ,

$$\begin{aligned}\|\Psi_h + D_0\Lambda\|_h^2 &= \|\Psi_h\|_h^2 + 2\,\mathrm{Re}\langle\Psi_h, D_0\Lambda\rangle_h + \|D_0\Lambda\|_h^2 \\ &= \|\Psi_h\|_h^2 + \|D_0\Lambda\|_h^2,\end{aligned}$$

because

$$D_0^{\dagger,h}\Psi_h = 0.$$

Thus

$$\|\Psi_h\|_h \leq \|\Psi'\|_h \tag{14.40}$$

for every representative Ψ' in the same gauge class. In particular, let Ψ_0 be the unweighted Coulomb representative,

$$D_0^{\dagger}\Psi_0 = 0.$$

Then

$$\boxed{\|\Psi_h\|_h^2 \leq \|\Psi_0\|_h^2} \tag{14.41}$$

Furthermore, because the curvature form is gauge invariant,

$$D_1(\Psi_0 - \Psi_h) = D_1 D_0 \Lambda = 0,$$

so

$$q[\Psi_0] = q[\Psi_h]. \tag{14.42}$$

Weighted Poincaré inequality

We now estimate the physical norm using Hölder's inequality:

$$\begin{aligned}\|\Psi_0\|_h^2 &= \int_X h\,|\Psi_0|^2\,dV_{\mathrm{St}} \\ &\leq \left(\int_X h^3\,dV_{\mathrm{St}}\right)^{1/3}\left(\int_X |\Psi_0|^3\,dV_{\mathrm{St}}\right)^{2/3} \\ &= \|h\|_{L^3}\|\Psi_0\|_{L^3}^2.\end{aligned}$$

Using the fixed-period Sobolev-Hodge estimate,

$$\|\Psi_0\|_{L^3}^2 \leq C_{\mathrm{AC}}\left\|\bar{\partial}\Psi_0\right\|_0^2,$$

we obtain

$$\|\Psi_h\|_h^2 \leq C_{\mathrm{AC}}\|h\|_{L^3}\left\|\bar{\partial}\Psi_0\right\|_0^2. \tag{14.43}$$

Gauge invariance of q gives

$$\left\|\bar{\partial}\Psi_0\right\|_0^2 = \left\|\bar{\partial}\Psi_h\right\|_0^2.$$

Hence

$$\|\Psi_h\|_h^2 \leq C_{\mathrm{AC}} H_3 \left\|\bar{\partial}\Psi_h\right\|_0^2 \tag{14.44}$$

Equivalently,

$$\left\|\bar{\partial}\Psi\right\|_0^2 \geq \Delta^2 \|\Psi\|_h^2 \tag{14.45}$$

where

$$\Delta^2 = \frac{1}{C_{\mathrm{AC}} \|h\|_{L^3}} > 0.$$

This is the desired global weighted Poincaré inequality.

14.5 Mass-gap theorem

Let K_h be the positive self-adjoint operator associated with

$$q[\Psi] = \left\|\bar{\partial}\Psi\right\|_0^2 \tag{14.46}$$

on the physical fixed-A-period warped Gauss slice.

Then

$$q[\Psi] = \langle \Psi, K_h \Psi \rangle_h.$$

The preceding estimate gives

$$\langle \Psi, K_h \Psi \rangle_h \geq \Delta^2 \langle \Psi, \Psi \rangle_h. \tag{14.47}$$

Therefore, in the sense of quadratic forms,

$$\boxed{K_h \geq \Delta^2 I.} \tag{14.48}$$

By the spectral theorem,

$$\boxed{\mathrm{spec}(K_h) \subset [\Delta^2, \infty).} \tag{14.49}$$

Consequently the planar master-field theory possesses a strictly positive mass gap,

$$\boxed{M_{\mathrm{gap}} = \inf \mathrm{spec}\, \sqrt{K_h} \geq \Delta > 0.} \tag{14.50}$$

Thus the positive gap does not rely on the numerical value of the lowest scalar eigenvalue. The proof uses only the global properties already established:

$$\boxed{\text{ellipticity of } \left(\ker \partial\,, \bar{\partial}\right) + \text{fixed } A\text{-period} + h \in L^3(X) \quad \Longrightarrow \quad K_h \geq \Delta^2 > 0.}$$

The exact primitive zero mode found earlier is consistent with this theorem: it lies precisely in the excluded direction that varies the compact A-period. Once Λ_{YM} is held fixed, that mode is not part of the physical fluctuation Hilbert space.

15 $\mathbb{R}^4$ Mass Gap Theorem and Proof

15.1 Exact Cochain Construction in the Decompactification Limit

The mass-gap theorem established above is formulated on the physical fixed-A-period collective Hilbert space associated with the deformed-conifold geometry. In order to pass from Yang–Mills theory on $S^3_R \times \mathbb{R}$ to the decompactified theory on $\mathbb{R}^4$, it is necessary to control not only the spectrum but also the manner in which the finite-radius Yang–Mills complex is embedded into the fixed master field complex.

The essential point is that the decompactification limit must not be implemented by deforming the ambient Dolbeault differential. The complex geometry of the deformed conifold remains fixed. All R-dependence is carried by the Yang–Mills differential, its embedding into the closed complex, and subsequently by the physical Riesz realization.

Introduce the dimensionless decompactification parameter

$$\varepsilon = \frac{1}{R\Lambda_{\mathrm{YM}}}.$$

Then

$$R \to \infty \qquad \Leftrightarrow \qquad \varepsilon \to 0.$$

Let

$$(\mathcal{V}^\bullet, d_\varepsilon)$$

denote the gauge-reduced planar Yang–Mills complex at finite R, and let

$$(\mathcal{C}^\bullet, D), \qquad D = \bar{\partial},$$

denote the fixed constrained master field complex,

$$\mathcal{C}^q = \ker \partial \cap \Omega^{2,q}(X).$$

The ambient differential therefore satisfies

$$D^2 = 0$$

independently of ε. The finite-radius differential satisfies

$$d_\varepsilon^2 = 0.$$

Write

$$d_\varepsilon = d_0 + \delta_\varepsilon,$$

where d_0 is the flat-limit differential and

$$\delta_\varepsilon \longrightarrow 0 \qquad (\varepsilon \to 0).$$

No assumption on the parity of δ_ε is required for the exact cochain construction. In a particular representation sector one may subsequently have

$$\delta_\varepsilon = \varepsilon d_1 + \varepsilon^2 d_2 + \cdots$$

or, if the first-order contribution vanishes by symmetry,

$$\delta_\varepsilon = \varepsilon^2 d_2 + O(\varepsilon^4).$$

The construction below applies to either case.

15.2 Flat-limit collective embedding

Let

$$\mathcal{C}_0^\bullet : (\mathcal{V}^\bullet, d_0) \longrightarrow (\mathcal{C}^\bullet, D)$$

be the flat-limit collective embedding. By construction it is a cochain map,

$$D_q C_0^q = C_0^{q+1} d_0^q. \tag{15.1}$$

The finite-R problem is to construct an embedding

$$\mathcal{C}_\varepsilon^\bullet : (\mathcal{V}^\bullet, d_\varepsilon) \longrightarrow (\mathcal{C}^\bullet, D)$$

such that

$$D_q C_\varepsilon^q = C_\varepsilon^{q+1} d_\varepsilon^q \tag{15.2}$$

holds exactly. The crucial feature is that $D = \bar{\partial}$ itself is not changed. This is the cochain formulation of the statement that the decompactification does not deform the complex structure of X. This is consistent with the physical construction developed above, in which the ambient Dolbeault differential remains fixed while the physical embedding and degree-dependent adjoints move.

15.3 Removal of the cohomological obstruction

The constrained complex contains one distinguished cohomological direction associated with variation of the compact A-period. As shown previously, this mode changes Λ_{YM} and therefore moves between different Yang–Mills theories rather than describing a physical excitation of a fixed theory. The physical fluctuation problem is consequently defined at fixed A-period. Let P_A denote the projection onto the removed period class. On the fixed-period physical complex we impose

$$P_A \Psi = 0.$$

The previous global cohomological analysis showed that, after removal of this class, every D-closed physical zero-energy fluctuation is exact and therefore gauge trivial. In particular the physical kernel of the one-sided master-field operator is trivial.

Let

$$H_{q+1} : \mathcal{C}^{q+1} \longrightarrow \mathcal{C}^q$$

be a contracting homotopy on this fixed-period complex, satisfying

$$D_q H_{q+1} + H_{q+2} D_{q+1} = I - P_A. \tag{15.3}$$

The required boundedness and graph-norm estimates for H will be established in the next subsection. At the present stage only the algebraic homotopy identity is required. The sole cohomological compatibility condition for the deformation is

$$\boxed{P_A C_\varepsilon^{q+1} \delta_\varepsilon^q = 0.} \tag{15.4}$$

Physically, equation (15.4) states that decompactification does not regenerate the excluded A-period variation.

15.4. Exact recursive construction

The closed $(2, q)$ complex terminates at degree three,

$$\mathcal{C}^0 \overset{D_0}{\to} \mathcal{C}^1 \overset{D_1}{\to} \mathcal{C}^2 \overset{D_2}{\to} \mathcal{C}^3.$$

We therefore construct C_ε recursively downward in degree. At the top degree set

$$C_\varepsilon^3 = C_0^3. \tag{15.5}$$

For $q = 2, 1, 0$, define

$$C_\varepsilon^q = C_0^q + H_{q+1} C_\varepsilon^{q+1} \delta_\varepsilon^q. \tag{15.6}$$

This is the exact compensating cochain deformation. The first correction has the familiar Green-homotopy form

$$C_\varepsilon^q - C_0^q = H_{q+1} C_\varepsilon^{q+1} \delta_\varepsilon^q.$$

Thus the compensator is not an independent master field normal mode. It is the unique horizontal correction required to preserve the chain condition when the open Yang–Mills differential changes. Iterating (15.6) gives the finite expression

$$\boxed{C_\varepsilon^q = C_0^q + \sum_{n=1}^{3-q} H_{q+1} \cdots H_{q+n}\, C_0^{q+n}\, \delta_\varepsilon^{q+n-1} \cdots \delta_\varepsilon^q.} \tag{15.7}$$

Because the complex has finite length, equation (15.7) is a finite sum rather than an infinite homological-perturbation series. This is the full cochain analogue of the exact construction obtained previously in the tensor sector.

15.5. Exact cochain theorem

Proposition 15.1 — Exact decompactifying cochain map.
Assume:

1. $d_\varepsilon^2 = 0$;

2. $D^2 = 0$;
3. C_0 satisfies $DC_0 = C_0 d_0$;
4. the fixed-period closed complex admits the homotopy (15.3);
5. the period obstruction (15.4) vanishes.

Then the maps defined recursively by (15.5)–(15.6) satisfy

$$DC_\varepsilon = C_\varepsilon d_\varepsilon \tag{15.8}$$

exactly.

Proof

The proof proceeds by descending induction in degree. At the highest nontrivial degree the statement follows directly from the definition. Assume that

$$D_{q+1} C_\varepsilon^{q+1} = C_\varepsilon^{q+2} d_\varepsilon^{q+1}. \tag{15.9}$$

Using (15.6),

$$D_q C_\varepsilon^q = D_q C_0^q + D_q H_{q+1} C_\varepsilon^{q+1} \delta_\varepsilon^q.$$

From the flat chain condition,

$$D_q C_0^q = C_0^{q+1} d_0^q.$$

Using the homotopy identity,

$$D_q H_{q+1} = I - P_A - H_{q+2} D_{q+1},$$

and the obstruction condition (15.4), we obtain

$$D_q C_\varepsilon^q = \&C_0^{q+1} d_0^q + C_\varepsilon^{q+1} \delta_\varepsilon^q - H_{q+2} D_{q+1} C_\varepsilon^{q+1} \delta_\varepsilon^q. \tag{15.10}$$

By the induction hypothesis,

$$D_{q+1} C_\varepsilon^{q+1} = C_\varepsilon^{q+2} d_\varepsilon^{q+1},$$

so

$$D_q C_\varepsilon^q = C_0^{q+1} d_0^q + C_\varepsilon^{q+1} \delta_\varepsilon^q - H_{q+2} C_\varepsilon^{q+2} d_\varepsilon^{q+1} \delta_\varepsilon^q. \tag{15.11}$$

Since

$$d_\varepsilon^2 = 0$$

and

$$d_0^2 = 0,$$

one has

$$d_\varepsilon^{q+1}\delta_\varepsilon^q = -\delta_\varepsilon^{q+1}d_0^q. \tag{15.12}$$

Therefore

$$\begin{aligned} D_q C_\varepsilon^q = {} & C_0^{q+1}d_0^q + C_\varepsilon^{q+1}\delta_\varepsilon^q \\ & + H_{q+2}C_\varepsilon^{q+2}\delta_\varepsilon^{q+1}d_0^q. \end{aligned}$$

Using the recursive definition at degree $q+1$,

$$C_\varepsilon^{q+1} = C_0^{q+1} + H_{q+2}C_\varepsilon^{q+2}\delta_\varepsilon^{q+1},$$

we obtain

$$D_q C_\varepsilon^q = C_\varepsilon^{q+1}d_0^q + C_\varepsilon^{q+1}\delta_\varepsilon^q.$$

Hence

$$D_q C_\varepsilon^q = C_\varepsilon^{q+1}\left(d_0^q + \delta_\varepsilon^q\right) = C_\varepsilon^{q+1}d_\varepsilon^q.$$

This proves (15.8). □

15.6. Consequences

The result has three immediate consequences. First, the ambient complex is independent of the compactification radius,

$$D = \bar{\partial}, \qquad D^2 = 0$$

for all R. Second, all finite-R dependence is transferred to the embedding

$$V_\varepsilon^q = \operatorname{Ran} C_\varepsilon^q \subset \mathcal{C}^q. \tag{15.13}$$

Thus decompactification changes how the Yang–Mills state space is realized inside the master field space rather than changing the master complex itself. Third, the construction identifies the origin of the compensating master field polarizations. If

$$\delta_\varepsilon = \varepsilon d_1 + \varepsilon^2 d_2 + \cdots,$$

then

$$C_\varepsilon = C_0 + \varepsilon C_1 + \varepsilon^2 C_2 + \cdots$$

with

$$C_1^q = H_{q+1}C_0^{q+1}d_1^q. \tag{15.14}$$

If the first-order deformation vanishes in a given physical sector, then instead

$$C_\varepsilon - C_0 = O(\varepsilon^2).$$

The precise graph-norm estimate will be established below once the boundedness of the physical homotopy is included. The exact cochain construction by itself does not imply that the physical adjoint is independent of ε. On the contrary, the physical degree-dependent Riesz metrics move with R, and therefore

$$D_\varepsilon^\sharp = G_{1,\varepsilon}^{-1} D^* G_{2,\varepsilon}$$

is ε-dependent even though $D = \bar{\partial}$ is fixed. The derivation and control of this moving adjoint will be the essential next step in the $\mathbb{R}^4$ mass-gap proof.

15.7. Bounded Homotopy and Graph-Norm Control

The exact construction of Section 15.1-15.6 is algebraic. For the decompactification theorem it must be supplemented by a uniform analytic estimate. In particular, the compensating operators H_q entering

$$C_\varepsilon^q = C_0^q + H_{q+1} C_\varepsilon^{q+1} \delta_\varepsilon^q$$

must not generate fields whose norm or first Dolbeault derivative diverges as $\varepsilon \to 0$. The required statement is a bounded contracting homotopy on the fixed-A-period constrained complex.

15.8. Fixed-period constrained complex

Recall

$$\mathcal{C}^q = \ker \partial \cap \Omega^{2,q}(X), \qquad D_q = \bar{\partial}|_{\mathcal{C}^q},$$

so that

$$D_{q+1} D_q = 0.$$

The physical fluctuation sector is obtained after removing the unique cohomological direction which varies the compact A-period. Denote by P_A the projection onto this direction and define

$$\mathcal{C}_{\mathrm{fp}}^q = (I - P_A)\mathcal{C}^q.$$

On this fixed-period complex the physical cohomology relevant to the master-field fluctuation problem is trivial. In particular, if

$$D_q \Psi = 0$$

and Ψ satisfies the fixed-period condition, then the corresponding cohomology class vanishes and

$$\Psi = D_{q-1} \Lambda$$

for an admissible Λ, modulo the usual stabilizer directions. The degree-one version of this statement was established above when proving the triviality of the physical kernel of the

master-field operator. The only nontrivial zero-energy cohomological direction is the variation of the A-period itself, which is absent at fixed Λ_{YM}.

15.9. Graph norms

For each degree define the graph norm

$$\|\Psi\|_{D_q}^2 = \|\Psi\|_q^2 + \left\|D_q\Psi\right\|_{q+1}^2,$$

where the degree-dependent norms are those appropriate to the fixed reference closed-field complex. For the physical degree-one field the relevant norm is the weighted norm,

$$\|\Psi\|_1^2 = \|\Psi\|_h^2,$$

whereas the curvature degree carries the unweighted norm,

$$\|\eta\|_2^2 = \|\eta\|_0^2.$$

Thus

$$\boxed{\|\Psi\|_{D_1}^2 = \|\Psi\|_h^2 + \left\|\bar{\partial}\Psi\right\|_0^2.} \tag{15.15}$$

This is precisely the graph norm naturally associated with the physical quadratic action.

15.10. Closed range and the reduced Green operator

Let

$$\square_q = D_{q-1}D_{q-1}^* + D_q^*D_q$$

be the Hodge operator associated with the constrained complex and the chosen reference degree-wise Hilbert structures. On the complement of the fixed-period cohomology define

$$\mathcal{H}_q^{\perp} = (I - P_A)\mathcal{H}_q.$$

The ellipticity of the constrained Dolbeault complex, together with the fixed-period condition and the Sobolev-Hodge estimates established in the mass-gap analysis, implies that zero is isolated from the physical spectrum. Equivalently, there exists

$$\delta_q > 0$$

such that

$$\langle u, \square_q u\rangle \geq \delta_q^2\|u\|^2, \qquad u \in \mathcal{H}_q^{\perp}. \tag{15.16}$$

Hence $\square_q$ has a bounded inverse on the cohomological complement,

$$G_q = \square_q^{-1}(I - P_A),$$

with

$$\left\|G_q\right\| \leq \delta_q^{-2}. \tag{15.17}$$

The operators

$$G_q$$

are the reduced Green operators of the fixed-period complex. Define

$$H_q = D_{q-1}^* G_q. \tag{15.18}$$

Then the standard Hodge identity gives

$$D_{q-1}H_q + H_{q+1}D_q = I - P_A.$$

Therefore

$$\boxed{DH + HD = I - P_A.} \tag{15.19}$$

This is exactly the contracting homotopy required in sections 15.1-15.6.

15.11. Boundedness of the homotopy

Let $u \in \mathcal{H}_q^{\perp}$. From (15.18),

$$H_q u = D_{q-1}^* G_q u.$$

Using positivity of the Hodge operator,

$$D_{q-1}D_{q-1}^* \leq \square_q$$

in the sense of quadratic forms. Consequently,

$$\begin{aligned} \|H_q u\|^2 &= \|D_{q-1}^* G_q u\|^2 \\ &= \langle G_q u, D_{q-1}D_{q-1}^* G_q u\rangle \\ &\leq \langle G_q u, \square_q G_q u\rangle \\ &= \langle G_q u, (I - P_A)u\rangle. \end{aligned}$$

Using (15.17),

$$\boxed{\|H_q u\| \leq \delta_q^{-1}\|u\|.} \tag{15.20}$$

The derivative of the compensator is controlled directly by the homotopy identity,

$$D_{q-1}H_q u = (I - P_A)u - H_{q+1}D_q u.$$

For $u \in \ker D_q$, which is precisely the case relevant when solving a cochain defect, this simplifies to

$$D_{q-1}H_q u = (I - P_A)u.$$

Hence

$$\|D_{q-1}H_q u\| \leq \|u\|.$$

Combining this with (15.16) gives

$$\left\|H_q u\right\|_{D_{q-1}}^2 \leq \left(1+\delta_q^{-2}\right)\|u\|^2, \qquad D_q u = 0. \tag{15.21}$$

Thus

$$\boxed{\left\|H_q u\right\|_{D_{q-1}} \leq c_{H,q}\|u\|, \qquad c_{H,q} = \sqrt{1+\delta_q^{-2}}.} \tag{15.22}$$

The compensating Green operator is therefore bounded from the curvature norm into the graph norm of the preceding degree. This is the functional-analytic statement required to promote the algebraic recursion of sections 15.1-15-6 to a controlled decompactification map.

15.12. Application to the physical degree-one field

For the physical field degree,

$$D_1 = \bar{\partial}: \mathcal{C}_{\mathrm{fp}}^1 \longrightarrow \mathcal{C}_{\mathrm{fp}}^2,$$

the graph norm is

$$\|\Psi\|_{D_1}^2 = \|\Psi\|_h^2 + \left\|\bar{\partial}\Psi\right\|_0^{2.}$$

The global weighted Poincaré estimate proved above states

$$\left\|\bar{\partial}\Psi\right\|_0^2 \geq \Delta^2\|\Psi\|_h^2 \tag{15.23}$$

on the fixed-period physical Gauss slice. It follows that the graph norm and curvature norm are equivalent there:

$$\|\Psi\|_{D_1}^2 \leq (1+\Delta^{-2})\left\|\bar{\partial}\Psi\right\|_0^{2.}$$

Therefore the physical right inverse satisfies

$$\boxed{\|H_2\eta\|_h \leq \Delta^{-1}\|\eta\|_0,} \tag{15.24}$$

and

$$\boxed{\|H_2\eta\|_{D_1} \leq \sqrt{1+\Delta^{-2}}\,\|\eta\|_{0.}} \tag{15.25}$$

These are the full-theory analogues of the bounded compensator estimates obtained explicitly in the tensor block (Section 7). Importantly, no individual $SO(4)$ representation enters (15.23)–(15.25). The estimate is global on the complete fixed-period physical master-field space.

15.13. Graph-norm convergence of the exact cochain

We now apply these estimates to the exact cochain map of sections 15.1-15.6.

Suppose

$$\left\|\delta_\varepsilon^q\right\| \leq c_q\,\varepsilon^\nu, \tag{15.26}$$

for some $\nu > 0$. For the physical even contraction relevant below,

$$\nu = 2,$$

so that

$$\delta_\varepsilon = O(\varepsilon^2).$$

From the recursion

$$C_\varepsilon^q - C_0^q = H_{q+1} C_\varepsilon^{q+1} \delta_\varepsilon^q$$

and the boundedness of H_{q+1},

$$\begin{aligned} \left\| C_\varepsilon^q - C_0^q \right\|_{\text{graph}} &\leq c_{H,q+1} \left\| C_\varepsilon^{q+1} \right\| \left\| \delta_\varepsilon^q \right\| \\ &\leq C_q \varepsilon^\nu. \end{aligned}$$

Since the complex has finite length, descending induction yields a uniform constant C such that

$$\| C_\varepsilon - C_0 \|_{\text{graph}} \leq C \varepsilon^\nu. \tag{15.27}$$

For the even decompactification branch,

$$\| C_\varepsilon - C_0 \|_{\text{graph}} \leq C \varepsilon^2. \tag{15.28}$$

Since

$$\varepsilon = (R \Lambda_{\text{YM}})^{-1},$$

this becomes

$$\| C_R - C_\infty \|_{\text{graph}} = O((R \Lambda_{\text{YM}})^{-2}). \tag{15.29}$$

Thus the finite-radius Yang–Mills complex approaches the flat-space collective embedding in graph norm.

15.14. Stability of the embedded physical subspace

Let

$$E_\varepsilon^q = \text{Ran } C_\varepsilon^q$$

and let

$$P_\varepsilon^q$$

denote the orthogonal projection onto E_ε^q with respect to a fixed reference Hilbert structure. Because C_ε remains injective for sufficiently small ε and

$$C_\varepsilon = C_0 + O(\varepsilon^2)$$

in graph norm, standard stability of closed subspaces gives

$$P_\varepsilon^q = P_0^q + O(\varepsilon^2). \tag{15.30}$$

Thus the physical collective image does not jump discontinuously in the decompactification limit. This observation will be important below because the physical Gauss representative is itself selected by an ε-dependent orthogonality condition.

Lemma 15.1— Fixed-period Hodge estimate. On the Lockhart–McOwen weighted domains chosen for the constrained complex

$$\mathcal{C}_{\mathrm{fp}}^q = (I - P_A)\mathcal{C}^q,$$

with weights away from the corresponding indicial spectrum, the constrained Hodge operators

$$\square_q = D_{q-1}D_{q-1}^* + D_q^* D_q$$

are Fredholm. After removal of the finite-dimensional cohomological sector, there exist constants $c_q > 0$ such that

$$\langle u, \square_q u \rangle \geq c_q \|u\|^2, \qquad u \perp \ker \square_q.$$

Consequently Ran D_q is closed and

$$\square_q^{-1} \left(I - P_{\ker \square_q} \right)$$

is bounded on the relevant weighted Hilbert space. Section 14 proved that at degree one the only kernel is the A-period class

15.15. Proposition

Proposition 15.2 — Bounded homotopy and controlled decompactification.
On the fixed-A-period constrained complex, assume that:

1. the only cohomological zero direction is the removed A-period class;
2. the constrained Hodge operator has a positive lower bound on its cohomological complement;
3. the finite-radius differential satisfies $d_\varepsilon = d_0 + \delta_\varepsilon, \qquad \|\delta_\varepsilon\| = O(\varepsilon^2).$

By the preceding fixed-period Hodge lemma and the estimate $\|\delta_\epsilon\| = O(\epsilon^2)$, the reduced Green operator defines a bounded contracting homotopy

$$H_q = D_{q-1}^* \square_q^{-1} (I - P_A)$$

satisfying

$$DH + HD = I - P_A,$$

and

$$\|H_q u\|_{\text{graph}} \le c_H \|u\|.$$

The exact cochain constructed in Section 15.1-15.6 therefore obeys

$$\boxed{C_\varepsilon = C_0 + O(\varepsilon^2)} \tag{15.31}$$

in graph norm. Equivalently,

$$\boxed{C_R = C_\infty + O((R\Lambda_{\text{YM}})^{-2}).} \tag{15.32}$$

The corresponding embedded collective subspaces satisfy

$$\boxed{P_\varepsilon = P_0 + O(\varepsilon^2).} \tag{15.33}$$

15.16. What has been established

Sections 15.1-15.16 establish the exact and analytically controlled transport of the complex itself:

$$(\mathcal{V}^\bullet, d_\varepsilon) \xrightarrow{C_\varepsilon} (\mathcal{C}^\bullet, \bar{\partial}),$$

with

$$\bar{\partial} C_\varepsilon = C_\varepsilon d_\varepsilon$$

exactly and

$$C_\varepsilon - C_0 = O(\varepsilon^2)$$

in graph norm. This is not yet the convergence of the physical kinetic master-field operator. The reason is essential: although

$$D = \bar{\partial}$$

remains fixed, the physical degree-wise inner products vary under decompactification. Consequently the physical adjoint $D_\varepsilon^\sharp$ is not fixed. The next step is therefore to construct the full degree-wise Riesz operators $G_{q,\varepsilon}$ from the quadratic 1PI theory and to prove

$$G_{q,\varepsilon} = G_{q,0} + O(\varepsilon^2).$$

Only after this step can the convergence of

$$D_\varepsilon^\sharp = G_{1,\varepsilon}^{-1} D^* G_{2,\varepsilon}$$

and of the kinetic master-field operator be established.

15.17. Quadratic master-field pencil

Let Ψ denote a linearized collective field in degree one. Before eliminating source-invisible master-field components, the quadratic action has the form

$$S_\varepsilon^{(2)} = \frac{1}{2}\int dt\,\left[\langle \Psi, G_{1,\varepsilon}^{\mathrm{full}}\Psi\rangle - \langle D\Psi, G_{2,\varepsilon}^{\mathrm{full}} D\Psi\rangle\right]. \tag{15.34}$$

Equivalently, after Fourier transformation in time,

$$S_\varepsilon^{(2)} = \frac{1}{2}\langle \Psi, \mathcal{A}_\varepsilon(z)\Psi\rangle,$$

where

$$\mathcal{A}_\varepsilon(z) = K_\varepsilon - zB_\varepsilon, \tag{15.35}$$

with

$$B_\varepsilon = G_{1,\varepsilon}^{\mathrm{full}},$$

and

$$K_\varepsilon = D^* G_{2,\varepsilon}^{\mathrm{full}} D.$$

Here z denotes the spectral variable, $z = M^2$ in Lorentzian signature or $z = -\omega_E^2$ after Euclidean continuation. The distinction between B_ε and K_ε is essential. The former is the kinetic Riesz metric, whereas the latter contains the curvature form. The physical spectrum is therefore determined by the generalized eigenvalue problem

$$K_\varepsilon \Psi = M^2 B_\varepsilon \Psi. \tag{15.36}$$

This is the full-complex analogue of the degree-wise Riesz structure already encountered in the tensor contraction. There the ambient $\bar{\partial}$ remained fixed while the physical adjoint moved through the Riesz operators.

15.18. Visible and complementary master-field directions

The exact cochain C_ε singles out the collective Yang–Mills image inside the larger master-field space. Let

$$E_\varepsilon = \operatorname{Ran} C_\varepsilon$$

denote this physical collective image and let F_ε denote a complementary master-field subspace containing source-invisible completion fields and orthogonal normalizable modes. Using the graph-continuous identification established in Section 15.7-15.16, all sufficiently small ε may be transported to a fixed reference splitting

$$\mathcal{H} = E \oplus F. \tag{15.37}$$

The complementary space F is chosen inside the same fixed-A-period constrained closed-field Hilbert space on which the global coercive estimate of Section 14 holds. It contains source-invisible completion directions but no additional cohomological or gauge-zero sector. Hence the endpoint coercive estimate restricts to F. With respect to this splitting the quadratic pencil takes the block form

$$\mathcal{A}_\varepsilon(z) = \begin{pmatrix} A_{EE,\varepsilon}(z) & A_{EF,\varepsilon}(z) \\ A_{FE,\varepsilon}(z) & A_{FF,\varepsilon}(z) \end{pmatrix}, \tag{15.38}$$

where

$$A_{IJ,\varepsilon}(z) = K_{IJ,\varepsilon} - zB_{IJ,\varepsilon}.$$

The E-component is the source-visible collective field. The F-component is not an additional low-energy degree of freedom of Yang–Mills theory. It is the master-field completion which must be eliminated to obtain the physical 1PI quadratic action.

15.19 Uniform invertibility of the complement

At the flat endpoint, the fixed-period coercive estimate obtained previously implies that the physical operator contains no zero-energy state. In particular, after removing the A-period direction and the gauge directions, the complementary block satisfies

$$K_{FF,0} \geq \gamma_F I \tag{15.39}$$

for some

$$\gamma_F > 0.$$

Hence

$$R_0 := K_{FF,0}^{-1}$$

exists and obeys

$$\|R_0\| \leq \gamma_F^{-1}. \tag{15.40}$$

The graph-norm convergence of Sections 15.7-15-16 and the smooth dependence of the quadratic master-field action imply

$$K_{FF,\varepsilon} = K_{FF,0} + O(\varepsilon^2) \tag{15.41}$$

on the even branch. Therefore for sufficiently small ε,

$$K_{FF,\varepsilon} \geq \frac{\gamma_F}{2} I,$$

and consequently

$$R_\varepsilon := K_{FF,\varepsilon}^{-1} \tag{15.42}$$

is uniformly bounded,

$$\boxed{\|R_\varepsilon\| \leq \frac{2}{\gamma_F}.} \tag{15.43}$$

This is the analytic input required for the exact Feshbach reduction. The uniform invertibility of the complementary block and the corresponding resolvent estimates are proved in Section 15.31 and 15.32 below.

15.20. Exact 1PI-Feshbach reduction

Write

$$\Psi = \phi + \chi, \qquad \phi \in E, \qquad \chi \in F.$$

The equation for the complementary field is

$$A_{FE,\varepsilon}(z)\phi + A_{FF,\varepsilon}(z)\chi = 0.$$

Hence

$$\chi = -A_{FF,\varepsilon}(z)^{-1}A_{FE,\varepsilon}(z)\phi. \tag{15.44}$$

Substituting back into the quadratic action gives the exact 1PI inverse propagator on the collective sector,

$$\boxed{\Gamma_\varepsilon(z) = A_{EE,\varepsilon}(z) - A_{EF,\varepsilon}(z)A_{FF,\varepsilon}(z)^{-1}A_{FE,\varepsilon}(z).} \tag{15.45}$$

Whenever $A_{FF,\epsilon}(z)$ is invertible, the Feshbach theorem gives the exact spectral equivalence

$$0 \in \sigma\big(\mathcal{A}_\epsilon(z)\big) \quad \Leftrightarrow \quad 0 \in \sigma\big(\Gamma_\epsilon(z)\big). \tag{15.45a}$$

Hence the exact physical poles in the source-visible sector are the zeros of the full z-dependent Schur kernel $\Gamma_\epsilon(z)$. The zero-energy quantities

$$K_{\mathrm{eff},\epsilon} = \Gamma_\epsilon(0), \qquad G_{1,\epsilon} = -\partial_z\Gamma_\epsilon(0)$$

determine the static stiffness and the induced low-energy Riesz metric. In general, however, $A_{FF,\epsilon}(z)^{-1}$ depends on z, so the generalized eigenvalue problem

$$K_{\mathrm{eff},\epsilon}\phi = zG_{1,\epsilon}\phi$$

is the first-order expansion of the exact Schur problem about $z = 0$, not the exact pole equation. Exact spectral statements below will therefore refer either to the full pencil $\mathcal{A}_\epsilon(z)$ or equivalently to the full Schur kernel $\Gamma_\epsilon(z)$.

Equation (15.45) is the Schur complement of the full closed-field quadratic pencil. At zero spectral parameter,

$$\boxed{K_{\mathrm{eff},\varepsilon} = K_{EE,\varepsilon} - K_{EF,\varepsilon}K_{FF,\varepsilon}^{-1}K_{FE,\varepsilon}.} \tag{15.46}$$

The second term is nonnegative before subtraction,

$$\begin{aligned}\langle v, K_{EF}K_{FF}^{-1}K_{FE}v\rangle &= \left\|K_{FF}^{-1/2}K_{FE}v\right\|^2 \\ &\geq 0.\end{aligned}$$

Thus

$$\boxed{K_{\mathrm{eff},\varepsilon} \leq K_{EE,\varepsilon}.} \tag{15.47}$$

This is the exact Green-Feshbach lowering already observed explicitly in the tensor sector. The earlier tensor calculation found precisely such a negative 1PI correction after integrating out complementary polarizations.

15.21. Effective degree-one Riesz metric

The zero-frequency physical Riesz metric induced by the exact 1PI pencil is obtained from its first derivative at $z = 0$. Define

$$G_{1,\varepsilon} = -\frac{\partial \Gamma_\varepsilon(z)}{\partial z}\bigg|_{z=0}. \tag{15.48}$$

Let

$$R_\varepsilon = K_{FF,\varepsilon}^{-1}.$$

Differentiating (15.45) gives

$$G_{1,\varepsilon} = B_{EE,\varepsilon} - B_{EF,\varepsilon} R_\varepsilon K_{FE,\varepsilon} - K_{EF,\varepsilon} R_\varepsilon B_{FE,\varepsilon} + K_{EF,\varepsilon} R_\varepsilon B_{FF,\varepsilon} R_\varepsilon K_{FE,\varepsilon}. \tag{15.49}$$

Thus the effective Riesz metric is not obtained by simply restricting the original kinetic metric to the cochain image. It contains the same nonlocal Green operator that occurs in the effective stiffness. Define the zero-energy Feshbach-horizontal lift

$$J_\varepsilon = \begin{pmatrix} I \\ -R_\varepsilon K_{FE,\varepsilon} \end{pmatrix}. \tag{15.50}$$

Then equations (15.46) and (15.49) can be written compactly as

$$K_{\mathrm{eff},\varepsilon} = J_\varepsilon^* K_\varepsilon J_\varepsilon, \tag{15.51}$$

and

$$G_{1,\varepsilon} = J_\varepsilon^* B_\varepsilon J_\varepsilon. \tag{15.52}$$

This induced metric controls the physical Hilbert norm and adjoint at zero frequency; the complete pole structure remains governed by the full kernel $\Gamma_\epsilon(z)$. This makes the geometric meaning transparent. The 1PI reduction first selects the horizontal closed-field representative

$$J_\varepsilon \phi = \phi - R_\varepsilon K_{FE,\varepsilon} \phi$$

and then pulls the full kinetic metric back to that representative. The effective Riesz metric is therefore induced dynamically by elimination of the complementary field space rather than inserted by hand. This is precisely the distinction required by the earlier tensor analysis, where a moving cochain inside a fixed metric was shown to be insufficient.

15.22. Degree-two Riesz transport

The same construction applies one degree higher.

Let

$$\mathcal{H}_2 = E_2 \oplus F_2$$

be the corresponding decomposition of the curvature space into the image generated by the physical cochain and its source-invisible complement. Write the full degree-two metric as

$$\mathbb{G}_{2,\varepsilon} = \begin{pmatrix} G_{2,EE,\varepsilon} & G_{2,EF,\varepsilon} \\ G_{2,FE,\varepsilon} & G_{2,FF,\varepsilon} \end{pmatrix}.$$

After horizontal elimination of the complementary curvature directions the physical degree-two Riesz operator is its Schur reduction,

$$G_{2,\varepsilon}^{\mathrm{eff}} = G_{2,EE,\varepsilon} - G_{2,EF,\varepsilon} G_{2,FF,\varepsilon}^{-1} G_{2,FE,\varepsilon}. \tag{15.53}$$

Equivalently, if $J_{2,\varepsilon}$ denotes the corresponding horizontal lift,

$$G_{2,\varepsilon}^{\mathrm{eff}} = J_{2,\varepsilon}^{*} \mathbb{G}_{2,\varepsilon} J_{2,\varepsilon}. \tag{15.54}$$

Below we suppress the superscript "eff" and denote the physical degree-wise metrics simply by

$$G_{1,\varepsilon}, \qquad G_{2,\varepsilon}.$$

The one-sided physical adjoint is consequently

$$D_{\varepsilon}^{\sharp} = G_{1,\varepsilon}^{-1} D^{*} G_{2,\varepsilon}. \tag{15.55}$$

This is the full-complex version of the tensor formula

$$\bar{\partial}_{\alpha}^{\sharp} = G_{21}(\alpha)^{-1} \bar{\partial}^{*} G_{22}(\alpha)$$

used previously.

15.23. Expansion at the flat endpoint

We now determine the size of the Riesz deformation.

Choose the flat splitting $E \oplus F$ to be orthogonal with respect to the endpoint quadratic theory. Then

$$K_{EF,0} = 0, \qquad B_{EF,0} = 0. \tag{15.56}$$

For the even physical decompactification branch the diagonal blocks satisfy

$$K_{EE,\varepsilon} = K_{EE,0} + \varepsilon^2 K_{EE}^{(2)} + O(\varepsilon^4),$$

$$B_{EE,\varepsilon} = B_{EE,0} + \varepsilon^2 B_{EE}^{(2)} + O(\varepsilon^4),$$

whereas the off-diagonal mixing may begin at first order,

$$K_{EF,\varepsilon} = \varepsilon X_K + O(\varepsilon^2),$$

$$B_{EF,\varepsilon} = \varepsilon X_B + O(\varepsilon^2). \tag{15.57}$$

This causes no linear correction to the effective operators because the Feshbach contribution is quadratic in the off-diagonal mixing.
Using

$$R_\varepsilon = R_0 + O(\varepsilon^2),$$

equation (15.46) gives

$$K_{\mathrm{eff},\varepsilon} = K_{\mathrm{eff},0} + \varepsilon^2 \left(K_{EE}^{(2)} - X_K R_0 X_K^* \right) + O(\varepsilon^3). \tag{15.58}$$

Likewise, equation (15.49) yields

$$G_{1,\varepsilon} = G_{1,0} + \varepsilon^2 \delta G_1^{(2)} + O(\varepsilon^3), \tag{15.59}$$

With

$$\delta G_1^{(2)} = \&B_{EE}^{(2)} - X_B R_0 X_K^* - X_K R_0 X_B^* + X_K R_0 B_{FF,0} R_0 X_K^*. \tag{15.60}$$

For a B_0-orthogonal endpoint splitting in which

$$X_B = 0,$$

and in the absence of an independent direct degree-one variation,

$$B_{EE}^{(2)} = 0,$$

this reduces to

$$\boxed{\delta G_1^{(2)} = X_K R_0 B_{FF,0} R_0 X_K^* \geq 0.} \tag{15.61}$$

The Riesz correction is therefore generated by the same complementary Green operator which produces the negative Feshbach correction to the stiffness. This is the operator-level origin of the simultaneous Green and horizontal-Riesz effects identified in the tensor calculation.

15.24. Uniform $O(\varepsilon^2)$ control

The complement resolvent is uniformly bounded by (15.43). The exact cochain satisfies

$$C_\varepsilon - C_0 = O(\varepsilon^2)$$

in graph norm, and the coefficients of the quadratic closed-field action vary smoothly under the $SO(4) \to ISO(3)$ contraction. Consequently all terms in the Schur formulas are uniformly controlled. There exist constants c_1, c_2 such that

$$\left\| G_{1,\varepsilon} - G_{1,0} \right\|_{\mathrm{rel}} \leq c_1 \varepsilon^2, \tag{15.62}$$

and

$$\left\| G_{2,\varepsilon} - G_{2,0} \right\|_{\mathrm{rel}} \leq c_2 \varepsilon^2. \tag{15.63}$$

Here the relative norm means

$$\|G_{q,\varepsilon} - G_{q,0}\|_{\mathrm{rel}} = \left\| G_{q,0}^{-1/2}(G_{q,\varepsilon} - G_{q,0})G_{q,0}^{-1/2} \right\|.$$

Equivalently,

$$G_{q,\varepsilon} = G_{q,0}^{1/2}(I + E_{q,\varepsilon})G_{q,0}^{1/2}, \qquad \|E_{q,\varepsilon}\| \le c_q \varepsilon^2. \tag{15.64}$$

For sufficiently small ε,

$$\|E_{q,\varepsilon}\| < \frac{1}{2},$$

and therefore

$$\boxed{\frac{1}{2}G_{q,0} \le G_{q,\varepsilon} \le \frac{3}{2}G_{q,0}.} \tag{15.65}$$

Thus the degree-wise physical norms remain uniformly equivalent throughout the decompactification region. In particular, no kinetic degeneracy develops as

$$R \to \infty.$$

The Schur formulas show that this estimate follows once the complementary resolvent is uniformly controlled. The full analytic proof is given in subsection 15.43.

15.25. Convergence of the inverse Riesz operator

Since

$$G_{1,\varepsilon} = G_{1,0}^{1/2}(I + E_{1,\varepsilon})G_{1,0}^{1/2},$$

we have

$$G_{1,\varepsilon}^{-1} = G_{1,0}^{-1/2}(I + E_{1,\varepsilon})^{-1}G_{1,0}^{-1/2}.$$

Using

$$(I + E)^{-1} - I = -(I + E)^{-1}E,$$

and $\|E_{1,\varepsilon}\| < 1/2$,

$$\left\|(I + E_{1,\varepsilon})^{-1} - I\right\| \le 2\|E_{1,\varepsilon}\|.$$

Therefore

$$\boxed{G_{1,\varepsilon}^{-1} = G_{1,0}^{-1} + O(\varepsilon^2).} \tag{15.66}$$

The inverse kinetic Riesz operator is thus as regular at the flat endpoint as the metric itself.

15.26. Proposition

Proposition 15.3 — 1PI Riesz transport.
Assume:

1. the exact cochain and graph-norm bounds of Sections 15.1-15.16;
2. a uniformly positive complementary block at the flat endpoint;
3. smooth dependence of the quadratic closed-field action on the contraction parameter;
4. the even physical decompactification branch.

Then the exact 1PI-Feshbach reduction defines positive degree-wise physical Riesz operators

$$G_{1,\varepsilon}, \qquad G_{2,\varepsilon},$$

such that

$$G_{q,\varepsilon} = G_{q,0} + O(\varepsilon^2), \qquad q = 1,2, \tag{15.67}$$

in relative operator norm.

Moreover,

$$G_{1,\varepsilon}^{-1} = G_{1,0}^{-1} + O(\varepsilon^2), \tag{15.68}$$

and the physical norms are uniformly equivalent,

$$c\, G_{q,0} \le G_{q,\varepsilon} \le C\, G_{q,0} \tag{15.69}$$

for sufficiently small ε. The physical adjoint is therefore

$$D_\varepsilon^\sharp = G_{1,\varepsilon}^{-1} D^* G_{2,\varepsilon}, \tag{15.70}$$

with all R-dependence contained in the degree-wise Riesz realization rather than in the ambient Dolbeault differential. The construction above defines the candidate physical Riesz transport. Its uniform $O(\varepsilon^2)$ regularity, positivity, and compatibility with the adjoint domains are established in Sections 15.28-15.38.

15.27. Interpretation

The decompactification has now been separated into three mathematically distinct structures:

$$\boxed{d_\varepsilon \longrightarrow C_\varepsilon \longrightarrow (G_{1,\varepsilon}, G_{2,\varepsilon}).}$$

The Yang–Mills differential carries the $SO(4) \to ISO(3)$ contraction. The exact cochain C_ε embeds this contracted complex into the fixed Dolbeault complex. Finally, the 1PI-Feshbach reduction determines the physical degree-wise Riesz metrics.

The ambient operator $D = \bar{\partial}$ does not move. The physical adjoint does:

$$D_\varepsilon^\sharp = G_{1,\varepsilon}^{-1} D^* G_{2,\varepsilon}.$$

This is the crucial distinction required for the $\mathbb{R}^4$ mass-gap theorem.

15.28 Full-complex 1PI Riesz regularity

The preceding section constructed the physical degree-wise Riesz operators through the exact 1PI-Feshbach reduction of the full closed-field quadratic theory. In order to use this construction in the decompactification theorem, three analytic points remain to be established. First, the full 1PI Riesz operators must inherit the $O(\varepsilon^2)$ regularity of the exact cochain and of the unreduced quadratic theory. Second, the complementary block entering the Feshbach reduction must remain uniformly invertible as $\varepsilon \to 0$. Third, the degree-two Riesz deformation must preserve the graph domain of the physical adjoint. We establish these statements in turn.

Let the unreduced quadratic closed-field pencil be

$$\mathbb{A}_\varepsilon(z) = \mathbb{K}_\varepsilon - z\mathbb{B}_\varepsilon.$$

The exact cochain construction and the graph-norm estimate of Sections 15.1-15.16 give

$$C_\varepsilon = C_0 + E_\varepsilon, \qquad \|E_\varepsilon\|_{\mathrm{graph}} = O(\varepsilon^2) \tag{15.71}$$

on the even decompactification branch. Since the ambient closed geometry and the Dolbeault operator remain fixed, the finite-ε dependence of the pulled-back quadratic forms is generated by the moving cochain together with the smooth finite-radius representation data. For the kinetic form,

$$\mathbb{B}_\varepsilon = C_\varepsilon^* \mathbb{B} C_\varepsilon.$$

Hence

$$\mathbb{B}_\varepsilon - \mathbb{B}_0 = E_\varepsilon^* \mathbb{B} C_0 + C_0^* \mathbb{B} E_\varepsilon + E_\varepsilon^* \mathbb{B} E_\varepsilon.$$

Since $\mathbb{B}$ is bounded with respect to the reference Hilbert norm,

$$\mathbb{B}_\varepsilon = \mathbb{B}_0 + O(\varepsilon^2). \tag{15.72}$$

Similarly,

$$\mathbb{K}_\varepsilon = (DC_\varepsilon)^* \mathbb{G}_2 (DC_\varepsilon).$$

Because

$$\|D(C_\varepsilon - C_0)\| = O(\varepsilon^2)$$

by graph-norm convergence,

$$\mathbb{K}_\varepsilon = \mathbb{K}_0 + O(\varepsilon^2) \tag{15.73}$$

in quadratic-form norm. Consequently,

$$\mathbb{A}_\varepsilon(z) = \mathbb{A}_0(z) + O(\varepsilon^2) \tag{15.74}$$

uniformly for z in compact subsets of the low-energy spectral region.

15.29 Fixed endpoint splitting

Let

$$\mathcal{H} = E \oplus F$$

be the collective/complementary splitting at the flat endpoint. Section 15.7-15.16 showed that the collective projectors satisfy

$$P_\varepsilon = P_0 + O(\varepsilon^2).$$

There is therefore a bounded near-identity identification

$$U_\varepsilon = I + O(\varepsilon^2)$$

which transports the moving decomposition $E_\varepsilon \oplus F_\varepsilon$ to the fixed decomposition $E \oplus F$.

After conjugation by U_ε, equation (15.74) remains valid. Thus every block of the quadratic pencil satisfies the corresponding form estimate.
In particular,

$$A_{EE,\varepsilon}(z) = A_{EE,0}(z) + O(\varepsilon^2), \tag{15.75}$$

and

$$A_{FF,\varepsilon}(z) = A_{FF,0}(z) + O(\varepsilon^2). \tag{15.76}$$

If a representation-dependent choice produces an off-diagonal term linear in ε,

$$A_{EF,\varepsilon}(z) = \varepsilon X(z) + O(\varepsilon^2), \tag{15.77}$$

its contribution to the Schur complement is nevertheless quadratic in ε.

15.30 Effective 1PI pencil

The exact 1PI inverse propagator is

$$\Gamma_\varepsilon(z) = A_{EE,\varepsilon}(z) - A_{EF,\varepsilon}(z)A_{FF,\varepsilon}(z)^{-1}A_{FE,\varepsilon}(z). \tag{15.78}$$

Once the uniform invertibility of $A_{FF,\varepsilon}(z)$ is established below, the resolvent identity gives

$$A_{FF,\varepsilon}(z)^{-1} = A_{FF,0}(z)^{-1} + O(\varepsilon^2). \tag{15.79}$$

Substitution into (15.78) then yields

$$\Gamma_\varepsilon(z) = \Gamma_0(z) + O(\varepsilon^2) \tag{15.80}$$

uniformly on compact low-energy z-sets. The effective degree-one Riesz operator is

$$G_{1,\varepsilon} = -\partial_z \Gamma_\varepsilon(z)|_{z=0}.$$

Therefore

$$G_{1,\varepsilon} = G_{1,0} + O(\varepsilon^2). \tag{15.81}$$

Applying the same argument one degree higher gives

$$G_{2,\varepsilon} = G_{2,0} + O(\varepsilon^2). \tag{15.82}$$

Thus the full-complex 1PI Riesz regularity required in Sections 15.17-15.28 follows from the exact cochain construction, provided the complementary inverse remains uniformly bounded. We prove this next.

15.31 Uniform complementary resolvent bound

Let $q_\varepsilon[\Psi]$ and $b_\varepsilon[\Psi]$ denote respectively the spatial and kinetic quadratic forms on the physical fixed-period space. At the endpoint the global coercive estimate gives

$$q_0[\Psi] \geq \Delta^2 b_0[\Psi], \qquad \Delta > 0. \tag{15.83}$$

Since this holds on the entire physical Hilbert space, it holds in particular on the complementary subspace F. Therefore

$$q_0[\chi] \geq \Delta^2 b_0[\chi], \qquad \chi \in F. \tag{15.84}$$

Equivalently,

$$K_{FF,0} \geq \Delta^2 B_{FF,0}. \tag{15.85}$$

Hence $K_{FF,0}$ is invertible and

$$\boxed{\left\|K_{FF,0}^{-1}\right\| \leq \Delta^{-2}} \tag{15.86}$$

in the $B_{FF,0}$-normalized Hilbert norm.

15.32. Stability for small ε

The unreduced quadratic-block regularity established above gives

$$|q_\varepsilon[\chi] - q_0[\chi]| \leq C_q \varepsilon^2 (q_0[\chi] + b_0[\chi]) \tag{15.87}$$

and

$$|b_\varepsilon[\chi] - b_0[\chi]| \leq C_b \varepsilon^2 b_0[\chi]. \tag{15.88}$$

Using

$$b_0[\chi] \leq \Delta^{-2} q_0[\chi],$$

we obtain

$$q_\varepsilon[\chi] \geq \left[1 - C_q \varepsilon^2 (1 + \Delta^{-2})\right] q_0[\chi]. \tag{15.89}$$

Therefore, for sufficiently small ε,

$$q_\varepsilon[\chi] \geq \frac{\Delta^2}{2} b_\varepsilon[\chi] \tag{15.90}$$

Consequently,

$$K_{FF,\varepsilon} \geq \frac{\Delta^2}{2} B_{FF,\varepsilon}. \tag{15.91}$$

Thus

$$\boxed{\left\| K_{FF,\varepsilon}^{-1} \right\| \leq \frac{2}{\Delta^2}} \tag{15.92}$$

uniformly for sufficiently small ε.

15.33. Spectral pencil

Now consider

$$A_{FF,\varepsilon}(z) = K_{FF,\varepsilon} - zB_{FF,\varepsilon}.$$

For real z,

$$\langle \chi, A_{FF,\varepsilon}(z)\chi \rangle \& = q_\varepsilon[\chi] - z b_\varepsilon[\chi] \geq \left(\frac{\Delta^2}{2} - z \right) b_\varepsilon[\chi]. \tag{15.93}$$

Choose

$$|z| \leq \frac{\Delta^2}{4}.$$

Then

$$A_{FF,\varepsilon}(z) \geq \frac{\Delta^2}{4} B_{FF,\varepsilon}. \tag{15.94}$$

Therefore

$$\boxed{\sup_{\substack{0 \leq \varepsilon < \varepsilon_* \\ |z| \leq \Delta^2/4}} \left\| A_{FF,\varepsilon}(z)^{-1} \right\| \leq \frac{4}{\Delta^2}.} \tag{15.95}$$

The complementary inverse required in the Feshbach construction is therefore uniformly bounded. Furthermore,

$$A_{FF,\varepsilon}^{-1} - A_{FF,0}^{-1} = -A_{FF,\varepsilon}^{-1} \left(A_{FF,\varepsilon} - A_{FF,0} \right) A_{FF,0}^{-1}.$$

Using (15.76) and (15.95),

$$A_{FF,\varepsilon}(z)^{-1} = A_{FF,0}(z)^{-1} + O(\varepsilon^2) \tag{15.96}$$

uniformly on the same low-energy neighborhood. This closes the assumption used in 15.29-15.31. It remains to prove that the degree-two Riesz deformation preserves the graph domain required for the physical adjoint. Recall

$$D_\varepsilon^\sharp = G_{1,\varepsilon}^{-1} D^* G_{2,\varepsilon}. \tag{15.97}$$

We seek the estimate

$$\left\|D^*\left(G_{2,\varepsilon}-G_{2,0}\right)u\right\| \leq C\varepsilon^2\left(\left\|D_0^\sharp u\right\|+\|u\|\right). \tag{15.98}$$

15.34. Endpoint adjoint graph norm

Since

$$D_0^\sharp = G_{1,0}^{-1}D^*G_{2,0},$$

and $G_{1,0}^{\pm 1}$ are bounded, the graph norm

$$\|u\|_{\sharp,0}^2 = \|u\|^2 + \left\|D_0^\sharp u\right\|^2 \tag{15.99}$$

is equivalent to

$$\|u\|^2 + \left\|D^*G_{2,0}u\right\|^2. \tag{15.100}$$

15.35. Green regularity

Let

$$R_F = K_{FF,0}^{-1}.$$

From Sections 15.32-15.34,

$$\|R_F\| \leq \Delta^{-2}.$$

More strongly,

$$R_F : \mathcal{H}_F \longrightarrow D\left(K_{FF,0}\right)$$

continuously. Indeed,

$$K_{FF,0}R_Ff = f,$$

and therefore

$$\|R_Ff\|_{D(K_F)} = \|R_Ff\| + \|K_FR_Ff\| \leq (\Delta^{-2}+1)\|f\|. \tag{15.101}$$

Thus

$$\boxed{\|R_Ff\|_{D(K_F)} \leq C_R\|f\|.} \tag{15.102}$$

The complementary Green operator therefore gains the regularity required by the strong elliptic domain.

15.36. Structure of the Riesz perturbation

The degree-two 1PI metric has the form

$$G_{2,\varepsilon} = G_{2,0} + \varepsilon^2L_2 + \varepsilon^2N_2 + O(\varepsilon^3), \tag{15.103}$$

where L_2 is a direct lower-order coefficient deformation and N_2 is the nonlocal Schur contribution. Schematically,

$$N_2 = XR_FX^*. \tag{15.104}$$

The mixing operator X originates from the same quadratic differential complex and exact cochain.

Lemma 15.2— Graph boundedness of the mixing operator. The off-diagonal first variation X extends continuously from the endpoint adjoint domain equipped with

$$\|u\|_{\sharp,0} = \left(\|u\|^2 + \left\|D_0^\sharp u\right\|^2\right)^{1/2}$$

to the complementary Hilbert space:

$$\|X^*u\| \le C_X\|u\|_{\sharp,0}.$$

The graph bound follows because the principal Dolbeault symbol is independent of ϵ; coefficient variations are lower-order bounded perturbations, while the nonlocal contribution factors through the bounded Green operator

$$R_F: \mathcal{H}_F \to D\big(K_{FF,0}\big).$$

Likewise, the principal-order variation of L_2 vanishes because the principal symbol is fixed, so

$$D^*L_2: D\big(D_0^\sharp\big) \to \mathcal{H}$$

is graph bounded.

$$\|X^*u\| \le C_X\big(\left\|D_0^\sharp u\right\| + \|u\|\big). \tag{15.105}$$

Because

$$R_FX^*u \in D(K_F),$$

and the remaining mixing operator is of lower differential order relative to K_F, elliptic graph regularity gives

$$\|D^*XR_FX^*u\| \le C\|R_FX^*u\|_{D(K_F)}.$$

Using (15.102) and (15.105),

$$\|D^*N_2u\| \le C_N\big(\left\|D_0^\sharp u\right\| + \|u\|\big). \tag{15.106}$$

For the local term L_2, smoothness of the coefficient deformation and invariance of the principal Dolbeault symbol imply

$$\|D^*L_2u\| \le C_L\big(\left\|D_0^\sharp u\right\| + \|u\|\big). \tag{15.107}$$

Combining (15.103), (15.106), and (15.107) gives

$$\boxed{\left\|D^*\big(G_{2,\varepsilon} - G_{2,0}\big)u\right\| \le C\varepsilon^2\big(\left\|D_0^\sharp u\right\| + \|u\|\big).} \tag{15.108}$$

This is the required graph-domain estimate.

15.37. Convergence of the adjoint

Using

$$D_\varepsilon^\sharp - D_0^\sharp = \left(G_{1,\varepsilon}^{-1} - G_{1,0}^{-1}\right) D^* G_{2,0} + G_{1,\varepsilon}^{-1} D^* \left(G_{2,\varepsilon} - G_{2,0}\right), \tag{15.109}$$

together with

$$G_{1,\varepsilon}^{-1} = G_{1,0}^{-1} + O(\varepsilon^2)$$

and (15.108), one obtains

$$\boxed{\left\| \left(D_\varepsilon^\sharp - D_0^\sharp\right) u \right\| \le C_\sharp \varepsilon^2 \left(\left\| D_0^\sharp u \right\| + \|u\| \right).} \tag{15.110}$$

Thus

$$\boxed{D_\varepsilon^\sharp = D_0^\sharp + O(\varepsilon^2)} \tag{15.111}$$

in graph-operator norm. The graph norms are therefore uniformly equivalent, and after identification through the fixed quotient the adjoint domains are stable for sufficiently small ε.

15.38. Analytic closure proposition

The results of this section can be collected into the following statement.

Proposition 15.4 — Analytic closure of the decompactifying 1PI reduction.
Let the exact cochain and bounded-homotopy hypotheses of Sections 15.1-15.16 hold on the fixed-A-period physical complex. Then, on the even decompactification branch,

$$C_\varepsilon = C_0 + O(\varepsilon^2)$$

in graph norm implies:

$$G_{q,\varepsilon} = G_{q,0} + O(\varepsilon^2), \qquad q = 1,2, \tag{15.112}$$

for the full 1PI-reduced Riesz operators;

$$\boxed{\sup_{\varepsilon < \varepsilon_*} \left\| K_{FF,\varepsilon}^{-1} \right\| < \infty;} \tag{15.113}$$

more generally,

$$\boxed{\sup_{\substack{\varepsilon < \varepsilon_* \\ |z| < \Delta^2/4}} \left\| \left(K_{FF,\varepsilon} - z B_{FF,\varepsilon}\right)^{-1} \right\| < \infty;} \tag{15.114}$$

and

$$\boxed{\left\| \left(D_\varepsilon^\sharp - D_0^\sharp\right) u \right\| \le C \varepsilon^2 \|u\|_{\sharp,0}.} \tag{15.115}$$

Thus the physical degree-wise Riesz structures, the Feshbach elimination, and the moving physical adjoint remain uniformly regular as

$$R \to \infty.$$

No singular complementary mode is generated by decompactification, and the physical adjoints remain uniformly controlled on the identified graph domain.

15.39 Moving Gauss slice

The physical degree-one field is not the unrestricted space $\mathcal{H}_1$. It is the warped Gauss slice determined by the preceding differential,

$$D_{0,\varepsilon}^{\sharp}\Psi = 0.$$

Since this adjoint is itself Riesz dependent, the physical horizontal representative moves with ε.

Let

$$\mathcal{G}_\varepsilon = \ker D_{0,\varepsilon}^{\sharp}$$

denote the finite-radius Gauss slice. The corresponding orthogonal projector is $P_\varepsilon^{\mathrm{hor}}$. From the same argument as above,

$$D_{0,\varepsilon}^{\sharp} = D_{0,0}^{\sharp} + O(\varepsilon^2)$$

in graph norm. Because the gauge range is closed on the fixed-period complex, the associated orthogonal projections vary continuously. Therefore

$$P_\varepsilon^{\mathrm{hor}} = P_0^{\mathrm{hor}} + O(\varepsilon^2). \tag{15.116}$$

Hence the finite-radius physical Hilbert spaces can be identified through the fixed quotient

$$\ker \partial \,/\, \overline{\mathrm{Ran}\, D_0},$$

while their horizontal representatives differ only by $O(\varepsilon^2)$. This avoids treating the moving Gauss slice as a change of the underlying gauge quotient.

15.40 Kinetic master-field operator

The physical one-sided spatial operator is

$$K_\varepsilon = D_\varepsilon^{\sharp} D \tag{15.117}$$

on the ε-dependent physical domain. After identifying the moving horizontal spaces through (15.116), the difference is

$$K_\varepsilon - K_0 = \left(D_\varepsilon^{\sharp} - D_0^{\sharp}\right) D. \tag{15.118}$$

Using (15.110),

$$\boxed{\|(K_\varepsilon - K_0)u\| \le C_K \varepsilon^2 (\|K_0 u\| + \|Du\| + \|u\|).} \tag{15.119}$$

Thus K_ε is a graph-small perturbation of K_0. For the spectral problem it is more natural to state the result at the level of quadratic forms.

15.41 Quadratic-form formulation

Define

$$b_\varepsilon[u, v] = \langle u, G_{1,\varepsilon} v \rangle_0 \tag{15.120}$$

and

$$q_\varepsilon[u, v] = \langle Du, G_{2,\varepsilon} Dv \rangle_0. \tag{15.121}$$

Then

$$q_\varepsilon[u, v] = b_\varepsilon[u, K_\varepsilon v]$$

on the strong domain. From Section 15.29-15.39,

$$G_{1,\varepsilon} = G_{1,0} + O(\varepsilon^2),$$

hence

$$\boxed{|b_\varepsilon[u] - b_0[u]| \le C_1 \varepsilon^2 b_0[u].} \tag{15.122}$$

Likewise,

$$G_{2,\varepsilon} = G_{2,0} + O(\varepsilon^2),$$

so

$$\boxed{|q_\varepsilon[u] - q_0[u]| \le C_2 \varepsilon^2 q_0[u].} \tag{15.123}$$

When the graph motion of the exact cochain representative is included explicitly, the more general estimate is

$$\boxed{|q_\varepsilon[u] - q_0[u]| \le C_3 \varepsilon^2 (q_0[u] + b_0[u]).} \tag{15.124}$$

Equations (15.122)–(15.124) are the central form estimates for the decompactification theorem.

15.42 Uniform closedness

Since

$$\frac{1}{2} G_{1,0} \le G_{1,\varepsilon} \le \frac{3}{2} G_{1,0},$$

the norms b_ε and b_0 are uniformly equivalent. Similarly, equation (15.124) implies that for sufficiently small ε,

$$q_\varepsilon[u] + b_\varepsilon[u]$$

is uniformly equivalent to

$$q_0[u] + b_0[u].$$

Therefore the form domains coincide after the quotient-horizontal identification,

$$\boxed{D(q_\varepsilon) = D(q_0)} \tag{15.125}$$

as sets with equivalent graph norms. Each q_ε is closed and lower semibounded. By the representation theorem for closed quadratic forms, there is consequently a unique positive self-adjoint operator associated with every pair

$$(q_\varepsilon, b_\varepsilon).$$

This operator is precisely the physical kinetic master-field operator K_ε.

15.43 Resolvent convergence

We compare the family on the fixed endpoint Hilbert space $(\mathcal{H}, b_0)$. Let

$$T_\epsilon = G_{1,\epsilon}^{-1/2} G_{1,0}^{1/2}.$$

Then $T_\epsilon : (\mathcal{H}, b_0) \to (\mathcal{H}, b_\epsilon)$ is unitary and

$$T_\epsilon = I + O(\epsilon^2).$$

Define the pulled-back operator

$$\widetilde{K}_\epsilon = T_\epsilon^{-1} K_\epsilon T_\epsilon.$$

The common-domain relative form estimates (15.122)–(15.124) then imply

$$\left\| \left(\widetilde{K}_\epsilon - z\right)^{-1} - (K_0 - z)^{-1} \right\| \le C_z \epsilon^2$$

for z in compact subsets of $\rho(K_0)$. From this point onward all K_ϵ are understood after the unitary identification by T_ϵ, and we suppress the tilde. The relative form estimates imply convergence of the closed forms,

$$(q_\varepsilon, b_\varepsilon) \longrightarrow (q_0, b_0).$$

In particular, they imply Mosco convergence. Therefore

$$K_\varepsilon \longrightarrow K_0$$

in strong resolvent sense. Because the form difference is uniformly $O(\varepsilon^2)$ and the form domains are common, the convergence can be strengthened to norm-resolvent convergence away from the spectrum. For any z in a compact subset of the resolvent set of K_0,

$$\|(K_\varepsilon - z)^{-1} - (K_0 - z)^{-1}\| \le C_z \varepsilon^2. \tag{15.126}$$

In particular,

$$\|(K_\varepsilon + 1)^{-1} - (K_0 + 1)^{-1}\| = O(\varepsilon^2). \tag{15.127}$$

Thus the entire physical kinetic master-field operator possesses a well-defined decompactification limit.

15.44 Euclidean master-field covariance

The Euclidean quadratic master-field operator is

$$\mathcal{A}_{E,\varepsilon} = -\partial_{t_E}^2 + K_\varepsilon. \tag{15.128}$$

Since the temporal operator is independent of ε, equation (15.127) immediately gives convergence of the full covariance,

$$\mathcal{A}_{E,\varepsilon}^{-1} \longrightarrow \mathcal{A}_{E,0}^{-1} \tag{15.129}$$

in the corresponding resolvent sense. Hence the Gaussian planar fluctuation measure has a well-defined flat-space limit at the level of its covariance operator. The glueball poles, which were identified previously with the spectral poles of this covariance, therefore converge to the spectrum of K_0.

15.45 Proposition

Proposition 15.5 — Convergence of the physical adjoint and kinetic master field. Assume the exact cochain construction and graph estimates of Sections 15.1-15.16 and the 1PI Riesz bounds of Sections 15.17-15.27. Then, after identification through the fixed gauge quotient,

$$D_\varepsilon^\sharp = D_0^\sharp + O(\varepsilon^2)$$

in graph-operator norm. The associated Gauss projectors satisfy

$$P_\varepsilon^{\mathrm{hor}} = P_0^{\mathrm{hor}} + O(\varepsilon^2).$$

The closed quadratic forms satisfy

$$|b_\varepsilon - b_0| \le C\varepsilon^2 b_0$$

and

$$|q_\varepsilon - q_0| \le C\varepsilon^2(q_0 + b_0).$$

Consequently the one-sided kinetic master-field operators obey

$$K_\varepsilon \longrightarrow K_0 \tag{15.130}$$

in norm-resolvent sense, with

$$\|(K_\varepsilon + 1)^{-1} - (K_0 + 1)^{-1}\| = O(\varepsilon^2). \tag{15.131}$$

Equivalently,

$$\boxed{K_R \longrightarrow K_\infty}$$

As $R \to \infty$.

15.46 Consequence for the mass-gap problem

At this point the decompactification problem has been reduced to a purely spectral question about the endpoint operator K_0. The previous global fixed-A-period estimate already gives

$$K_0 \geq \Delta^2 I, \qquad \Delta > 0.$$

The remaining step is therefore to combine this lower bound with the uniform form convergence above and prove that the positive spectral gap cannot close as

$$\varepsilon \to 0.$$

That final argument gives the $\mathbb{R}^4$ mass-gap theorem.

15.47 $\mathbb{R}^4$ Mass-Gap Theorem

We have now established all ingredients required for the decompactification of the planar kinetic master field. The finite-radius Yang–Mills complex admits an exact embedding

$$\mathcal{C}_\varepsilon^\bullet : (\mathcal{V}^\bullet, d_\varepsilon) \longrightarrow (\mathcal{C}^\bullet, D),$$

satisfying

$$\bar{\partial} C_\varepsilon = C_\varepsilon d_\varepsilon$$

and

$$C_\varepsilon = C_0 + O(\varepsilon^2)$$

in graph norm. The corresponding 1PI reduction produces degree-wise physical Riesz operators

$$G_{1,\varepsilon}, \qquad G_{2,\varepsilon},$$

with

$$G_{q,\varepsilon} = G_{q,0} + O(\varepsilon^2), \qquad q = 1,2,$$

and therefore

$$D_\varepsilon^\sharp = G_{1,\varepsilon}^{-1} D^* G_{2,\varepsilon} \longrightarrow D_0^\sharp.$$

The one-sided kinetic master-field operators

$$K_\varepsilon = D_\varepsilon^\sharp D$$

converge in norm-resolvent sense to the flat-space operator K_0. It remains to show that the positive lower spectral bound survives the limit.

15.48 Endpoint coercivity

At the flat endpoint the physical Hilbert space is the fixed-A-period gauge-reduced collective space

$$\mathcal{H}_{\mathrm{phys},0} = \left\{\Psi : \partial\Psi = 0,\ D_{0,0}^{\sharp}\Psi = 0,\ \delta\Pi_A[\Psi] = 0\right\}_{L^2_{G_{1,0}}}.$$

The physical energy form is

$$q_0[\Psi] = \left\langle D\Psi, G_{2,0} D\Psi \right\rangle_0,$$

and the kinetic norm is

$$b_0[\Psi] = \left\langle \Psi, G_{1,0}\Psi \right\rangle_0.$$

Let J_0 denote the zero-energy Feshbach-horizontal lift,

$$J_0 = \begin{pmatrix} I \\ -K_{FF,0}^{-1} K_{FE,0} \end{pmatrix}.$$

The endpoint effective forms are pullbacks of the full physical quadratic forms:

$$q_0[\phi] = \mathbb{q}_0[J_0\phi], \qquad b_0[\phi] = \mathbb{b}_0[J_0\phi].$$

Since Chapter 14 establishes

$$\mathbb{q}_0[U] \geq \Delta^2 \mathbb{b}_0[U]$$

on the complete fixed-A-period physical Hilbert space, substitution $U = J_0\phi$ gives

$$q_0[\phi] \geq \Delta^2 b_0[\phi].$$

The global fixed-period Sobolev-Hodge estimate established previously gives

$$\boxed{q_0[\Psi] \geq \Delta^2 b_0[\Psi], \qquad \Delta > 0.} \tag{15.132}$$

Equivalently,

$$\boxed{K_0 \geq \Delta^2 I} \tag{15.133}$$

in the sense of quadratic forms. This is a statement on the complete physical master-field Hilbert space. It does not depend on a decomposition into scalar, tensor, vector, or higher-spin sectors. The only cohomological zero direction is the variation of the compact A-period. At fixed Λ_{YM}, this mode is excluded from the physical fluctuation space. Thus the physical kernel is trivial and the lower bound in (15.132) is strictly positive.

15.49 Finite-radius form estimates

From Sections 15.28-15.43,

$$|b_\varepsilon[\Psi] - b_0[\Psi]| \leq C_b \varepsilon^2 b_0[\Psi] \tag{15.134}$$

and

$$|q_\varepsilon[\Psi] - q_0[\Psi]| \leq C_q \varepsilon^2 (q_0[\Psi] + b_0[\Psi]). \tag{15.135}$$

Therefore

$$b_\varepsilon[\Psi] \le (1 + C_b\varepsilon^2) b_0[\Psi], \tag{15.136}$$

while

$$q_\varepsilon[\Psi] \ge q_0[\Psi] - C_q\varepsilon^2 (q_0[\Psi] + b_0[\Psi]).$$

Using (15.132),

$$b_0[\Psi] \le \Delta^{-2} q_0[\Psi],$$

so

$$q_\varepsilon[\Psi] \ge \left[1 - C_q\varepsilon^2(1 + \Delta^{-2})\right] q_0[\Psi]. \tag{15.137}$$

Applying (15.132) once more,

$$q_\varepsilon[\Psi] \ge \Delta^2 \left[1 - C_q\varepsilon^2(1 + \Delta^{-2})\right] b_0[\Psi]. \tag{15.138}$$

Together with (15.136),

$$\boxed{\frac{q_\varepsilon[\Psi]}{b_\varepsilon[\Psi]} \ge \Delta^2 \frac{1 - C_q\varepsilon^2(1 + \Delta^{-2})}{1 + C_b\varepsilon^2}.} \tag{15.139}$$

The right-hand side converges to Δ^2 as $\varepsilon \to 0$. Hence there exists $\varepsilon_* > 0$ such that

$$0 \le \varepsilon < \varepsilon_*$$

implies

$$\boxed{\frac{q_\varepsilon[\Psi]}{b_\varepsilon[\Psi]} \ge \frac{\Delta^2}{2}} \tag{15.140}$$

for every nonzero physical Ψ. Consequently,

$$\boxed{K_\varepsilon \ge \frac{\Delta^2}{2} I} \tag{15.141}$$

for all sufficiently large R. Consequently the exact full quadratic pencil has no zero for

$$0 \le z < \frac{\Delta^2}{2}. \tag{15.141a}$$

In the low-energy region where $A_{FF,\epsilon}(z)$ is uniformly invertible, the exact Feshbach equivalence (15.45a) therefore implies

$$0 \notin \sigma\big(\Gamma_\epsilon(z)\big), \qquad 0 \le z < \frac{\Delta^2}{4}. \tag{15.141b}$$

Thus the exact source-visible 1PI propagator has no pole approaching $z = 0$. This conclusion uses the complete z-dependent Schur kernel and does not rely on its zero-frequency truncation.

15.50 Stability of the bottom of the spectrum

By the Rayleigh-Ritz characterization,

$$\lambda_{\mathrm{gap}}(\varepsilon) = \inf_{\Psi \neq 0} \frac{q_\varepsilon[\Psi]}{b_\varepsilon[\Psi]}. \tag{15.142}$$

Equation (15.139) therefore gives

$$\lambda_{\mathrm{gap}}(\varepsilon) \geq \Delta^2 \frac{1 - C_q \varepsilon^2 (1 + \Delta^{-2})}{1 + C_b \varepsilon^2}. \tag{15.143}$$

In particular,

$$\boxed{\liminf_{\varepsilon \to 0} \lambda_{\mathrm{gap}}(\varepsilon) \geq \Delta^2 > 0.} \tag{15.144}$$

Norm-resolvent convergence gives the complementary spectral continuity statement. Since

$$K_\varepsilon \longrightarrow K_0$$

in norm-resolvent sense, isolated spectral data below the essential threshold converge to the corresponding spectral data of K_0. Thus, when the bottom of the spectrum is an isolated eigenvalue,

$$\boxed{\lambda_{\mathrm{gap}}(\varepsilon) \longrightarrow \lambda_{\mathrm{gap}}(0).} \tag{15.145}$$

Even without assuming isolation of the lowest spectral point, the uniform lower bound (15.140), equivalently (15.143), is already sufficient to establish a strictly positive mass gap.

15.51 Decompactified master-field covariance

The Euclidean planar fluctuation operator is

$$\mathcal{A}_{E,\varepsilon} = -\partial_{t_E}^2 + K_\varepsilon.$$

From (15.141),

$$\mathcal{A}_{E,\varepsilon} \geq -\partial_{t_E}^2 + \frac{\Delta^2}{2}.$$

Therefore its inverse has no pole at zero Euclidean momentum. In the decompactification limit,

$$\mathcal{A}_{E,\varepsilon}^{-1} \longrightarrow \left(-\partial_{t_E}^2 + K_0\right)^{-1}.$$

The covariance restricted to the source-visible sector is equivalently obtained by exact Feshbach reduction:

$$P_E \mathcal{A}_{E,\epsilon}^{-1} P_E = \Gamma_\epsilon(-\omega_E^2)^{-1}.$$

Hence its poles are precisely the zeros of the full Schur kernel. The lower bound above excludes such zeros in a neighbourhood of zero Euclidean or Minkowski mass. The limiting planar covariance is therefore massive. Equivalently, every physical single-trace two-point function

obtained from the collective Gaussian field has a spectral representation whose support is bounded away from zero by the physical mass gap.

15.52 Restoration of the Yang–Mills scale

The operators used above are dimensionless operators in units fixed by the deformed-conifold geometry. The compact A-period determines the unique physical scale of the planar theory. Writing

$$\Pi_A = c_A(\alpha'\Lambda_{\mathrm{YM}})^3, \tag{15.146}$$

the dimensionful flat-space kinetic operator has the form

$$\boxed{K_{\mathbb{R}^4}^{\mathrm{phys}} = \Lambda_{\mathrm{YM}}^2 \widehat{K}_0} \tag{15.147}$$

up to the fixed convention-dependent normalization already absorbed into the definition of the dimensionless operator $\widehat{K}_0$. Let

$$c_{\mathrm{gap}}^2 = \inf \sigma\big(\widehat{K}_0\big). \tag{15.148}$$

Equation (15.133) implies

$$c_{\mathrm{gap}}^2 \geq \Delta^2 > 0.$$

Hence

$$\boxed{M_{\mathrm{gap}}^{\mathbb{R}^4} = c_{\mathrm{gap}}\Lambda_{\mathrm{YM}}, \qquad c_{\mathrm{gap}} > 0.} \tag{15.149}$$

This is the desired dimensionful statement.

15.53 Large-R asymptotics

Since

$$\varepsilon = (R\Lambda_{\mathrm{YM}})^{-1},$$

and the decompactification branch is even. The common-domain relative form estimates and the norm-resolvent bound give an $O(\epsilon^2)$ perturbation of the endpoint spectral problem. Consequently, if the bottom endpoint spectral value is isolated,

$$\lambda_{\mathrm{gap}}(\epsilon) = \lambda_{\mathrm{gap}}(0) + O(\epsilon^2).$$

Therefore the bottom of the spectrum satisfies

$$\lambda_{\mathrm{gap}}(R) = \lambda_{\mathrm{gap}}(\infty) + O((R\Lambda_{\mathrm{YM}})^{-2}), \tag{15.150}$$

whenever the endpoint eigenvalue is isolated. Taking square roots gives

$$\boxed{M_{\mathrm{gap}}(R) = c_{\mathrm{gap}}\Lambda_{\mathrm{YM}}[1 + O((R\Lambda_{\mathrm{YM}})^{-2})].} \tag{15.151}$$

Thus the finite-volume mass gap tends to a nonzero flat-space limit rather than scaling to zero as $1/R$. The $M_{\mathrm{gap}}(R)\sim 1/R$ behavior belongs to the opposite, ultraviolet finite-volume regime. In the decompactification regime $R\Lambda_{\mathrm{YM}} \to \infty$, the A-period fixes the remaining dimensionful scale and the spectrum approaches constants in units of Λ_{YM}.

15.54 Theorem

Theorem 15.1 — $\mathbb{R}^4$ planar mass gap.

Under the planar master-field identification established in the so far and the fixed-A-period physical construction, the decompactifying family of kinetic master-field operators constructed in Chapter 15 converges in norm-resolvent sense to the flat-space operator $K_{\mathbb{R}^4}$. Since the latter satisfies the global coercive estimate

$$K_{\mathbb{R}^4} \geq \Delta^2 I, \qquad \Delta > 0,$$

the decompactified planar theory possesses a strictly positive mass gap. Moreover, there exists $R_* > 0$ such that for all

$$R > R_*$$

the finite-radius planar kinetic master-field operator has a strictly positive uniform lower bound,

$$\boxed{K_R \geq \frac{\Delta^2}{2}\Lambda_{\mathrm{YM}}^2 I.} \tag{15.152}$$

Moreover, the $R \to \infty$ limit exists in norm-resolvent sense and defines a positive self-adjoint operator on the physical planar Hilbert space over $\mathbb{R}^4$,

$$K_{\mathbb{R}^4} = \lim_{R\to\infty} K_R,$$

with

$$\boxed{\operatorname{spec} K_{\mathbb{R}^4} \subset [c_{\mathrm{gap}}^2\Lambda_{\mathrm{YM}}^2, \infty), \qquad c_{\mathrm{gap}} > 0.} \tag{15.153}$$

Consequently the decompactified planar theory possesses a nonzero mass gap,

$$\boxed{M_{\mathrm{gap}}^{N=\infty,\mathbb{R}^4} = c_{\mathrm{gap}}\Lambda_{\mathrm{YM}} > 0.} \tag{15.154}$$

If the lowest endpoint spectral value is isolated, then

$$\boxed{M_{\mathrm{gap}}(R) = c_{\mathrm{gap}}\Lambda_{\mathrm{YM}}[1 + O((R\Lambda_{\mathrm{YM}})^{-2})].} \tag{15.155}$$

Proof

Sections 15.1-15.16 give an exact family of cochain embeddings with graph-norm convergence

$$C_\varepsilon = C_0 + O(\varepsilon^2).$$

Sections 15.18-15.28 show that the physical degree-wise Riesz structures satisfy

$$G_{q,\varepsilon} = G_{q,0} + O(\varepsilon^2)$$

and remain uniformly positive. Section 15.28-15.43 then gives

$$K_\varepsilon \longrightarrow K_0$$

in norm-resolvent sense and the relative form bounds (15.134)–(15.135). These bounds are applied to the exact physical quadratic pencil. They imply a uniform positive lower bound on the full pencil for $0 \le z < \Delta^2/2$. Wherever the complementary block is invertible, exact Feshbach equivalence transfers this exclusion directly to the full source-visible Schur kernel $\Gamma_\epsilon(z)$. Thus the result does not depend on truncating the energy-dependent 1PI kernel at $z = 0$. The endpoint operator satisfies

$$q_0[\Psi] \ge \Delta^2 b_0[\Psi].$$

Combining this inequality with the relative form estimates gives (15.139), and therefore, for sufficiently small ε,

$$K_\varepsilon \ge \frac{\Delta^2}{2} I.$$

Thus zero cannot enter the physical spectrum during decompactification.
Passing to the norm-resolvent limit gives

$$K_0 \ge \Delta^2 I.$$

Finally, restoring dimensions through the fixed A-period yields

$$K_{\mathbb{R}^4}^{\mathrm{phys}} = \Lambda_{\mathrm{YM}}^2 \widehat{K}_0,$$

and therefore

$$\boxed{M_{\mathrm{gap}}^{N=\infty,\mathbb{R}^4} = \Lambda_{\mathrm{YM}} \sqrt{\inf \sigma\big(\widehat{K}_0\big)} = c_{\mathrm{gap}} \Lambda_{\mathrm{YM}},}$$

with

$$c_{\mathrm{gap}} > 0.$$

This proves the theorem. □

15.55 Relation to the scalar singlet and tensor sectors

The theorem is stronger than a channel-by-channel calculation. The primitive scalar singlet is contained in the same fixed-period physical Hilbert space. Its formal zero mode is the excluded A-period variation and therefore does not represent a massless excitation. Likewise, the tensor sector considered earlier is one invariant subspace of the full master-field problem. Its nontrivial $SO(4) \to ISO(3)$ contraction provides an explicit spectral test of the general

mechanism, but it is not required separately for the existence of the gap once Theorem 15.1 has been established.

15.56 Explicit construction of the flat-space physical kinetic operator

So far, we have established the existence of the decompactified collective operator and its convergence under the $SO(4) \to ISO(3)$ contraction. We now identify its explicit differential realization and derive the associated flat-space propagator. The central observation is that the fixed endpoint Dolbeault complex, together with the physical Stenzel–Riesz pairing, determines the quadratic operator without introducing independent degree-wise metric functions.

Let

$$X = T^*S^3$$

be the deformed conifold equipped with the Stenzel metric g_{St}, and let

$$D = \bar{\partial}$$

be its fixed Dolbeault differential. The endpoint pairing identification is

$$\begin{aligned} b_0[u,v]\& &= \int_X \hat{h}\,(\tau)\, u \wedge *_6\, \overline{v}, \\ q_0[u,v]\& &= \int_X \bar{\partial}\, u \wedge *_6\, \overline{\bar{\partial} v}. \end{aligned} \tag{15.156}$$

Here $*_6$ is the Hodge operator of the six-dimensional Stenzel metric, and

$$\hat{h}(\tau) = \frac{h(\tau)}{h(0)}$$

is the normalized radial solution of the RG-generated Poisson equation. The degree-one field belongs to the constrained complex

$$\mathcal{C}^1 = \ker \partial \cap \Omega^{2,1}(X).$$

Its physical realization additionally satisfies the weighted Gauss condition

$$\bar{\partial}^*\big(\hat{h}\, u\big) = 0$$

and the fixed-background harmonic constraint. The quadratic forms are defined on this physical domain. In particular, the operator is not obtained by acting with $\hat{h}^{-1}\bar{\partial}^*\bar{\partial}$ on arbitrary unconstrained $(2,1)$-forms. Define

$$L = \partial : \Omega^{2,1}(X) \longrightarrow \Omega^{3,1}(X)$$

and

$$F = \bar{\partial}^*\bar{\partial}.$$

All adjoints in this subsection refer to the unweighted Stenzel pairing. Introduce the positive constraint operator

$$\mathcal{A} = LM_{\hat{h}^{-1}}L^*. \tag{15.157}$$

For a sufficiently regular field u, let λ_u solve

$$\mathcal{A}\lambda_u = L\big(\hat{h}^{-1}Fu\big). \tag{15.158}$$

Whenever the appropriate range inverse exists, the resulting constrained operator is

$$K_0 u = \hat{h}^{-1}\big(\bar{\partial}^*\bar{\partial}u - \partial^*\lambda_u\big). \tag{15.159}$$

Equivalently,

$$K_0 = \big[M_{\hat{h}^{-1}} - M_{\hat{h}^{-1}}L^*\mathcal{A}^\dagger LM_{\hat{h}^{-1}}\big]\bar{\partial}^*\bar{\partial}. \tag{15.160}$$

The inverse $\mathcal{A}^\dagger$ is understood on the admissible constraint range. Formula (15.160) is a strong-domain representation, rather than a bounded-operator identity on unrestricted unweighted L^2. Indeed, the multiplier equation implies

$$LK_0 u = 0.$$

Moreover, for every admissible constrained test field v,

$$\begin{aligned} b_0[v, K_0 u] &= \langle v, Fu - L^*\lambda_u\rangle_0 \\ &= \big\langle \bar{\partial}v, \bar{\partial}u\big\rangle_0 \\ &= q_0[v, u]. \end{aligned}$$

Thus (15.159) represents precisely the operator associated with the quadratic action. For every admissible gauge direction $D_0\eta$,

$$b_0[D_0\eta, K_0 u] = q_0[D_0\eta, u] = 0,$$

because

$$D_1 D_0 = 0.$$

Consequently, the operator maps its physical strong domain into the weighted Gauss slice. Let a denote the harmonic A-period modulus. Since

$$D_1 a = 0,$$

we also have

$$b_0[a, K_0 u] = 0.$$

Hence the orthogonal complement of the harmonic kernel is invariant under the associated self-adjoint operator. The definitive definition of K_0 is the quadratic-form realization

$$q_0[v, u] = b_0[v, K_0 u], \qquad v \in \mathcal{V}_{\text{phys}}. \tag{15.161}$$

The multiplier expression supplies its explicit differential–Green representation on the corresponding strong domain. Under the endpoint pairing identification (15.156), this

eliminates the need to specify independent Riesz operators $G_{1,0}$ and $G_{2,0}$. Their physical roles are contained in the weighted kinetic form, the unweighted curvature form and the constrained projection. This identification concerns the complete endpoint physical sector. It is stronger than merely knowing the zero-frequency coefficients of an independently truncated Feshbach pencil and must be compatible with the exact collective embedding constructed earlier.

15.57. Physical normalization and the flat-space mass operator

The operator K_0 in (15.160) is dimensionless when the Stenzel geometry and warp function are expressed in the normalized variables used in the spectral calculation. The compact period supplies the intrinsic Yang–Mills scale,

$$\Pi_A = c_A(\alpha'\Lambda_{\mathrm{YM}})^3.$$

The physical dimensional normalization is determined by the relative normalization of the quadratic stiffness and kinetic pairings. Writing this normalization as $\kappa > 0$, define

$$\mathcal{M}_0^2 = \frac{\Lambda_{\mathrm{YM}}^2}{\kappa} K_0. \tag{15.162}$$

This definition is compatible with the dimensional restoration discussed in §15.52, with the normalization convention now displayed explicitly. The physical mass-squared operator acts on the reduced Hilbert space

$$\mathcal{H}_0 = \mathcal{H}_{\mathrm{Gauss}} \ominus \ker K_0.$$

Suppose the endpoint spectral problem admits normalized physical eigenfields

$$K_0\Psi_n = \lambda_n\Psi_n,$$

with

$$b_0[\Psi_m, \Psi_n] = \delta_{mn}.$$

Then

$$M_n^2 = \frac{\Lambda_{\mathrm{YM}}^2}{\kappa}\lambda_n. \tag{15.163}$$

The corresponding energy at vanishing spatial momentum is

$$E_n(\mathbf{0}) = M_n.$$

If the physical resolvent is compact and the harmonic background modulus is the entire zero-energy kernel, the reduced eigenvalues obey

$$0 < \lambda_0 \le \lambda_1 \le \cdots, \qquad \lambda_n \to \infty.$$

In that case the spectrum is a discrete tower of massive collective excitations.

The compact-resolvent and kernel statements are independent analytic requirements; the explicit formula (15.160) alone does not establish them. The primitive scalar sector supplies an explicit example, with positive dimensionless eigenvalues

$$\lambda_{0^{++},n} \simeq 0.422006,\ 1.035338,\ 1.874017,\ 2.935003, \ldots$$

on the harmonic spectral complement. The same construction applies to the tensor, vector and other constrained sectors, with their respective Stenzel angular representations and physical projections.

15.58. The flat-space Euclidean propagator

We now pass from the rest-frame mass operator to the four-dimensional propagator. The decompactification

$$S_R^3 \times \mathbb{R} \longrightarrow \mathbb{R}^{3,1}$$

replaces the compact spatial $SO(4)$ symmetry by the Euclidean spatial group $ISO(3)$. As discussed below in Chapter 16, the collective map must intertwine the physical translation generators so that the momentum $\mathbf{p}$ carried by the flat collective field is the ordinary spatial momentum of Yang–Mills theory. The spatial contraction by itself fixes the momentum representation, but not the complete relativistic dispersion. For the following formula we use the transported four-dimensional quadratic dynamics: after canonical normalization, the physical collective modes have Euclidean kinetic operator $-\Delta_{\mathbb{R}^4} + \mathcal{M}_0^2$. Thus, writing

$$x = (t_E, \mathbf{x}) \in \mathbb{R}^4,$$

the canonically normalized quadratic action takes the form

$$S_{E,0}^{(2)}[\Phi] = \frac{1}{2}\int_{\mathbb{R}^4}\left[\sum_{\mu=0}^{3} b_0\left[\partial_\mu\Phi, \partial_\mu\Phi\right] + \frac{\Lambda_{\mathrm{YM}}^2}{\kappa} q_0[\Phi,\Phi]\right] d^4x. \tag{16.164}$$

Here

$$\Phi\colon \mathbb{R}^4 \longrightarrow \mathcal{H}_0$$

is a field taking values in the internal physical collective Hilbert space. The internal Stenzel coordinates and the physical spacetime coordinates play different roles. The operator K_0 acts on the internal collective variables, whereas $-\Delta_{\mathbb{R}^4}$ acts on physical spacetime.

The Euclidean quadratic operator is therefore

$$\mathcal{D}_{E,0} = -\Delta_{\mathbb{R}^4} \otimes I + I \otimes \mathcal{M}_0^2. \tag{15.165}$$

Its inverse is the planar collective propagator,

$$\mathfrak{C}_{E,0} = \left(-\Delta_{\mathbb{R}^4} + \frac{\Lambda_{\mathrm{YM}}^2}{\kappa} K_0\right)^{-1}. \tag{15.166}$$

This is the central propagator formula. Fourier transforming with respect to physical spacetime gives

$$\Phi(x) = \int_{\mathbb{R}^4} \frac{d^4 p_E}{(2\pi)^4} e^{ip_E \cdot x} \widetilde{\Phi}(p_E).$$

The quadratic action becomes

$$S^{(2)}_{E,0} = \frac{1}{2} \int_{\mathbb{R}^4} \frac{d^4 p_E}{(2\pi)^4} b_0 \big[\widetilde{\Phi}(-p_E), (p_E^2 + \mathcal{M}_0^2) \widetilde{\Phi}(p_E) \big].$$

Therefore the operator-valued momentum-space propagator is

$$\widetilde{\mathfrak{C}}_{E,0}(p_E) = \left(p_E^2 I + \frac{\Lambda_{\mathrm{YM}}^2}{\kappa} K_0 \right)^{-1}, \tag{15.167}$$

where

$$p_E^2 = p_{E,0}^2 + |\mathbf{p}|^{2.}$$

Substituting (15.160) gives the explicit Stenzel realization

$$\widetilde{\mathfrak{C}}_{E,0}(p_E) = \left[p_E^2 I + \frac{\Lambda_{\mathrm{YM}}^2}{\kappa} (M_{\hat{h}^{-1}} - M_{\hat{h}^{-1}} \partial^* (\partial M_{\hat{h}^{-1}} \partial^*)^\dagger \partial M_{\hat{h}^{-1}}) \bar{\partial}^* \bar{\partial} \right]^{-1}_{phys}. \tag{15.168}$$

The subscript indicates the self-adjoint physical realization, including the weighted Gauss condition, the ∂-closed constraint and removal of the harmonic modulus. Equation (15.168) is a formal strong-domain representation. Its rigorous meaning is the resolvent of the closed quadratic form (15.164), not an assertion that the displayed individual unbounded factors possess inverses on unrestricted L^2. The operator contains no independent phenomenological mass parameter. Once the endpoint pairing, collective domain and normalization κ are fixed, its spectral content is determined by the Stenzel differential geometry and the RG-generated function $\hat{h}$.

Let $E_{K_0}(\lambda)$ denote the projection-valued spectral measure of K_0. The spectral theorem gives

$$\widetilde{\mathfrak{C}}_{E,0}(p_E) = \int_{\sigma(K_0)} \frac{dE_{K_0}(\lambda)}{p_E^2 + \frac{\Lambda_{\mathrm{YM}}^2}{\kappa} \lambda}. \tag{16.169}$$

This formula does not require the spectrum to be discrete. If K_0 has compact resolvent, it reduces to

$$\widetilde{\mathfrak{C}}_{E,0}(p_E) = \sum_n \frac{|\Psi_n\rangle\langle\Psi_n|_{b_0}}{p_E^2 + M_n^2}. \tag{16.170}$$

The notation $\langle\Psi_n|_{b_0}$ denotes the dual defined by the physical kinetic pairing. Thus every normalized eigenmode contributes a propagator of the standard massive form

$$\frac{1}{p_E^2 + M_n^2}.$$

The eigenvalues of the constrained Stenzel operator determine the pole positions,

$$p_E^2 = -M_n^2.$$

The residues are determined by the physical overlap of the source with the corresponding eigenmode. At fixed spatial momentum, define

$$\omega(\mathbf{p}) = \sqrt{|\mathbf{p}|^2 + \mathcal{M}_0^2}.$$

Then

$$\widetilde{\mathfrak{C}}_{E,0}(p_{E,0}, \mathbf{p}) = \left(p_{E,0}^2 + \omega(\mathbf{p})^2\right)^{-1}.$$

Its inverse Fourier transform in Euclidean time is

$$\mathfrak{C}_{E,0}(t_E, \mathbf{p}) = \frac{1}{2\omega(\mathbf{p})} e^{-|t_E|\omega(\mathbf{p})}. \tag{15.172}$$

In the discrete spectral representation,

$$\mathfrak{C}_{E,0}(t_E, \mathbf{p}) = \sum_n \frac{e^{-|t_E|\sqrt{|\mathbf{p}|^2 + M_n^2}}}{2\sqrt{|\mathbf{p}|^2 + M_n^2}} |\Psi_n\rangle\langle\Psi_n|_{b_0}. \tag{15.173}$$

The resulting dispersion relations are

$$E_n(\mathbf{p})^2 = |\mathbf{p}|^2 + M_n^2. \tag{15.174}$$

Thus the internal Stenzel operator determines the rest masses, while the physical spacetime kinetic term determines their relativistic propagation.

15.59. Gauge-invariant sources and planar glueball residues

To identify the collective propagator with Yang–Mills glueball propagation, one must specify how gauge-invariant Yang–Mills operators couple to the physical collective field. Let

$$\mathcal{O}_A(x)$$

be a canonically normalized, centered gauge-invariant single-trace operator of pure $SU(N)$ Yang–Mills theory. Examples include the scalar and pseudoscalar operators

$$\mathcal{O}_{0^{++}}(x) \propto \mathrm{Tr}\, F_{\mu\nu} F^{\mu\nu}(x),$$

$$\mathcal{O}_{0^{-+}}(x) \propto \mathrm{Tr}\, F_{\mu\nu} \tilde{F}^{\mu\nu}(x),$$

and suitable symmetric traceless tensor operators coupling to the 2^{++} channel. The overall normalization of these operators is chosen so that their planar connected two-point functions have finite nontrivial limits.

Their microscopic generating functional is

$$Z_{\mathrm{YM}}[J] = \left\langle \exp\left(\int_{\mathbb{R}^4} J_A(x) \mathcal{O}_A(x)\, d^4x \right) \right\rangle.$$

The connected two-point function is

$$G_{AB}^{\mathrm{YM}}(x-y) = \left. \frac{\delta^2 \log Z_{\mathrm{YM}}[J]}{\delta J_A(x) \delta J_B(y)} \right|_{J=0}.$$

At strict planar order, the leading fluctuations of appropriately normalized single-trace operators are Gaussian, provided their planar limit exists and the connected higher cumulants are suppressed. On the collective side, let j_A be the physical source vector associated with $\mathcal{O}_A$. The source coupling is

$$S_J[\Phi] = -\int_{\mathbb{R}^4} J_A(x)\, b_0[j_A, \Phi(x)]\, d^4x.$$

For regular Hilbert-space sources, the corresponding connected covariance is

$$G_{AB}^{\mathrm{coll}}(p_E) = b_0[j_A, (p_E^2 + \mathcal{M}_0^2)^{-1} j_B]. \tag{15.175}$$

Local composite Yang–Mills operators generally require distributional source vectors and ultraviolet renormalization. In that case the expression is understood through smearing and, where necessary, contact-term subtraction. Inserting the spectral decomposition yields

$$G_{AB}^{\mathrm{coll}}(p_E) = \sum_n \frac{Z_{AB,n}}{p_E^2 + M_n^2} + G_{AB}^{\mathrm{contact}}(p_E), \tag{15.176}$$

where

$$Z_{AB,n} = b_0[j_A, \Psi_n]\, b_0[\Psi_n, j_B]. \tag{15.177}$$

If a mass eigenvalue is degenerate, the product is replaced by the corresponding finite-rank spectral projector. For a diagonal Hermitian source,

$$Z_{AA,n} = |b_0[j_A, \Psi_n]|^2 \geq 0. \tag{15.178}$$

The residue matrices are consequently positive semidefinite. A given source need not overlap with every eigenmode. Thus a physical glueball eigenvalue may be absent from a particular two-point function even though it is present in the collective spectrum. The source map determines which masses appear in which gauge-invariant correlators and with what residues. This is the distinction between determining a mass spectrum and determining the full planar glueball propagator. For the complete physical spectral resolution, define the matrix-valued spectral measure

$$d\mu_{AB}(\lambda) = b_0\big[j_A, dE_{K_0}(\lambda) j_B\big].$$

Then

$$G_{AB}^{\mathrm{coll}}(p_E) = \int_{\sigma(K_0)} \frac{d\mu_{AB}(\lambda)}{p_E^2 + \frac{\Lambda_{\mathrm{YM}}^2}{\kappa}\lambda} + G_{AB}^{\mathrm{contact}}(p_E). \tag{15.179}$$

Introducing the physical mass variable

$$s = \frac{\Lambda_{\mathrm{YM}}^2}{\kappa}\lambda,$$

this becomes

$$G_{AB}^{\mathrm{coll}}(p_E) = \int_0^\infty \frac{d\rho_{AB}(s)}{p_E^2 + s} + G_{AB}^{\mathrm{contact}}(p_E). \tag{15.180}$$

For discrete spectral support,

$$d\rho_{AB}(s) = \sum_n Z_{AB,n}\, \delta(s - M_n^2)\, ds.$$

Hence the planar glueball spectral density consists of isolated massive poles whenever the reduced collective operator has compact resolvent. The positivity of the spectral measure is inherited from the positive physical Riesz pairing. This representation is the Källén–Lehmann form expected for gauge-invariant two-point functions of a unitary relativistic quantum field theory, subject to the usual ultraviolet subtractions required by local composite operators.

15.60. Spin, parity and charge-conjugation classification

The flat-space contraction

$$SO(4) \longrightarrow ISO(3)$$

identifies the physical spatial rotation group acting on the collective excitations. Consequently, provided the collective map preserves the corresponding symmetry representations, the reduced physical Hilbert space decomposes into angular-momentum sectors. For pure Yang–Mills theory at $\theta = 0$, spatial parity P and charge conjugation C additionally classify the gauge-invariant glueball states. The physical Hilbert space may therefore be decomposed into sectors

$$\mathcal{H}_0 = \bigoplus_{J^{PC}} \mathcal{H}_{J^{PC}}, \tag{15.181}$$

with the understanding that this requires the parity and charge-conjugation actions to be implemented and intertwined by the collective map. Because the physical mass operator is invariant under these symmetries,

$$[\mathcal{M}_0^2, J_i] = 0,$$

$$[\mathcal{M}_0^2, P] = 0, \qquad [\mathcal{M}_0^2, C] = 0,$$

the propagator decomposes accordingly:

$$\widetilde{\mathfrak{C}}_{E,0} = \bigoplus_{J^{PC}} \left(p_E^2 + \mathcal{M}_{0,J^{PC}}^2\right)^{-1}. \tag{15.182}$$

Each block is associated with a definite glueball quantum-number channel. For scalar sources,

$$G_{0^{++}}(p_E) = \sum_n \frac{Z_{0^{++},n}}{p_E^2 + M_{0^{++},n}^2} + \text{contact terms}. \tag{15.183}$$

For tensor sources, the covariance carries the corresponding spin-two polarization structure:

$$G_{\mu\nu,\rho\sigma}^{(2^{++})}(p_E) = \sum_n \frac{Z_{2^{++},n}\, \mathcal{P}_{\mu\nu,\rho\sigma}^{(2)}(p_E; M_n)}{p_E^2 + M_{2^{++},n}^2} + \text{regular tensor structures}.$$

Here $\mathcal{P}^{(2)}$ denotes the appropriate covariant physical polarization numerator; its precise off-shell form depends on the chosen tensor interpolating operator. The tensor contraction studied in Chapter 7 is therefore interpreted as determining the rest-mass spectrum of the planar 2^{++} glueball channel, provided the contracted tensor module is identified with that physical Yang–Mills representation. The primitive scalar sector gives the corresponding 0^{++} tower. The complete Stenzel complex organizes the remaining glueball sectors through its allowed constrained representation content. The correspondence between the $SO(4)$ internal representations and flat-space J^{PC} is not merely a relabeling of angular harmonics. It requires the physical collective intertwiner and the appropriate parity and charge-conjugation actions.

15.61. Exact Feshbach propagator and planar self-energy

Until now we have employed the complete self-adjoint endpoint operator K_0. It is useful to connect this propagator with the exact Feshbach–Schur formulation developed in §15.20. Suppose the collective fluctuation space admits a source-visible decomposition

$$\mathcal{H} = E \oplus F,$$

where E contains the fields coupled directly to the chosen Yang–Mills sources and F contains complementary polarizations. Write the full quadratic pencil as

$$\mathcal{A}(z) = Q - zB,$$

and decompose it into blocks:

$$\mathcal{A}(z) = \begin{pmatrix} A_{EE}(z) & A_{EF}(z) \\ A_{FE}(z) & A_{FF}(z) \end{pmatrix}.$$

Whenever the complementary block is invertible, integrating out the F-fields gives

$$\Gamma_E(z) = A_{EE}(z) - A_{EF}(z) A_{FF}(z)^{-1} A_{FE}(z). \tag{15.185}$$

Define the exact collective self-energy

$$\Sigma_E(z) = A_{EF}(z) A_{FF}(z)^{-1} A_{FE}(z). \tag{15.186}$$

Then

$$\Gamma_E(z) = A_{EE}(z) - \Sigma_E(z).$$

This is the operator-valued Dyson form of the effective inverse propagator. Let

$$\rho = \frac{\Lambda_{\mathrm{YM}}^2}{\kappa}$$

and use the Euclidean continuation

$$z = -\frac{p_E^2}{\rho}.$$

For the full quadratic form $p_E^2 B + \rho Q$, exact block inversion gives the source-visible propagator

$$\tilde{G}_E(p_E) = \frac{1}{\rho}\Gamma_E\left(-\frac{p_E^2}{\rho}\right)^{-1}. \tag{15.187}$$

This is the exact propagator after elimination of the complementary polarizations. In particular, its poles are determined by

$$0 \in \sigma\left(\Gamma_E\left(\frac{M_n^2}{\rho}\right)\right). \tag{15.188}$$

The full spectral dependence of the Feshbach kernel must be retained.Indeed,

$$\Gamma_E(z) = K_{\mathrm{eff}} - zB_{\mathrm{eff}} + O(z^2)$$

near $z = 0$, but the linearized pencil need not reproduce the exact poles away from the expansion point. The full constrained operator K_0 and the exact Schur kernel provide equivalent descriptions when both are constructed from the same complete quadratic theory with compatible domains. The Feshbach representation changes the coordinates in which the inverse propagator is evaluated; it does not alter the exact physical pole spectrum. This establishes the mathematical analogy with the Schwinger–Dyson description of the planar Yang–Mills two-point function. In a microscopic formulation, the connected two-point function satisfies a Dyson equation involving a proper self-energy. The complementary Schur term Σ_E supplies its collective counterpart. However, the equality of Σ_E with a particular microscopic Yang–Mills 1PI self-energy requires the full source and quadratic-action identification. A Schur complement is not automatically the same object as the microscopic gluon self-energy. The propagator considered here is a gauge-invariant glueball propagator, not the gauge-dependent elementary gluon propagator.

15.62. Identification with the planar Yang–Mills glueball propagator

We can now state the precise conditions under which the spectral propagator derived above represents the planar glueball two-point function of Yang–Mills theory on $\mathbb{R}^4$. Let

$$\mathcal{H}_{\mathrm{YM}}^{(1)}$$

denote the gauge-invariant one-particle Hilbert space arising from the strict-planar Yang–Mills fundamental modular region at fixed Λ_{YM}. Let

$$\mathcal{C}_\infty : \mathcal{H}_{\mathrm{YM}}^{(1)} \longrightarrow \mathcal{H}_0$$

be the decompactified collective map. The required identification consists of three related structures. First, the collective map must preserve the physical quadratic pairing:

$$b_{\mathrm{YM}}(u, v) = b_0(\mathcal{C}_\infty u, \mathcal{C}_\infty v). \tag{15.189}$$

Second, it must preserve the physical curvature Hessian:

$$b_{\mathrm{YM}}(u, v) = b_0(\mathcal{C}_\infty u, \mathcal{C}_\infty v). \tag{15.190}$$

These identities must hold on the appropriate form domains, not merely on formal smooth fields. Third, the map must transport gauge-invariant sources and physical spacetime translations, so that the microscopic and collective source couplings describe the same observables. Under these conditions, the corresponding positive self-adjoint quadratic operators are unitarily equivalent on the source-generated one-particle sector:

$$\mathcal{C}_\infty K_{\mathrm{YM}}^{\mathrm{phys}} \mathcal{C}_\infty^{-1} = \mathcal{M}_0^2. \tag{15.191}$$

Here the Yang–Mills operator denotes the physical quadratic mass operator in the strict-planar Gaussian fluctuation representation, not the unreduced gauge-field Hessian. It follows from the functional calculus that

$$\mathcal{C}_\infty \left(p_E^2 + K_{\mathrm{YM}}^{\mathrm{phys}}\right)^{-1} \mathcal{C}_\infty^{-1} = (p_E^2 + \mathcal{M}_0^2)^{-1}.$$

Let $\mathcal{O}_A$ and $\mathcal{O}_B$ be gauge-invariant single-trace sources whose centered planar fluctuations are represented by vectors $j_A, j_B \in \mathcal{H}_0$, or by corresponding distributional source functionals. Then their connected planar propagator is

$$G_{AB}^{\mathrm{YM},\,N=\infty}(p_E)\& = b_0[j_A, (p_E^2 + \mathcal{M}_0^2)^{-1} j_B] + \text{contact terms}. \tag{15.192}$$

In a discrete spectral realization,

$$G_{AB}^{\mathrm{YM},\,N=\infty}(p_E) = \sum_n \frac{b_0[j_A, \Psi_n]\, b_0[\Psi_n, j_B]}{p_E^2 + \frac{\Lambda_{\mathrm{YM}}^2}{\kappa}\lambda_n} + \text{contact terms}. \tag{15.193}$$

This is the planar glueball spectral representation. It identifies the massive poles with physical gauge-invariant single-trace excitations of flat-space Yang–Mills theory.

The correspondence also fixes their source residues, provided the normalization and source transport are established.

Proposition 15.6 — Conditional identification of the flat-space collective propagator

Assume that:

1. The endpoint quadratic forms are given by (15.156), with the physical constraints and self-adjoint domains specified above.
2. The reduced endpoint operator K_0 is positive and self-adjoint, and its nonphysical harmonic modulus has been removed.
3. The decompactified collective map implements (15.189)–(15.190), intertwines physical translations and rotations, and identifies the renormalized gauge-invariant source functionals.
4. The strict-planar Yang–Mills fluctuation sector has a centered Gaussian limit whose quadratic action is transported by this map, with the Euclidean spacetime kinetic term normalized as in (15.164).

Then the connected planar Yang–Mills two-point functions in that source sector coincide, modulo local contact terms, with the source-projected resolvent of the constrained Stenzel operator:

$$\boxed{G_{AB}^{\mathrm{YM},N=\infty}(p_E) = b_0\left[j_A, \left(p_E^2 + \frac{\Lambda_{\mathrm{YM}}^2}{\kappa}K_0\right)^{-1} j_B\right] + \text{contact terms.}} \tag{15.194}$$

Their spectral poles are the physical collective masses

$$M_n^2 = \frac{\Lambda_{\mathrm{YM}}^2}{\kappa}\lambda_n,$$

and their residues are determined by the corresponding source overlaps.

If K_0 has compact resolvent and its kernel has been removed, the source-visible spectrum consists of isolated massive glueball poles, with no massless pole.

Proof

The pairing and stiffness identities (15.189)–(15.190) imply equivalence of the corresponding closed quadratic forms under the collective map. By uniqueness of the self-adjoint operators associated with those forms, the quadratic mass operators are unitarily intertwined. Functional calculus then intertwines their resolvents. The transported Gaussian quadratic action has inverse covariance

$$p_E^2 + \mathcal{M}_0^{2.}$$

Taking two derivatives of its connected generating functional with respect to the transported physical sources gives (15.194). The spectral theorem supplies the pole decomposition and the positive-semidefinite residue matrices. Compact resolvent gives discrete spectral support, while positivity of the reduced operator excludes massless poles.

This proves the proposition. □

The proposition makes explicit the difference between constructing a collective propagator and proving that it coincides with a particular microscopic Yang–Mills correlator. The latter conclusion depends on the asserted physical source and Hessian transport, not only on the formal cochain identity.

15.63. Interpretation and relation to the planar Yang–Mills master field

The preceding construction has a direct interpretation in terms of the fundamental modular region and the large-N master field. At finite N, Yang–Mills theory is described by gauge-field configurations modulo local gauge transformations. The physical fundamental modular region supplies a representative of each gauge orbit, up to its boundary identifications. In the strict planar limit, the collective construction seeks to reorganize the gauge-invariant fluctuations into a reduced master-field Hilbert space. Under the proposed open–closed map, this collective space is represented by constrained $(2,1)$-forms on the deformed conifold. The complex structure and its compact period encode the RG-invariant background data, while the radial function $\hat{h}$ supplies the physical kinetic Riesz metric. The one-sided Dolbeault differential determines the curvature stiffness. The resulting quadratic mass operator is

$$\boxed{\mathcal{M}_0^2 = \frac{\Lambda_{\mathrm{YM}}^2}{\kappa}\left[M_{\hat{h}^{-1}} - M_{\hat{h}^{-1}}\partial^*(\partial M_{\hat{h}^{-1}}\partial^*)^\dagger \partial M_{\hat{h}^{-1}}\right]\bar{\partial}^*\bar{\partial}} \tag{15.195}$$

on its specified physical strong domain. Its four-dimensional propagator is

$$\boxed{\widetilde{\mathfrak{C}}_{E,0}(p_E) = (p_E^2 + \mathcal{M}_0^2)^{-1}.} \tag{15.196}$$

In the source-visible sector, the equivalent exact representation is the inverse energy-dependent Feshbach kernel developed in §15.61. The spectral decomposition organizes the strict-planar collective excitations into glueball channels characterized by their masses, spatial spins, parity, charge conjugation and gauge-invariant source residues. The massive spectrum is generated by the interaction of the fixed Stenzel differential geometry, the physical constrained Hilbert space and the RG-produced warp profile. No independent glueball mass term is introduced into the differential operator. The nontrivial Yang–Mills scale enters through the A-period and the relative kinetic normalization. The associated Euclidean covariance is positive whenever the reduced mass operator is positive self-adjoint; a strictly positive mass lower bound additionally yields infrared regularity and the massive reflection-positive structure used in Chapter 16. These observations provide the connection between three descriptions:

Planar Yang--Mills FMR fluctuations
$\updownarrow \quad \mathcal{C}_\infty$
Constrained Stenzel master-field fluctuations
$\updownarrow$ spectral decomposition
Massive planar glueball propagators.

The first correspondence requires the physical open–closed source and quadratic-form identifications. The second follows mathematically from the self-adjoint spectral resolution of the constrained operator. Thus the flat-space propagator is completely characterized at the operator level by the Stenzel geometry, the warp factor and the physical constraints. Establishing that characterization as an exact statement about the microscopic planar Yang–Mills two-point functions requires completing the source and Riesz matching assumptions stated in Proposition 15.6. Chapter 16 now studies the Euclidean covariance, its reflection positivity and its reconstruction as a positive-energy quantum field theory.

16 Existence Theorem

16.1 Reflection positivity of the Gaussian planar master-field process

The chapter 12 constructed the strict planar Yang–Mills fluctuation measure on the gauge-reduced fundamental modular region and identified its geometric representative with the centered Gaussian measure $\gamma_{\mathfrak{C}}$ whose Euclidean covariance is

$$\mathfrak{C} = \mathcal{A}_E^{-1} = (-\partial_\tau^2 + K)^{-1}.$$

Here K denotes the positive self-adjoint physical master-field operator acting on the one-particle Hilbert space $\mathcal{H}_{\mathrm{phys}}^{(1)}$. The mass-gap theorem established above gives

$$K \geq \Delta^2 I, \qquad \Delta > 0. \tag{16.1}$$

In particular, K is strictly positive. Define

$$\omega = K^{1/2}. \tag{16.2}$$

Then

$$\omega \geq \Delta I.$$

The purpose of this section is to prove that the Gaussian planar master-field measure is reflection positive with respect to Euclidean time reflection [41].

16.2. Euclidean covariance kernel

By the spectral theorem for the positive self-adjoint operator K,

$$\mathfrak{C} = (-\partial_\tau^2 + \omega^2)^{-1}.$$

Fourier transformation in Euclidean time gives

$$\mathfrak{C}(p_0) = (p_0^2 + \omega^2)^{-1}. \tag{16.3}$$

The inverse Fourier transform is the operator-valued kernel

$$\mathfrak{C}(\tau,\sigma)=\frac{1}{2\omega}e^{-|\tau-\sigma|\omega}. \tag{16.4}$$

Indeed, by functional calculus this formula is obtained by applying the scalar identity

$$\int_{\mathbb{R}}\frac{dp_0}{2\pi}\frac{e^{ip_0t}}{p_0^2+\lambda^2}=\frac{1}{2\lambda}e^{-\lambda|t|}$$

to the spectral resolution of ω. Because

$$\omega\geq\Delta I,$$

the inverse powers occurring in (16.3) are bounded:

$$\|\omega^{-1}\|\leq\Delta^{-1}.$$

Thus the covariance is infrared regular.

16.3. Time reflection

Let

$$\theta\colon\tau\mapsto-\tau$$

denote Euclidean time reflection. For an $\mathcal{H}^{(1)}_{\text{phys}}$-valued test function f, define

$$(\Theta f)(\tau)=f(-\tau)$$

for real fields, and

$$(\Theta f)(\tau)=\overline{f(-\tau)}$$

in the complexified formulation.

Let

$$\mathcal{E}_+=\{f\in\mathcal{E}\colon\operatorname{supp}f\subset(0,\infty)\} \tag{16.5}$$

denote the positive-time test-function subspace. Reflection positivity of the covariance means that

$$\langle\Theta f,\mathfrak{C}f\rangle\geq0\qquad\forall f\in\mathcal{E}_+. \tag{16.6}$$

We now prove this directly.

16.4. Factorization of the reflected covariance

For $\tau,\sigma>0$,

$$|-\tau-\sigma|=\tau+\sigma.$$

Therefore

$$\mathfrak{C}(-\tau,\sigma)=\frac{1}{2\omega}e^{-(\tau+\sigma)\omega}. \tag{16.7}$$

Since ω commutes with every bounded Borel function of itself,

$$\frac{1}{2\omega}e^{-(\tau+\sigma)\omega} = \frac{12}{(\omega^{-1/2}e^{-\tau\omega})}\left(e^{-\sigma\omega}\omega^{-1/2}\right). \tag{16.8}$$

Hence, for $f \in \mathcal{E}_+$,

$$\begin{aligned}\langle \Theta f, f\rangle &= \int_0^\infty d\tau \int_0^\infty d\,\sigma\, \langle f(\tau), \mathbb{C}(-\tau,\sigma) f(\sigma)\rangle \\ &= \frac{1}{2}\int_0^\infty d\tau \int_0^\infty d\,\sigma\, \left\langle \omega^{-1/2}e^{-\tau\omega} f(\tau), \omega^{-1/2}e^{-\sigma\omega} f(\sigma)\right\rangle.\end{aligned}$$

Therefore

$$\langle \Theta f, \mathbb{C} f\rangle = \frac{1}{2}\left\| \int_0^\infty d\tau\, \omega^{-1/2}e^{-\tau\omega} f(\tau)\right\|^2. \tag{16.9}$$

It follows immediately that

$$\boxed{\langle \Theta f, Cf\rangle \geq 0.} \tag{16.10}$$

Thus the covariance of the planar Gaussian master-field process is reflection positive.

16.5. Reflection positivity of the full Gaussian measure

For a centered Gaussian measure, all Schwinger functions are determined by the two-point covariance through Wick's theorem. The characteristic functional is

$$\chi(f) = \exp\left[-\frac{1}{2}\langle f, \mathbb{C} f\rangle\right].$$

Let F be a cylindrical functional depending only on fields at positive Euclidean times,

$$F(\Phi) = F\big(\Phi(f_1), \dots, \Phi(f_n)\big), \qquad f_i \in \mathcal{E}_+. \tag{16.11}$$

The reflected functional is

$$(\Theta F)(\Phi) = \overline{F(\Theta\Phi)}.$$

For Gaussian measures, positivity of the reflected covariance kernel implies positivity of all finite-dimensional Gaussian distributions generated by positive-time test functions. Consequently,

$$\int_{\mathcal{E}'} (\Theta F)\,(\Phi) F(\Phi)\, d\gamma_{\mathbb{C}}(\Phi) \geq 0. \tag{16.12}$$

Thus γ_C is Osterwalder–Schrader reflection positive. Since the planar FMR fluctuation measure was identified in the preceding construction with this Gaussian measure,

$$\gamma_{\mathrm{FMR}} \simeq \gamma_{\mathbb{C}} \tag{16.13}$$

under the planar collective map, reflection positivity transports to the planar Yang–Mills master-field measure. Hence

$$\boxed{\gamma_{\mathrm{FMR}}^{N=\infty} \text{ is reflection positive.}} \tag{16.14}$$

16.6 Spectral interpretation

The proof may also be written directly in the spectral basis of K.

Let

$$K\Psi_\alpha = M_\alpha^2 \Psi_\alpha, \qquad M_\alpha \geq \Delta.$$

Then

$$\mathfrak{C}(\tau,\sigma) = \sum_\alpha \frac{e^{-M_\alpha|\tau-\sigma|}}{2M_\alpha} |\Psi_\alpha\rangle\langle\Psi_\alpha|, \tag{16.15}$$

with the sum replaced by the appropriate spectral integral whenever necessary. For positive Euclidean times,

$$\mathfrak{C}(-\tau,\sigma) = \sum_\alpha \frac{e^{-M_\alpha(\tau+\sigma)}}{2M_\alpha} |\Psi_\alpha\rangle\langle\Psi_\alpha|. \tag{16.16}$$

Therefore

$$\langle \Theta f, \mathfrak{C} f\rangle = \frac{1}{2}\sum_\alpha \frac{1}{M_\alpha} \left| \int_0^\infty e^{-M_\alpha \tau} f_\alpha(\tau)\, d\tau \right|^2 \geq 0. \tag{16.17}$$

Thus every physical glueball mode contributes a separately positive term to the reflected covariance. The positivity of the Euclidean master-field measure is therefore equivalent to the positivity of the physical spectral measure.

16.7. Relation to the previously constructed Hilbert space

The physical one-particle Hilbert space was already constructed from the gauge-reduced master-field complex as

$$\mathcal{H}_{\mathrm{phys}}^{(1)}.$$

Bosonic quantization gave the symmetric Fock space

$$\mathcal{F}_{\mathrm{phys}} = \Gamma_s\left(\mathcal{H}_{\mathrm{phys}}^{(1)}\right) = \bigoplus_{n=0}^{\infty} \mathrm{Sym}^n \mathcal{H}_{\mathrm{phys}}^{(1)}. \tag{16.18}$$

The present argument does not introduce a second Hilbert space. Rather, reflection positivity shows that the Euclidean Gaussian FMR measure is compatible with precisely this physical Hilbert-space structure. At the one-particle level the positive Euclidean-time semigroup appearing in (16.4) is

$$e^{-\tau\omega}, \qquad \omega = K^{1/2}.$$

Hence the one-particle Hamiltonian is

$$H^{(1)} = K^{1/2}. \tag{16.19}$$

On the full bosonic space,

$$H = d\Gamma\big(K^{1/2}\big), \tag{16.20}$$

up to the conventional vacuum-energy normalization. Thus the Hilbert and Fock structures introduced previously from canonical quantization are recovered from the Euclidean master-field measure itself.

16.8. Reflection-positivity theorem

The argument above may be summarized as follows.

Theorem 16.1 — Reflection positivity of the planar Yang–Mills master-field measure

Let $\mathcal{H}^{(1)}_{\mathrm{phys}}$ be the physical planar one-particle Hilbert space and let K be the positive self-adjoint master-field operator obtained from the gauge-reduced quadratic Yang–Mills action. Assume the mass-gap estimate

$$K \geq \Delta^2 I, \qquad \Delta > 0.$$

Let

$$\mathfrak{C} = (-\partial_\tau^2 + K)^{-1}$$

be the covariance of the centered strict-planar Gaussian FMR fluctuation process.

Then:

1. the covariance admits the representation

$$\mathfrak{C}(\tau,\sigma) = \frac{1}{2K^{1/2}} e^{-|\tau-\sigma|K^{1/2}};$$

2. for every positive-time test function f,

$$\langle \Theta f, \mathfrak{C} f\rangle = \frac{1}{2}\left\| \int_0^\infty K^{-1/4}\, e^{-\tau K^{1/2}} f(\tau)\, d\tau \right\|^2 \geq 0;$$

3. the associated Gaussian measure γ_C is Osterwalder–Schrader reflection positive;

4. under the previously established identification of $\gamma_{\mathfrak{C}}$ with the strict-planar FMR fluctuation measure, the planar Yang–Mills master-field measure is reflection positive;

5. the corresponding positive-time generator on the one-particle sector is

$$H^{(1)} = K^{1/2},$$

and the bosonic quantum Hamiltonian on the previously constructed Fock space is

$$H = d\Gamma\big(K^{1/2}\big).$$

In particular,

$$\boxed{\operatorname{spec} H^{(1)} \subset [\Delta, \infty).} \tag{16.21}$$

Thus the strictly positive mass gap of the physical master-field operator is compatible with a reflection-positive Euclidean planar Yang–Mills measure and with the previously constructed positive-energy quantum Hilbert space.

16.9 Equivariance of the collective map

Let $\mathcal{H}_{\mathrm{YM},R}^{\mathrm{phys}}$ be the physical Yang–Mills tangent space on S_R^3, and let $\mathcal{H}_{\mathrm{coll},R}$ be its collective realization on the reconstructed geometry. The collective map is

$$C_R : \mathcal{H}_{\mathrm{YM},R}^{\mathrm{phys}} \longrightarrow \mathcal{H}_{\mathrm{coll},R}. \tag{16.22}$$

The spatial isometry group of S_R^3 is

$$G_R = SO(4).$$

Let

$$U_R(g)$$

denote its action on Yang–Mills fluctuations, and let

$$\widetilde{U}_R(g)$$

denote the induced action on the collective fields.

We want to prove

$$C_R U_R(g) = \widetilde{U}_R(g) C_R \tag{16.23}$$

for every $g \in SO(4)$.

16.10 The $SO(4)$ action on the Yang–Mills physical space

For $g \in SO(4)$, let

$$\varphi_g : S_R^3 \to S_R^3$$

be the corresponding isometry. A spatial Yang–Mills connection transforms by pullback,

$$A \mapsto \varphi_g^* A.$$

At the linearized level,

$$a \mapsto U_R(g)a := \varphi_g^* a. \tag{16.24}$$

Because φ_g is an isometry,

$$G_{\mathrm{HS}}(U_R(g)a, U_R(g)b) = G_{\mathrm{HS}}(a,b). \tag{16.25}$$

Thus

$$U_R(g)$$

is unitary with respect to the physical Yang–Mills Riesz metric. Moreover, pullback commutes with the exterior differential:

$$d\,\varphi_g^* = \varphi_g^* d. \tag{16.26}$$

Since the Yang–Mills covariant differential transforms naturally,

$$D_{\varphi_g^* A}\varphi_g^* = \varphi_g^* D_A.$$

Hence the gauge directions are preserved. The FMR itself is defined through the norm minimization condition

$$\|A\|_{\mathrm{HS}} \leq \|A^u\|_{\mathrm{HS}} \qquad \forall u \in \mathcal{G}.$$

Because both the norm and the gauge action are natural under spatial isometries, if A lies in the FMR, then so does $\varphi_g^* A$, modulo the same boundary identifications. Therefore $SO(4)$ acts on the gauge-reduced configuration space and hence on its tangent space

$$\mathcal{H}^{\mathrm{phys}}_{\mathrm{YM},R}.$$

This is consistent with the manuscript's construction of the FMR as the metric realization of the physical gauge quotient.

16.11 The $SO(4)$ action on the collective space

The reconstructed geometry is

$$X_R \simeq T^* S^3_R.$$

The $SO(4)$ action on S^3_R lifts canonically to the cotangent bundle,

$$g: T^* S^3_R \to T^* S^3_R. \tag{16.27}$$

The Stenzel structure used in the collective theory is $SO(4)$-invariant. Consequently its geometric ingredients satisfy

$$g^* J = J, \qquad g^* \Omega = \Omega, \qquad g^* g_{\mathrm{St}} = g_{\mathrm{St}}. \tag{16.28}$$

The Riesz weight is radial,

$$h = h(\tau),$$

and τ is $SO(4)$-invariant. Hence

$$g^* h = h. \tag{16.29}$$

Therefore

$$\widetilde{U}_R(g)\Psi = g^*\Psi \tag{16.30}$$

acts unitarily on the weighted collective Hilbert space:

$$G_h\big(\widetilde{U}_R(g)\Psi, \widetilde{U}_R(g)\Phi\big) = G_h(\Psi, \Phi). \tag{16.31}$$

Because the complex structure is invariant,

$$\bar{\partial}\, g^* = g^*\bar{\partial}.$$

Similarly,

$$\partial\, g^* = g^*\partial.$$

Thus the constrained physical collective space

$$\ker \partial \cap \ker \bar{\partial}^\sharp$$

is invariant under $SO(4)$.

16.12 Conjugating the collective map

Now define, for fixed $g \in SO(4)$,

$$C_R^{(g)} := \widetilde{U}_R(g) C_R U_R(g)^{-1}. \tag{16.32}$$

We show that $C_R^{(g)}$ satisfies exactly the same defining properties as C_R.

Metric transport

The collective map was constructed to transport the Yang–Mills physical Riesz metric:

$$G_{\mathrm{YM}}(a, b) = G_h(C_R a, C_R b).$$

Then

$$\begin{aligned} G_h\left(C_R^{(g)} a, C_R^{(g)} b\right) &= G_h\big(\widetilde{U}_R(g) C_R U_R(g)^{-1} a, \widetilde{U}_R(g) C_R U_R(g)^{-1} b\big) \\ &= G_h(C_R U_R(g)^{-1} a, C_R U_R(g)^{-1} b) \\ &= G_{\mathrm{YM}}(U_R(g)^{-1} a, U_R(g)^{-1} b) \\ &= G_{\mathrm{YM}}(a, b). \end{aligned} \tag{16.33}$$

Hence $C_R^{(g)}$ transports exactly the same Riesz structure.

16.13 Differential intertwining

The collective realization also transports the physical differential:

$$D C_R = C_R Q_R, \tag{16.34}$$

where Q_R denotes the Yang–Mills physical linear differential and

$$D = \bar{\partial}$$

on the collective side. Since both differentials are symmetry-equivariant,

$$Q_R U_R(g) = U_R(g) Q_R,$$

and

$$D\widetilde{U}_R(g) = \widetilde{U}_R(g) D.$$

Therefore

$$\begin{aligned} DC_R^{(g)} &= D\widetilde{U}_R(g) C_R U_R(g)^{-1} \\ &= \widetilde{U}_R(g) D C_R U_R(g)^{-1} \\ &= \widetilde{U}_R(g) C_R Q_R U_R(g)^{-1} \\ &= \widetilde{U}_R(g) C_R U_R(g)^{-1} Q_R \\ &= C_R^{(g)} Q_R. \end{aligned} \tag{16.35}$$

Thus $C_R^{(g)}$ obeys the same chain-map relation as C_R.

16.14 Source and period normalization

There is one more piece, because metric preservation and the chain-map relation alone could leave an internal automorphism. The collective map is fixed by the same source-period normalization used in its construction. The fixed A-period direction is $SO(4)$-invariant: the A-cycle is the zero-section

$$S_R^3 \subset T^* S_R^3,$$

and $SO(4)$ preserves it as a cycle.

Hence

$$\int_A g^* \Omega = \int_{gA} \Omega = \int_A \Omega. \tag{16.36}$$

The RG source normalization and fixed-A-period condition are therefore unchanged by g. Likewise, the asymptotic and regularity conditions entering the collective map depend only on $SO(4)$-invariant geometric data. Therefore $C_R^{(g)}$ satisfies:

$$\begin{cases} \text{the same chain condition,} \\ \text{the same Riesz-isometry condition,} \\ \text{the same fixed-period condition,} \\ \text{the same regularity/asymptotic conditions.} \end{cases}$$

16.15 Use uniqueness

The exact collective-cochain construction is unique once the physical horizontal convention, period normalization, and boundary conditions have been fixed. Therefore

$$C_R^{(g)} = C_R. \tag{16.37}$$

Using the definition (16.32),

$$\widetilde{U}_R(g) C_R U_R(g)^{-1} = C_R. \tag{16.38}$$

Multiplying on the right by $U_R(g)$,

$$\boxed{\widetilde{U}_R(g) C_R = C_R U_R(g).} \tag{16.39}$$

Thus the collective map is $SO(4)$-equivariant.

16.16 Proposition

Proposition 16.1— $SO(4)$-equivariance of the collective master-field realization

Let

$$C_R : \mathcal{H}_{\mathrm{YM},R}^{\mathrm{phys}} \to \mathcal{H}_{\mathrm{coll},R}$$

be the collective realization of planar Yang–Mills on S_R^3, uniquely fixed by

$$D C_R = C_R Q_R, \tag{16.40}$$

preservation of the physical Riesz pairing, the fixed-A-period normalization, and the physical regularity and boundary conditions. Let U_R and $\widetilde{U}_R$ denote respectively the natural $SO(4)$ actions on the gauge-reduced Yang–Mills tangent space and on the collective field space.

Then

$$C_R U_R(g) = \widetilde{U}_R(g) C_R, \qquad g \in SO(4). \tag{16.41}$$

Hence the collective realization preserves the complete $SO(4)$ representation decomposition. In particular, if

$$a \in \mathcal{H}_{\mathrm{YM},R}^{\mathrm{phys}}$$

transforms in an irreducible representation I, then

$$C_R a$$

transforms in the same representation I. This supplies the justification for the $SO(4)$ channel decomposition used in the preceding spectral analysis.

16.17 Decompactification gives translations

At finite R,

$$SO(4) \simeq SU(2)_L \times SU(2)_R.$$

Introduce rotation and contracted translation generators schematically as

$$J_i = J_i^{(L)} + J_i^{(R)} \tag{16.42}$$

$$P_i^{(R)} = \frac{1}{R}\left(J_i^{(L)} - J_i^{(R)}\right). \tag{16.43}$$

Then

$$\left[P_i^{(R)}, P_j^{(R)}\right] = O(R^{-2})J_k \longrightarrow 0, \tag{16.44}$$

whereas

$$\left[J_i, P_j\right] = i\epsilon_{ijk}P_k.$$

Therefore

$$SO(4) \longrightarrow ISO(3).$$

Since

$$C_R U_R(g) = \widetilde{U}_R(g) C_R$$

at every finite R, and we have already established graph convergence

$$C_R \to C_\infty,$$

the limiting map inherits the contracted intertwining relation:

$$\boxed{C_\infty U_\infty(a, \mathcal{R}) = \widetilde{U}_\infty(a, \mathcal{R}) C_\infty.} \tag{16.45}$$

In particular,

$$\boxed{C_\infty P_i^{\mathrm{YM}} = P_i^{\mathrm{coll}} C_\infty.} \tag{16.46}$$

So the spatial translations of the decompactified master-field theory are not independently postulated. They are the contraction of the spatial $SO(4)$ representation that was already transported by the master-field realization.

16.18 Momentum decomposition of the decompactified master-field Hilbert space

By the equivariance result of the preceding subsection, the decompactified collective map

$$C_\infty : \mathcal{H}_{\mathrm{YM}}^{\mathrm{phys}} \longrightarrow \mathcal{H}_{\mathrm{coll}}^{\mathrm{phys}}$$

intertwines the spatial Euclidean group:

$$C_\infty U_{\mathrm{YM}}(a, R) = U_{\mathrm{coll}}(a, R) C_\infty, \qquad (a, R) \in ISO(3). \tag{16.47}$$

In particular, for the translation subgroup,

$$C_\infty U_{\mathrm{YM}}(a) = U_{\mathrm{coll}}(a) C_\infty, \qquad a \in \mathbb{R}^3. \tag{16.48}$$

We now derive the momentum-space decomposition of the physical one-particle Hilbert space.

16.19 Strongly continuous translation representation

The decompactified spatial manifold is $\mathbb{R}^3$. Translations act on a collective field Ψ by pullback,

$$(U(a)\Psi)(x) = \Psi(x - a).$$

Because the limiting Riesz metric is spatially translation invariant,

$$\langle U(a)\Psi, U(a)\Phi\rangle_{\text{phys}} = \langle \Psi, \Phi\rangle_{\text{phys}}. \tag{16.49}$$

Thus $U(a)$ is unitary. Moreover,

$$\lim_{a\to 0} \|U(a)\Psi - \Psi\|_{\text{phys}} = 0 \tag{16.50}$$

for smooth compactly supported physical fields, and hence by density for every

$$\Psi \in \mathcal{H}^{(1)}_{\text{phys}}.$$

Therefore

$$a \mapsto U(a)$$

is a strongly continuous unitary representation of the Abelian group $\mathbb{R}^3$.

16.20 Momentum generators

By the multi-parameter Stone theorem, there exist three mutually strongly commuting self-adjoint operators

$$P_1, \qquad P_2, \qquad P_3$$

such that

$$U(a) = e^{-ia\cdot P}. \tag{16.51}$$

Because the representation is Abelian,

$$[P_i, P_j] = 0$$

in the strong spectral sense. The joint spectral theorem therefore gives a projection-valued measure

$$E_P(d\mathbf{p}) \tag{16.52}$$

on $\mathbb{R}^3$ such that

$$U(a) = \int_{\mathbb{R}^3} e^{-ia\cdot\mathbf{p}}\, E_P(d\mathbf{p}). \tag{16.53}$$

Consequently, $\mathcal{H}^{(1)}_{\text{phys}}$ admits a direct-integral decomposition

$$\boxed{\mathcal{H}^{(1)}_{\text{phys}} \simeq \int_{\mathbb{R}^3}^{\oplus} \mathcal{H}\,(\mathbf{p})\, d\mu(\mathbf{p}),} \tag{16.54}$$

where $d\mu$ is the joint spectral measure of the translation generators. This statement follows solely from the $ISO(3)$ representation.

16.21 Why the measure is ordinary momentum measure

We can sharpen (16.54) in the present realization. After decompactification, a physical collective field has ordinary spatial dependence

$$\Psi = \Psi(\mathbf{x}; \xi),$$

where

$$\mathbf{x} \in \mathbb{R}^3$$

is the spatial center-of-mass coordinate and ξ denotes the remaining internal collective variables. Before imposition of the physical constraints, the limiting field Hilbert space therefore has the form

$$\mathcal{H}_{\text{kin}} = L^2(\mathbb{R}^3_{\mathbf{x}}; \mathcal{K}), \tag{16.55}$$

for an internal Hilbert space $\mathcal{K}$ carrying the radial, polarization, and collective degrees of freedom. Translations act as

$$(U(a)\Psi)(\mathbf{x}, \xi) = \Psi(\mathbf{x} - a, \xi).$$

The spatial Fourier transform

$$(\mathcal{F}\Psi)(\mathbf{p}, \xi) = \frac{1}{(2\pi)^{3/2}} \int_{\mathbb{R}^3} e^{-i\mathbf{p}\cdot\mathbf{x}}\, \Psi(\mathbf{x}, \xi)\, d^3x \tag{16.56}$$

is unitary by the vector-valued Plancherel theorem,

$$\|\Psi\|^2 = \int_{\mathbb{R}^3} \left\|\widehat{\Psi}(\mathbf{p})\right\|^2_{\mathcal{K}}\, d^3p. \tag{16.57}$$

Hence

$$L^2(\mathbb{R}^3; \mathcal{K}) \simeq \int_{\mathbb{R}^3}^{\oplus} \mathcal{K}\, d^3p. \tag{16.58}$$

In this representation,

$$\left(\widehat{U}(a)\widehat{\Psi}\right)(\mathbf{p}) = e^{-ia\cdot\mathbf{p}}\widehat{\Psi}(\mathbf{p}),$$

and

$$\left(P_i\widehat{\Psi}\right)(\mathbf{p}) = p_i\widehat{\Psi}(\mathbf{p}).$$

Thus the spectral measure of translations is absolutely continuous with respect to Lebesgue measure and the direct integral may be written with d^3p.

16.22 Physical constraints decompose fiberwise

The physical Hilbert space is not the whole kinematical space. It is obtained by imposing the closed-string-Yang–Mills constraints. In the limiting collective realization these have the form

$$\partial\Psi = 0, \qquad D^{\sharp}\Psi = 0,$$

together with the fixed-period and regularity conditions. Because the collective map is $ISO(3)$-equivariant, the physical constraint operators commute with translations:

$$[\partial, U(a)] = 0, \qquad \left[D^{\sharp}, U(a)\right] = 0. \tag{16.59}$$

Hence they are decomposable in momentum space,

$$\partial = \int_{\mathbb{R}^3}^{\oplus} \partial(\mathbf{p})\, d^3p \tag{16.60}$$

$$D^{\sharp} = \int_{\mathbb{R}^3}^{\oplus} D^{\sharp}(\mathbf{p})\, d^3p. \tag{16.61}$$

Therefore the physical subspace is itself decomposable:

$$\mathcal{H}^{(1)}_{\text{phys}} \simeq \int_{\mathbb{R}^3}^{\oplus} \mathcal{H}_{\text{phys}}\,(\mathbf{p})\, d^3p, \tag{16.62}$$

with

$$\boxed{\mathcal{H}_{\text{phys}}(\mathbf{p}) = \ker \partial\,(\mathbf{p}) \cap \ker D^{\sharp}\,(\mathbf{p})}$$

together with the fiberwise fixed-period and regularity conditions. This is the precise momentum-space decomposition of the physical planar master-field Hilbert space.

16.23 Decomposition of the master-field operator

The physical mass/kinetic operator K is obtained from the translation-invariant quadratic master-field action.
Hence

$$KU(a) = U(a)K.$$

Any self-adjoint operator commuting with the joint spectral measure of the translations is decomposable. Therefore

$$K = \int_{\mathbb{R}^3}^{\oplus} K(\mathbf{p})\, d^3p. \tag{16.63}$$

Each fibre operator

$$K(\mathbf{p}) \colon \mathcal{H}_{\text{phys}}(\mathbf{p}) \to \mathcal{H}_{\text{phys}}(\mathbf{p})$$

is self-adjoint. The spectral equation therefore becomes

$$K(\mathbf{p})\Psi_{n,\mathbf{p}} = E_n(\mathbf{p})^2\Psi_{n,\mathbf{p}}. \tag{16.64}$$

At zero momentum,

$$K(0)\Psi_{n,0} = M_n^2\Psi_{n,0},$$

so the masses calculated from the rest-frame collective spectral problem are precisely

$$M_n^2 = E_n(\mathbf{0})^2. \tag{16.65}$$

16.24 Compatibility with the Yang–Mills realization

The momentum generators on the collective and Yang–Mills sides satisfy

$$C_\infty P_i^{\mathrm{YM}} = P_i^{\mathrm{coll}} C_\infty. \tag{16.66}$$

Therefore their joint spectral measures are intertwined:

$$C_\infty E_{\mathrm{YM}}(\Omega) = E_{\mathrm{coll}}(\Omega) C_\infty \tag{16.67}$$

for every Borel set

$$\Omega \subset \mathbb{R}^{3.}$$

Indeed, (16.67) follows from the functional calculus applied to (16.66). Thus C_∞ acts fiberwise almost everywhere:

$$C_\infty = \int_{\mathbb{R}^3}^{\oplus} C_\infty(\mathbf{p})\, d^3p\,. \tag{16.68}$$

Consequently,

$$C_\infty(\mathbf{p}) : \mathcal{H}_{\mathrm{YM}}^{\mathrm{phys}}(\mathbf{p}) \longrightarrow \mathcal{H}_{\mathrm{coll}}^{\mathrm{phys}}(\mathbf{p})$$

is the momentum-resolved master-field realization. So the momentum $\mathbf{p}$ appearing in the collective theory is not an auxiliary closed-string momentum. It is exactly the ordinary spatial momentum of the underlying planar Yang–Mills state.

Proposition 16.2 — Momentum decomposition of the planar master-field Hilbert space

Let

$$C_\infty : \mathcal{H}_{\mathrm{YM}}^{\mathrm{phys}} \to \mathcal{H}_{\mathrm{coll}}^{\mathrm{phys}}$$

be the $ISO(3)$-equivariant decompactified collective realization. Then the translation subgroup defines three mutually strongly commuting self-adjoint momentum generators P_i, and

$$\boxed{\mathcal{H}_{\mathrm{phys}}^{(1)} \simeq \int_{\mathbb{R}^3}^{\oplus} \mathcal{H}_{\mathrm{phys}}(\mathbf{p})\, d^3p.}$$

The collective map decomposes fiberwise,

$$\boxed{C_\infty = \int_{\mathbb{R}^3}^{\oplus} C_\infty\,(\mathbf{p})\,d^3p,}$$

and the physical master-field operator is decomposable,

$$\boxed{K = \int_{\mathbb{R}^3}^{\oplus} K\,(\mathbf{p})\,d^3p.}$$

Its fiber spectra define the planar dispersion relations,

$$K(\mathbf{p})\Psi_{n,\mathbf{p}} = E_n(\mathbf{p})^2\Psi_{n,\mathbf{p}},$$

with

$$\boxed{E_n(\mathbf{0}) = M_n.}$$

16.25 Relativistic dispersion of the planar master-field modes

From subsection 16.22 we have

$$\mathcal{H}^{(1)}_{\text{phys}} \simeq \int_{\mathbb{R}^3}^{\oplus} \mathcal{H}_{\text{phys}}\,(\mathbf{p})\,d^3p,$$

and

$$K = \int_{\mathbb{R}^3}^{\oplus} K\,(\mathbf{p})\,d^3p.$$

We now prove that on every isolated one-particle branch,

$$K_n(\mathbf{p}) = |\mathbf{p}|^2 + M_n^2 \tag{16.69}$$

16.26 Decompactified Yang–Mills Hessian

The starting point is not the closed geometry but the Yang–Mills master-field realization. At finite radius the collective quadratic action is the transported physical Hessian of Yang–Mills on

$$\mathbb{R}_t \times S^3_R.$$

After the decompactification limit,

$$S^3_R \longrightarrow \mathbb{R}^3,$$

the underlying Yang–Mills theory is defined on flat spacetime

$$\mathbb{R}^{1,3}$$

or, after Wick rotation, $\mathbb{R}^4_E$. The Euclidean Yang–Mills action is invariant under the Euclidean group $E(4)$. Consequently its gauge-reduced Hessian at the translation-invariant vacuum commutes with the Euclidean action. Let

$$\mathcal{A}_{\mathrm{YM},E}$$

denote this physical Hessian. Then

$$\left[\mathcal{A}_{\mathrm{YM},E}, U_{\mathrm{YM}}(g)\right] = 0, \qquad g \in E(4). \tag{16.70}$$

The collective map transports the physical Hessian,

$$\mathcal{A}_{\mathrm{coll},E} = C_\infty \mathcal{A}_{\mathrm{YM},E} C_\infty^{-1}. \tag{16.71}$$

Therefore the collective Hessian carries the same Euclidean symmetry. Indeed, defining

$$U_{\mathrm{coll}}(g) = C_\infty U_{\mathrm{YM}}(g) C_\infty^{-1}, \tag{16.72}$$

one finds

$$\begin{aligned} U_{\mathrm{coll}}(g)\mathcal{A}_{\mathrm{coll},E}U_{\mathrm{coll}}(g)^{-1} &= C_\infty U_{\mathrm{YM}}(g)\mathcal{A}_{\mathrm{YM},E}U_{\mathrm{YM}}(g)^{-1}C_\infty^{-1} \\ &= C_\infty \mathcal{A}_{\mathrm{YM},E} C_\infty^{-1} \\ &= \mathcal{A}_{\mathrm{coll},E}. \end{aligned}$$

Hence

$$\left[\mathcal{A}_{\mathrm{coll},E}, U_{\mathrm{coll}}(g)\right] = 0, \qquad g \in E(4). \tag{16.73}$$

This is stronger than the $ISO(3)$ result derived above: the latter controls spatial momentum, while (16.73) also relates the temporal and spatial momentum dependence.

16.27 Momentum-space form of the quadratic kernel

Translation invariance implies that the Euclidean Hessian is diagonal in four-momentum:

$$\mathcal{A}_{\mathrm{coll},E} = \int_{\mathbb{R}^4}^{\oplus} \mathcal{A}_E(p)\, d^4p.$$

Let

$$p = (p_0, \mathbf{p}).$$

Because the vacuum is $E(4)$-invariant, a scalar coefficient of the inverse propagator can depend on p only through the Euclidean invariant

$$p_E^2 = p_0^2 + |\mathbf{p}|^{2.}$$

For a one-particle branch n, after projection onto its physical spin/polarization space,

$$\Gamma_n(p) = F_n(p_E^2). \tag{16.74}$$

For nonzero spin there may additionally be the appropriate covariant polarization projector, but the pole position is still governed only by p_E^2.

16.28 Gaussian master-field normalization

The strict planar master-field theory is Gaussian, and the physical quadratic action has already been normalized so that the Euclidean-time kinetic term is

$$\|\partial_\tau \Phi\|^2.$$

Thus on the n-th canonically normalized one-particle branch the quadratic kernel has unit coefficient of p_0^2. Near its one-particle pole it therefore takes the form

$$\Gamma_n(p) = p_0^2 + \mathcal{E}_n(\mathbf{p})^2. \tag{16.75}$$

But by $E(4)$ invariance the same kernel must depend on momentum through

$$p_0^2 + |\mathbf{p}|^{2.}$$

Therefore there is a function f_n such that

$$\Gamma_n(p) = f_n(p_0^2 + |\mathbf{p}|^2).$$

For the Gaussian Hessian constructed in the master-field theory the one-particle pole is simple. Let its zero after analytic continuation occur at

$$p_E^2 = -M_n^2.$$

Canonical normalization fixes the quadratic factor to

$$\Gamma_n(p) = p_E^2 + M_n^2 \tag{16.76}$$

on the corresponding elementary Gaussian branch. Equivalently,

$$\Gamma_n(p_0, \mathbf{p}) = p_0^2 + |\mathbf{p}|^2 + M_n^2. \tag{16.77}$$

At $\mathbf{p} = 0$,

$$\Gamma_n(p_0, \mathbf{0}) = p_0^2 + M_n^2 \tag{16.78}$$

which is precisely the rest-frame spectral problem solved previously.

16.29 Fiber Hamiltonian

On the other hand, from the Hamiltonian decomposition we have

$$\Gamma_n(p_0, \mathbf{p}) = p_0^2 + K_n(\mathbf{p}). \tag{16.79}$$

Comparing (16.77) and (16.79),

$$K_n(\mathbf{p}) = |\mathbf{p}|^2 + M_n^2. \tag{16.80}$$

Thus

$$K(\mathbf{p}) = |\mathbf{p}|^2 I + K(\mathbf{0}) \tag{16.81}$$

on each spectral branch, or more generally in the mass spectral representation,

$$K(\mathbf{p}) = |\mathbf{p}|^2 I + M^2,$$

where M^2 is the rest-frame master-field mass operator. The positive one-particle Hamiltonian is therefore

$$H^{(1)}(\mathbf{p}) = K(\mathbf{p})^{1/2} = \sqrt{|\mathbf{p}|^2 + M^2}. \tag{16.82}$$

For an eigenstate

$$M^2\Psi_n = M_n^2\Psi_n,$$

one obtains

$$E_n(\mathbf{p}) = \sqrt{|\mathbf{p}|^2 + M_n^2}. \tag{16.83}$$

Hence

$$E_n(\mathbf{p})^2 - |\mathbf{p}|^2 = M_n^2. \tag{16.84}$$

So the masses calculated from the collective spectral problem are Lorentz-invariant rest masses.

16.30 Covariance

The Euclidean covariance on branch n is consequently

$$\mathfrak{C}_n(p_0, \mathbf{p}) = \frac{P_n(p)}{p_0^2 + |\mathbf{p}|^2 + M_n^2}, \tag{16.85}$$

where P_n is the physical polarization projector; for a scalar branch,

$$P_n = 1.$$

The denominator depends only on

$$p_E^2 = p_0^2 + |\mathbf{p}|^2.$$

Therefore the pole structure is explicitly $E(4)$-invariant. For a discrete rest spectrum,

$$\mathfrak{C}_E(p) = \sum_n \frac{Z_n P_n(p)}{p_E^2 + M_n^2}, \tag{18.86}$$

with positive residues Z_n in the reflection-positive physical sectors.

16.31 Why this is not an additional assumption

Importantly, relativistic covariance of the closed collective theory is not assumed independently. The logical chain is

Yang--Mills on $S^3_R \times \mathbb{R}$
$\downarrow$
physical FMR/master-field Hessian
$\downarrow C_R$
collective Hessian
$\downarrow R \to \infty$
Yang--Mills on $\mathbb{R}^4$
$\downarrow$
$E(4)$-invariant physical Hessian
$\downarrow C_\infty$
$E(4)$-invariant collective Gaussian process.

The relativistic dispersion relation is therefore inherited from the original Yang–Mills theory through the already-established master-field equivalence. The internal Stenzel-RG geometry determines the spectrum

$$M_n^{2.}$$

The ordinary spacetime symmetry determines how each such rest mass propagates:

$$M_n^2 \quad \longrightarrow \quad p_E^2 + M_n^2.$$

Those are two conceptually different pieces of information.

Proposition 16.3 — Relativistic dispersion of the planar master field

Let the decompactified collective map C_∞ transport the physical Yang–Mills quadratic Hessian to the collective master-field Hessian. Then the $E(4)$ invariance of Yang–Mills on $\mathbb{R}^4$ is inherited by the collective quadratic theory.
If $M^2 = K(\mathbf{0})$ denotes the rest-frame physical mass operator, then

$$K(\mathbf{p}) = |\mathbf{p}|^2 I + M^2$$

Consequently,

$$\boxed{H^{(1)} = \sqrt{\mathbf{P}^2 + M^2},}$$

and for each mass eigenstate,

$$\boxed{E_n(\mathbf{p})^2 = |\mathbf{p}|^2 + M_n^{2.}}$$

The Euclidean covariance is therefore

$$\boxed{\mathfrak{C}_E(p) = (p_E^2 + M^2)^{-1},}$$

with the appropriate physical polarization projectors in non-scalar channels.

16.32 Regularity and Temperedness of the Planar Schwinger Functions

The preceding sections established that the strict planar master-field fluctuation measure is Gaussian, reflection positive, and $E(4)$-covariant after decompactification. Its Euclidean two-point covariance is

$$\mathfrak{C}_E = (-\Delta_{\mathbb{R}^4} + M^2)^{-1}, \tag{16.87}$$

where M^2 is the positive self-adjoint rest-frame mass operator on the physical one-particle space and satisfies

$$M^2 \geq \Delta^2 I, \qquad \Delta > 0. \tag{16.88}$$

We now prove that the corresponding Schwinger functions define tempered distributions and satisfy the regularity properties required for Osterwalder–Schrader reconstruction [41,42].

16.33 The operator-valued Euclidean covariance

In four-momentum space,

$$p = (p_0, \mathbf{p}) \in \mathbb{R}^4,$$

the covariance is

$$\widehat{\mathfrak{C}}_E(p) = (p_E^2 + M^2)^{-1}, \tag{16.89}$$

with

$$p_E^2 = p_0^2 + |\mathbf{p}|^{2.}$$

By the spectral theorem,

$$M^2 = \int_{\sigma(M^2)} \lambda \, dE_{M^2}(\lambda), \tag{16.90}$$

so

$$\widehat{\mathfrak{C}}_E(p) = \int_{\sigma(M^2)} \frac{1}{p_E^2 + \lambda} \, dE_{M^2}(\lambda). \tag{16.91}$$

Since

$$\lambda \geq \Delta^2,$$

one obtains the uniform operator bound

$$\left\| \widehat{\mathfrak{C}}_E(p) \right\| \leq \frac{1}{p_E^2 + \Delta^2}. \tag{16.92}$$

Thus the Euclidean propagator is both infrared regular and polynomially bounded in momentum space.

16.34 Continuity on Schwartz test functions

Let

$$\mathcal{S}\left(\mathbb{R}^4;\mathcal{H}_{\text{phys}}^{(1)}\right)$$

denote the Schwartz space of rapidly decreasing $\mathcal{H}_{\text{phys}}^{(1)}$-valued test functions.

For

$$f,g\in\mathcal{S}\left(\mathbb{R}^4;\mathcal{H}_{\text{phys}}^{(1)}\right),$$

define

$$\mathfrak{C}_E(f,g)=\int_{\mathbb{R}^4}\left\langle\hat{f}(p),\hat{C}_E(p)\hat{g}(p)\right\rangle\,d^4p. \tag{16.93}$$

Using (16.92),

$$|\mathfrak{C}_E(f,g)|\le\int_{\mathbb{R}^4}\frac{\left\|\hat{f}(p)\right\|\,\|\hat{g}(p)\|}{p_E^2+\Delta^2}\,d^4p. \tag{16.94}$$

Choose an integer $N>2$. Then

$$\left\|\hat{f}(p)\right\|\le p_N(f)(1+|p|)^{-N},$$

and similarly

$$\|\hat{g}(p)\|\le p_N(g)(1+|p|)^{-N},$$

where p_N is a standard Schwartz seminorm. Hence

$$|\mathfrak{C}_E(f,g)|\le p_N(f)p_N(g)\int_{\mathbb{R}^4}\frac{(1+|p|)^{-2N}}{p_E^2+\Delta^2}\,d^4p. \tag{16.95}$$

The integral is finite for sufficiently large N. Therefore there exists a constant C_N such that

$$|\mathfrak{C}_E(f,g)|\le C_N p_N(f)p_N(g). \tag{16.96}$$

Thus C_E is a continuous bilinear form on Schwartz space. Equivalently,

$$\mathfrak{C}_E\in\mathcal{S}'\left(\mathbb{R}^4;\operatorname{End}\mathcal{H}_{\text{phys}}^{(1)}\right). \tag{16.97}$$

So the planar two-point Schwinger function is a tempered operator-valued distribution.

16.35 Position-space covariance

The inverse Fourier transform defines

$$\mathfrak{C}_E(x-y)$$

as a translation-invariant tempered distribution. Using the spectral resolution of M^2,

$$\mathfrak{C}_E(x)=\int_{\sigma(M^2)}\Delta_E\left(x;\sqrt{\lambda}\right)dE_{M^2}(\lambda), \tag{16.98}$$

where

$$\Delta_E(x;m) = \int_{\mathbb{R}^4} \frac{d^4p}{(2\pi)^4} \frac{e^{ip\cdot x}}{p_E^2 + m^2}$$

is the ordinary massive Euclidean scalar Green kernel. For discrete mass spectrum this reduces to

$$\mathfrak{C}_E(x) = \sum_n \Delta_E\,(x;M_n)P_n, \tag{16.99}$$

where P_n denotes the corresponding spectral projector, with the obvious replacement by a spectral integral whenever continuous multiplicity is present. Thus the nontrivial Yang–Mills information is encoded in the mass spectral measure and physical polarization structure, while the spacetime propagation is the standard massive Euclidean propagation.

16.36 Gaussian Schwinger functions

Because the strict planar master-field measure is centered and Gaussian, all Schwinger functions are determined by $\mathfrak{C}_E$. The odd Schwinger functions vanish,

$$S_{2n+1} = 0. \tag{16.100}$$

The even Schwinger functions satisfy Wick's rule,

$$S_{2n}(x_1, \dots, x_{2n}) = \sum_{\mathcal{P}} \prod_{(i,j)\in\mathcal{P}} \mathfrak{C}_E\,(x_i - x_j), \tag{16.101}$$

where the sum runs over all pairings $\mathcal{P}$ of the $2n$ arguments. More generally, for test functions $f_1, \dots, f_{2n}$,

$$S_{2n}(f_1, \dots, f_{2n}) = \sum_{\mathcal{P}} \prod_{(i,j)\in\mathcal{P}} \mathfrak{C}_E\,(f_i, f_j).$$

Because C_E is continuous on Schwartz space, every finite product appearing in (16.101) is continuous in the corresponding Schwartz topology. Hence

$$S_n \in \mathcal{S}'((\mathbb{R}^4)^n) \tag{16.102}$$

with the appropriate physical polarization indices understood. Thus every planar Schwinger function is tempered.

16.37 Symmetry

The covariance is symmetric,

$$C_E(x-y) = C_E(y-x)^*.$$

For real bosonic master-field variables,

$$C_E(x-y) = C_E(y-x).$$

Consequently Wick's formula implies complete permutation symmetry,

$$S_n\big(x_{\pi(1)},\dots,x_{\pi(n)}\big)=S_n(x_1,\dots,x_n) \tag{16.103}$$

for every permutation

$$\pi\in S_n.$$

Thus the Schwinger functions satisfy the bosonic symmetry axiom.

16.38 Euclidean covariance

Section 16.26 established that the decompactified master-field Hessian is $E(4)$-invariant. Hence for every Euclidean transformation

$$g\in E(4),$$

the two-point function obeys

$$C_E(gx-gy)=U(g)C_E(x-y)U(g)^{-1},$$

where $U(g)$ is the appropriate representation on the physical polarization space. For scalar sectors this reduces simply to

$$C_E(gx-gy)=C_E(x-y).$$

Since all higher Schwinger functions are Wick polynomials in C_E, they inherit the same transformation law. Therefore

$$S_n(gx_1,\dots,gx_n)=U_n(g)S_n(x_1,\dots,x_n), \tag{16.104}$$

with $U_n(g)$ the induced tensor-product representation. Thus the full family of planar Schwinger functions is $E(4)$-covariant.

16.39 Positivity and continuity of the Gaussian characteristic functional

The centered Gaussian measure is characterized by

$$\chi(f)=\exp\left[-\frac{1}{2}\mathfrak{C}_E(f,f)\right].$$

Since $\mathfrak{C}_E$ is positive,

$$\mathfrak{C}_E(f,f)\geq 0,$$

and continuous in the Schwartz topology, χ is continuous and positive definite. Hence the Gaussian probability measure constructed previously is supported on the tempered-distribution completion

$$\mathcal{S}'\left(\mathbb{R}^4;\mathcal{H}^{(1)}_{\mathrm{phys}}\right).$$

The present argument therefore strengthens the earlier nuclear-space construction by showing explicitly that the decompactified planar master-field measure defines ordinary tempered Euclidean Schwinger distributions on four-dimensional spacetime.

16.40 Connected Schwinger functions

For a centered Gaussian process, the connected generating functional is quadratic:

$$W[J] = \frac{1}{2}\mathfrak{C}_E(J,J).$$

Therefore

$$S_2^c = \mathfrak{C}_E, \tag{16.105}$$

whereas

$$S_n^c = 0, \qquad n \geq 3. \tag{16.106}$$

This is the Schwinger-function expression of strict planar Gaussianity. It does not imply that the underlying microscopic finite-N Yang–Mills theory is free. The nontrivial dynamics is encoded in the planar mass operator M^2, its spectral measure, and the $1/N$-suppressed corrections to the strict planar fluctuation theory.

Proposition 16.4 — Tempered planar Schwinger functions

Let the decompactified strict-planar Yang–Mills master-field measure be the centered Gaussian measure with covariance

$$\mathfrak{C}_E = (-\Delta_{\mathbb{R}^4} + M^2)^{-1},$$

where M^2 is positive self-adjoint and satisfies

$$M^2 \geq \Delta^2 I > 0.$$

Then:

$$\mathfrak{C}_E \in \mathcal{S}'\left(\mathbb{R}^4; \operatorname{End}\mathcal{H}_{\text{phys}}^{(1)}\right); \tag{16.107}$$

the Gaussian Schwinger functions defined by Wick's rule satisfy

$$S_n \in \mathcal{S}'((\mathbb{R}^4)^n); \tag{16.108}$$

they are bosonically symmetric and $E(4)$-covariant; and their connected components satisfy

$$S_2^c = C_E, \qquad S_n^c = 0 \quad (n \geq 3).$$

Hence the strict planar master-field process defines a regular family of tempered Euclidean Schwinger distributions.

16.41 Clustering and Uniqueness of the Vacuum

The previous sections established that the strict planar master-field measure defines a centered, reflection-positive, $E(4)$-covariant Gaussian process with covariance

$$\mathfrak{C}_E = (-\Delta_{\mathbb{R}^4} + M^2)^{-1},$$

where the rest-frame mass operator satisfies

$$M^2 \geq \Delta^2 I, \qquad \Delta > 0. \tag{16.109}$$

We now show that the positive mass gap implies clustering of the Euclidean Schwinger functions. For the reconstructed quantum theory, this implies uniqueness of the translation-invariant vacuum.

16.42 Spectral representation and the mass threshold

Let

$$M = (M^2)^{1/2}.$$

Then

$$M \geq \Delta I.$$

By the spectral theorem,

$$M = \int_{\sigma(M)} m \, dE_M(m), \qquad \sigma(M) \subset [\Delta, \infty). \tag{16.110}$$

The covariance can be written as

$$\mathfrak{C}_E(x) = \int_{\Delta}^{\infty} \Delta_E(x; m) \, dE_M(m), \tag{16.111}$$

where

$$\Delta_E(x; m) = \int_{\mathbb{R}^4} \frac{d^4p}{(2\pi)^4} \frac{e^{ip\cdot x}}{p_E^2 + m^2}$$

is the massive Euclidean Green kernel. Thus every contribution to the covariance carries mass at least Δ.

16.43 Exponential decay in Euclidean time

For fixed spatial momentum,

$$\widehat{\mathfrak{C}}_E(\tau, \mathbf{p}) = \frac{e^{-|\tau|\sqrt{\mathbf{p}^2 + M^2}}}{2\sqrt{\mathbf{p}^2 + M^2}}. \tag{16.112}$$

Since

$$\sqrt{\mathbf{p}^2 + M^2} \geq M \geq \Delta,$$

we obtain

$$\left\| e^{-|\tau|\sqrt{\mathbf{p}^2+M^2}} \right\| \leq e^{-\Delta|\tau|}. \tag{16.113}$$

Moreover,

$$\left\| \left(2\sqrt{\mathbf{p}^2+M^2}\right)^{-1} \right\| \leq \frac{1}{2\Delta}. \tag{16.114}$$

Therefore

$$\left\| \widehat{\mathfrak{C}}_E(\tau,\mathbf{p}) \right\| \leq \frac{1}{2\Delta} e^{-\Delta|\tau|}. \tag{16.115}$$

In particular, for suitably smeared test functions f, g,

$$|\mathfrak{C}_E(f,\tau_a g)| \leq C_{f,g} e^{-\Delta|a|} \tag{16.116}$$

for sufficiently large Euclidean-time separation a. Here

$$(\tau_a g)(x) = g(x - ae_0).$$

Thus the two-point function clusters exponentially in Euclidean time with rate bounded below by the mass gap.

16.44 Full Euclidean separation

Because the covariance is $E(4)$-invariant, the same conclusion holds for arbitrary Euclidean separation. For a scalar mass $m > 0$, the four-dimensional Euclidean Green kernel is

$$\Delta_E(x;m) = \frac{m}{4\pi^2|x|} K_1(m|x|), \tag{16.117}$$

where K_1 is the modified Bessel function. Its large-distance asymptotics is

$$K_1(r) \sim \sqrt{\frac{\pi}{2r}} e^{-r}.$$

Hence

$$\Delta_E(x;m) = O\left(|x|^{-3/2} e^{-m|x|}\right).$$

Since every mass in the spectral resolution satisfies

$$m \geq \Delta,$$

the operator-valued covariance satisfies, after smearing in the internal variables,

$$\mathfrak{C}_E(x) = O\left(|x|^{-3/2} e^{-\Delta|x|}\right) \tag{16.118}$$

in the large-separation sense. More invariantly, if f and g have compact separated supports and $g_a(x) = g(x-a)$, then

$$\mathfrak{C}_E(f, g_a) \longrightarrow 0 \tag{16.119}$$

as

$$|a| \to \infty,$$

with exponential rate controlled by Δ.

16.45 Clustering of the Gaussian field

Let

$$\Phi(f)$$

denote the Gaussian master field smeared with f. Since the measure is centered,

$$\langle \Phi(f) \rangle = 0.$$

Therefore

$$\langle \Phi(f)\Phi(g_a) \rangle_c = \mathfrak{C}_E(f, g_a).$$

By (16.119),

$$\lim_{|a|\to\infty} \langle \Phi(f)\Phi(g_a) \rangle_c = 0. \tag{16.120}$$

Indeed,

$$|\langle \Phi(f)\Phi(g_a) \rangle_c| \leq C_{f,g} e^{-\Delta|a|} \tag{16.121}$$

up to the polynomial prefactor associated with the four-dimensional massive Green kernel. Thus the basic planar master field clusters.

16.46 Clustering of polynomial observables

Consider two polynomial cylindrical observables

$$F(\Phi)$$

and

$$G(\Phi).$$

Translate the second by a,

$$(\tau_a G)(\Phi) = G(\tau_{-a}\Phi). \tag{16.122}$$

Because the measure is Gaussian, their expectation is determined by Wick contractions. The product

$$\langle F\, \tau_a G \rangle$$

contains two classes of contractions:

1. contractions entirely within F or entirely within G;
2. contractions joining a field appearing in F to a field appearing in $\tau_a G$.

The first class gives exactly

$$\langle F\rangle\langle G\rangle.$$

Every remaining term contains at least one cross-contraction

$$C_E\big(f_i, \tau_a g_j\big).$$

Each such cross-contraction tends to zero exponentially by (16.119). Therefore

$$\lim_{|a|\to\infty}[\langle F\ \tau_a G\rangle - \langle F\rangle\langle G\rangle] = 0. \tag{16.123}$$

For finite-degree polynomial observables one obtains an estimate of the form

$$|\langle F\ \tau_a G\rangle - \langle F\rangle\langle G\rangle| \le C_{F,G}P(|a|)e^{-\Delta|a|}, \tag{16.124}$$

where P is at most a polynomial determined by the spacetime dimension, smearing functions, and tensor structure. Thus the strict planar master-field measure satisfies the Euclidean cluster property.

16.47 Connected Schwinger functions

For the elementary centered Gaussian field,

$$S_n^c = 0, \qquad n \ge 3.$$

Hence the only elementary connected Schwinger function is

$$S_2^c = C_E.$$

The cluster property therefore reduces immediately to the decay of the covariance. For composite polynomial observables, connected correlators need not vanish identically, because Wick contractions can connect the composite insertions. Nevertheless every connected diagram contains at least one propagator joining separated clusters. Since every such propagator decays with mass at least Δ, all connected composite correlators obey the same cluster limit. Thus

$$\boxed{\text{mass gap} \quad \Longrightarrow \quad \text{exponential clustering.}}$$

16.48 Cluster property and the vacuum sector

Let

$$\mathcal{H}_{\mathrm{OS}}$$

be the Hilbert space obtained from the reflection-positive Euclidean measure, and let

$$\Omega$$

denote the vector represented by the constant functional 1. Euclidean translations induce, after reconstruction, the physical translation representation

$$U(a).$$

The vacuum satisfies

$$U(a)\Omega = \Omega. \tag{16.125}$$

Suppose there were a second normalized translation-invariant vector

$$\Omega' \perp \Omega.$$

Then the translation-invariant subspace would have dimension greater than one. The Euclidean cluster property excludes this. Indeed, clustering implies for observables A, B,

$$\lim_{|a|\to\infty} \langle \Omega, A\, U(a) B U(a)^{-1} \Omega \rangle = \langle \Omega, A\Omega \rangle \langle \Omega, B\Omega \rangle. \tag{16.126}$$

Equivalently, on the cyclic subspace generated by local observables,

$$* w - lim_{|a|\to\infty} U(a) = |\Omega\rangle\langle\Omega|. \tag{16.127}$$

The projection onto the translation-invariant subspace is therefore rank one. Hence

$$\dim \ker P_\mu = 1 \tag{16.128}$$

in the joint translation-invariant sector, or equivalently

$$\mathcal{H}_{\text{inv}} = \mathbb{C}\Omega. \tag{16.129}$$

Thus the vacuum is unique up to phase.

16.49 Direct spectral proof in the Gaussian Fock realization

In the present theory this conclusion can also be seen directly from the Fock-space representation already constructed earlier. The physical Hilbert space is

$$\mathcal{F}_{\text{phys}} = \Gamma_s\left(\mathcal{H}^{(1)}_{\text{phys}}\right).$$

The Hamiltonian is

$$H = d\Gamma\left(H^{(1)}\right),$$

where

$$H^{(1)} = \sqrt{\mathbf{P}^2 + M^2}.$$

Because

$$M \geq \Delta > 0,$$

one has

$$H^{(1)} \geq \Delta I.$$

On the n-particle sector,

$$H^{(n)} = \sum_{j=1}^{n} H_j^{(1)}, \tag{16.130}$$

and therefore

$$H^{(n)} \geq n\Delta I. \tag{16.131}$$

Thus no nonzero vector in any sector with $n \geq 1$ can have zero energy. The only zero-energy state is the Fock vacuum,

$$|0\rangle.$$

Hence

$$\ker H = \mathbb{C}|0\rangle. \tag{16.132}$$

This gives a second proof of vacuum uniqueness and agrees with the result inferred from clustering.

16.50 Spectral gap above the vacuum

The same argument shows that the reconstructed Hamiltonian satisfies

$$\operatorname{spec} H \subset \{0\} \cup [\Delta, \infty). \tag{16.133}$$

More precisely, the one-particle sector begins at

$$M_{\text{gap}} = \inf \sigma(M) \geq \Delta,$$

while the n-particle continuum begins no lower than

$$nM_{\text{gap}}.$$

Therefore

$$\boxed{\operatorname{dist}(0, \sigma(H) \setminus \{0\}) = M_{\text{gap}} > 0.} \tag{16.134}$$

Thus the Euclidean clustering scale and the spectral mass gap of the reconstructed Hamiltonian are the same physical quantity.

Proposition 16.5 — Clustering and uniqueness of the planar vacuum

Let the strict planar Yang–Mills master-field process be the centered Gaussian process with $E(4)$-covariant covariance

$$\mathfrak{C}_E = (-\Delta_{\mathbb{R}^4} + M^2)^{-1},$$

where

$$M^2 \geq \Delta^2 I > 0.$$

Then:

1. the two-point covariance decays exponentially at large Euclidean separation, with exponential rate no smaller than Δ;
2. for polynomial cylindrical observables F, G,

$$\boxed{\lim_{|a|\to\infty}[\langle F\,\tau_a G\rangle - \langle F\rangle\langle G\rangle] = 0;}$$

3. hence the planar Schwinger functions satisfy the Euclidean cluster property;
4. under Osterwalder–Schrader reconstruction, the translation-invariant subspace is one-dimensional,

$$\boxed{\mathcal{H}_{\mathrm{inv}} = \mathbb{C}\Omega;}$$

5. equivalently, in the Gaussian Fock realization,

$$\boxed{\ker H = \mathbb{C}|0\rangle;}$$

6. the reconstructed Hamiltonian has spectral gap

$$\boxed{\operatorname{spec} H \subset \{0\} \cup [M_{\mathrm{gap}}, \infty), \qquad M_{\mathrm{gap}} > 0.}$$

Therefore the strict planar Yang–Mills master-field theory possesses a unique vacuum and exponentially clustering Euclidean correlations.

16.51 Osterwalder–Schrader Reconstruction

The preceding sections established that the decompactified strict-planar Yang–Mills master-field measure is a centered Gaussian probability measure

$$\gamma_{\mathfrak{C}}$$

whose Schwinger functions are tempered, symmetric, $E(4)$-covariant, reflection positive and clustering. Its covariance is

$$\mathfrak{C}_E = (-\Delta_{\mathbb{R}^4} + M^2)^{-1},$$

where the positive self-adjoint rest-mass operator satisfies

$$M^2 \geq \Delta^2 I, \qquad \Delta > 0.$$

We now apply Osterwalder–Schrader reconstruction and identify the resulting quantum theory with the Hilbert-space realization already obtained from canonical quantization of the planar master field.

16.52 Positive-time functionals and the OS inner product

Let

$$\mathcal{E}_+$$

denote the space of test functions supported in positive Euclidean time,

$$\operatorname{supp} f \subset \{\tau > 0\}.$$

Let

$$\mathcal{D}_+$$

be the algebra of polynomial cylindrical functionals generated by the smeared fields

$$\Phi(f), \qquad f \in \mathcal{E}_+.$$

Euclidean time reflection is

$$\theta(\tau, \mathbf{x}) = (-\tau, \mathbf{x}),$$

and induces the antilinear operation

$$\Theta F(\Phi) = \overline{F(\theta\Phi)}.$$

Define the Osterwalder–Schrader sesquilinear form

$$(F, G)_{\mathrm{OS}} = \int (\Theta F)(\Phi)\, G(\Phi)\, d\gamma_{\mathbb{C}}(\Phi). \tag{16.135}$$

Reflection positivity proved in Section 16.4 gives

$$(F, F)_{\mathrm{OS}} \geq 0.$$

Hence

$$\mathcal{N} = \{F \in \mathcal{D}_+ : (F, F)_{\mathrm{OS}} = 0\} \tag{16.136}$$

is a null subspace. The reconstructed Hilbert space is

$$\mathcal{H}_{\mathrm{OS}} = \overline{\mathcal{D}_+ / \mathcal{N}}^{\|\cdot\|_{\mathrm{OS}}}. \tag{16.137}$$

The equivalence class of the constant functional 1 defines the vacuum vector

$$\Omega = [1]. \tag{16.138}$$

16.53 The one-particle OS space

The one-particle sector may be read off directly from the reflected covariance.

Let

$$\omega = \sqrt{\mathbf{P}^2 + M^2}.$$

For $f \in \mathcal{E}_+$, define

$$j(f) = \frac{1}{\sqrt{2}} \int_0^\infty e^{-\tau\omega}\, \omega^{-1/2} f(\tau)\, d\tau. \tag{16.139}$$

Section 16.4 showed that

$$\langle \Theta f, \mathfrak{C}_E f \rangle = \|j(f)\|^2.$$

More generally,

$$\langle \Theta f, \mathfrak{C}_E g \rangle = \langle j(f), j(g) \rangle. \tag{16.140}$$

Thus quotienting the positive-time linear fields by their OS-null vectors and completing gives precisely the one-particle Hilbert space determined by ω.

Consequently,

$$\mathcal{H}_{\mathrm{OS}}^{(1)} \simeq \mathcal{H}_{\mathrm{phys}}^{(1)}. \tag{16.141}$$

The OS reconstruction therefore does not introduce a new one-particle space. It recovers from the Euclidean master-field measure the physical one-particle Hilbert space constructed previously from the gauge-reduced Yang–Mills-master-field complex.

16.54 Reconstruction of Euclidean time evolution

For $s \geq 0$, positive Euclidean-time translation acts on test functions by

$$(T_E(s)f)(\tau, \mathbf{x}) = f(\tau - s, \mathbf{x}),$$

whenever the translated support remains in positive time. Reflection positivity implies that this operation descends to a contraction semigroup on $\mathcal{H}_{\mathrm{OS}}$.

Thus

$$T_E(s) = e^{-sH}, \qquad s \geq 0, \tag{16.142}$$

for a unique positive self-adjoint operator H,

$$H \geq 0.$$

On the one-particle space, (16.139) gives

$$T_E^{(1)}(s) = e^{-s\omega}.$$

Therefore

$$H^{(1)} = \omega = \sqrt{\mathbf{P}^2 + M^2}. \tag{16.143}$$

For an eigenstate of the rest-mass operator,

$$M^2 \Psi_n = M_n^2 \Psi_n,$$

the reconstructed one-particle energy is

$$E_n(\mathbf{p}) = \sqrt{\mathbf{p}^2 + M_n^2}. \tag{16.144}$$

Hence the OS reconstruction reproduces exactly the relativistic dispersion relation obtained in Section 16.25–16.29

16.55 Spatial translations and rotations

Spatial Euclidean transformations preserve the positive-time half-space and therefore act directly on the OS quotient.

For

$$a \in \mathbb{R}^3,$$

spatial translations define unitary operators

$$U(\mathbf{a}) = e^{-i\mathbf{a}\cdot\mathbf{P}},$$

where

$$P_1, P_2, P_3$$

are the self-adjoint momentum generators constructed in Section 16.20. Likewise, spatial rotations act unitarily. Thus the reconstructed one-particle representation has joint energy-momentum spectrum

$$p_0 = \sqrt{\mathbf{p}^2 + m^2}, \qquad m \in \sigma(M). \tag{16.145}$$

In particular,

$$p_0 \geq |\mathbf{p}|,$$

so the spectrum lies in the closed forward light cone after analytic continuation to Minkowski signature.

16.56 Gaussian reconstruction and Fock space

The Euclidean measure is Gaussian. Therefore the OS Hilbert space generated by its polynomial field algebra is the symmetric bosonic Fock space over its one-particle OS space.

Using (16.141),

$$\mathcal{H}_{\mathrm{OS}} \simeq \Gamma_s\left(\mathcal{H}^{(1)}_{\mathrm{phys}}\right). \tag{16.146}$$

Explicitly,

$$\mathcal{H}_{\mathrm{OS}} \simeq \mathbb{C}\Omega \oplus \mathcal{H}^{(1)}_{\mathrm{phys}} \oplus \mathrm{Sym}^2\,\mathcal{H}^{(1)}_{\mathrm{phys}} \oplus \cdots.$$

This is precisely the Fock-space structure already obtained from canonical quantization of the quadratic planar master-field action. The OS construction therefore gives an independent Euclidean reconstruction of the previously derived Hilbert space. The one-particle sector describes single planar glueballs, while the n-particle sectors describe the corresponding n-glueball states.

16.57 Reconstructed Hamiltonian

For a Gaussian theory, the full Hamiltonian is the second quantization of the one-particle Hamiltonian:

$$H = d\Gamma(\omega) = d\Gamma\left(\sqrt{\mathbf{P}^2 + M^2}\right). \tag{16.147}$$

On the n-particle sector,

$$H^{(n)} = \sum_{j=1}^{n} \sqrt{\mathbf{P}_j^2 + M_j^2}.$$

Since

$$M \geq M_{\text{gap}} > 0,$$

one has

$$H^{(1)} \geq M_{\text{gap}}.$$

For every $n \geq 1$,

$$H^{(n)} \geq n M_{\text{gap}}.$$

Therefore

$$\operatorname{spec} H \subset \{0\} \cup [M_{\text{gap}}, \infty). \tag{16.148}$$

The vacuum is the unique zero-energy state,

$$\ker H = \mathbb{C}\Omega. \tag{16.149}$$

This agrees with the independent clustering argument of Section 16.45.

16.58 Reconstruction of the field

The Euclidean random field determines, through OS reconstruction, an operator-valued quantum field

$$\widehat{\Phi}(x)$$

on a dense invariant domain of

$$\mathcal{H}_{\text{OS}}.$$

Its vacuum correlation functions are obtained by analytic continuation of the Euclidean Schwinger functions. Because the latter are Gaussian, the reconstructed strict-planar field is a generalized free bosonic field whose nontrivial spectral data are determined by the Yang–Mills master-field mass operator M^2. For the two-point function one obtains the spectral representation

$$\langle \Omega, \widehat{\Phi}(x)\widehat{\Phi}(y)\Omega \rangle = \int_{\sigma(M)} dE_M(m)\, \Delta_+^{(m)}(x-y), \tag{16.150}$$

with the appropriate polarization structures in non-scalar sectors. The term "generalized free" here refers to the strict-planar fluctuation field. The nonperturbative Yang–Mills information remains encoded in the mass spectrum, residues, polarization content, and master-field geometry, while interactions among canonically normalized glueballs enter at subleading orders in $1/N$.

16.59 Compatibility with the planar FMR realization

The Euclidean measure used in the reconstruction is not an independently postulated closed-string measure. By the preceding master-field construction, it is the geometric representative of the centered and canonically rescaled planar FMR fluctuation measure. Thus the reconstruction chain is

$$\boxed{\begin{array}{c}\text{Yang--Mills FMR quotient}\\ \downarrow\\ \text{planar master field}\\ \downarrow C_\infty\\ \text{Gaussian collective measure } \gamma_C\\ \downarrow \text{ reflection positivity}\\ \mathcal{H}_{\mathrm{OS}}\\ \simeq\\ \Gamma_s\left(\mathcal{H}^{(1)}_{\mathrm{phys}}\right).\end{array}}$$

The OS Hilbert space is therefore the quantum Hilbert space of the same planar Yang–Mills master field whose configuration-space realization and spectral operator were constructed previously.

Theorem 16.2 — Osterwalder–Schrader reconstruction of the planar master field

Let the strict-planar FMR fluctuation measure be represented by the centered Gaussian master-field measure with covariance

$$\mathfrak{C}_E = (-\Delta_{\mathbb{R}^4} + M^2)^{-1},$$

where

$$M^2 \geq M^2_{\mathrm{gap}} I > 0.$$

Assume the master-field identification and decompactification results established in the preceding chapters. Then the Euclidean Schwinger functions satisfy reflection positivity, $E(4)$ covariance, permutation symmetry, temperedness and clustering. Their Osterwalder–Schrader reconstruction yields a quantum Hilbert space

$$\boxed{\mathcal{H}_{\mathrm{OS}} \simeq \Gamma_s\left(\mathcal{H}^{(1)}_{\mathrm{phys}}\right),}$$

vacuum

$$\boxed{\Omega = |0\rangle,}$$

one-particle Hamiltonian

$$\boxed{H^{(1)} = \sqrt{\mathbf{P}^2 + M^2},}$$

and full Hamiltonian

$$\boxed{H = d\Gamma\left(\sqrt{\mathbf{P}^2 + M^2}\right).}$$

Moreover,

$$\boxed{\ker H = \mathbb{C}\Omega}$$

and

$$\boxed{\operatorname{spec} H \subset \{0\} \cup [M_{\text{gap}}, \infty).}$$

Thus the Euclidean planar FMR master-field measure reconstructs the same positive-energy quantum Hilbert space obtained previously by canonical quantization.

16.60 Existence Theorem for Planar Yang–Mills on $\mathbb{R}^4$

We may now collect the preceding results. The earlier chapters established:

(i) the planar gauge-reduced FMR/master-field realization
(ii) the Gaussian strict-$N = \infty$ fluctuation measure
(iii) the decompactification $S_R^3 \times \mathbb{R} \longrightarrow \mathbb{R}^4$

(iv) $M^2 \geq M_{\text{gap}}^2 > 0.$

Chapter 16 established so far:

(v) reflection positivity
(vi) $E(4)$ covariance
(vii) tempered Schwinger distributions
(viii) clustering and vacuum uniqueness
(ix) OS reconstruction.

These results give the following theorem.

Theorem 16.3 — Existence and mass gap of the strict-planar Yang–Mills master-field theory

Under the planar FMR-master-field identification established in the preceding construction, the $N = \infty$ limit of pure $SU(N)$ Yang–Mills theory admits on $\mathbb{R}^4$ a Euclidean master-field measure

$$\gamma_{\mathfrak{C}}$$

whose covariance is

$$\boxed{\mathfrak{C}_E = (-\Delta_{\mathbb{R}^4} + M^2)^{-1}.}$$

Its Schwinger functions are tempered, bosonically symmetric, $E(4)$-covariant, reflection positive and clustering. Consequently they admit Osterwalder–Schrader reconstruction to a positive-energy quantum theory with Hilbert space

$$\boxed{\mathcal{H} = \Gamma_s\left(\mathcal{H}^{(1)}_{\text{phys}}\right),}$$

unique vacuum

$$\boxed{\ker H = \mathbb{C}\Omega,}$$

and Hamiltonian

$$\boxed{H = d\Gamma\left(\sqrt{\mathbf{P}^2 + M^2}\right).}$$

The physical mass operator obeys

$$\boxed{M^2 \geq M^2_{\text{gap}} I, \qquad M_{\text{gap}} > 0,}$$

and therefore

$$\boxed{\text{spec}\, H \subset \{0\} \cup [M_{\text{gap}}, \infty).}$$

With the Yang–Mills scale restored,

$$\boxed{M_{\text{gap}} = c_{\text{gap}} \Lambda_{\text{YM}}, \qquad c_{\text{gap}} > 0.}$$

Hence, within the established planar master-field realization, the decompactified $N = \infty$ $SU(N)$ Yang–Mills theory exists as a reflection-positive Euclidean quantum field theory and possesses a strictly positive mass gap.

17 String theory perspective

17.1 The Extended Open String-Field Theory

The purpose of this section is to explain the string-field-theoretic origin of the master-field action derived in the preceding chapters. The construction does not begin by postulating a closed-string dual. Instead, one first regards Yang–Mills theory as a particular one-dimensional field theory whose target is an open string-field algebra. Only after taking the planar limit and passing to the cyclic trace sector does the closed-string description emerge.

The starting point is an open string-field algebra

$$\mathcal{A} \tag{17.1}$$

equipped with the standard homotopy-algebraic data of open string field theory. At the classical level these consist of a graded state space, a BRST differential

$$Q: \mathcal{A}^\bullet \to \mathcal{A}^{\bullet+1}, \qquad Q^2 = 0, \tag{17.2}$$

a cyclic pairing, and an A_∞ product structure

$$m_k: \mathcal{A}^{\otimes k} \to \mathcal{A}. \tag{17.3}$$

For a strictly associative truncation one may write

$$m_1 = Q, \qquad m_2(A, B) = A \star B,$$

while the general formulation is naturally A_∞. The important step for the present construction is to distinguish the internal string-field differential from physical time evolution. The ordinary open string field is therefore enlarged to a time-dependent family by tensoring the string-field algebra with the de Rham algebra of the physical time line:

$$\hat{\mathcal{A}} = \mathcal{A} \,\widehat{\otimes}\, \Omega^\bullet(\mathbb{R}_t). \tag{17.4}$$

This is the basic extended string-field algebra of the construction. The total grading is

$$|\cdot|_{\mathrm{tot}} = |\cdot|_{\mathcal{A}} + \deg_{\mathbb{R}}. \tag{17.5}$$

The differential becomes

$$\hat{Q} = Q + d_t, \tag{17.6}$$

with

$$d_t = dt\, \partial_t.$$

Since

$$Q^2 = d_t^2 = 0, \qquad [Q, d_t] = 0,$$

one has

$$\hat{Q}^2 = 0. \tag{17.7}$$

Thus the physical time direction is introduced without modifying the intrinsic BRST complex of the underlying open string theory. This is the central distinction from ordinary target-space string field theory: the parameter t appearing in the present construction is the physical Yang–Mills time, whereas the algebra $\mathcal{A}$ supplies the internal off-shell string degrees of freedom.

17.2 The extended string field

A total degree-one field may be written as

$$\hat{\Psi} = \Psi(t) + dt\, \Phi(t), \tag{17.8}$$

where

$$\Psi(t) \in \mathcal{A}^1, \qquad \Phi(t) \in \mathcal{A}^0.$$

The degree-zero component Φ plays the role of the temporal gauge field. For an associative presentation, define the extended product by the usual graded tensor-product rule,

$$(A \otimes \alpha) \hat{\star} (B \otimes \beta) = (-1)^{|\alpha||B|}(A \star B) \otimes (\alpha \wedge \beta).$$

The extended curvature is

$$\hat{\mathcal{F}} = \hat{Q}\hat{\Psi} + \frac{1}{2}[\hat{\Psi}, \hat{\Psi}]. \tag{17.9}$$

Separating the dt-independent and dt-dependent pieces gives

$$\hat{\mathcal{F}} = \mathcal{F}(\Psi) + dt\, D_t \Psi,$$

where

$$\mathcal{F}(\Psi) = Q\Psi + \frac{1}{2}[\Psi, \Psi]$$

and

$$D_t \Psi = \partial_t \Psi - Q\Phi - [\Psi, \Phi]. \tag{17.10}$$

Thus the temporal component of the extended string curvature is precisely the gauge-covariant time derivative. This is one of the basic structural reasons the extension is useful: the same algebraic curvature contains both the intrinsic string-field curvature and the physical time evolution.

17.3 Gauge symmetry

Let

$$\hat{\Lambda}$$

be a total degree-zero gauge parameter. The infinitesimal gauge transformation is

$$\delta_{\hat{\Lambda}} \hat{\Psi} = \hat{Q}\hat{\Lambda} + [\hat{\Psi}, \hat{\Lambda}]. \tag{17.11}$$

For a time-dependent degree-zero parameter $\Lambda(t)$,

$$\delta\Psi = Q\Lambda + [\Psi, \Lambda],$$

while

$$\delta\Phi = \partial_t \Lambda + [\Phi, \Lambda]. \tag{17.12}$$

Consequently,

$$D_t \Psi$$

transforms covariantly. This is exactly the algebraic structure required later for the Yang–Mills pair

$$(A_i, A_0).$$

17.4 Riesz structure on string-field space

The construction requires more than the cyclic BV pairing alone. Yang–Mills dynamics is second order in physical time and therefore requires a positive physical Riesz structure on the ghost-number-one configuration space.

Let G_1 denote the physical metric on degree one and G_2 the corresponding metric on the curvature sector. The important point, established in the main body of this work, is that these degree-wise Riesz structures determine the physical adjoint

$$Q^\sharp = G_1^{-1} Q^* G_2. \tag{17.13}$$

They should not be identified with a universal world-sheet zero-mode formula. In particular, the physical kinetic operator is the one-sided operator

$$K = Q^\sharp Q \tag{17.14}$$

on the physical Gauss slice. This replaces the stronger and generally incorrect claim made in earlier formulations that the relevant operator is automatically the full BRST Laplacian

$$QQ^\sharp + Q^\sharp Q.$$

The preceding gauge direction has already been removed by Gauss' law; the physical spectral problem therefore contains only the one-sided quadratic form.

17.5 The extended second-order action

The natural second-order action on the extended string-field space is

$$S_{\text{ext}} = \frac{1}{2g_s^2} \int_{\mathbb{R}} dt \left[G_1(D_t\Psi, D_t\Psi) - G_2\big(\mathcal{F}(\Psi), \mathcal{F}(\Psi)\big) \right]. \tag{17.15}$$

In Euclidean signature,

$$S_{\text{ext},E} = \frac{1}{2g_s^2} \int d\tau \left[G_1(D_\tau\Psi, D_\tau\Psi) + G_2\big(\mathcal{F}(\Psi), \mathcal{F}(\Psi)\big) \right]. \tag{17.16}$$

This action is not the ordinary first-order cubic open-string field-theory action. Rather, it is a one-dimensional sigma model whose target is the off-shell open string-field space.

The string-field algebra supplies:

$$Q, \qquad m_k, \qquad \mathcal{F},$$

while the physical Yang–Mills dynamics is introduced through

$$t, \qquad D_t, \qquad G_q.$$

The same extension,

$$\hat{\mathcal{A}} = \mathcal{A} \otimes \Omega^\bullet(I)$$

yields the dt-component of the curvature as the string-field covariant time derivative.

17.6 Linearized physical operator

Linearize around a background $\Psi_\star$ satisfying

$$\mathcal{F}(\Psi_\star) = 0.$$

The linearized differential is

$$Q_\star = Q + [\Psi_\star, \cdot]. \tag{17.17}$$

For

$$\Psi = \Psi_\star + \psi,$$

the Euclidean quadratic action is

$$S_E^{(2)}[\psi] = \frac{1}{2g_s^2}\int d\tau \left[\|\partial_\tau \psi - Q_\star \phi\|_{G_1}^2 + \|Q_\star \psi\|_{G_2}^2\right]. \tag{17.18}$$

Variation with respect to ϕ gives the linearized Gauss constraint

$$Q_\star^\sharp \partial_\tau \psi = 0. \tag{17.18}$$

After choosing temporal gauge only after imposing this constraint, the physical fluctuation belongs to

$$\mathcal{H}_{\text{phys}}^{(1)} = \ker Q_\star^\sharp \tag{17.19}$$

together with any additional algebraic constraints of the specific string-field realization. The physical quadratic action becomes

$$S_E^{(2)} = \frac{1}{2g_s^2}\int d\tau \left[\|\partial_\tau \psi\|_{G_1}^2 + \|Q_\star \psi\|_{G_2}^2\right]. \tag{17.20}$$

Thus the physical operator is

$$K_\star = Q_\star^\sharp Q_\star. \tag{17.21}$$

The Euclidean covariance is

$$\mathfrak{C}_E = (-\partial_\tau^2 + K_\star)^{-1}. \tag{17.22}$$

This is exactly the abstract form of the planar master-field covariance derived in the main construction.

17.7 Topological A-model specialization

We now specialize the input open string theory to the topological A-model.

Let

$$X = T^*Y$$

with Y a three-dimensional Lagrangian submanifold wrapped by N coincident A-branes. For the case relevant below,

$$Y = S^{3.}$$

The classical open A-model string-field algebra [43] is quasi-isomorphic to

$$\mathcal{A}_A \simeq \Omega^\bullet(Y, \operatorname{End} E). \tag{17.23}$$

For a stack of N branes,

$$\operatorname{End} E \simeq \operatorname{Mat}_N.$$

The string-field operations reduce to

$$Q \longrightarrow d_Y \tag{17.24}$$

$$m_2 \longrightarrow \wedge \tag{17.25}$$

with matrix multiplication in the Chan–Paton factors. The degree-one field becomes

$$\Psi \longrightarrow A(t) \in \Omega^1\big(Y, \mathfrak{su}(N)\big). \tag{17.26}$$

The degree-zero temporal field becomes

$$\Phi \longrightarrow A_0(t) \in \Omega^0\big(Y, \mathfrak{su}(N)\big). \tag{17.27}$$

Therefore

$$D_t\Psi \longrightarrow \partial_t A - D_A A_0 = F_{0i}\, dx^i,$$

while

$$\mathcal{F}(\Psi) \longrightarrow dA + A \wedge A = F_A.$$

This realizes the topological A-model algebra as the de Rham complex and the degree-one string field as a matrix-valued one-form.

17.8 Yang–Mills as the extended A-model string-field theory

The physical Riesz metric reduces to the Hodge–Hilbert–Schmidt pairing

$$G_1(a, b) = \frac{1}{\lambda(\mu)} \int_Y \operatorname{tr}_N(a \wedge *_3\, b), \tag{17.28}$$

with the corresponding degree-two pairing

$$G_2(F_1, F_2) = \frac{1}{\lambda(\mu)} \int_Y \operatorname{tr}_N(F_1 \wedge *_3\, F_2) \tag{17.29}$$

before the large-N geometric reorganization discussed later. Substituting (17.23)–(17.29) into the extended action (17.15) gives

$$S = \frac{N^2}{2\lambda}\int dt\big[\|\partial_t A - D_A A_0\|^2_{\mathrm{HS}} - \|F_A\|^2_{\mathrm{HS}}\big].$$

Equivalently,

$$S = -\frac{1}{4g^2_{\mathrm{YM}}}\int_{\mathbb{R}\times Y} d^4x\,\sqrt{|g|}\ \mathrm{Tr}\,F_{\mu\nu}F^{\mu\nu}. \tag{17.30}$$

Thus

$$\text{pure Yang--Mills on } Y\times\mathbb{R}$$

is obtained as the physical degree-one sector of the extended topological open A-model string-field theory. For

$$Y = S^3,$$

this gives precisely the Yang–Mills theory used throughout the preceding construction.

17.9 What is—and is not—being claimed

The statement is not that ordinary topological open-string field theory itself is four-dimensional Yang–Mills. Ordinary open A-model string field theory on Y gives the familiar topological gauge theory associated with the A-model. The new ingredient is the physical-time extension

$$\mathcal{A}_A \longrightarrow \mathcal{A}_A \otimes \Omega^\bullet(\mathbb{R}_t)$$

together with the second-order Riesz action (1.19).Thus

$$\boxed{\text{open A-model string-field algebra + physical time + Riesz metric} \Longrightarrow \text{Yang--Mills.}}$$

The topological string provides the algebraic field content,

$$d, \qquad \wedge, \qquad \mathrm{Tr},$$

while the extended sigma-model construction supplies the physical second-order dynamics. This distinction is important because it allows the topological open–closed algebra to survive while the resulting master-field theory possesses ordinary propagating Yang–Mills degrees of freedom.

Proposition 17.1 — Yang–Mills realization of the extended open A-model algebra

Let Y be a three-dimensional Lagrangian A-brane and let

$$\mathcal{A}_A \simeq \Omega^\bullet(Y, \mathrm{End}\,E)$$

be the classical open A-model string-field algebra. Extend it by physical time,

$$\hat{\mathcal{A}}_A = \mathcal{A}_A \otimes \Omega^\bullet(\mathbb{R}_t),$$

and equip the degree-one and degree-two sectors with the Hodge–Hilbert–Schmidt Riesz pairings. Then the extended curvature of the total degree-one field

$$\hat{A} = A + dt\, A_0$$

is the ordinary four-dimensional Yang–Mills curvature,

$$\hat{\mathcal{F}} = F_A + dt \wedge (\partial_t A - D_A A_0),$$

and the second-order string-field-space action

$$S_{\text{ext}} = \frac{1}{2g_{\text{YM}}^2} \int dt [\|D_t A\|^2 - \|F_A\|^2]$$

is exactly pure Yang–Mills theory on

$$Y \times \mathbb{R}.$$

For

$$Y = S^3,$$

the extended topological open A-model therefore provides the string-field algebra underlying the Yang–Mills master-field construction developed in this work.

17.10 The Large-N Trace Sector and the Open–Closed String-Field Algebra

The previous section showed that pure Yang–Mills theory on

$$Y \times \mathbb{R}_t$$

may be realized as the physical degree-one sector of an extended open A-model string-field algebra,

$$\hat{\mathcal{A}}_A = \mathcal{A}_A \otimes \Omega^\bullet(\mathbb{R}_t),$$

with

$$\mathcal{A}_A \simeq \Omega^\bullet(Y, \operatorname{End} E).$$

For the application of interest,

$$Y = S^3,$$

and a stack of N A-branes gives

$$\operatorname{End} E \simeq \operatorname{Mat}_N.$$

We now explain why the strict planar trace sector of this open theory admits a natural closed-string description. The relevant mechanism has two logically distinct layers.

First, at the level of topological field theory, cyclic and Hochschild constructions associate a closed sector to the open A_∞ algebra.

Second, in the present Yang–Mills realization, the large-N Chan–Paton algebra is geometrized explicitly and the resulting closed sector is realized as the B-model/BCOV field algebra on

$$X = T^*S^{3.}$$

The second statement is more concrete than the first and is the one ultimately used in the master-field construction.

17.11 Cyclic words from open-string Chan–Paton factors

Let

$$A_1, \dots, A_k \in \mathcal{A}_A \otimes \mathrm{Mat}_N. \tag{17.31}$$

The open-string product composes Chan–Paton indices,

$$(A_1A_2)^i{}_k = (A_1)^i{}_j (A_2)^j{}_k.$$

Closing the chain by taking the normalized color trace produces

$$\mathrm{tr}_N(A_1 \cdots A_k).$$

This expression is invariant under cyclic permutations,

$$\mathrm{tr}_N(A_1A_2 \cdots A_k) = \mathrm{tr}_N(A_2 \cdots A_kA_1).$$

Thus a gauge-invariant single-trace observable is naturally not an ordered tensor word but a cyclic word. At the algebraic level, these objects belong to the cyclic chain complex of the open-string algebra,

$$CC_\bullet(\mathcal{A}_A). \tag{17.32}$$

With the S^1-equivariant refinement appropriate to closed-string descendants, one may use the negative cyclic complex

$$CC_\bullet^-(\mathcal{A}_A). \tag{17.33}$$

The chain differential contains the Hochschild differential b together with the Connes operator B. Schematically,

$$d_{\mathrm{cyc}} = b + uB.$$

The precise grading conventions will not be important below. What matters is that the cyclic complex is built functorially from the open-string A_∞ algebra and precisely captures cyclic gauge-invariant combinations of open-string fields.

17.12 Single traces and multi-traces

A single trace corresponds to one cyclic chain.

A product of r traces,

$$\prod_{a=1}^{r} \mathrm{tr}_N\left(A_{a,1}\cdots A_{a,k_a}\right),$$

corresponds to a symmetric product of cyclic chains,

$$\mathrm{Sym}^r\, CC_\bullet^-(\mathcal{A}_A). \tag{17.34}$$

Therefore the full polynomial trace algebra is naturally organized as

$$Sym^\bullet CC_\bullet^-(\mathcal{A}_A). \tag{17.35}$$

This already anticipates the closed-string Fock structure:

single trace ↔ single closed-string excitation,

while

multi-trace ↔ multi-closed-string excitation.

The attachment makes this same identification explicitly: multi-trace operators live in the symmetrized product of cyclic chains. In the Yang–Mills interpretation, the corresponding quantum states are respectively one-glueball and multi-glueball states.

17.13 Open–closed topological field theory

A central result of open–closed topological field theory is that the closed sector generated by an open A_∞ algebra is controlled by its Hochschild or cyclic deformation complex. Schematically,

$$CC_\bullet^-\left(\mathcal{A}_{\mathrm{open}}\right) \rightsquigarrow \mathcal{A}_{\mathrm{closed}}. \tag{17.36}$$

The arrow here should not be interpreted as a literal equality of chain complexes in every presentation. Rather, it denotes the open–closed map provided by the TCFT structure, possibly after passing to an appropriate quasi-isomorphic model. The physical closed-string states are then obtained from the cohomology of the closed complex. Thus the algebraic content of the planar trace sector already has the correct structure to reorganize into closed-string degrees of freedom. This is the abstract origin of the open–closed correspondence in the present construction.

17.14 Why the present construction is stronger than the abstract open–closed statement

The abstract Hochschild-cyclic correspondence alone does not determine the physical target-space realization of the closed sector. In the present case, however, the preceding Yang–Mills construction supplies additional geometric data:

large-N color algebra

RG scale

Hilbert--Schmidt/Riesz metric

and

the physical differential complex.

These data determine a concrete geometric realization rather than merely an abstract closed TCFT. The color algebra is resolved into fuzzy spherical harmonics and, in the planar limit, becomes a classical algebra of functions or Hamiltonian vector fields on the emergent color sphere. Including the RG scale enlarges this internal space to a three-dimensional fibre

$$W.$$

Locally,

$$W \simeq S^2 \times \mathbb{R}_+ \tag{17.37}$$

away from the infrared cap, with the S^2 collapsing smoothly at the tip. The large-N gauge algebra is then represented by volume-preserving vector fields,

$$\mathfrak{su}(N) \longrightarrow \mathfrak{sdiff}(W). \tag{17.38}$$

This is the concrete geometrization of the Chan–Paton sector used by the master-field map.

17.15 From the fibre algebra to the deformed conifold

The physical base of the open theory is S^3. The emergent three-dimensional internal fibre W combines with this base to produce the six-dimensional space

$$X = T^*S^3.$$

Locally,

$$X \simeq S^3 \times W.$$

The large-N Yang–Mills master field may therefore be represented as an S^3-form valued in vector fields on W. Schematically,

$$A \in \Omega^1\big(S^3, \mathfrak{su}(N)\big)$$

Becomes

$$A_\infty \in \Omega^1\big(S^3, \mathfrak{sdiff}(W)\big).$$

Using the local product structure of X, such fields may be rewritten [44] as polyvector fields on X. Thus the large-N color geometrization turns the open-string Chan–Paton directions into actual directions of the closed-string geometry. This is the physical realization of the open–closed algebraic mechanism.

17.16 The closed B-model field algebra

The closed B-model field space is naturally described in terms of polyvector fields

$$PV^{p,q}(X) = \Omega^{0,q}(X, \wedge^p T^{1,0}X).$$

The differential is

$$Q_{\mathrm{B}} = \bar{\partial}. \tag{17.39}$$

The nonlinear bracket is the Schouten bracket,

$$[\cdot,\cdot]_{\mathrm{Sch}}.$$

The corresponding Maurer–Cartan equation is

$$\bar{\partial}\mu + \frac{1}{2}[\mu,\mu]_{\mathrm{Sch}} = 0. \tag{17.40}$$

The physical deformation relevant to the Yang–Mills master field lies in

$$\mu \in PV^{1,1}(X).$$

Let Ω be the holomorphic volume form of the deformed conifold. Contraction with Ω gives the standard identification

$$PV^{1,1}(X) \longrightarrow \Omega^{2,1}(X),$$

$$\mathcal{G} = \mu \lrcorner \Omega.$$

Hence the master field may equivalently be represented as

$$\mathcal{G} \in \Omega^{2,1}(X).$$

This is the closed B-model field used throughout the main body of the manuscript.

17.17 The master-field curvature

Under contraction with Ω, the B-model Maurer–Cartan curvature becomes

$$\mathcal{F}(\mathcal{G}) = \bar{\partial}\mathcal{G} + \frac{1}{2}[\mathcal{G},\mathcal{G}]_{\Omega}. \tag{17.41}$$

Here the bracket on forms is the image of the Schouten bracket under the holomorphic-volume-form identification. Thus the nonlinear physical curvature of the Yang–Mills master field is precisely the integrability curvature of the corresponding closed B-model complex-structure deformation. The condition

$$\mathcal{F}(\mathcal{G}) = 0$$

is the integrability condition for the deformed complex structure. This explains why the master-field differential appearing in the main construction is

$$D = \bar{\partial}, \tag{17.42}$$

rather than an arbitrary operator introduced independently on the closed side.

17.18 The constrained closed complex

The physical Yang–Mills master field does not occupy the unrestricted BCOV field space. The open–closed map selects the constrained subcomplex

$$\mathcal{C}^q = \ker \partial \cap \Omega^{2,q}(X).$$

On this complex,

$$D = \bar{\partial}, \qquad D^2 = 0.$$

The degree-one physical field is

$$\mathcal{G} \in \mathcal{C}^{1.}$$

The ∂-constraint is essential: it is part of the physical master-field image and is preserved by the exact cochain map developed in the decompactification analysis. Thus the concrete closed algebra appearing in the Yang–Mills realization is not the entire formal BCOV complex, but the particular constrained sector selected by the planar open–closed map.

17.19 Time-dependent open–closed map

The extension by physical time commutes with the open–closed reorganization. At every time t,

$$A(t)$$

is an open A-model-Yang–Mills string field. The large-N collective map sends it to a closed B-model field,

$$C_\infty : A(t) \longmapsto \mathcal{G}(t). \tag{17.43}$$

Therefore

$$t \mapsto A(t)$$

becomes

$$t \mapsto \mathcal{G}(t).$$

The open–closed correspondence is thus promoted from a relation between static topological string algebras to a relation between one-dimensional sigma models on those algebras.

Schematically,

open A-model string-field space	$\xrightarrow{N\to\infty}$	closed B-model string-field space
$A(t)$		$\mathcal{G}(t)$
↓		↓
Yang-Mills		master-field theory.

This is the dynamical version of the open–closed mechanism used in the present work. Thus time-dependent open-string fields reproducing Yang–Mills reorganize at large N into time-dependent closed B-model fields.

17.20 Transport of the algebraic operations

The collective map does not merely identify field labels.

It intertwines the physical differential structures,

$$DC_\infty = C_\infty Q_{\mathrm{YM}},$$

and transports the nonlinear bracket,

$$C_\infty[A,B]_{\mathrm{YM}} = [C_\infty A, C_\infty B]_\Omega.$$

Consequently,

$$C_\infty\left(Q_{\mathrm{YM}}A + \frac{1}{2}[A,A]_{\mathrm{YM}}\right) = \bar{\partial}\mathcal{G} + \frac{1}{2}[\mathcal{G},\mathcal{G}]_\Omega.$$

Thus

$$C_\infty F_{\mathrm{YM}} = \mathcal{F}(\mathcal{G}).$$

This is the nonlinear algebraic statement underlying the master-field action. The exact cochain construction developed earlier in the manuscript is the linearized and finite-radius-controlled version of this intertwining relation.

17.21 Transport of the Riesz structure

The open–closed algebraic map alone is not enough to determine the physical Yang–Mills dynamics.

The crucial additional structure is the Riesz metric.

On the open side, the physical metric is the running Yang–Mills metric

$$G_{\mathrm{YM}}(\mu) = \frac{1}{\lambda(\mu)}G_{\mathrm{HS}}.$$

The master-field realization geometrizes this scale dependence into a radial weight on X,

$$G_h(\delta\mathcal{G}_1, \delta\mathcal{G}_2) = \int_X h(\tau)\,\delta\mathcal{G}_1 \wedge *_6 \overline{\delta\mathcal{G}_2}.$$

Thus

$$G_{\mathrm{YM}}(a,b) = G_h(C_\infty a, C_\infty b).$$

The radial coordinate of the closed-string geometry therefore records the Yang–Mills RG scale. This is the point at which the physical Yang–Mills theory goes beyond the purely topological open–closed correspondence. The algebraic closed sector comes from the topological string

structure, whereas the nontrivial running physical inner product comes from the Yang–Mills renormalization group.

17.22 The closed master-field action

Combining the transported differential, bracket, and Riesz metric gives the closed master-field action

$$S[\mathcal{G}] = \frac{1}{2g_s^2}\int dt\left[\|D_t\mathcal{G}\|_h^2 - \|\mathcal{F}(\mathcal{G})\|_0^2\right].$$

Here

$$\mathcal{F}(\mathcal{G}) = \bar{\partial}\mathcal{G} + \frac{1}{2}[\mathcal{G},\mathcal{G}]_\Omega.$$

At the strict planar level,

$$g_s \sim \frac{1}{N}.$$

Thus

$$g_s \to 0$$

is the classical closed-string master-field limit. The nonlinear closed action should therefore be interpreted as the large-N geometrization of the Yang–Mills FMR-master-field action, not as an independently postulated B-model target-space theory.

17.23 Single strings, glueballs and traces

The operator interpretation now becomes transparent. A single collective normal mode

$$\Psi_n \in \mathcal{H}^{(1)}_{\text{phys}}$$

is simultaneously:

a planar single-trace excitation

a one-glueball state

and

a single closed B-model master-field excitation.

Products of traces correspond to symmetric products of these modes,

$$\mathrm{Sym}^k\,\mathcal{H}^{(1)}_{\text{phys}},$$

that is, multi-glueball or multi-closed-string states.

This matches the Fock-space construction already established in the main body of the manuscript.

17.24 Relation to BCOV theory

The closed B-model is described off shell by BCOV-Kodaira–Spencer-type polyvector fields. The master-field construction uses the same underlying algebraic ingredients:

$$PV^{\cdot,\cdot}(X), \qquad \bar{\partial}, \qquad [\cdot,\cdot]_{\mathrm{Sch}},$$

together with the Calabi–Yau volume form Ω. However, the physical Yang–Mills realization introduces two additional structures:

$$\boxed{\text{the fixed-period constraint}}$$

and

$$\boxed{\text{the warped physical Riesz metric } G_h.}$$

Therefore the master-field theory should be viewed as a particular physically weighted and constrained realization of the closed B-model field algebra. Its physical kinetic operator is not the unrestricted BCOV Laplacian. Rather, on the physical degree-one Gauss slice it is

$$K_h = h^{-1}\bar{\partial}^{\dagger}\bar{\partial}.$$

This is precisely the one-sided operator derived in the main body of the paper.

Proposition 17.2 — Open–closed realization of the planar Yang–Mills trace sector

Let

$$\mathcal{A}_A \simeq \Omega^{\bullet}(S^3, \mathrm{Mat}_N)$$

be the open A-model string-field algebra underlying the extended Yang–Mills construction.

Then:

1. single-trace gauge-invariant observables define cyclic chains in
$$CC_{\bullet}^{-}(\mathcal{A}_A);$$
2. multi-trace observables belong to the symmetric algebra
$$Sym^{\bullet}CC_{\bullet}^{-}(\mathcal{A}_A);$$
3. the abstract open–closed TCFT construction associates to this cyclic sector a closed-string deformation complex;
4. in the planar Yang–Mills realization, the large-N Chan–Paton algebra together with the RG direction is geometrized into the fibre of
$$X = T^{*}S^3;$$
5. the resulting collective field is represented by

$$\mu \in PV^{1,1}(X),$$

or equivalently

$$\mathcal{G} = \mu \llcorner \Omega \in \ker \partial \cap \Omega^{2,1}(X);$$

6. the collective map intertwines the nonlinear curvatures,

$$C_\infty F_{\mathrm{YM}} = \bar{\partial}\mathcal{G} + \frac{1}{2}[\mathcal{G}, \mathcal{G}]_\Omega;$$

7. the Yang–Mills Riesz metric is transported to the warped closed-field metric

$$G_h.$$

Therefore the strict planar single-trace sector of Yang–Mills is realized as a constrained closed B-model string-field algebra on the deformed conifold, while the multi-trace sector is its symmetric multi-string completion. The abstract cyclic-Hochschild open–closed map and the concrete collective map C_∞ should therefore be distinguished. The former identifies, at the level of topological string-field algebras, the natural closed sector associated with the open A-model algebra. The latter is the physically normalized realization relevant to planar Yang–Mills: it carries not only the differential and bracket structure, but also the fixed-period condition, the RG-scale dependence, and the physical Riesz metric. In this sense, C_∞ may be viewed as the Yang–Mills refinement of the abstract open–closed correspondence, specialized to the deformed-conifold B-model and completed by the geometric data required for the physical master-field theory.

Data Availability Statement

No new experimental or observational data were generated or analyzed in this study. All results are derived from the analytical and numerical methods described in the manuscript. Any numerical data underlying the reported spectra can be reproduced from the equations, boundary conditions, and procedures given in the text.

Conflict of Interest Statement

The author declares that there are no conflicts of interest associated with this work.

Bibliography

1. **A. Jaffe and E. Witten**, "Quantum Yang–Mills Theory," in *The Millennium Prize Problems*, Clay Mathematics Institute / American Mathematical Society, 2006.

2. **C. N. Yang and R. L. Mills**, "Conservation of Isotopic Spin and Isotopic Gauge Invariance," *Physical Review* **96** (1954), 191–195. DOI: 10.1103/PhysRev.96.191. APS Journals

3. **D. J. Gross and F. Wilczek**, "Ultraviolet Behavior of Non-Abelian Gauge Theories," *Physical Review Letters* **30** (1973), 1343–1346. DOI: 10.1103/PhysRevLett.30.1343. APS Journals

4. **H. D. Politzer**, "Reliable Perturbative Results for Strong Interactions?," *Physical Review Letters* **30** (1973), 1346–1349. DOI: 10.1103/PhysRevLett.30.1346. APS Journals

5. **K. G. Wilson**, "Confinement of Quarks," *Physical Review D* **10** (1974), 2445–2459. DOI: 10.1103/PhysRevD.10.2445. APS Journals

6. **C. J. Morningstar and M. Peardon**, "The Glueball Spectrum from an Anisotropic Lattice Study," *Physical Review D* **60** (1999), 034509. DOI: 10.1103/PhysRevD.60.034509. APS Journals

7. **G. 't Hooft**, "A Planar Diagram Theory for Strong Interactions," *Nuclear Physics B* **72** (1974), 461–473. DOI: 10.1016/0550-3213(74)90154-0. ScienceDirect

8. **M. Bershadsky, S. Cecotti, H. Ooguri and C. Vafa**, "Kodaira-Spencer Theory of Gravity and Exact Results for Quantum String Amplitudes," *Communications in Mathematical Physics* **165** (1994), 311–427. DOI: 10.1007/BF02099774. DOI

9. **E. Witten**, "Chern-Simons Gauge Theory as a String Theory," in *The Floer Memorial Volume*, Progress in Mathematics **133** (1995), 637–678, arXiv:hep-th/9207094. Mathematics Department

10. **K. Costello**, "Topological Conformal Field Theories and Calabi–Yau Categories," *Advances in Mathematics* **210** (2007), 165–214. DOI: 10.1016/j.aim.2006.06.004. ScienceDirect

11. **K. Osterwalder and R. Schrader**, "Axioms for Euclidean Green's Functions," *Communications in Mathematical Physics* **31** (1973), 83–112. DOI: 10.1007/BF01645738. DOI

12. **T. Kato**, *Perturbation Theory for Linear Operators*, 2nd ed., Springer, Berlin, 1976.

13. **M. Stenzel**, "Ricci-flat metrics on the complexification of a compact rank one symmetric space," *Manuscripta Mathematica* **80** (1993), 151–163.

14. **P. Candelas and X. C. de la Ossa**, "Comments on Conifolds," *Nuclear Physics B* **342** (1990), 246–268.

15. **R. O. Wells**, *Differential Analysis on Complex Manifolds*, Springer.

16. **E. Inönü and E. P. Wigner**, "On the Contraction of Groups and Their Representations," *Proceedings of the National Academy of Sciences of the United States of America* **39** (1953), 510–524.

17. **H. Feshbach**, "Unified Theory of Nuclear Reactions," *Annals of Physics* **5** (1958), 357–390.

18. **V. Bach, J. Fröhlich and I. M. Sigal**, "Quantum Electrodynamics of Confined Nonrelativistic Particles," *Advances in Mathematics* **137** (1998), 299–395.

19. **V. N. Gribov**, "Quantization of Non-Abelian Gauge Theories," *Nuclear Physics B* **139** (1978), 1–19.

20. **I. M. Singer**, "Some Remarks on the Gribov Ambiguity," *Communications in Mathematical Physics* **60** (1978), 7–12.

21. **D. Zwanziger**, "Fundamental Modular Region, Boltzmann Factor and Area Law in Lattice Gauge Theory," *Nuclear Physics B* **412** (1994), 657–730.

22. **G. Dell'Antonio and D. Zwanziger**, "Every Gauge Orbit Passes Inside the Gribov Horizon," *Communications in Mathematical Physics* **138** (1991), 291–299.

23. **J. Hoppe**, "Diffeomorphism Groups, Quantization, and $SU(\infty)$," *International Journal of Modern Physics A* **4** (1989), 5235–5248.

24. **B. de Wit, J. Hoppe and H. Nicolai**, "On the Quantum Mechanics of Supermembranes," *Nuclear Physics B* **305** (1988), 545–581.

25. **J. Madore**, "The Fuzzy Sphere," *Classical and Quantum Gravity* **9** (1992), 69–87.

26. **E. G. Floratos, J. Iliopoulos and G. Tiktopoulos**, "A Note on $SU(\infty)$ Classical Yang–Mills Theories," *Physics Letters B* **217** (1989), 285–288.

27. **D. J. Gross and F. Wilczek**, "Ultraviolet Behavior of Non-Abelian Gauge Theories," *Phys. Rev. Lett.* **30** (1973), 1343–1346.

28. **H. D. Politzer**, "Reliable Perturbative Results for Strong Interactions?," *Phys. Rev. Lett.* **30** (1973), 1346–1349.

29. **C. Voisin**, *Hodge Theory and Complex Algebraic Geometry I*, Cambridge University Press, 2002.

30. **J. Milnor**, *Singular Points of Complex Hypersurfaces*, Princeton University Press, 1968.

31. **D. Huybrechts**, *Complex Geometry: An Introduction*, Springer, 2005.

32. **P. A. Griffiths**, "Periods of Integrals on Algebraic Manifolds, I," *American Journal of Mathematics* **90** (1968), 568–626.

33. **O. Aharony, S. S. Gubser, J. Maldacena, H. Ooguri and Y. Oz**, "Large N Field Theories, String Theory and Gravity," *Physics Reports* **323** (2000), 183–386.

34. **E. Witten**, "The $1/N$ Expansion in Atomic and Particle Physics," in *Recent Developments in Gauge Theories*, NATO Advanced Study Institute Series B, Vol. 59, Plenum Press, 1980, pp. 403–419.

35. **R. A. Minlos**, "Generalized Random Processes and Their Extension to a Measure," *Trudy Moskovskogo Matematicheskogo Obshchestva* **8** (1959), 497–518; English translation in *Selected Translations in Mathematical Statistics and Probability* **3** (1963), 291–313.

36. **I. M. Gel'fand and N. Ya. Vilenkin**, *Generalized Functions, Vol. 4: Applications of Harmonic Analysis*, Academic Press, 1964.

37. **G. Källén**, "On the Definition of the Renormalization Constants in Quantum Electrodynamics," *Helvetica Physica Acta* **25** (1952), 417–434.

38. **H. Lehmann**, "Über Eigenschaften von Ausbreitungsfunktionen und Renormierungskonstanten quantisierter Felder," *Nuovo Cimento* **11** (1954), 342–357.

39. **K. G. Wilson**, "Non-Lagrangian Models of Current Algebra," *Physical Review* **179** (1969), 1499–1512.

40. **R. B. Lockhart and R. C. McOwen**, "Elliptic Differential Operators on Noncompact Manifolds," *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze* **12** (1985), 409–447.

41. **K. Osterwalder and R. Schrader**, "Axioms for Euclidean Green's Functions II," *Communications in Mathematical Physics* **42** (1975), 281–305.

42. **R. F. Streater and A. S. Wightman**, *PCT, Spin and Statistics, and All That*, Princeton University Press.

43. **E. Witten**, "Chern-Simons Gauge Theory as a String Theory," in *The Floer Memorial Volume*, Progress in Mathematics **133** (1995), 637–678, arXiv:hep-th/9207094.

44. **R. Gopakumar and C. Vafa**, "On the Gauge Theory/Geometry Correspondence," *Advances in Theoretical and Mathematical Physics* **3** (1999), 1415–1443, arXiv:hep-th/9811131.